Supplemental Material for Mutualism can constrain host diversity and evolution

2026-03-19

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Chapter 1

Introduction

This is the supplemental material our 2026 Artificial Life Conference submission, "Mutualism can constrain host diversity and evolution". This is not intended as a stand-alone document, but as a companion to our main manuscript.

1.1 About our supplemental material

Our supplemental material is hosted using GitHub pages. We compiled our data analyses and supplemental documentation into this web-accessible book using bookdown.

The source code and configuration files for this supplemental material can be found in this GitHub repository.

Our supplemental material includes the following:

- Data availability (Section 2)
- Local compilation instructions (Section 3)
- Experiment analyses
  - Flat rewards experiment (Section 5)
  - Differential rewards experiments (Section @ref())
  - Experiments characterizing when mutualism constrains host complexity:
    - Varying environmental rewards for complex functions (Section 7)
    - Varying mutualist donate proportion (Section 8)
    - Varying symbiont interaction synergy factor (Section 9)
    - Varying function restrictions (Section 10)

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Chapter 2

Data availability

2.1 Source code

The source code for his work is publicly accessible on GitHub: https://github.com/amlalejini/. This repository has also been archived on Zenodo: TODO

2.2 Experiment results

Data generated from our experiments used in analyses are available online, archived in an OSF repository: https://osf.io/j3ezd/

Chapter 3

Compilation instructions

Instructions for compiling and running the software used in this study on your local machine. All experiments were run on Mac or Linux-based operating systems.

You will need a C++ compiler that supports C++20. We used g++14 for all local compilations.

You will also need Python to run graph generation and analysis. Python dependencies are listed in the `requirements.txt` at the root of this repository.

Statistical analyses and data visualizations were conducted using R.

Experiments were conducted using the Symbulation software platform. The specific version of Symbulation used for these experiments can be found here: https://github.com/amlalejini/SymbulationEmp/tree/sgp-mode-refactor (commit hash: `476bd8a4c0e6390942ede46ee9cd1bbe5ab8ec70`). This supplemental repository has the correct Symbulation version pinned as a git submodule (in `third-party/`).

3.1 Instructions

First, clone the `2026-alife-mutualism-constrains-evolution` repository (https://github.com/amlalejini/2026-alife-mutualism-constrains-evolution) to your local machine. Then, initialize and update the git submodules inside the repository. From inside the repository, run:

```
git submodule update --init --recursive
```

This will download the correct version of Symbulation along with all of its necessary dependencies.

To compile Symbulation, navigate to the `third-party/SymbulationEmp` directory, and run `make sgp-mode`. The compiled executable will be placed inside the `third-party/SymbulationEmp` directory.

The Symbulation configuration files used for each of experiments can be found in the `experiments/` directory (within the `hpc/config/` subdirectory for any given experiment).

Chapter 4

Virtual hardware for digital organisms

4.1 Virtual hardware components

Each digital organism comprises a linear program (its genome) and a set of virtual hardware components for executing its genome. Genomes use the SignalGP-lite representation (https://github.com/mmore500/signalgp-lite).

In this work, the virtual hardware contains: - 8 registers: these are used as "working memory", used as source operands and destination operands for instructions - 2 memory stacks: serve as additional memory for programs. Programs can push values from registers to a stack or pop values from a stack into a register. Each stack has a maximum size of 16 values.

4.2 Instruction set

SignalGP instruction set used in digital organism genomes in this work.

Abbreviations:

- EOP: End of program
- Reg: local register
  - Reg[0] indicates the value at the register specified by an instruction's first *argument*, Reg[1] indicates the value at the register specified by an instruction's second argument, and Reg[2] indicates the value at the register specified by the instruction's third argument.
  - Reg[0], Reg[1], *etc*: Register 0, Register 1, *etc.*

- Arg: Instruction argument
  - Arg[i] indicates the i'th instruction argument (an integer encoded in the genome)
  - E.g., Arg[0] is an instruction's first argument

Instructions that would produce undefined behavior (e.g., division by zero) are treated as no operations.

Instruction	Arguments Used	Description
`Nop`	0	No operation
`Increment`	1	Reg[0] = Reg[0] + 1
`Decrement`	1	Reg[0] = Reg[0] - 1
`ShiftLeft`	1	Reg[0] = Reg[0] « 1
`ShiftRight`	1	Reg[0] = Reg[0] » 1
`Subtract`	3	Reg[0] = Reg[1] - Reg[2]
`Nand`	3	Reg[0] = ~(Reg[1] & Reg[2])
`Push`	1	Push Reg[0] to active memory stack
`Pop`	1	Pop top of active memory stack. Reg[0] = popped value
`SwapStack`	0	Change active stack.
`Swap`	2	Swap values between Reg[0] and Reg[1]
`Reproduce`	0	Triggers organism reproduction attempt if organism has sufficient resources.
`IO`	1	Output value in Reg[0], and set Reg[0] = new input value.
`JumpIfNEq`	2	if (Reg[0] != Reg[1]) Jump to Anchor specified by instruction tag (tag matching determines target Anchor).

Instruction	Arguments Used	Description
`JumpIfLess`	2	if (Reg[0] < Reg[1]) Jump to Anchor specified by instruction tag (tag matching determines target Anchor).
`JumpIfEq`	2	if (Reg[0] == Reg[1]) Jump to Anchor specified by instruction tag (tag matching determines target Anchor).
`Anchor`	0	Tag-based code label that can be jumped to with a jump instruction.

Chapter 5

Equal function rewards

All logic functions yield equal resources when performed by digital organisms.
No "abiotic" selective pressure to evolve more complex functions.

5.1 Dependencies and setup

Load R library dependencies.

```
library(tidyverse)
library(cowplot)
library(khroma)
library(rstatix)
library(knitr)
library(kableExtra)
```

Define relevant file paths.

```
# Check if Rmd is being compiled using bookdown
bookdown <- exists("bookdown_build")

# Build relative path from R's working directory to what is actually the working
# directory for this experiment.
experiment_slug <- "2025-11-24-health-flat-rewards"
supp_file_prefix <- "flat-rewards"
exp_directory <- paste(
  "experiments",
  experiment_slug,
  "analysis",
```

```
  sep = "/"
)

# Get path to the supplemental material analysis directory
supp_directory <- paste(
  "experiments",
  "manuscript-analyses",
  sep = "/"
)

# Adjust experiment directory if being knitted for bookdown build
if (bookdown) {
  exp_directory <- paste0(
    bookdown_wd_prefix,
    exp_directory
  )
  supp_directory <- paste0(
    bookdown_wd_prefix,
    supp_directory
  )
}
```

Define default plot theme, plot output directory.

```
# Use the cowplot theme as our default for ggplots
theme_set(theme_cowplot())
# Create directory to save plots
plot_dir <- paste(
  supp_directory,
  "plots",
  supp_file_prefix,
  sep = "/"
)
dir.create(
  plot_dir,
  showWarnings = FALSE
)

# Use vibrant theme for plot color palettes
vibrant_colors <- color("vibrant")(7)
color_pal <- c(
  vibrant_colors[["blue"]],
  vibrant_colors[["orange"]],
  vibrant_colors[["magenta"]]
)
```

5.2 Load and preprocess data

```
tasks <- c("NOT", "NAND", "OR_NOT", "AND", "OR", "AND_NOT", "NOR", "XOR", "EQU")
max_pop_size <- 100 * 100
threshold <- 0.025 * max_pop_size
```

Load summary data.

```
summary_data_path <- paste(
  exp_directory,
  "data",
  "summary.csv",
  sep = "/"
)
summary_data <- read_csv(summary_data_path)
# Do any universal preprocessing
summary_data <- summary_data %>%
  mutate(
    SEED = as.factor(SEED),
    task_credit = paste0(
      "H", HOST_ONLY_FIRST_TASK_CREDIT,
      "S", SYM_ONLY_FIRST_TASK_CREDIT
    ),
    sym_condition = case_when(
      START_MOI == "0" ~ "no-syms",
      START_MOI == "1" & HEALTH_TYPE == "parasite" ~ "parasite",
      START_MOI == "1" & HEALTH_TYPE == "mutualist" ~ "mutualist",
      .default = "unknown" # precaution against unexpected parameterizations
    ),
    interaction_multiplier = as.factor(PARASITE_CYCLE_STEAL_MULTIPLIER)
  ) %>%
  mutate(
    task_credit = as.factor(task_credit),
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist")
    )
  ) %>%
  # Filter to just IM=4 data => that's our baseline in the paper.
  filter(interaction_multiplier == "4")
```

Load time series data.

```
time_data_path <- paste(
  exp_directory,
  "data",
  "time_series.csv",
  sep = "/"
)
time_data <- read_csv(time_data_path)

# Do any universal preprocessing
time_data <- time_data %>%
  mutate(
    SEED = as.factor(SEED),
    sym_condition = case_when(
      START_MOI == "0" ~ "no-syms",
      START_MOI == "1" & HEALTH_TYPE == "parasite" ~ "parasite",
      START_MOI == "1" & HEALTH_TYPE == "mutualist" ~ "mutualist",
      .default = "unknown" # precaution against unexpected parameterizations
    ),
    interaction_multiplier = as.factor(PARASITE_CYCLE_STEAL_MULTIPLIER)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist")
    )
  ) %>%
  # Filter to just IM=4 data => that's our baseline in the paper.
  filter(interaction_multiplier == "4")
```

5.3 Extinctions

Did host or symbiont populations go extinct in any runs?

```
sym_min_pop_sizes <- time_data %>%
  group_by(SEED) %>%
  slice_min(
    OrgCounts_hosted_sym_count,
    n = 1
  ) %>%
  slice_min(
    update,
    n = 1
  )
```

```
reps_with_extinct_syms <- sym_min_pop_sizes %>%
  group_by(
    sym_condition
  ) %>%
  summarize(
    reps_with_extinct_syms = sum(OrgCounts_hosted_sym_count == 0),
    total_reps = n()
  )

host_min_pop_sizes <- time_data %>%
  group_by(SEED) %>%
  slice_min(
    OrgCounts_host_count,
    n = 1
  ) %>%
  slice_min(
    update,
    n = 1
  )

reps_with_extinct_hosts <- host_min_pop_sizes %>%
  group_by(
    sym_condition
  ) %>%
  summarize(
    reps_with_extinct_hosts = sum(OrgCounts_host_count == 0),
    total_reps = n()
  )

extinctions <- full_join(
  reps_with_extinct_syms,
  reps_with_extinct_hosts
)
extinctions <- extinctions %>%
  arrange(
    desc(reps_with_extinct_hosts), desc(reps_with_extinct_syms)
  )
extinction_table <- kable(extinctions) %>%
  kable_styling(latex_options = "striped")
save_kable(
  extinction_table,
  paste(
    plot_dir,
    "extinctions.pdf",
    sep = "/"
```

```
  )
)

extinctions
```

```
## # A tibble: 3 x 4
##   sym_condition reps_with_extinct_syms total_reps reps_with_extinct_hosts
##   <fct>                          <int>      <int>                   <int>
## 1 no-syms                           50         50                       0
## 2 parasite                           0         50                       0
## 3 mutualist                          0         50                       0
```

5.4 Host function performance

First, quantify which functions have been performed over the course of an entire run.

```
# Pivot task data long, selecting only relevant columns
host_over_time_parent_task_data <- time_data %>%
  select(
    SEED,
    sym_condition,
    interaction_multiplier,
    update,
    CurUpdate_AND_NOT_in_host_parent_org_counts,
    CurUpdate_AND_in_host_parent_org_counts,
    CurUpdate_EQU_in_host_parent_org_counts,
    CurUpdate_NAND_in_host_parent_org_counts,
    CurUpdate_NOR_in_host_parent_org_counts,
    CurUpdate_NOT_in_host_parent_org_counts,
    CurUpdate_OR_NOT_in_host_parent_org_counts,
    CurUpdate_OR_in_host_parent_org_counts,
    CurUpdate_XOR_in_host_parent_org_counts
  ) %>%
  pivot_longer(
    c(
      CurUpdate_AND_NOT_in_host_parent_org_counts,
      CurUpdate_AND_in_host_parent_org_counts,
      CurUpdate_EQU_in_host_parent_org_counts,
      CurUpdate_NAND_in_host_parent_org_counts,
      CurUpdate_NOR_in_host_parent_org_counts,
      CurUpdate_NOT_in_host_parent_org_counts,
      CurUpdate_OR_NOT_in_host_parent_org_counts,
```

```
      CurUpdate_OR_in_host_parent_org_counts,
      CurUpdate_XOR_in_host_parent_org_counts
    ),
    names_to = "task",
    values_to = "task_in_parent_count"
  ) %>%
  mutate(
    task = str_replace(task, "CurUpdate_", "")
  ) %>%
  mutate(
    task = str_replace(task, "_in_host_parent_org_counts", "")
  ) %>%
  mutate(
    task_completed_by_parent = task_in_parent_count >= threshold,
    task = factor(
      task,
      levels = tasks,
      labels = tasks
    )
  ) %>%
  mutate(
    task_complexity = case_when(
      task == "NOT" ~ 1,
      task == "NAND" ~ 1,
      task == "OR_NOT" ~ 2,
      task == "AND" ~ 2,
      task == "OR" ~ 3,
      task == "AND_NOT" ~ 3,
      task == "NOR" ~ 4,
      task == "XOR" ~ 4,
      task == "EQU" ~ 5,
      .default = -1
    )
  )

host_parent_tasks_all_time_by_task <- host_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier,
    task
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
```

```
  ) %>%
  ungroup() %>%
  group_by(
    sym_condition,
    interaction_multiplier,
    task
  ) %>%
  summarize(
    n = n(),
    parent_tasks_completed_ever = sum(parent_task_completed_ever == TRUE)
  )
```

Next, summarize function complexity level.

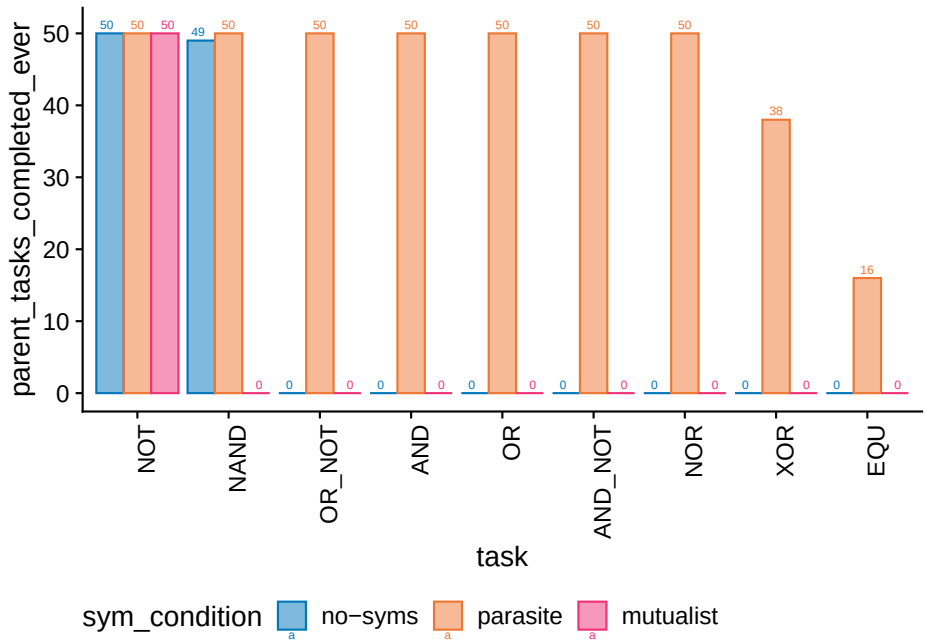
```
host_max_task_complexity_data <- host_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier,
    task,
    task_complexity
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  # This filter works fine because all replicates start with a task
  filter(
    parent_task_completed_ever
  ) %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier
  ) %>%
  summarize(
    max_task_complexity = max(task_complexity)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c(
        "no-syms",
        "parasite",
```

```
        "mutualist"
      )
    )
  )
```

5.4.1 Host function performance - all time

```
host_parent_tasks_all_time_by_task_plt <- host_parent_tasks_all_time_by_task %>%
  ggplot(
    aes(
      x = task,
      color = sym_condition,
      fill = sym_condition,
      y = parent_tasks_completed_ever
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = parent_tasks_completed_ever),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(
      angle = 90,
      hjust = 1
    )
  )
ggsave(
  file = paste(
    plot_dir,
    "host_parent_tasks_all_time_by_task_plt.pdf",
```

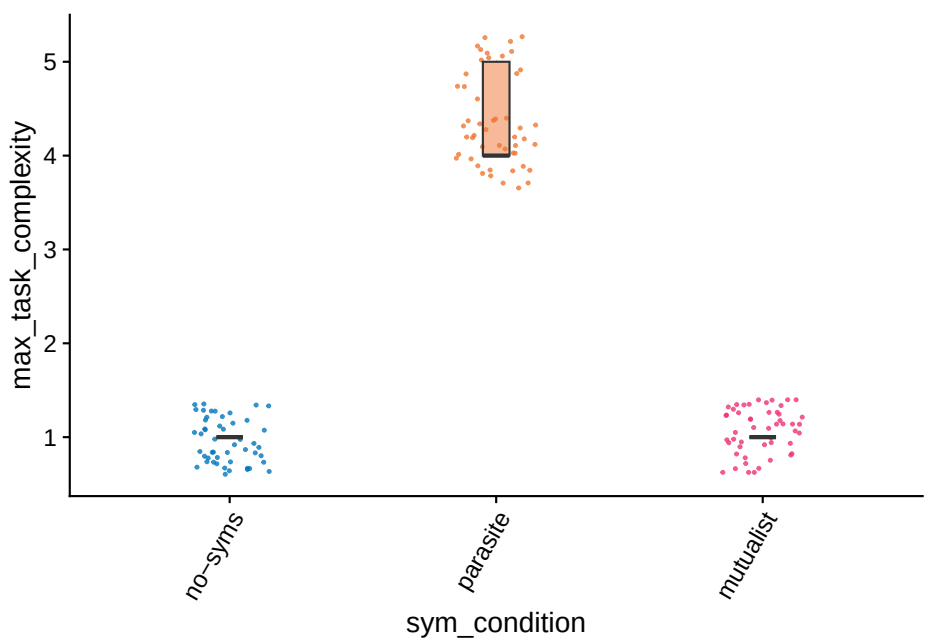
```
    sep = "/"
  ),
  plot = host_parent_tasks_all_time_by_task_plt,
  width = 8,
  height = 8
)
host_parent_tasks_all_time_by_task_plt
```



5.4.2 Host maximum function complexity evolved

```
host_max_task_complexity_plt <- host_max_task_complexity_data %>%
  ggplot(
    aes(x = sym_condition, fill = sym_condition, y = max_task_complexity)
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
```

```
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_max_task_complexity_plt.pdf", sep = "/"),
  plot = host_max_task_complexity_plt,
  width = 4,
  height = 4
)
host_max_task_complexity_plt
```



Kruskal-Wallis test to check significance among distribution differences.

```
kw_test <- host_max_task_complexity_data %>%
  group_by(interaction_multiplier) %>%
  rstatix::kruskal_test(max_task_complexity ~ sym_condition)
kw_test
```

```
## # A tibble: 1 x 7
##   interaction_multiplier .y.                   n statistic    df        p method
## * <fct>                  <chr>             <int>     <dbl> <int>    <dbl> <chr>
## 1 4                      max_task_complex~   150      144.     2 5.99e-32 Krusk~
```

Kruskal-Wallis test is significant, so we run Bonferroni-corrected pairwise Wilcoxon rank-sum tests to determine significant differences between pairs of treatments.

```
pairwise.wilcox.test(
  x = host_max_task_complexity_data$max_task_complexity,
  g = host_max_task_complexity_data$sym_condition,
  p.adjust.method = "bonferroni",
  exact = FALSE
)
```

```
##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  host_max_task_complexity_data$max_task_complexity and host_max_task_complexi
##
##           no-syms parasite
## parasite  <2e-16  -
## mutualist -       <2e-16
##
## P value adjustment method: bonferroni
```

5.5 Symbiont function performance

```
# Pivot task data long, selecting only relevant columns
sym_over_time_parent_task_data <- time_data %>%
  select(
    SEED,
    sym_condition,
    interaction_multiplier,
    update,
```

```
    CurUpdate_AND_NOT_in_sym_parent_org_counts,
    CurUpdate_AND_in_sym_parent_org_counts,
    CurUpdate_EQU_in_sym_parent_org_counts,
    CurUpdate_NAND_in_sym_parent_org_counts,
    CurUpdate_NOR_in_sym_parent_org_counts,
    CurUpdate_NOT_in_sym_parent_org_counts,
    CurUpdate_OR_NOT_in_sym_parent_org_counts,
    CurUpdate_OR_in_sym_parent_org_counts,
    CurUpdate_XOR_in_sym_parent_org_counts
  ) %>%
  pivot_longer(
    c(
      CurUpdate_AND_NOT_in_sym_parent_org_counts,
      CurUpdate_AND_in_sym_parent_org_counts,
      CurUpdate_EQU_in_sym_parent_org_counts,
      CurUpdate_NAND_in_sym_parent_org_counts,
      CurUpdate_NOR_in_sym_parent_org_counts,
      CurUpdate_NOT_in_sym_parent_org_counts,
      CurUpdate_OR_NOT_in_sym_parent_org_counts,
      CurUpdate_OR_in_sym_parent_org_counts,
      CurUpdate_XOR_in_sym_parent_org_counts
    ),
    names_to = "task",
    values_to = "task_in_parent_count"
  ) %>%
  mutate(
    task = str_replace(task, "CurUpdate_", "")
  ) %>%
  mutate(
    task = str_replace(task, "_in_sym_parent_org_counts", "")
  ) %>%
  mutate(
    task_completed_by_parent = task_in_parent_count >= threshold,
    task = factor(
      task,
      levels = tasks,
      labels = tasks
    )
  ) %>%
  mutate(
    task_complexity = case_when(
      task == "NOT" ~ 1,
      task == "NAND" ~ 1,
      task == "OR_NOT" ~ 2,
      task == "AND" ~ 2,
```

```
      task == "OR" ~ 3,
      task == "AND_NOT" ~ 3,
      task == "NOR" ~ 4,
      task == "XOR" ~ 4,
      task == "EQU" ~ 5,
      .default = -1
    )
  )

sym_parent_tasks_all_time_by_task <- sym_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier,
    task
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  group_by(
    sym_condition,
    interaction_multiplier,
    task
  ) %>%
  summarize(
    n = n(),
    parent_tasks_completed_ever = sum(parent_task_completed_ever == TRUE)
  )
```

Next, summarize function complexity level.

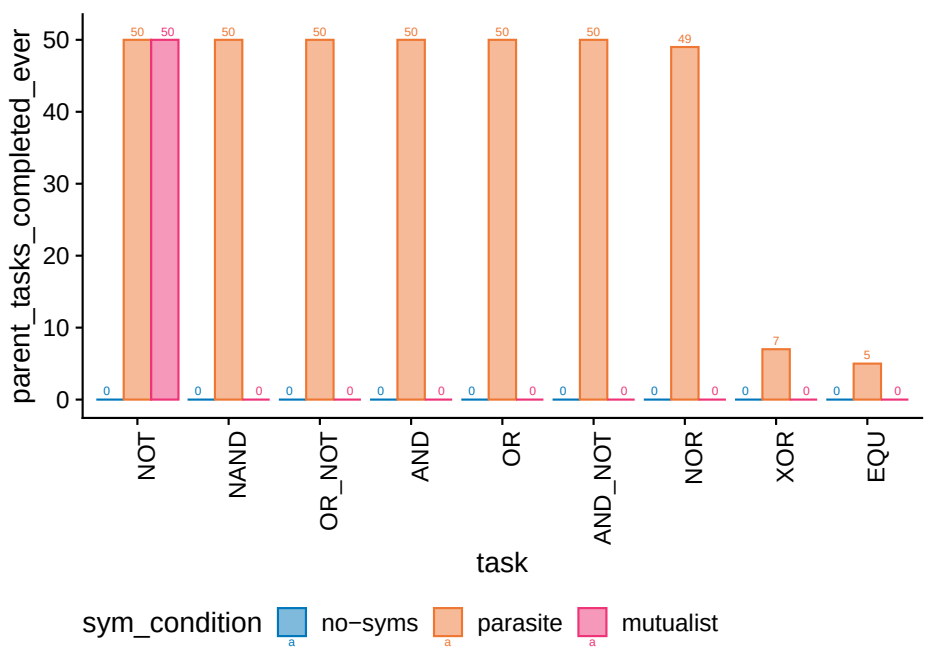
```
sym_max_task_complexity_data <- sym_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier,
    task,
    task_complexity
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
```

```
  ungroup() %>%
  # This filter works fine because all replicates start with a task
  filter(
    parent_task_completed_ever
  ) %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier
  ) %>%
  summarize(
    max_task_complexity = max(task_complexity)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c(
        "no-syms",
        "parasite",
        "mutualist"
      )
    )
  )
```

5.5.1 Symbiont function performance - all time

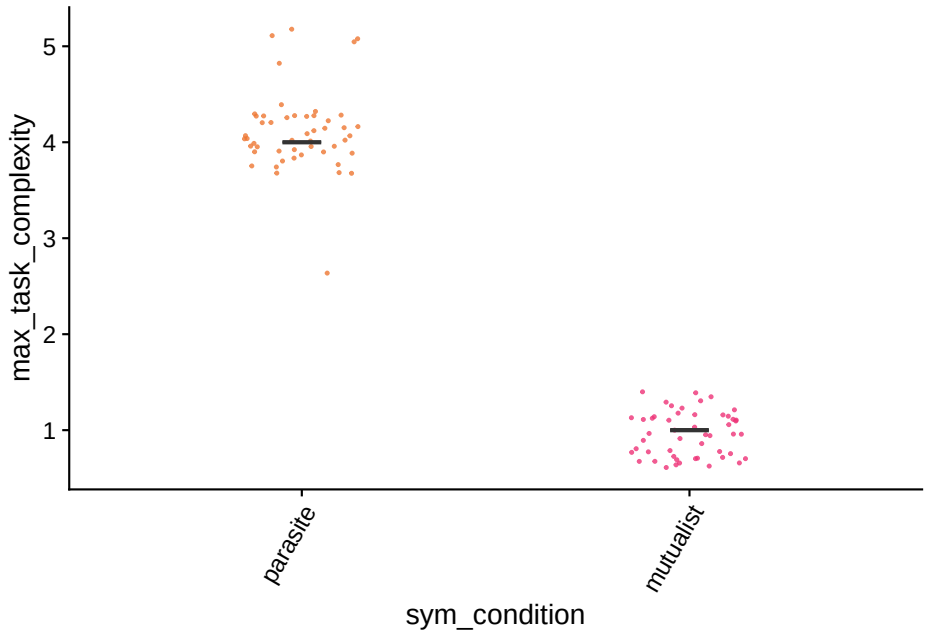
```
sym_parent_tasks_all_time_by_task_plt <- sym_parent_tasks_all_time_by_task %>%
  ggplot(
    aes(
      x = task,
      color = sym_condition,
      fill = sym_condition,
      y = parent_tasks_completed_ever
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = parent_tasks_completed_ever),
    position = position_dodge(width = 1),
    vjust = -0.5,
```

```
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(
      angle = 90,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_parent_tasks_all_time_by_task_plt.pdf", sep = "/"),
  plot = sym_parent_tasks_all_time_by_task_plt,
  width = 8,
  height = 8
)
sym_parent_tasks_all_time_by_task_plt
```



5.5.2 Symbiont maximum function complexity evolved

```
sym_max_task_complexity_plt <- sym_max_task_complexity_data %>%
  ggplot(
    aes(x = sym_condition, fill = sym_condition, y = max_task_complexity)
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = c(color_pal[2], color_pal[3])) +
  scale_fill_manual(values = c(color_pal[2], color_pal[3])) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_max_task_complexity_plt.pdf", sep = "/"),
  plot = sym_max_task_complexity_plt,
  width = 4,
  height = 4
)
sym_max_task_complexity_plt
```



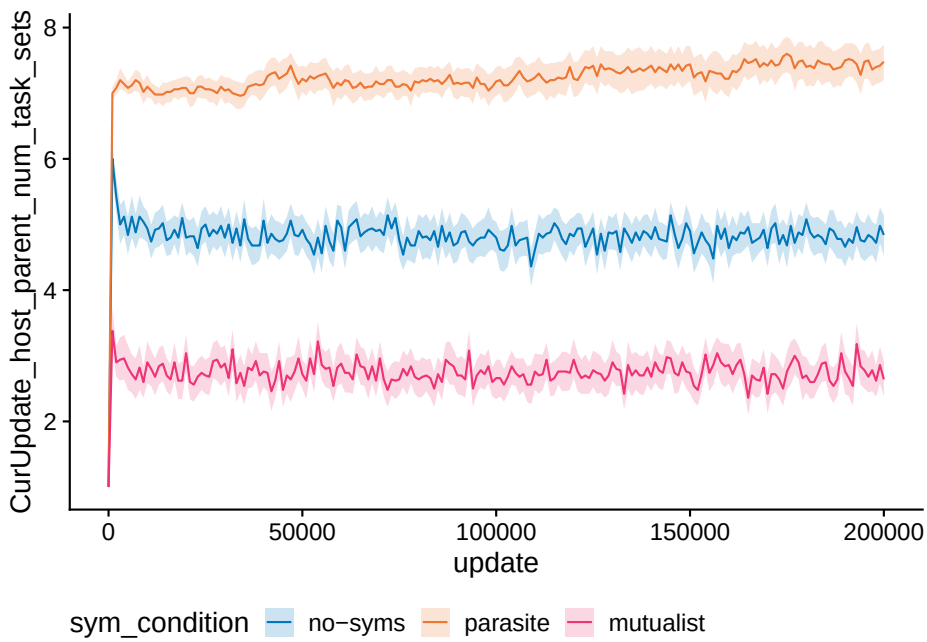
5.6 Host phenotypic diversity

5.6.1 Number of distinct function profiles

Over time:

```
host_parent_num_task_sets_plt <- time_data %>%
  # sample to make plotting faster
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_parent_num_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
```

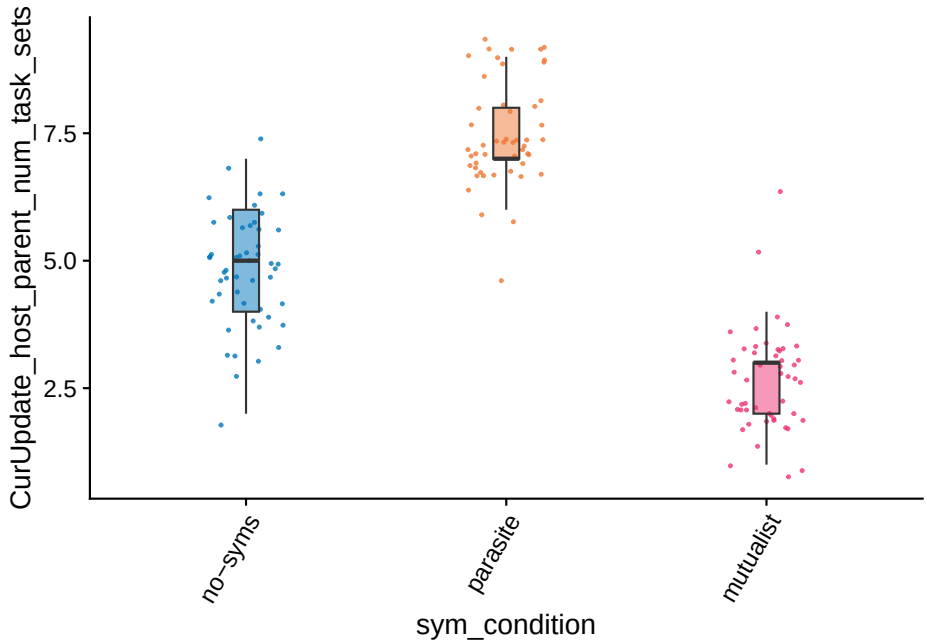
```
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_parent_num_task_sets_plt.pdf", sep = "/"),
  plot = host_parent_num_task_sets_plt,
  width = 8,
  height = 8
)
host_parent_num_task_sets_plt
```



At the end of the experiment:

```
host_final_num_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_parent_num_task_sets
    )
  ) +
```

```
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_final_num_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = host_final_num_task_sets_elapsed_plt,
  width = 4,
  height = 4
)
host_final_num_task_sets_elapsed_plt
```

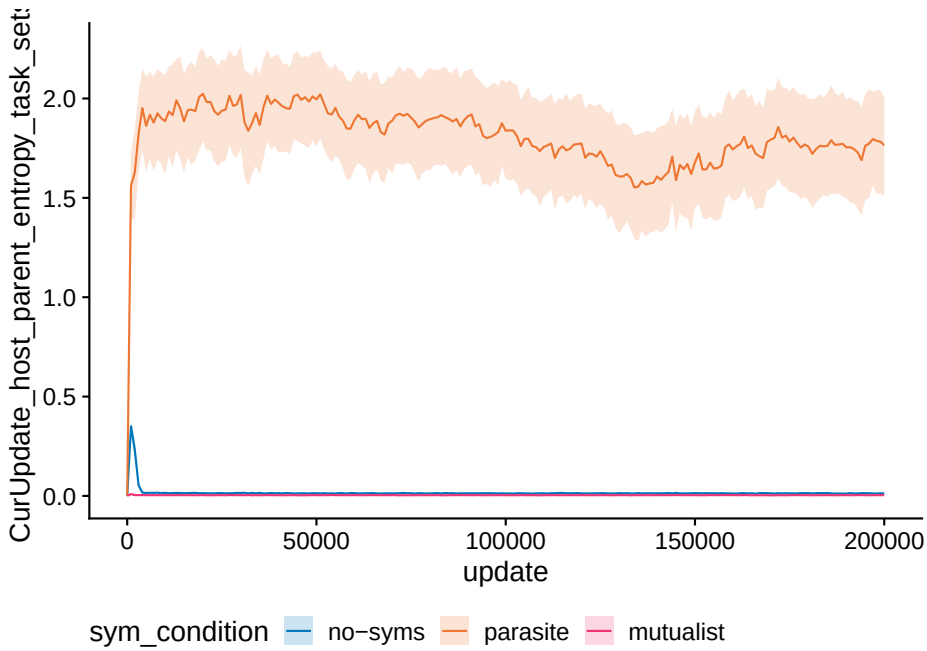


5.6.2 Shannon diversity of function profiles

Over time:

```
host_parent_entropy_task_sets_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
```

```
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_parent_entropy_task_sets_plt.pdf", sep = "/"),
  plot = host_parent_entropy_task_sets_plt,
  width = 8,
  height = 8
)
host_parent_entropy_task_sets_plt
```



At the end of the experiment:

```
host_final_entopy_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_parent_entropy_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
```

```
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_final_entopy_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = host_final_entopy_task_sets_elapsed_plt,
  width = 10,
  height = 10
)
```

Statistical analysis

```
kw_test <- summary_data %>%
  group_by(interaction_multiplier) %>%
  rstatix::kruskal_test(CurUpdate_host_parent_entropy_task_sets ~ sym_condition)
kw_test
```

```
## # A tibble: 1 x 7
##   interaction_multiplier .y.                    n statistic    df        p method
## * <fct>                  <chr>              <int>     <dbl> <int>    <dbl> <chr>
## 1 4                      CurUpdate_host_p~   150      128.     2 1.38e-28 Krusk~
```

```
pairwise.wilcox.test(
  x = summary_data$CurUpdate_host_parent_entropy_task_sets,
  g = summary_data$sym_condition,
  p.adjust.method = "bonferroni",
  exact = FALSE
)
```

```
##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
```

```
##
## data:  summary_data$CurUpdate_host_parent_entropy_task_sets and summary_data$sym_co
##
##           no-syms parasite
## parasite  < 2e-16 -
## mutualist 2.5e-15 < 2e-16
##
## P value adjustment method: bonferroni
```

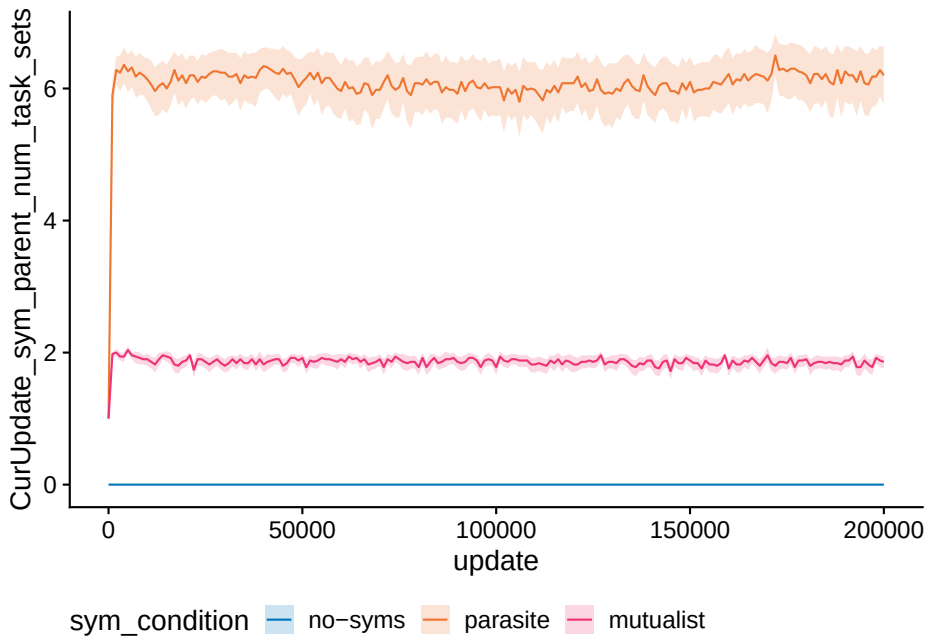
5.7 Symbiont phenotypic diversity

5.7.1 Number of distinct function profiles

Over time:

```
sym_parent_num_task_sets_plt <- time_data %>%
  # sample to make plotting faster
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_sym_parent_num_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "sym_parent_num_task_sets_plt.pdf", sep = "/"),
  plot = sym_parent_num_task_sets_plt,
  width = 8,
  height = 8
```

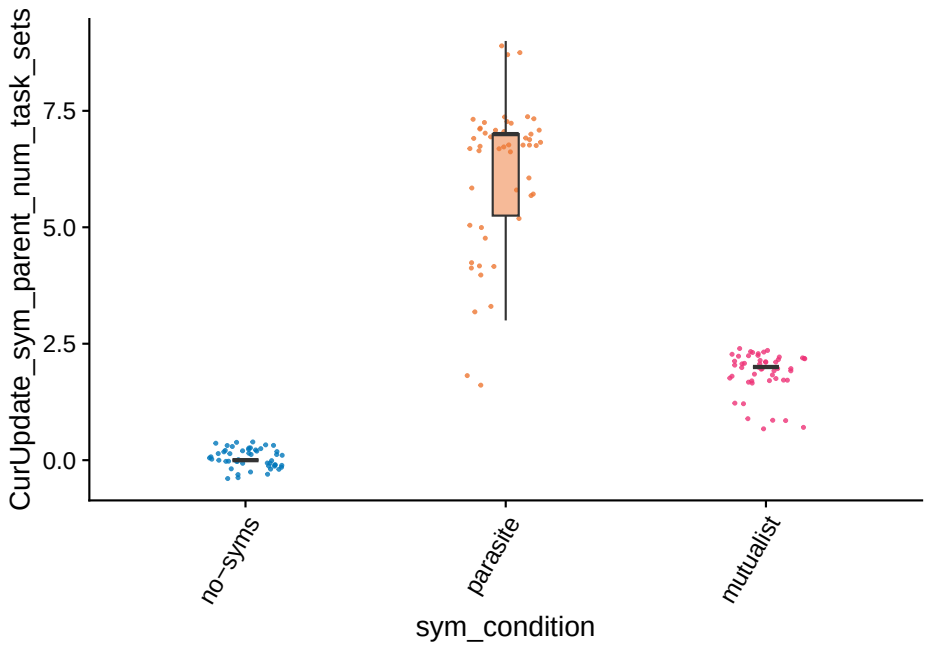
```
)
sym_parent_num_task_sets_plt
```



At the end of the experiment:

```
sym_final_num_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_parent_num_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
```

```
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_final_num_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = sym_final_num_task_sets_elapsed_plt,
  width = 4,
  height = 4
)
sym_final_num_task_sets_elapsed_plt
```

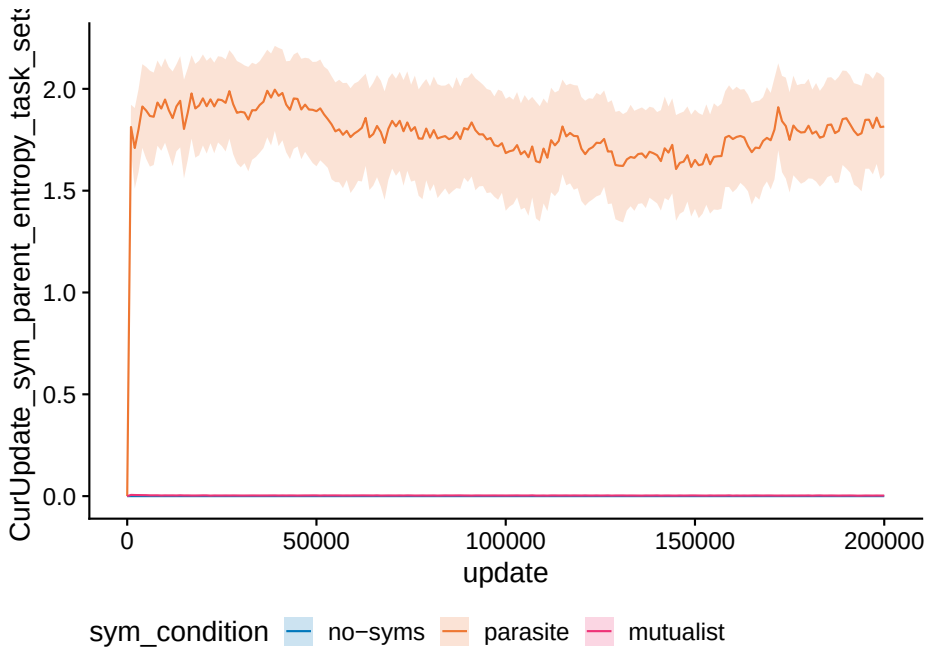


5.7.2 Shannon diversity of function profiles

Over time:

```
sym_parent_entropy_task_sets_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
```

```
  ggplot(
    aes(
      x = update,
      y = CurUpdate_sym_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "sym_parent_entropy_task_sets_plt.pdf", sep = "/"),
  plot = sym_parent_entropy_task_sets_plt,
  width = 8,
  height = 8
)
sym_parent_entropy_task_sets_plt
```



At the end of the experiment:

```
sym_final_entopy_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_parent_entropy_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
```

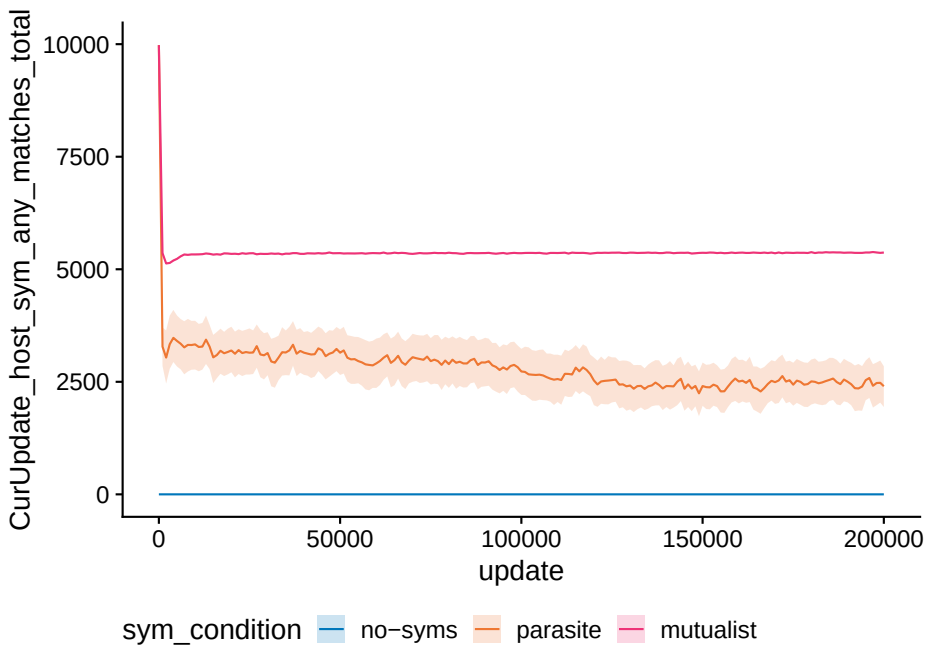
```
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_final_entopy_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = sym_final_entopy_task_sets_elapsed_plt,
  width = 10,
  height = 10
)
```

5.8 Trait matches over time

Counts of host-symbiont pairs with any trait matches. Note that organisms that were just born will not have had time to express any Boolean logic functions, so they will appear as not matching.

```
host_sym_any_matches_total_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_sym_any_matches_total,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_sym_any_matches_total_plt.pdf", sep = "/"),
  plot = host_sym_any_matches_total_plt,
```

```
  width = 8,
  height = 8
)
host_sym_any_matches_total_plt
```

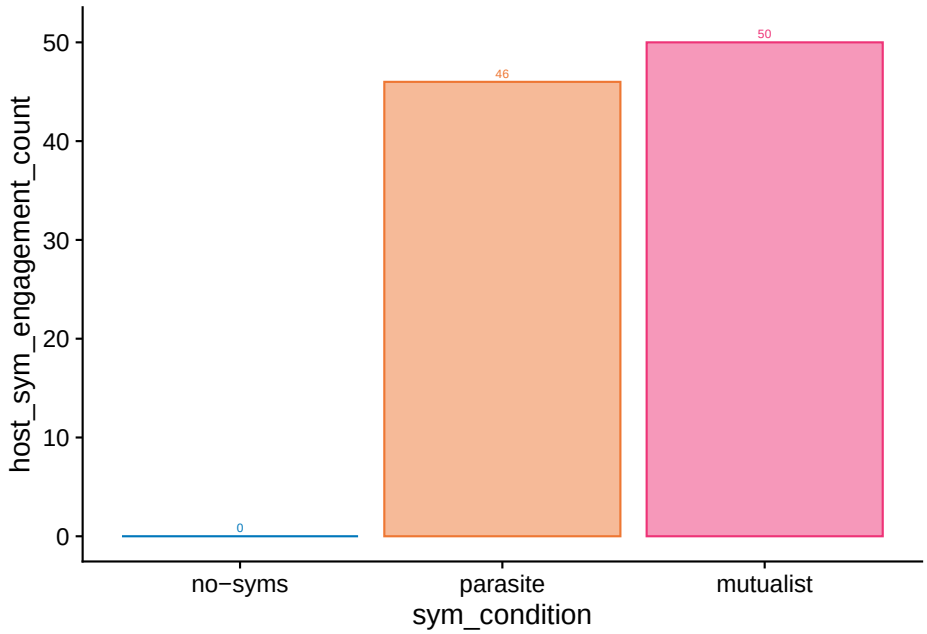


5.9 Evolutionary engagement

Use trait matching metrics to determine if host-symbiont populations are evolutionarily engaged.

```
engagement <- summary_data %>%
  mutate(
    host_sym_engagement = CurUpdate_host_sym_any_matches_total > threshold,
  ) %>%
  group_by(
    sym_condition
  ) %>%
  summarise(
    host_sym_engagement_count = sum(host_sym_engagement)
  )
```

```
host_sym_engagement_count_plt <- engagement %>%
  ggplot(
    aes(
      x = sym_condition,
      color = sym_condition,
      fill = sym_condition,
      y = host_sym_engagement_count
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = host_sym_engagement_count),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  theme(
    legend.position = "none"
  )

ggsave(
  file = paste(plot_dir, "host_sym_engagement_count_plt.pdf", sep = "/"),
  plot = host_sym_engagement_count_plt,
  width = 4,
  height = 4
)
host_sym_engagement_count_plt
```

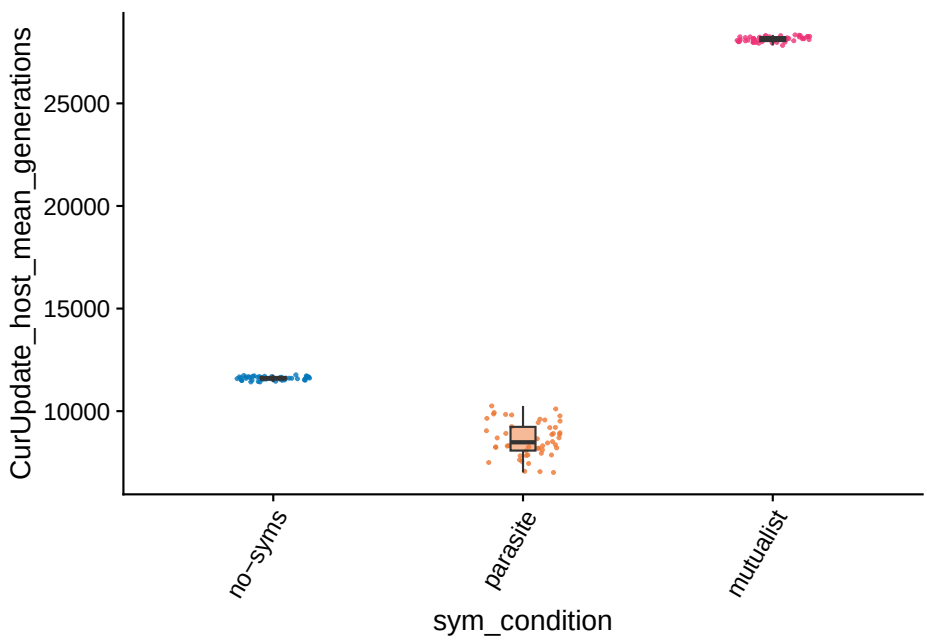


5.10 Generations of evolution elapsed

5.10.1 Hosts

```
host_avg_gens_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_mean_generations
    )
  ) +
  geom_point(
    mapping = aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
```

```
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_avg_gens_elapsed_plt.pdf", sep = "/"),
  plot = host_avg_gens_elapsed_plt,
  width = 4,
  height = 4
)
host_avg_gens_elapsed_plt
```




```
summary_data %>%
  group_by(
    sym_condition
  ) %>%
  summarise(
```

```
    mean_gens = mean(CurUpdate_host_mean_generations),
    median_gens = median(CurUpdate_host_mean_generations),
    var_gens = var(CurUpdate_host_mean_generations),
    sdev_gens = sd(CurUpdate_host_mean_generations)
  )
```

```
## # A tibble: 3 x 5
##   sym_condition mean_gens median_gens var_gens sdev_gens
##   <fct>             <dbl>       <dbl>    <dbl>     <dbl>
## 1 no-syms          11600.      11600     6960.      83.4
## 2 parasite          8621.       8484.  722611.     850.
## 3 mutualist        28125.      28122.   14533.     121.
```

```
kw_test <- summary_data %>%
  rstatix::kruskal_test(CurUpdate_host_mean_generations ~ sym_condition)
kw_test
```

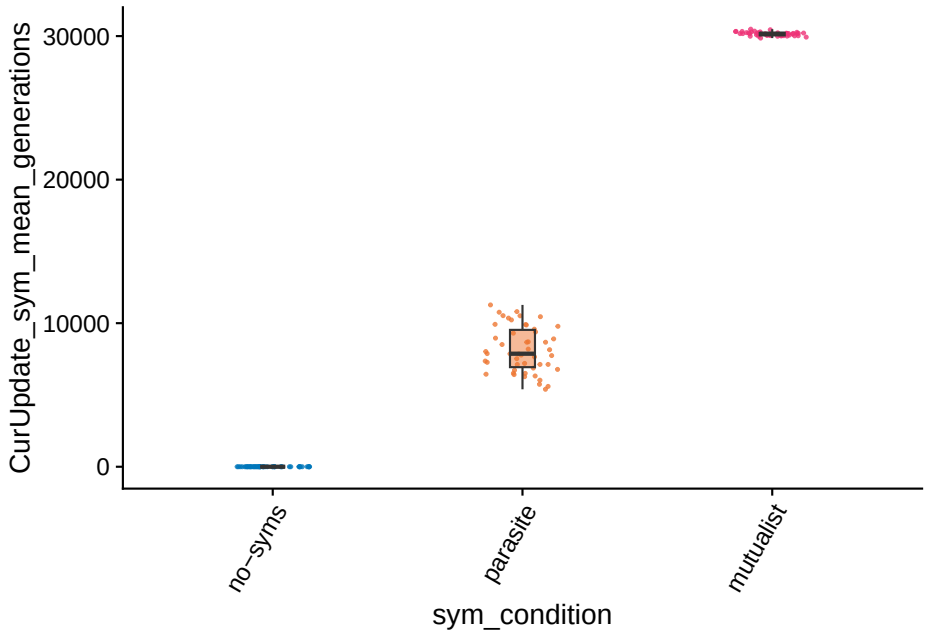
```
## # A tibble: 1 x 6
##   .y.                                n statistic    df        p method
## * <chr>                            <int>     <dbl> <int>    <dbl> <chr>
## 1 CurUpdate_host_mean_generations   150      132.     2 1.73e-29 Kruskal-Wallis
```

```
pairwise.wilcox.test(
  x = summary_data$CurUpdate_host_mean_generations,
  g = summary_data$sym_condition,
  p.adjust.method = "bonferroni",
  exact = FALSE
)
```

```
##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  summary_data$CurUpdate_host_mean_generations and summary_data$sym_condition
##
##           no-syms parasite
## parasite  <2e-16  -
## mutualist <2e-16  <2e-16
##
## P value adjustment method: bonferroni
```

5.10.2 Symbionts

```
sym_avg_gens_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_mean_generations
    )
  ) +
  geom_point(
    mapping = aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_avg_gens_elapsed_plt.pdf", sep = "/"),
  plot = sym_avg_gens_elapsed_plt,
  width = 4,
  height = 4
)
sym_avg_gens_elapsed_plt
```



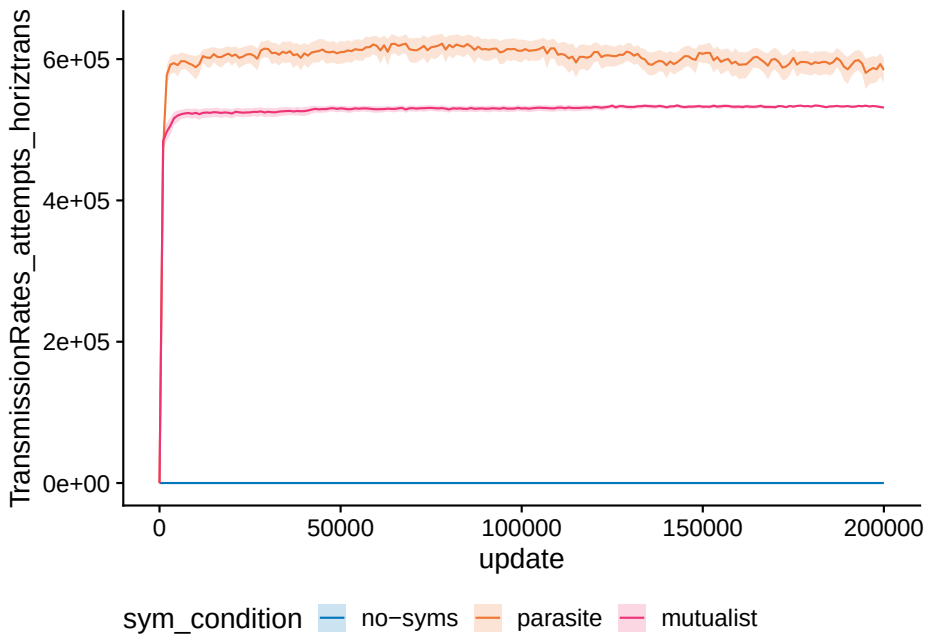

```
summary_data %>%
  group_by(
    sym_condition
  ) %>%
  summarise(
    mean_gens = mean(CurUpdate_sym_mean_generations),
    median_gens = median(CurUpdate_sym_mean_generations),
    var_gens = var(CurUpdate_sym_mean_generations),
    sdev_gens = sd(CurUpdate_sym_mean_generations)
  )
```

```
## # A tibble: 3 x 5
##   sym_condition mean_gens median_gens var_gens sdev_gens
##   <fct>             <dbl>       <dbl>    <dbl>     <dbl>
## 1 no-syms               0           0        0         0
## 2 parasite           8173.       7874. 2510452.     1584.
## 3 mutualist         30153.      30164.   16103.      127.
```

5.11 Symbiont transmission rates

5.11.1 Horizontal transmission attempts

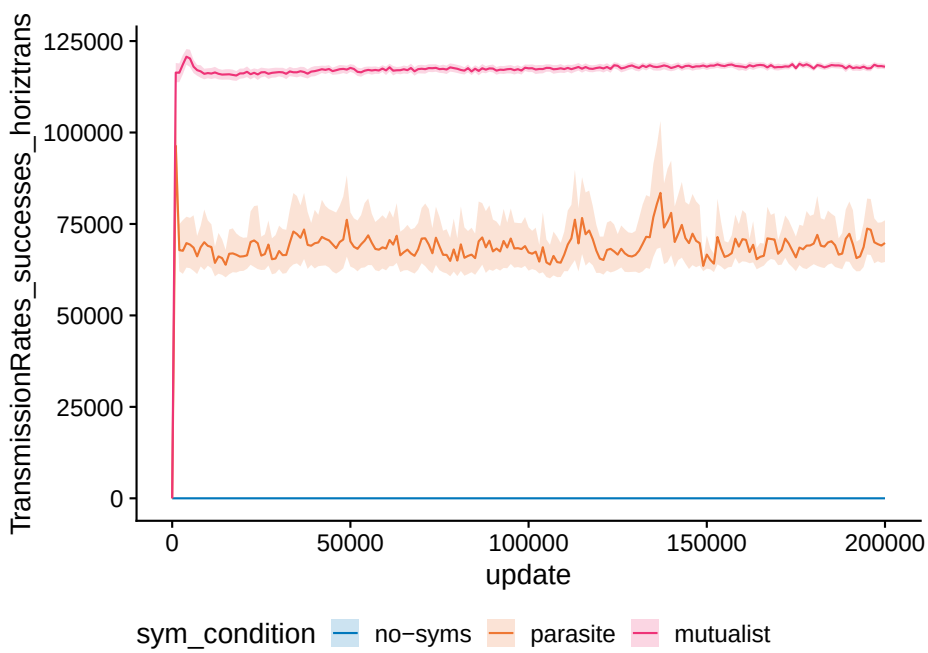
```
TransmissionRates_attempts_horiztrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_attempts_horiztrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_attempts_horiztrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_attempts_horiztrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_attempts_horiztrans_plt
```



5.11.2 Horizontal transmission successes

```
TransmissionRates_successes_horiztrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_successes_horiztrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
```

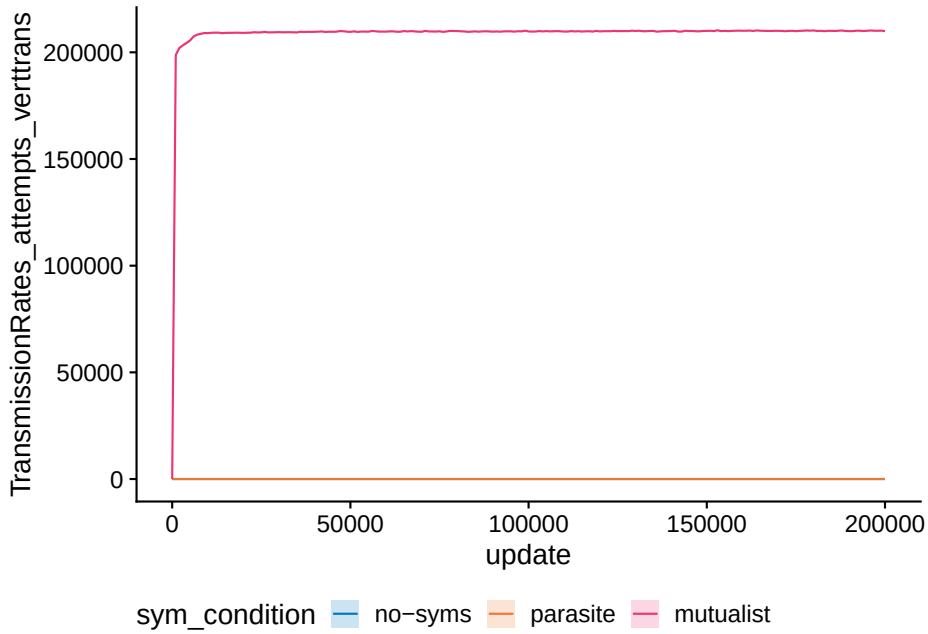
```
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_successes_horiztrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_successes_horiztrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_successes_horiztrans_plt
```



5.11.3 Vertical transmission attempts

```
TransmissionRates_attempts_verttrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_attempts_verttrans,
```

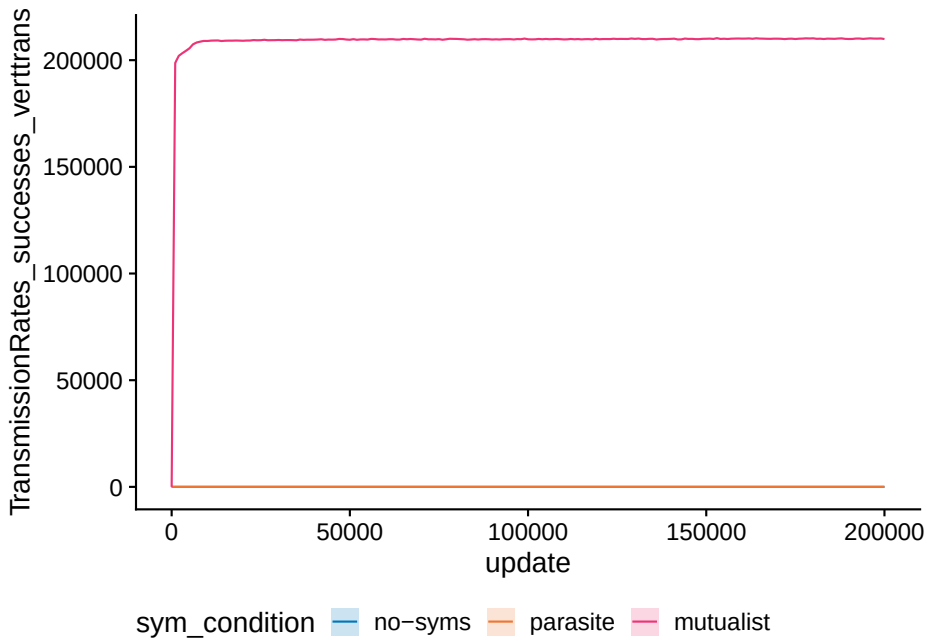
```
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_attempts_verttrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_attempts_verttrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_attempts_verttrans_plt
```



5.11.4 Successes vertical transmission

```
TransmissionRates_successes_verttrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_successes_verttrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
```

```
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_successes_verttrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_successes_verttrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_successes_verttrans_plt
```



5.12 Population size

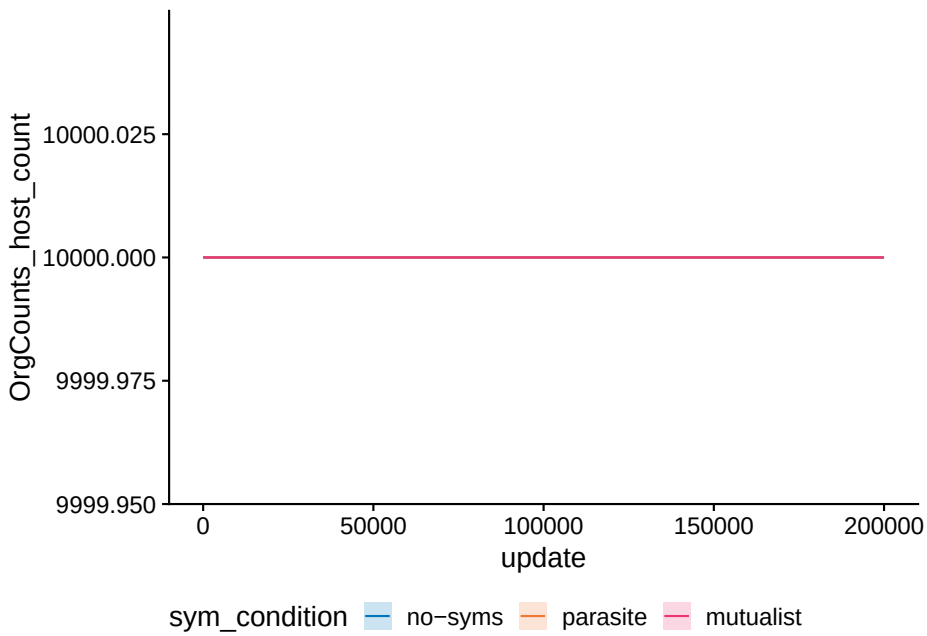
```
pop_size_summary <- time_data %>%
  group_by(
    SEED,
    sym_condition
  ) %>%
```

```
  summarize(
    host_pop_size_avg = mean(OrgCounts_host_count),
    host_pop_size_median = median(OrgCounts_host_count),
    host_pop_size_variance = var(OrgCounts_host_count),
    sym_pop_size_avg = mean(OrgCounts_hosted_sym_count),
    sym_pop_size_median = median(OrgCounts_hosted_sym_count),
    sym_pop_size_variance = var(OrgCounts_hosted_sym_count)
  )
```

5.12.1 Host population size

```
host_pop_size_ot_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = OrgCounts_host_count,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "host_pop_size_ot_plt.pdf",
    sep = "/"
  ),
  plot = host_pop_size_ot_plt,
  width = 8,
  height = 8
```

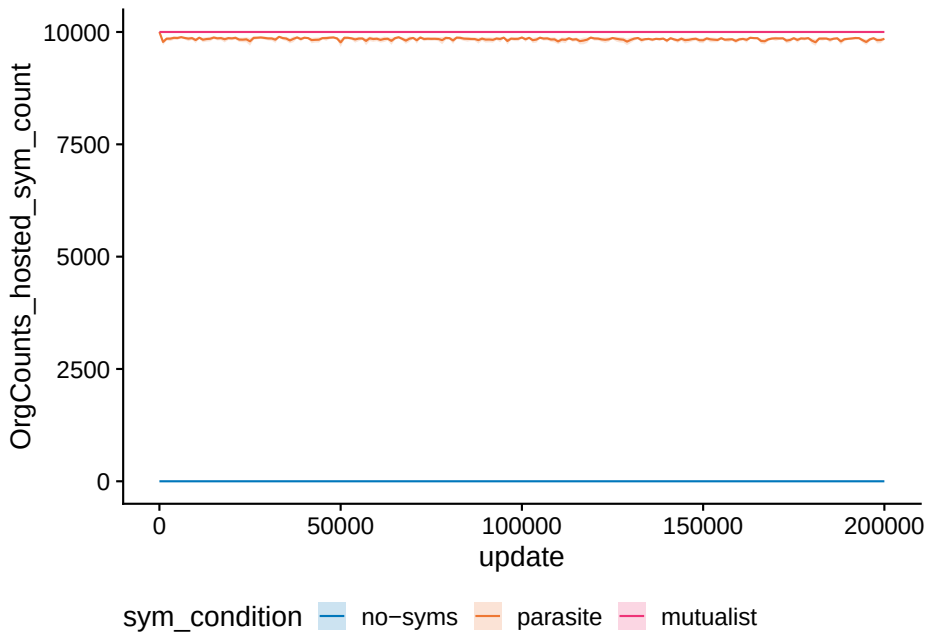
```
)
host_pop_size_ot_plt
```



5.12.2 Symbiont population size

```
sym_pop_size_ot_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = OrgCounts_hosted_sym_count,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
```

```
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "sym_pop_size_ot_plt.pdf",
    sep = "/"
  ),
  plot = sym_pop_size_ot_plt,
  width = 8,
  height = 8
)
sym_pop_size_ot_plt
```

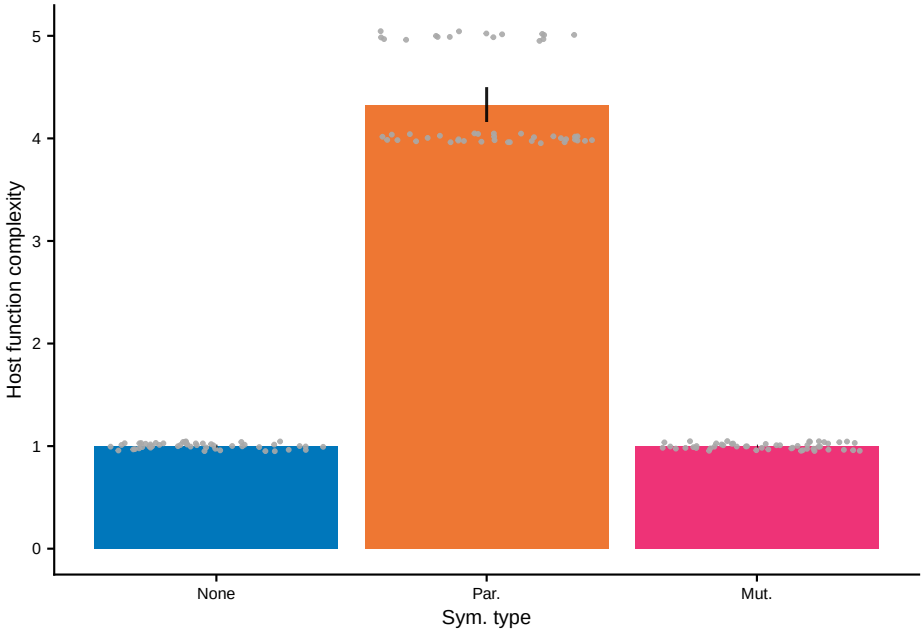


5.13 Manuscript figures

```
host_max_task_complexity_data_ci <- host_max_task_complexity_data %>%
  group_by(
```

```
    sym_condition
  ) %>%
  summarize(
    mean_max_task_complexity = mean(max_task_complexity),
    min_max_task_complexity = min(max_task_complexity),
    max_max_task_complexity = max(max_task_complexity),
    ci = list(
      mean_cl_boot(
        max_task_complexity,
        conf.int = 0.99,
        B = 10000,
      ) %>% rename(mean = y, lwr = ymin, upr = ymax)
    )
  ) %>%
  unnest(cols = c(ci)) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist"),
      labels = c("No symbionts", "Parasites", "Mutualists")
    )
  )

dist_data <- host_max_task_complexity_data %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist"),
      labels = c("No symbionts", "Parasites", "Mutualists")
    )
  )

manuscript_host_task_complexity_plt <- host_max_task_complexity_data_ci %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = mean_max_task_complexity
    )
  ) +
  geom_bar(stat = "identity") +
  geom_point(
    data = dist_data,
    mapping = aes(x = sym_condition, y = max_task_complexity),
    position = position_jitter(width = .4, height = 0.05),
```

```
    size = .7,
    alpha = 0.9,
    color = "darkgray",
    fill = "darkgray"
  ) +
  geom_linerange(
    aes(
      ymin = lwr,
      ymax = upr
    ),
    alpha = 0.9,
    linewidth = 0.7,
    color = "black"
  ) +
  scale_x_discrete(
    limits = c("No symbionts", "Parasites", "Mutualists"),
    labels = c("None", "Par.", "Mut.")
  ) +
  scale_fill_manual(values = color_pal) +
  labs(
    x = "Sym. type",
    y = "Host function complexity"
  ) +
  theme(
    legend.position = "none",
    axis.title.y = element_text(size = 10),
    axis.text.y = element_text(size = 8),
    axis.title.x = element_text(size = 10),
    axis.text.x = element_text(size = 8)
  )


ggsave(
  file = paste(
    plot_dir,
    "manuscript_flat_rewards_host_task_complexity_plt.pdf",
    sep = "/"
  ),
  plot = manuscript_host_task_complexity_plt,
  width = 3,
  height = 3
)

manuscript_host_task_complexity_plt
```


Host function complexity
0
1
2
3
4
5
None
Par.
Mut.
Sym. type

Chapter 6

Differential function rewards

Logic functions have rewards proportional to their complexity (i.e., the minimum number of nand operations needed to encode them).

Environment configuration:

```
{"name": "NOT", "value": 1, "reward_mode": "add"},
{"name": "NAND", "value": 1, "reward_mode": "add"},
{"name": "OR_NOT", "value": 2, "reward_mode": "add"},
{"name": "AND", "value": 2, "reward_mode": "add"},
{"name": "OR", "value": 4, "reward_mode": "add"},
{"name": "AND_NOT", "value": 4, "reward_mode": "add"},
{"name": "NOR", "value": 8, "reward_mode": "add"},
{"name": "XOR", "value": 8, "reward_mode": "add"},
{"name": "EQU", "value": 16, "reward_mode": "add"}
```

6.1 Dependencies and setup

Load R library dependencies.

```
library(tidyverse)
library(cowplot)
library(khroma)
library(rstatix)
library(knitr)
library(kableExtra)
```

Define relevant file paths.

```
# Check if Rmd is being compiled using bookdown
bookdown <- exists("bookdown_build")

# Build relative path from R's working directory to what is actually the working
# directory for this experiment.
experiment_slug <- "2025-11-25-health-diff-rewards"
supp_file_prefix <- "diff-rewards"
exp_directory <- paste(
  "experiments",
  experiment_slug,
  "analysis",
  sep = "/"
)

# Get path to the supplemental material analysis directory
supp_directory <- paste(
  "experiments",
  "manuscript-analyses",
  sep = "/"
)

# Adjust experiment directory if being knitted for bookdown build
if (bookdown) {
  exp_directory <- paste0(
    bookdown_wd_prefix,
    exp_directory
  )
  supp_directory <- paste0(
    bookdown_wd_prefix,
    supp_directory
  )
}
```

Define default plot theme, plot output directory.

```
# Use the cowplot theme as our default for ggplots
theme_set(theme_cowplot())
# Create directory to save plots
plot_dir <- paste(
  supp_directory,
  "plots",
  supp_file_prefix,
  sep = "/"
```

```
)
dir.create(
  plot_dir,
  showWarnings = FALSE
)

# Use vibrant theme for plot color palettes
vibrant_colors <- color("vibrant")(7)
color_pal <- c(
  vibrant_colors[["blue"]],
  vibrant_colors[["orange"]],
  vibrant_colors[["magenta"]]
)
```

6.2 Load and preprocess data

```
tasks <- c("NOT", "NAND", "OR_NOT", "AND", "OR", "AND_NOT", "NOR", "XOR", "EQU")
max_pop_size <- 100 * 100
threshold <- 0.025 * max_pop_size
```

Load summary data.

```
summary_data_path <- paste(
  exp_directory,
  "data",
  "summary.csv",
  sep = "/"
)
summary_data <- read_csv(summary_data_path)
# Do any universal preprocessing
summary_data <- summary_data %>%
  mutate(
    SEED = as.factor(SEED),
    task_credit = paste0(
      "H", HOST_ONLY_FIRST_TASK_CREDIT,
      "S", SYM_ONLY_FIRST_TASK_CREDIT
    ),
    sym_condition = case_when(
      START_MOI == "0" ~ "no-syms",
      START_MOI == "1" & HEALTH_TYPE == "parasite" ~ "parasite",
      START_MOI == "1" & HEALTH_TYPE == "mutualist" ~ "mutualist",
      .default = "unknown" # precaution against unexpected parameterizations
```

```
    ),
    interaction_multiplier = as.factor(PARASITE_CYCLE_STEAL_MULTIPLIER)
  ) %>%
  mutate(
    task_credit = as.factor(task_credit),
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist")
    )
  ) %>%
  # Filter to just IM=4 data => that's our baseline in the paper.
  filter(interaction_multiplier == "4")
```

Load time series data.

```
time_data_path <- paste(
  exp_directory,
  "data",
  "time_series.csv",
  sep = "/"
)
time_data <- read_csv(time_data_path)

# Do any universal preprocessing
time_data <- time_data %>%
  mutate(
    SEED = as.factor(SEED),
    sym_condition = case_when(
      START_MOI == "0" ~ "no-syms",
      START_MOI == "1" & HEALTH_TYPE == "parasite" ~ "parasite",
      START_MOI == "1" & HEALTH_TYPE == "mutualist" ~ "mutualist",
      .default = "unknown" # precaution against unexpected parameterizations
    ),
    interaction_multiplier = as.factor(PARASITE_CYCLE_STEAL_MULTIPLIER)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist")
    )
  ) %>%
  # Filter to just IM=4 data => that's our baseline in the paper.
  filter(interaction_multiplier == "4")
```

6.3 Extinctions

Did host or symbiont populations go extinct in any runs?

```
sym_min_pop_sizes <- time_data %>%
  group_by(SEED) %>%
  slice_min(
    OrgCounts_hosted_sym_count,
    n = 1
  ) %>%
  slice_min(
    update,
    n = 1
  )

reps_with_extinct_syms <- sym_min_pop_sizes %>%
  group_by(
    sym_condition
  ) %>%
  summarize(
    reps_with_extinct_syms = sum(OrgCounts_hosted_sym_count == 0),
    total_reps = n()
  )

host_min_pop_sizes <- time_data %>%
  group_by(SEED) %>%
  slice_min(
    OrgCounts_host_count,
    n = 1
  ) %>%
  slice_min(
    update,
    n = 1
  )

reps_with_extinct_hosts <- host_min_pop_sizes %>%
  group_by(
    sym_condition
  ) %>%
  summarize(
    reps_with_extinct_hosts = sum(OrgCounts_host_count == 0),
    total_reps = n()
  )

extinctions <- full_join(
```

```
  reps_with_extinct_syms,
  reps_with_extinct_hosts
)
extinctions <- extinctions %>%
  arrange(
    desc(reps_with_extinct_hosts), desc(reps_with_extinct_syms)
  )
extinction_table <- kable(extinctions) %>%
  kable_styling(latex_options = "striped")
save_kable(
  extinction_table,
  paste(
    plot_dir,
    "extinctions.pdf",
    sep = "/"
  )
)

extinctions
```

```
## # A tibble: 3 x 4
##   sym_condition reps_with_extinct_syms total_reps reps_with_extinct_hosts
##   <fct>                          <int>      <int>                   <int>
## 1 no-syms                           50         50                       0
## 2 parasite                          36         50                       0
## 3 mutualist                          0         50                       0
```

6.4 Host function performance

First, quantify which functions have been performed over the course of an entire run.

```
# Pivot task data long, selecting only relevant columns
host_over_time_parent_task_data <- time_data %>%
  select(
    SEED,
    sym_condition,
    interaction_multiplier,
    update,
    CurUpdate_AND_NOT_in_host_parent_org_counts,
    CurUpdate_AND_in_host_parent_org_counts,
    CurUpdate_EQU_in_host_parent_org_counts,
    CurUpdate_NAND_in_host_parent_org_counts,
```

```
    CurUpdate_NOR_in_host_parent_org_counts,
    CurUpdate_NOT_in_host_parent_org_counts,
    CurUpdate_OR_NOT_in_host_parent_org_counts,
    CurUpdate_OR_in_host_parent_org_counts,
    CurUpdate_XOR_in_host_parent_org_counts
  ) %>%
  pivot_longer(
    c(
      CurUpdate_AND_NOT_in_host_parent_org_counts,
      CurUpdate_AND_in_host_parent_org_counts,
      CurUpdate_EQU_in_host_parent_org_counts,
      CurUpdate_NAND_in_host_parent_org_counts,
      CurUpdate_NOR_in_host_parent_org_counts,
      CurUpdate_NOT_in_host_parent_org_counts,
      CurUpdate_OR_NOT_in_host_parent_org_counts,
      CurUpdate_OR_in_host_parent_org_counts,
      CurUpdate_XOR_in_host_parent_org_counts
    ),
    names_to = "task",
    values_to = "task_in_parent_count"
  ) %>%
  mutate(
    task = str_replace(task, "CurUpdate_", "")
  ) %>%
  mutate(
    task = str_replace(task, "_in_host_parent_org_counts", "")
  ) %>%
  mutate(
    task_completed_by_parent = task_in_parent_count >= threshold,
    task = factor(
      task,
      levels = tasks,
      labels = tasks
    )
  ) %>%
  mutate(
    task_complexity = case_when(
      task == "NOT" ~ 1,
      task == "NAND" ~ 1,
      task == "OR_NOT" ~ 2,
      task == "AND" ~ 2,
      task == "OR" ~ 3,
      task == "AND_NOT" ~ 3,
      task == "NOR" ~ 4,
      task == "XOR" ~ 4,
```

```
      task == "EQU" ~ 5,
      .default = -1
    )
  )

host_parent_tasks_all_time_by_task <- host_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier,
    task
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  group_by(
    sym_condition,
    interaction_multiplier,
    task
  ) %>%
  summarize(
    n = n(),
    parent_tasks_completed_ever = sum(parent_task_completed_ever == TRUE)
  )
```

Next, summarize function complexity level.

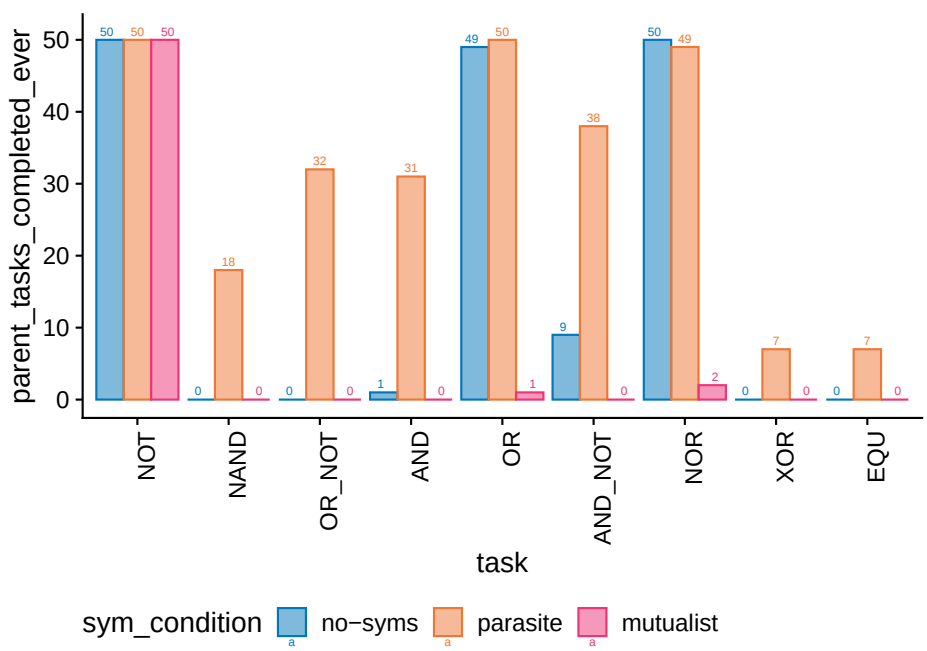
```
host_max_task_complexity_data <- host_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier,
    task,
    task_complexity
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  # This filter works fine because all replicates start with a task
  filter(
    parent_task_completed_ever
```

```
  ) %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier
  ) %>%
  summarize(
    max_task_complexity = max(task_complexity)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c(
        "no-syms",
        "parasite",
        "mutualist"
      )
    )
  )
```

6.4.1 Host function performance - all time

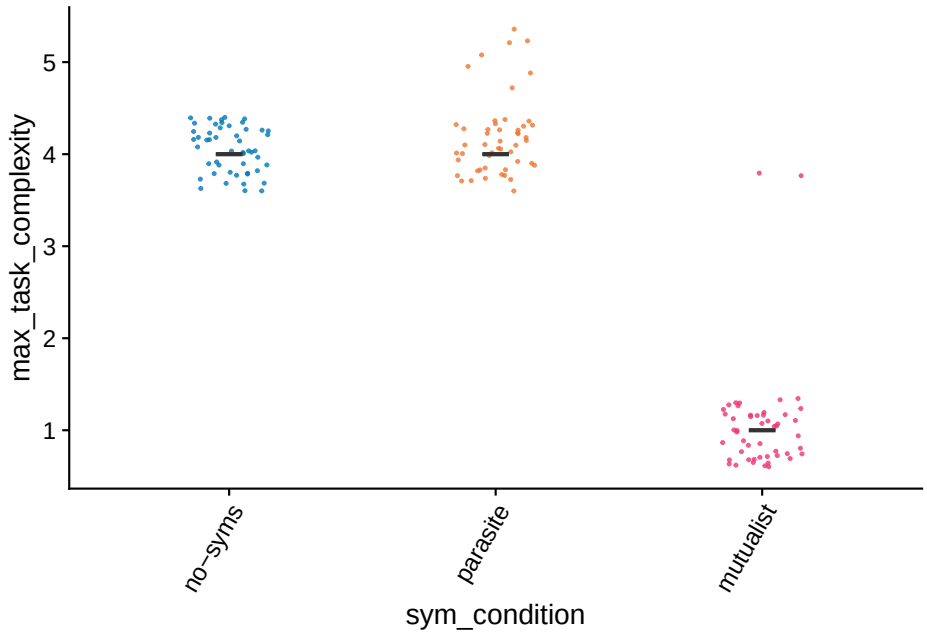
```
host_parent_tasks_all_time_by_task_plt <- host_parent_tasks_all_time_by_task %>%
  ggplot(
    aes(
      x = task,
      color = sym_condition,
      fill = sym_condition,
      y = parent_tasks_completed_ever
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = parent_tasks_completed_ever),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
```

```
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(
      angle = 90,
      hjust = 1
    )
  )
ggsave(
  file = paste(
    plot_dir,
    "host_parent_tasks_all_time_by_task_plt.pdf",
    sep = "/"
  ),
  plot = host_parent_tasks_all_time_by_task_plt,
  width = 8,
  height = 8
)
host_parent_tasks_all_time_by_task_plt
```



6.4.2 Host maximum function complexity evolved

```
host_max_task_complexity_plt <- host_max_task_complexity_data %>%
  ggplot(
    aes(x = sym_condition, fill = sym_condition, y = max_task_complexity)
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_max_task_complexity_plt.pdf", sep = "/"),
  plot = host_max_task_complexity_plt,
  width = 4,
  height = 4
)
host_max_task_complexity_plt
```



Kruskal-Wallis test to check significance among distribution differences.

```
kw_test <- host_max_task_complexity_data %>%
  group_by(interaction_multiplier) %>%
  rstatix::kruskal_test(max_task_complexity ~ sym_condition)
kw_test
```

```
## # A tibble: 1 x 7
##   interaction_multiplier .y.                   n statistic    df        p method
## * <fct>                  <chr>             <int>     <dbl> <int>    <dbl> <chr>
## 1 4                      max_task_complex~   150      130.     2 5.68e-29 Krusk~
```

Kruskal-Wallis test is significant, so we run Bonferroni-corrected pairwise Wilcoxon rank-sum tests to determine significant differences between pairs of treatments.

```
pairwise.wilcox.test(
  x = host_max_task_complexity_data$max_task_complexity,
  g = host_max_task_complexity_data$sym_condition,
  p.adjust.method = "bonferroni",
  exact = FALSE
)
```

```
##
```

```
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  host_max_task_complexity_data$max_task_complexity and host_max_task_complexity_data$sym
##
##           no-syms parasite
## parasite  0.019   -
## mutualist <2e-16  <2e-16
##
## P value adjustment method: bonferroni
```

6.5 Symbiont function performance

```
# Pivot task data long, selecting only relevant columns
sym_over_time_parent_task_data <- time_data %>%
  select(
    SEED,
    sym_condition,
    interaction_multiplier,
    update,
    CurUpdate_AND_NOT_in_sym_parent_org_counts,
    CurUpdate_AND_in_sym_parent_org_counts,
    CurUpdate_EQU_in_sym_parent_org_counts,
    CurUpdate_NAND_in_sym_parent_org_counts,
    CurUpdate_NOR_in_sym_parent_org_counts,
    CurUpdate_NOT_in_sym_parent_org_counts,
    CurUpdate_OR_NOT_in_sym_parent_org_counts,
    CurUpdate_OR_in_sym_parent_org_counts,
    CurUpdate_XOR_in_sym_parent_org_counts
  ) %>%
  pivot_longer(
    c(
      CurUpdate_AND_NOT_in_sym_parent_org_counts,
      CurUpdate_AND_in_sym_parent_org_counts,
      CurUpdate_EQU_in_sym_parent_org_counts,
      CurUpdate_NAND_in_sym_parent_org_counts,
      CurUpdate_NOR_in_sym_parent_org_counts,
      CurUpdate_NOT_in_sym_parent_org_counts,
      CurUpdate_OR_NOT_in_sym_parent_org_counts,
      CurUpdate_OR_in_sym_parent_org_counts,
      CurUpdate_XOR_in_sym_parent_org_counts
    ),
    names_to = "task",
    values_to = "task_in_parent_count"
```

```
  ) %>%
  mutate(
    task = str_replace(task, "CurUpdate_", "")
  ) %>%
  mutate(
    task = str_replace(task, "_in_sym_parent_org_counts", "")
  ) %>%
  mutate(
    task_completed_by_parent = task_in_parent_count >= threshold,
    task = factor(
      task,
      levels = tasks,
      labels = tasks
    )
  ) %>%
  mutate(
    task_complexity = case_when(
      task == "NOT" ~ 1,
      task == "NAND" ~ 1,
      task == "OR_NOT" ~ 2,
      task == "AND" ~ 2,
      task == "OR" ~ 3,
      task == "AND_NOT" ~ 3,
      task == "NOR" ~ 4,
      task == "XOR" ~ 4,
      task == "EQU" ~ 5,
      .default = -1
    )
  )

sym_parent_tasks_all_time_by_task <- sym_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier,
    task
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  group_by(
    sym_condition,
    interaction_multiplier,
```

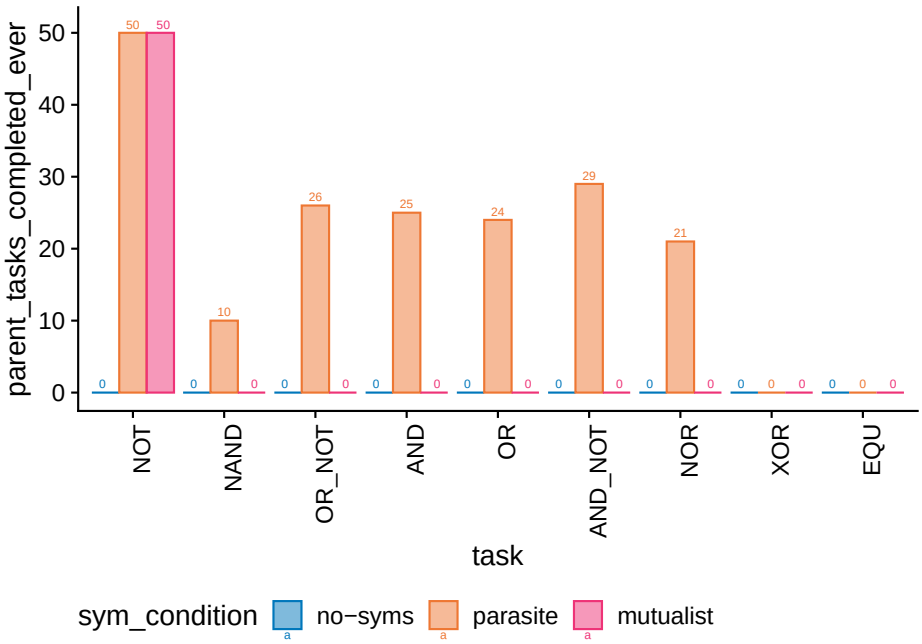
```
    task
  ) %>%
  summarize(
    n = n(),
    parent_tasks_completed_ever = sum(parent_task_completed_ever == TRUE)
  )
```

Next, summarize function complexity level.

```
sym_max_task_complexity_data <- sym_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier,
    task,
    task_complexity
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  # This filter works fine because all replicates start with a task
  filter(
    parent_task_completed_ever
  ) %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier
  ) %>%
  summarize(
    max_task_complexity = max(task_complexity)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c(
        "no-syms",
        "parasite",
        "mutualist"
      )
    )
  )
```

6.5.1 Symbiont function performance - all time

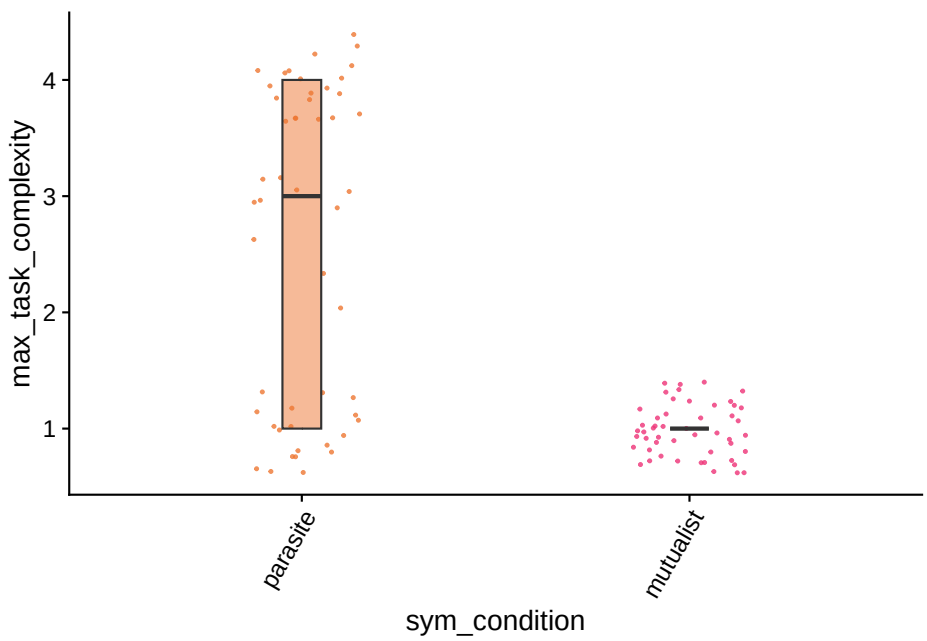
```
sym_parent_tasks_all_time_by_task_plt <- sym_parent_tasks_all_time_by_task %>%
  ggplot(
    aes(
      x = task,
      color = sym_condition,
      fill = sym_condition,
      y = parent_tasks_completed_ever
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = parent_tasks_completed_ever),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(
      angle = 90,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_parent_tasks_all_time_by_task_plt.pdf", sep = "/"),
  plot = sym_parent_tasks_all_time_by_task_plt,
  width = 8,
  height = 8
)
sym_parent_tasks_all_time_by_task_plt
```



6.5.2 Symbiont maximum function complexity evolved

```
sym_max_task_complexity_plt <- sym_max_task_complexity_data %>%
  ggplot(
    aes(x = sym_condition, fill = sym_condition, y = max_task_complexity)
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = c(color_pal[2], color_pal[3])) +
  scale_fill_manual(values = c(color_pal[2], color_pal[3])) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
```

```
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_max_task_complexity_plt.pdf", sep = "/"),
  plot = sym_max_task_complexity_plt,
  width = 4,
  height = 4
)
sym_max_task_complexity_plt
```



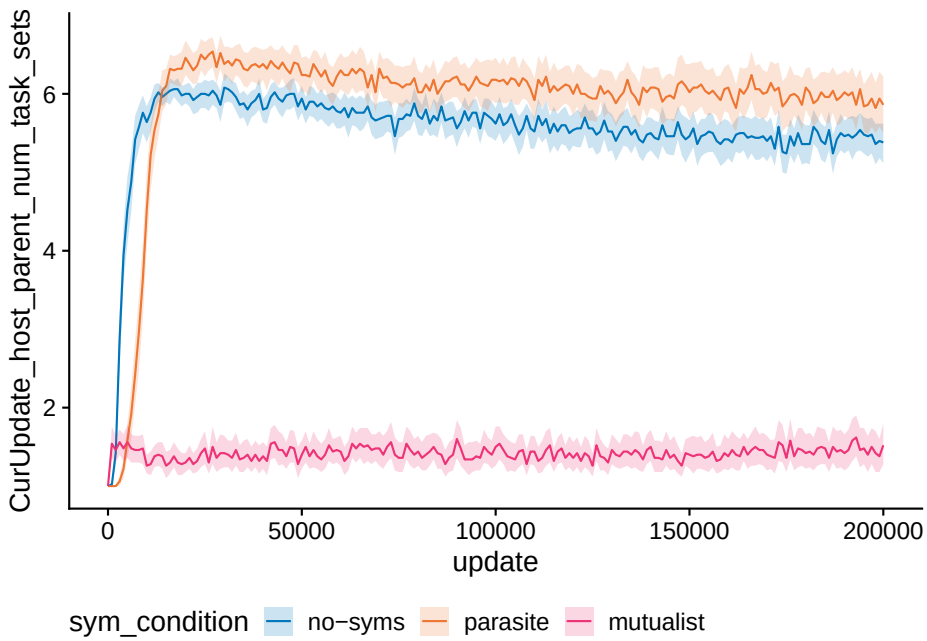
6.6 Host phenotypic diversity

6.6.1 Number of distinct function profiles

Over time:

```
host_parent_num_task_sets_plt <- time_data %>%
  # sample to make plotting faster
  filter(update %% 1000 == 0) %>%
  ggplot(
```

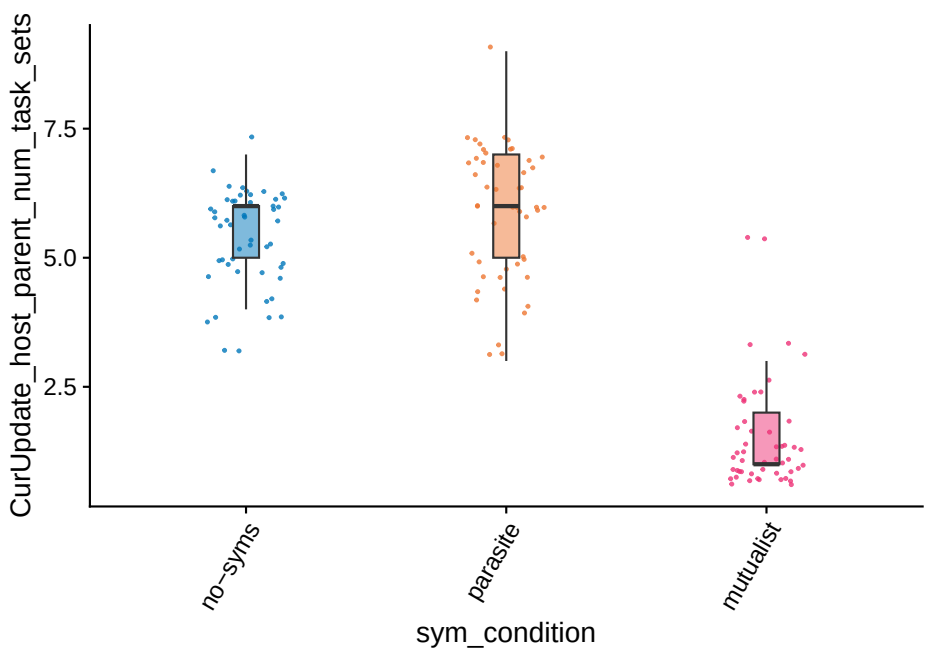
```
    aes(
      x = update,
      y = CurUpdate_host_parent_num_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_parent_num_task_sets_plt.pdf", sep = "/"),
  plot = host_parent_num_task_sets_plt,
  width = 8,
  height = 8
)
host_parent_num_task_sets_plt
```



At the end of the experiment:

```
host_final_num_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_parent_num_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
```

```
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_final_num_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = host_final_num_task_sets_elapsed_plt,
  width = 4,
  height = 4
)
host_final_num_task_sets_elapsed_plt
```

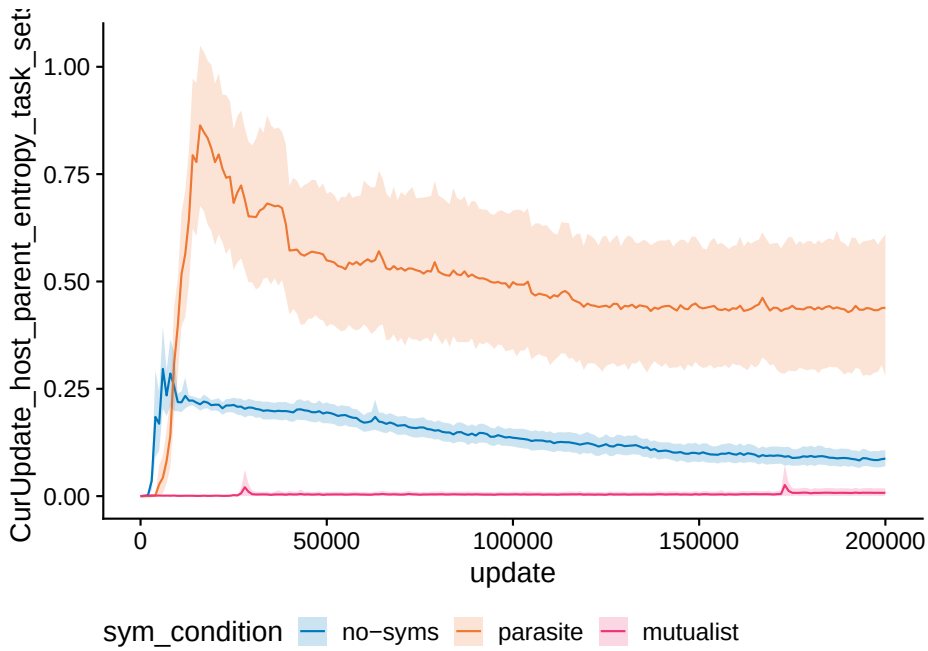


6.6.2 Shannon diversity of function profiles

Over time:

```
host_parent_entropy_task_sets_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
```

```
      y = CurUpdate_host_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_parent_entropy_task_sets_plt.pdf", sep = "/"),
  plot = host_parent_entropy_task_sets_plt,
  width = 8,
  height = 8
)
host_parent_entropy_task_sets_plt
```



At the end of the experiment:

```
host_final_entopy_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_parent_entropy_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
```

```
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_final_entopy_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = host_final_entopy_task_sets_elapsed_plt,
  width = 10,
  height = 10
)
```

Statistical analysis

```
kw_test <- summary_data %>%
  group_by(interaction_multiplier) %>%
  rstatix::kruskal_test(CurUpdate_host_parent_entropy_task_sets ~ sym_condition)
kw_test
```

```
## # A tibble: 1 x 7
##   interaction_multiplier .y.                    n statistic    df        p method
## * <fct>                  <chr>              <int>     <dbl> <int>    <dbl> <chr>
## 1 4                      CurUpdate_host_p~   150      97.3     2 7.48e-22 Krusk~
```

```
pairwise.wilcox.test(
  x = summary_data$CurUpdate_host_parent_entropy_task_sets,
  g = summary_data$sym_condition,
  p.adjust.method = "bonferroni",
  exact = FALSE
)
```

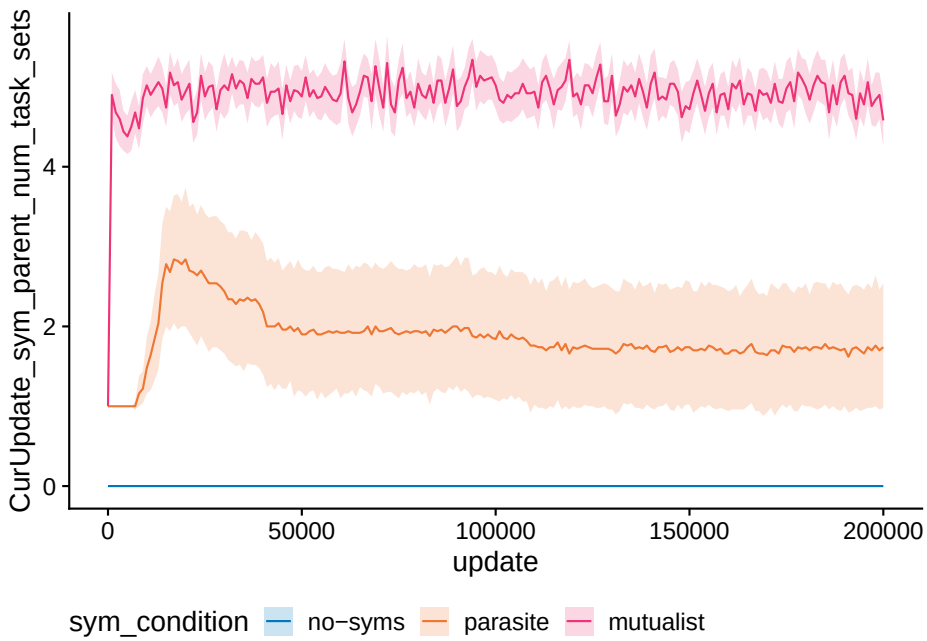
```
##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  summary_data$CurUpdate_host_parent_entropy_task_sets and summary_data$sym_co
##
##           no-syms parasite
## parasite  4e-04   -
## mutualist 6.9e-16 < 2e-16
##
## P value adjustment method: bonferroni
```

6.7 Symbiont phenotypic diversity

6.7.1 Number of distinct function profiles

Over time:

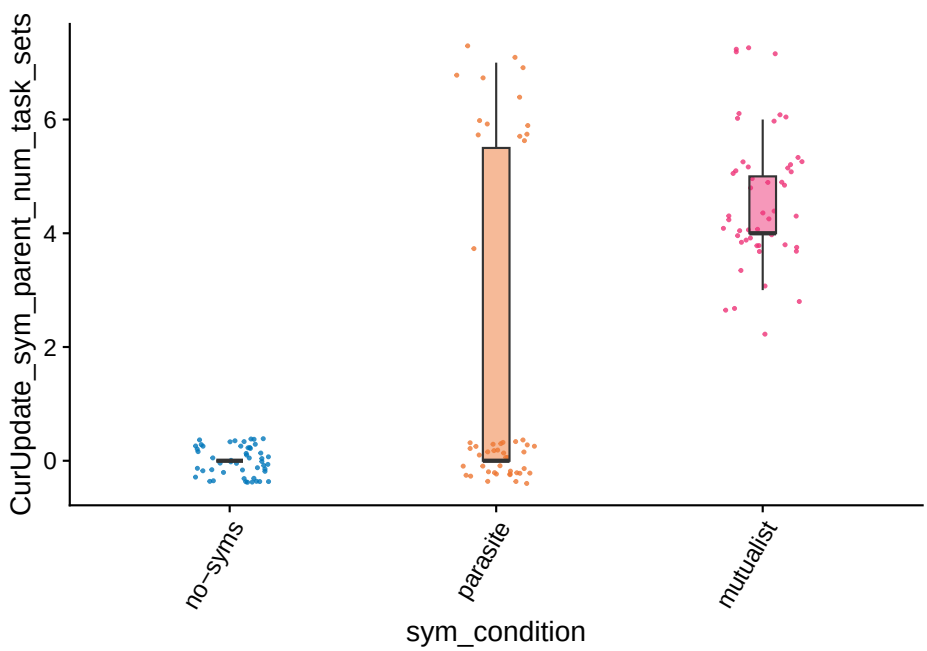
```
sym_parent_num_task_sets_plt <- time_data %>%
  # sample to make plotting faster
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_sym_parent_num_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "sym_parent_num_task_sets_plt.pdf", sep = "/"),
  plot = sym_parent_num_task_sets_plt,
  width = 8,
  height = 8
)
sym_parent_num_task_sets_plt
```



At the end of the experiment:

```
sym_final_num_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_parent_num_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
```

```
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_final_num_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = sym_final_num_task_sets_elapsed_plt,
  width = 4,
  height = 4
)
sym_final_num_task_sets_elapsed_plt
```

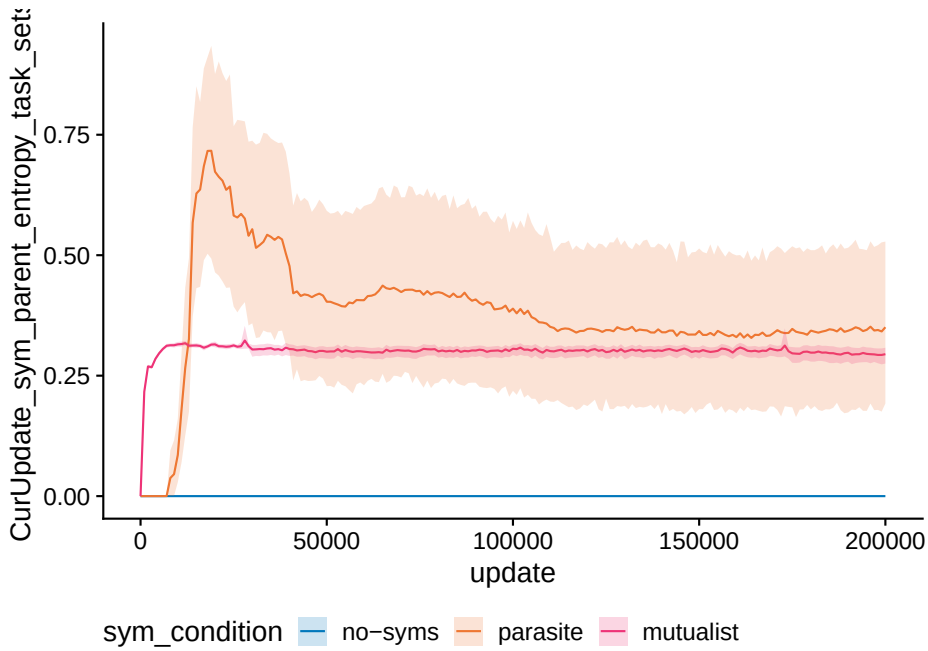


6.7.2 Shannon diversity of function profiles

Over time:

```
sym_parent_entropy_task_sets_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
```

```
      y = CurUpdate_sym_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "sym_parent_entropy_task_sets_plt.pdf", sep = "/"),
  plot = sym_parent_entropy_task_sets_plt,
  width = 8,
  height = 8
)
sym_parent_entropy_task_sets_plt
```



At the end of the experiment:

```
sym_final_entopy_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_parent_entropy_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
```

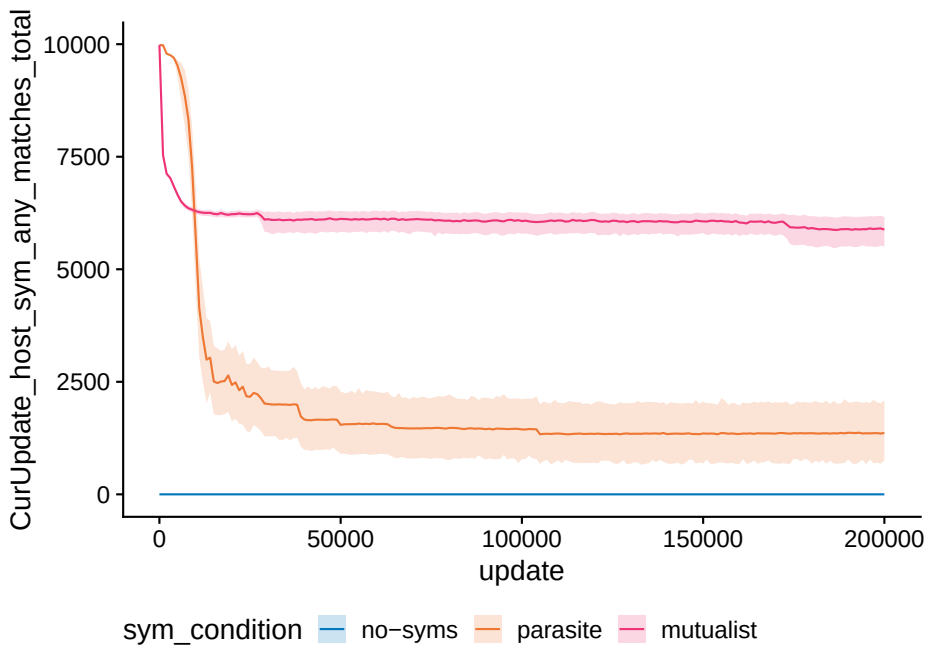
```
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_final_entopy_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = sym_final_entopy_task_sets_elapsed_plt,
  width = 10,
  height = 10
)
```

6.8 Trait matches over time

Counts of host-symbiont pairs with any trait matches. Note that organisms that were just born will not have had time to express any Boolean logic functions, so they will appear as not matching.

```
host_sym_any_matches_total_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_sym_any_matches_total,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_sym_any_matches_total_plt.pdf", sep = "/"),
  plot = host_sym_any_matches_total_plt,
```

```
  width = 8,
  height = 8
)
host_sym_any_matches_total_plt
```

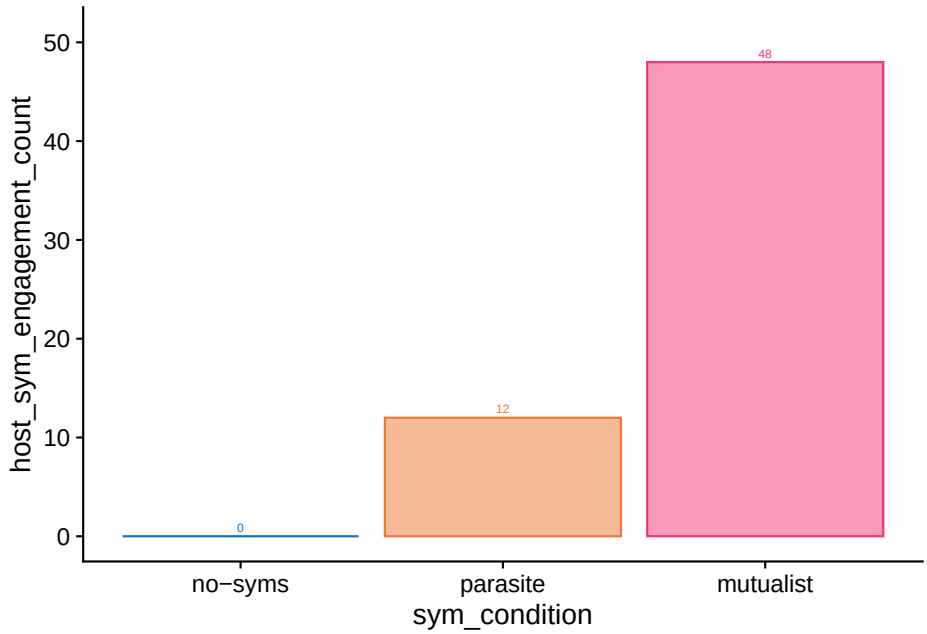


6.9 Evolutionary engagement

Use trait matching metrics to determine if host-symbiont populations are evolutionarily engaged.

```
engagement <- summary_data %>%
  mutate(
    host_sym_engagement = CurUpdate_host_sym_any_matches_total > threshold,
  ) %>%
  group_by(
    sym_condition
  ) %>%
  summarise(
    host_sym_engagement_count = sum(host_sym_engagement)
  )
```

```
host_sym_engagement_count_plt <- engagement %>%
  ggplot(
    aes(
      x = sym_condition,
      color = sym_condition,
      fill = sym_condition,
      y = host_sym_engagement_count
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = host_sym_engagement_count),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  theme(
    legend.position = "none"
  )

ggsave(
  file = paste(plot_dir, "host_sym_engagement_count_plt.pdf", sep = "/"),
  plot = host_sym_engagement_count_plt,
  width = 4,
  height = 4
)
host_sym_engagement_count_plt
```

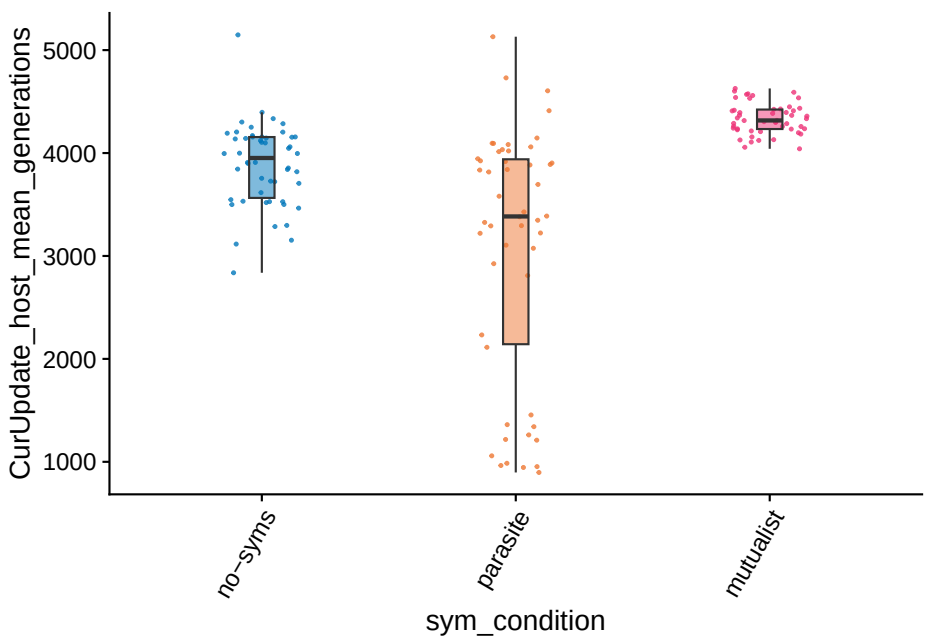


6.10 Generations of evolution elapsed

6.10.1 Hosts

```
host_avg_gens_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_mean_generations
    )
  ) +
  geom_point(
    mapping = aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
```

```
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_avg_gens_elapsed_plt.pdf", sep = "/"),
  plot = host_avg_gens_elapsed_plt,
  width = 4,
  height = 4
)
host_avg_gens_elapsed_plt
```




```
summary_data %>%
  group_by(
    sym_condition
  ) %>%
  summarise(
```

```
    mean_gens = mean(CurUpdate_host_mean_generations),
    median_gens = median(CurUpdate_host_mean_generations),
    var_gens = var(CurUpdate_host_mean_generations),
    sdev_gens = sd(CurUpdate_host_mean_generations)
  )
```

```
## # A tibble: 3 x 5
##   sym_condition mean_gens median_gens var_gens sdev_gens
##   <fct>             <dbl>       <dbl>    <dbl>     <dbl>
## 1 no-syms           3887.       3952.  160311.      400.
## 2 parasite          3069.       3384. 1496216.     1223.
## 3 mutualist         4333.       4317.   22501.      150.
```

```
kw_test <- summary_data %>%
  rstatix::kruskal_test(CurUpdate_host_mean_generations ~ sym_condition)
kw_test
```

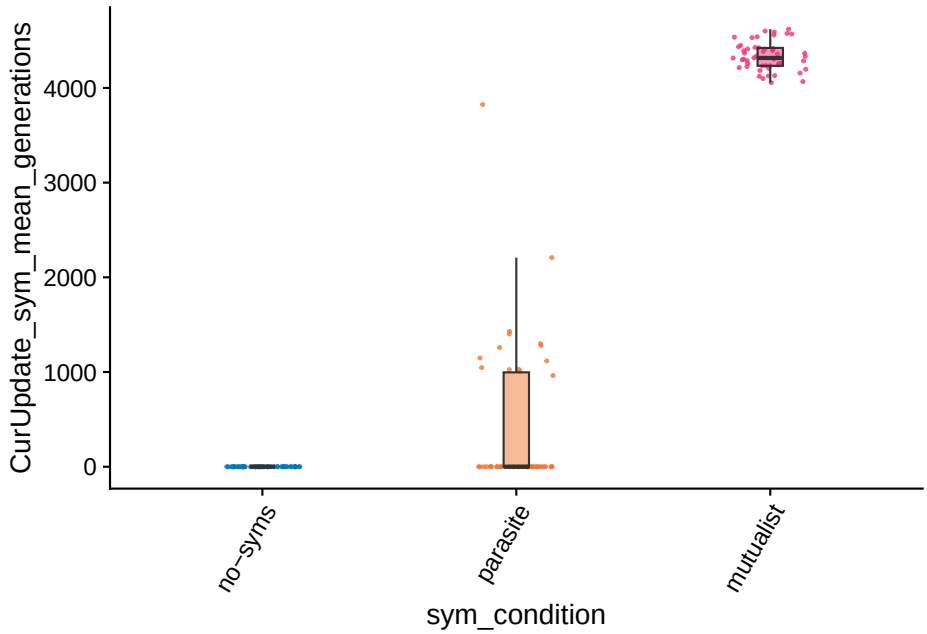
```
## # A tibble: 1 x 6
##   .y.                                n statistic    df        p method
## * <chr>                            <int>     <dbl> <int>    <dbl> <chr>
## 1 CurUpdate_host_mean_generations   150      73.6     2 1.06e-16 Kruskal-Wallis
```

```
pairwise.wilcox.test(
  x = summary_data$CurUpdate_host_mean_generations,
  g = summary_data$sym_condition,
  p.adjust.method = "bonferroni",
  exact = FALSE
)
```

```
##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  summary_data$CurUpdate_host_mean_generations and summary_data$sym_condition
##
##           no-syms parasite
## parasite  0.00048 -
## mutualist 2.1e-11 1.3e-12
##
## P value adjustment method: bonferroni
```

6.10.2 Symbionts

```
sym_avg_gens_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_mean_generations
    )
  ) +
  geom_point(
    mapping = aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_avg_gens_elapsed_plt.pdf", sep = "/"),
  plot = sym_avg_gens_elapsed_plt,
  width = 4,
  height = 4
)
sym_avg_gens_elapsed_plt
```




```
summary_data %>%
  group_by(
    sym_condition
  ) %>%
  summarise(
    mean_gens = mean(CurUpdate_sym_mean_generations),
    median_gens = median(CurUpdate_sym_mean_generations),
    var_gens = var(CurUpdate_sym_mean_generations),
    sdev_gens = sd(CurUpdate_sym_mean_generations)
  )
```

```
## # A tibble: 3 x 5
##   sym_condition mean_gens median_gens var_gens sdev_gens
##   <fct>             <dbl>       <dbl>    <dbl>     <dbl>
## 1 no-syms               0           0        0         0
## 2 parasite           401.           0   573648.      757.
## 3 mutualist         4333.        4317.   22198.      149.
```

6.11 Symbiont transmission rates

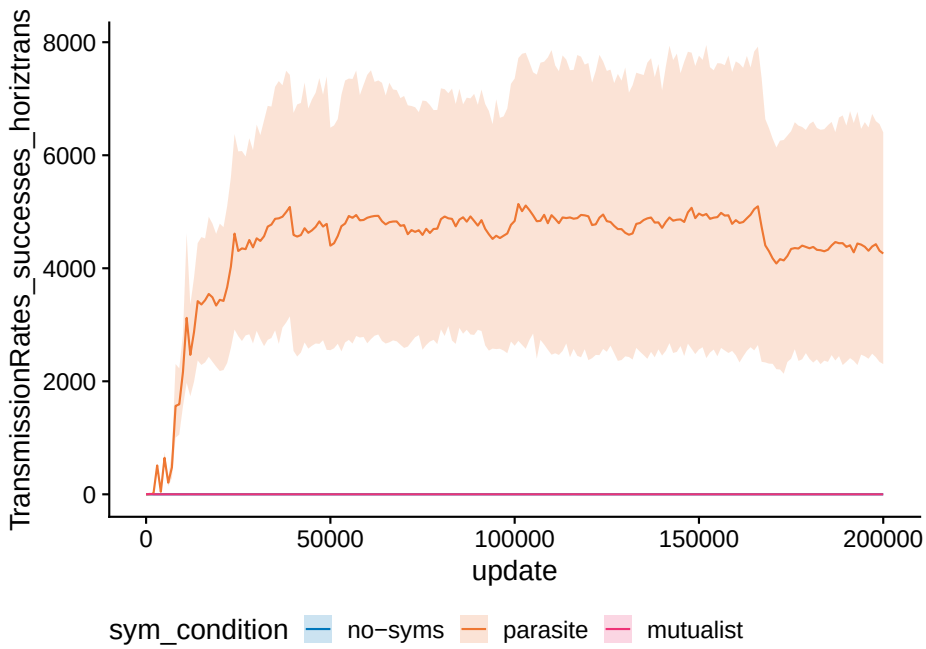
6.11.1 Horizontal transmission attempts

```
TransmissionRates_attempts_horiztrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_attempts_horiztrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_attempts_horiztrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_attempts_horiztrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_attempts_horiztrans_plt
```

6.11.2 Horizontal transmission successes

```
TransmissionRates_successes_horiztrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_successes_horiztrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
```

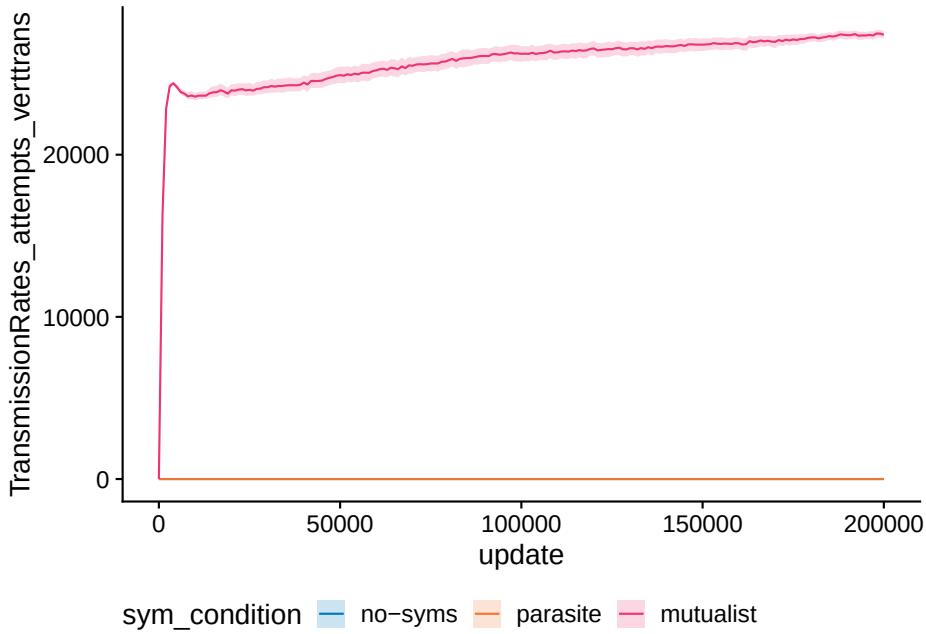
```
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_successes_horiztrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_successes_horiztrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_successes_horiztrans_plt
```



6.11.3 Vertical transmission attempts

```
TransmissionRates_attempts_verttrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_attempts_verttrans,
```

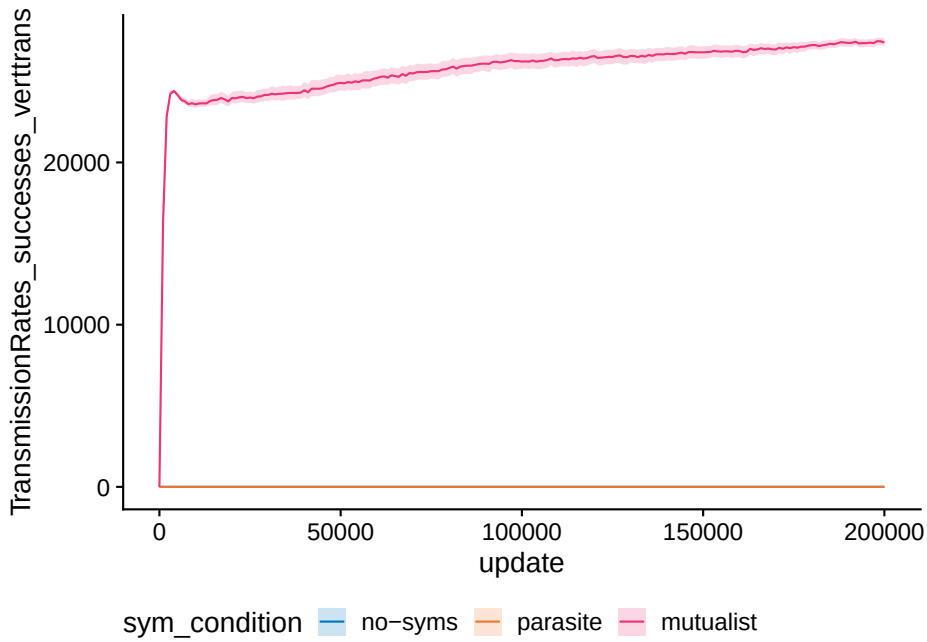
```
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_attempts_verttrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_attempts_verttrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_attempts_verttrans_plt
```



6.11.4 Successes vertical transmission

```
TransmissionRates_successes_verttrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_successes_verttrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
```

```
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_successes_verttrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_successes_verttrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_successes_verttrans_plt
```



6.12 Population size

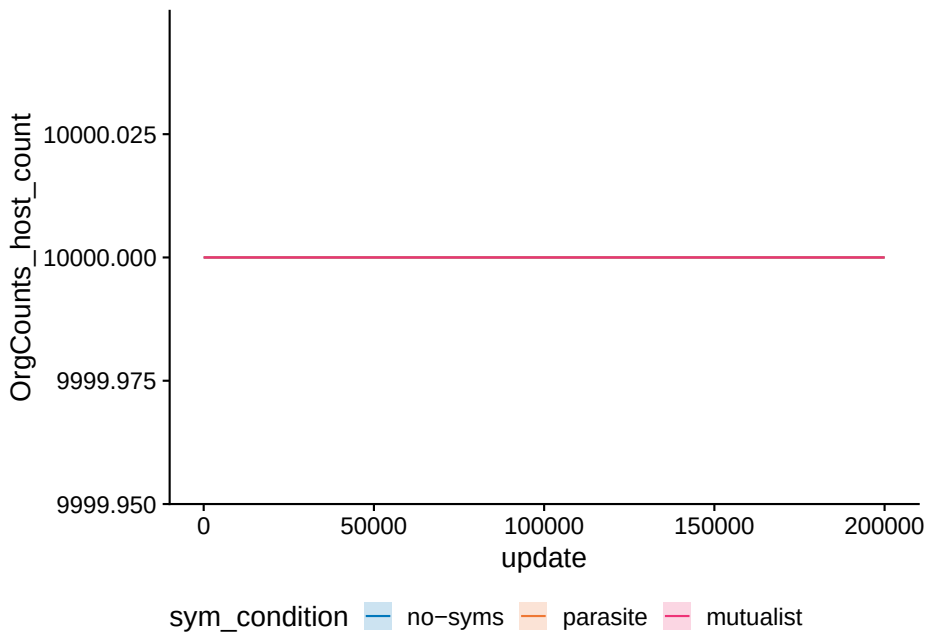
```
pop_size_summary <- time_data %>%
  group_by(
    SEED,
    sym_condition
  ) %>%
```

```
  summarize(
    host_pop_size_avg = mean(OrgCounts_host_count),
    host_pop_size_median = median(OrgCounts_host_count),
    host_pop_size_variance = var(OrgCounts_host_count),
    sym_pop_size_avg = mean(OrgCounts_hosted_sym_count),
    sym_pop_size_median = median(OrgCounts_hosted_sym_count),
    sym_pop_size_variance = var(OrgCounts_hosted_sym_count)
  )
```

6.12.1 Host population size

```
host_pop_size_ot_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = OrgCounts_host_count,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "host_pop_size_ot_plt.pdf",
    sep = "/"
  ),
  plot = host_pop_size_ot_plt,
  width = 8,
  height = 8
```

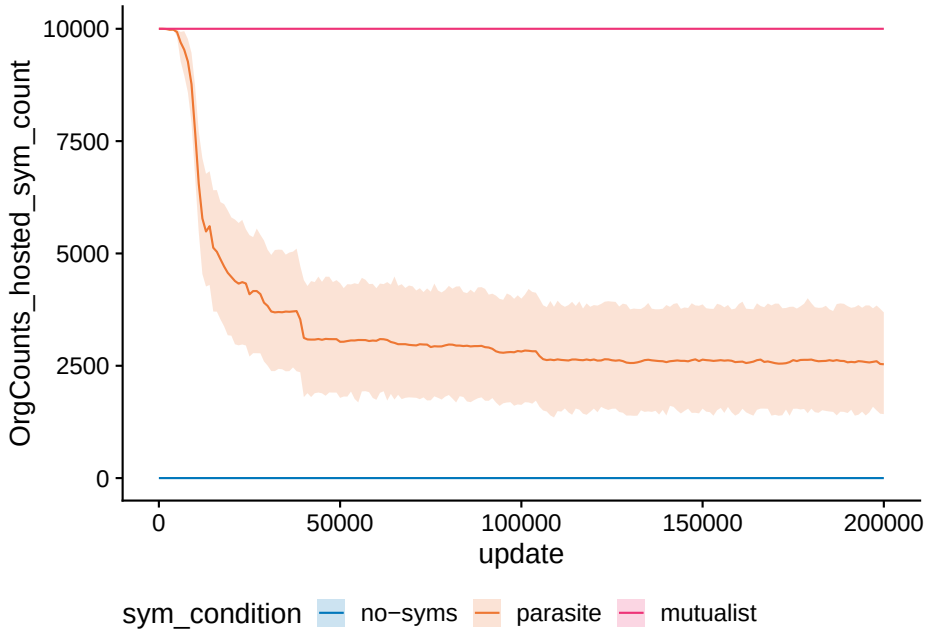
```
)
host_pop_size_ot_plt
```



6.12.2 Symbiont population size

```
sym_pop_size_ot_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = OrgCounts_hosted_sym_count,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
```

```
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "sym_pop_size_ot_plt.pdf",
    sep = "/"
  ),
  plot = sym_pop_size_ot_plt,
  width = 8,
  height = 8
)
sym_pop_size_ot_plt
```



6.13 Manuscript figures

Read cure data in for combined manuscript figure.

```
cure_max_task_complexity_data <- read_csv(
  file = paste(
    exp_directory,
    "data",
    "cure_host_max_task_complexity_over_time.csv",
    sep = "/"
  )
)
cure_max_task_complexity_data <- cure_max_task_complexity_data %>%
  mutate(
    cure_condition = factor(
      cure_condition,
      levels = c("No symbionts", "25%", "50%", "75%", "Not cured"),
      labels = c("No sym.", "25%", "50%", "75%", "Never")
    )
  )
```

```
host_max_task_complexity_data_ci <- host_max_task_complexity_data %>%
  group_by(
    sym_condition,
    interaction_multiplier
  ) %>%
  summarize(
    mean_max_task_complexity = mean(max_task_complexity),
    min_max_task_complexity = min(max_task_complexity),
    max_max_task_complexity = max(max_task_complexity),
    ci = list(
      mean_cl_boot(
        max_task_complexity,
        conf.int = 0.99,
        B = 10000,
      ) %>% rename(mean=y, lwr=ymin, upr=ymax)
    )
  ) %>%
  unnest(cols = c(ci)) %>%
  filter(
    interaction_multiplier == "4"
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist"),
      labels = c("No symbionts", "Parasites", "Mutualists")
    )
  )
```

```
dist_data <- host_max_task_complexity_data %>%
  filter(interaction_multiplier == "4") %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist"),
      labels = c("No symbionts", "Parasites", "Mutualists")
    )
  )

manuscript_host_task_complexity_plt <- host_max_task_complexity_data_ci %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = mean
    )
  ) +
  geom_bar(stat = "identity") +
  geom_point(
    data = dist_data,
    mapping = aes(x = sym_condition, y = max_task_complexity),
    position = position_jitter(width = .4, height = 0.05),
    size = .7,
    alpha = 0.9,
    color = "darkgray",
    fill = "darkgray"
  ) +
  geom_linerange(
    aes(
      ymin = lwr,
      ymax = upr
    ),
    alpha = 0.9,
    linewidth = 0.7,
    color = "black"
  ) +
  scale_x_discrete() +
  scale_fill_manual(values=color_pal) +
  scale_y_continuous(
    limits = c(0, 5.1)
  ) +
  labs(
    x = "Symbiont type",
    y = "Mean function complexity"
```

```
  ) +
  theme(
    legend.position = "none"
  )
```

```
manuscript_host_parent_entropy_task_sets_plt <- time_data %>%
  filter(
    # update %% 10000 == 0 &
    interaction_multiplier == "4"
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist"),
      labels = c("No symbionts", "Parasites", "Mutualists")
    )
  ) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.99),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_fill_manual(
    values = color_pal,
    limits = c("No symbionts", "Parasites", "Mutualists"),
    labels = c("None", "Par.", "Mut.")
  ) +
  scale_color_manual(
    values = color_pal,
    limits = c("No symbionts", "Parasites", "Mutualists"),
    labels = c("None", "Par.", "Mut.")
  ) +
  labs(
    x = "Updates",
```

```
    y = "Shannon diversity",
    color = "Symbiont type",
    fill = "Symbiont type"
  ) +
  theme(
    legend.position = "bottom"
  )
```

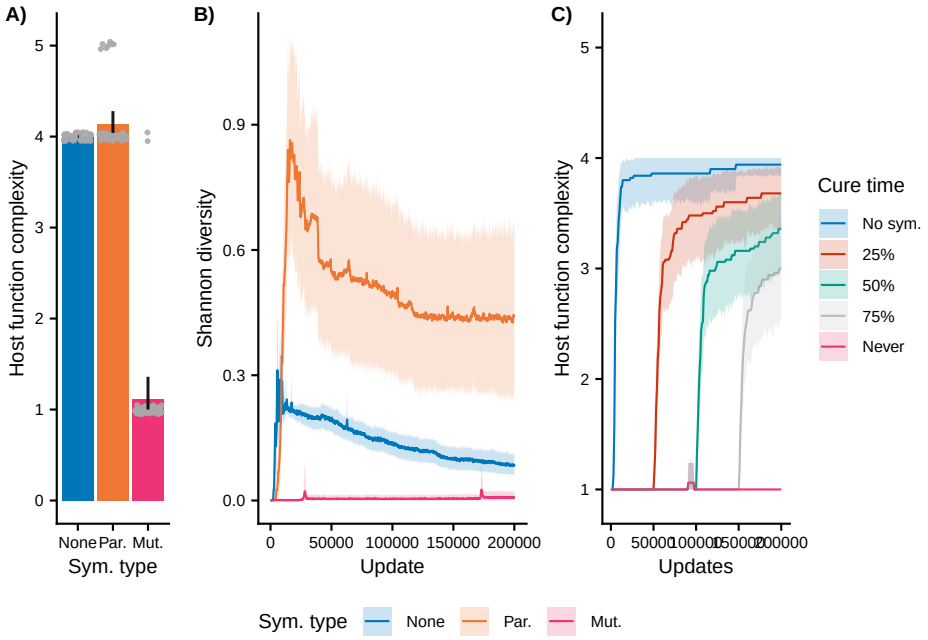
```
# Define cure color palette
# - no-syms -> blue
# - magenta -> mutualist no cure
# - 25% -> red
# - 50% -> teal
# - 75% -> grey
cure_color_pal <- c(
  vibrant_colors[["blue"]],
  vibrant_colors[["red"]],
  vibrant_colors[["teal"]],
  vibrant_colors[["grey"]],
  vibrant_colors[["magenta"]]
)
cure_host_max_task_complexity_over_time_plt <- cure_max_task_complexity_data %>%
  filter(
    update %% 1000 == 0
  ) %>%
  ggplot(
    aes(
      x = update,
      y = max_task_complexity,
      color = cure_condition,
      fill = cure_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.99),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_fill_manual(
    values = cure_color_pal
  ) +
  scale_color_manual(values = cure_color_pal) +
```

```
  scale_y_continuous(
    limits = c(0.9, 5.1)
  ) +
  labs(
    x = "Updates",
    y = "Host function complexity",
    color = "Cure time",
    fill = "Cure time"
  ) +
  theme(
    legend.position = "bottom"
  )

ggsave(
  file = paste(
    plot_dir,
    "cure_host_max_task_complexity_over_time_plt.pdf",
    sep = "/"
  ),
  plot = cure_host_max_task_complexity_over_time_plt,
  width = 6,
  height = 4
)
```

6.13.1 Combine into single panel

```
plt_grid_with_cure <- plot_grid(
  manuscript_host_task_complexity_plt +
    labs(
      x = "Sym. type",
      y = "Host function complexity",
      color = "Sym. type",
      fill = "Sym. type"
    ) +
    scale_x_discrete(
      limits = c("No symbionts", "Parasites", "Mutualists"),
      labels = c("None", "Par.", "Mut.")
    ) +
    theme(
      legend.position = "none",
      axis.title.y = element_text(size = 10),
      axis.text.y = element_text(size = 8),
      axis.title.x = element_text(size = 10),
```

```
      axis.text.x = element_text(size = 8)
    ),
  manuscript_host_parent_entropy_task_sets_plt +
    labs(
      x = "Update",
      y = "Shannon diversity",
      color = "Sym. type",
      fill = "Sym. type"
    ) +
    theme(
      legend.position = "bottom",
      legend.title = element_text(size = 10),
      legend.text = element_text(size = 8),
      axis.title.y = element_text(size = 10),
      axis.text.y = element_text(size = 8),
      axis.title.x = element_text(size = 10),
      axis.text.x = element_text(size = 8)
    ),
  cure_host_max_task_complexity_over_time_plt +
    theme(
      legend.position = "right",
      legend.title = element_text(size = 10),
      legend.text = element_text(size = 8),
      axis.title.y = element_text(size = 10),
      axis.text.y = element_text(size = 8),
      axis.title.x = element_text(size = 10),
      axis.text.x = element_text(size = 8)
    ),
  nrow = 1,
  ncol = 3,
  align = "h",
  labels = c("A)", "B)", "C)"),
  label_size = 10,
  rel_widths = c(0.2, 0.38, 0.42)
)
plt_grid_with_cure
```




```
save_plot(
  filename = paste(
    plot_dir,
    "fig-diff-rewards-panel-with-cure.pdf",
    sep = "/"
  ),
  plot = plt_grid_with_cure,
  base_width = 8,
  base_height = 3.5,
  dpi = 600
)
```

Chapter 7

Environment strength

Repeat differential function rewards experiment, except vary the reward level of complex functions.

7.1 Dependencies and setup

Load R library dependencies.

```
library(tidyverse)
library(cowplot)
library(khroma)
library(rstatix)
library(knitr)
library(kableExtra)
```

Define relevant file paths.

```
# Check if Rmd is being compiled using bookdown
bookdown <- exists("bookdown_build")

# Build relative path from R's working directory to what is actually the working
# directory for this experiment.
experiment_slug <- "2026-02-23-health-task-rewards"
supp_file_prefix <- "task-rewards"
exp_directory <- paste(
  "experiments",
  experiment_slug,
  "analysis",
```

```
  sep = "/"
)

# Get path to the supplemental material analysis directory
supp_directory <- paste(
  "experiments",
  "manuscript-analyses",
  sep = "/"
)

# Adjust experiment directory if being knitted for bookdown build
if (bookdown) {
  exp_directory <- paste0(
    bookdown_wd_prefix,
    exp_directory
  )
  supp_directory <- paste0(
    bookdown_wd_prefix,
    supp_directory
  )
}
```

Define default plot theme, plot output directory.

```
# Use the cowplot theme as our default for ggplots
theme_set(theme_cowplot())
# Create directory to save plots
plot_dir <- paste(
  supp_directory,
  "plots",
  supp_file_prefix,
  sep = "/"
)
dir.create(
  plot_dir,
  showWarnings = FALSE
)

# Use vibrant theme for plot color palettes
vibrant_colors <- color("vibrant")(7)
color_pal <- c(
  vibrant_colors[["blue"]],
  vibrant_colors[["magenta"]]
)
```

7.2 Load and preprocess data

```
tasks <- c("NOT", "NAND", "OR_NOT", "AND", "OR", "AND_NOT", "NOR", "XOR", "EQU")
max_pop_size <- 100 * 100
threshold <- 0.025 * max_pop_size
```

Load summary data.

```
summary_data_path <- paste(
  exp_directory,
  "data",
  "summary.csv",
  sep = "/"
)
summary_data <- read_csv(summary_data_path)
# Do any universal preprocessing
summary_data <- summary_data %>%
  mutate(
    SEED = as.factor(SEED),
    task_credit = paste0(
      "H", HOST_ONLY_FIRST_TASK_CREDIT,
      "S", SYM_ONLY_FIRST_TASK_CREDIT
    ),
    sym_condition = case_when(
      START_MOI == "0" ~ "no-syms",
      START_MOI == "1" & HEALTH_TYPE == "parasite" ~ "parasite",
      START_MOI == "1" & HEALTH_TYPE == "mutualist" ~ "mutualist",
      .default = "unknown" # precaution against unexpected parameterizations
    ),
    environment = case_when(
      TASK_ENV_CFG_PATH == "environment-rewards-p5.json" ~ "x0.5",
      TASK_ENV_CFG_PATH == "environment-rewards-1.json" ~ "x1",
      TASK_ENV_CFG_PATH == "environment-rewards-2.json" ~ "x2",
      TASK_ENV_CFG_PATH == "environment-rewards-4.json" ~ "x4",
      TASK_ENV_CFG_PATH == "environment-rewards-8.json" ~ "x8",
      TASK_ENV_CFG_PATH == "environment-rewards-16.json" ~ "x16",
      .default = "unknown"
    )
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist")
    ),
```

```
    environment = factor(
      environment,
      levels = c("x0.5", "x1", "x2", "x4", "x8", "x16")
    )
  )
```

Load time series data.

```
time_data_path <- paste(
  exp_directory,
  "data",
  "time_series.csv",
  sep = "/"
)
time_data <- read_csv(time_data_path)

# Do any universal preprocessing
time_data <- time_data %>%
  mutate(
    SEED = as.factor(SEED),
    sym_condition = case_when(
      START_MOI == "0" ~ "no-syms",
      START_MOI == "1" & HEALTH_TYPE == "parasite" ~ "parasite",
      START_MOI == "1" & HEALTH_TYPE == "mutualist" ~ "mutualist",
      .default = "unknown" # precaution against unexpected parameterizations
    ),
    environment = case_when(
      TASK_ENV_CFG_PATH == "environment-rewards-p5.json" ~ "x0.5",
      TASK_ENV_CFG_PATH == "environment-rewards-1.json" ~ "x1",
      TASK_ENV_CFG_PATH == "environment-rewards-2.json" ~ "x2",
      TASK_ENV_CFG_PATH == "environment-rewards-4.json" ~ "x4",
      TASK_ENV_CFG_PATH == "environment-rewards-8.json" ~ "x8",
      TASK_ENV_CFG_PATH == "environment-rewards-16.json" ~ "x16",
      .default = "unknown"
    )
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist")
    ),
    environment = factor(
      environment,
      levels = c("x0.5", "x1", "x2", "x4", "x8", "x16")
    )
```

```
  )
```

7.3 Extinctions

Did host or symbiont populations go extinct in any runs?

```
sym_min_pop_sizes <- time_data %>%
  group_by(SEED) %>%
  slice_min(
    OrgCounts_hosted_sym_count,
    n = 1
  ) %>%
  slice_min(
    update,
    n = 1
  )

reps_with_extinct_syms <- sym_min_pop_sizes %>%
  group_by(
    sym_condition,
    environment
  ) %>%
  summarize(
    reps_with_extinct_syms = sum(OrgCounts_hosted_sym_count == 0),
    total_reps = n()
  )

host_min_pop_sizes <- time_data %>%
  group_by(SEED) %>%
  slice_min(
    OrgCounts_host_count,
    n = 1
  ) %>%
  slice_min(
    update,
    n = 1
  )

reps_with_extinct_hosts <- host_min_pop_sizes %>%
  group_by(
    sym_condition,
    environment
  ) %>%
```

```
  summarize(
    reps_with_extinct_hosts = sum(OrgCounts_host_count == 0),
    total_reps = n()
  )

extinctions <- full_join(
  reps_with_extinct_syms,
  reps_with_extinct_hosts
)
extinctions <- extinctions %>%
  arrange(
    desc(reps_with_extinct_hosts), desc(reps_with_extinct_syms)
  )
extinction_table <- kable(extinctions) %>%
  kable_styling(latex_options = "striped")
save_kable(
  extinction_table,
  paste(
    plot_dir,
    "extinctions.pdf",
    sep = "/"
  )
)

extinctions
```

```
## # A tibble: 12 x 5
## # Groups:   sym_condition [2]
##    sym_condition environment reps_with_extinct_syms total_reps
##    <fct>         <fct>                        <int>      <int>
##  1 no-syms       x0.5                            50         50
##  2 no-syms       x1                              50         50
##  3 no-syms       x2                              50         50
##  4 no-syms       x4                              50         50
##  5 no-syms       x8                              50         50
##  6 no-syms       x16                             50         50
##  7 mutualist     x0.5                             0         50
##  8 mutualist     x1                               0         50
##  9 mutualist     x2                               0         50
## 10 mutualist     x4                               0         50
## 11 mutualist     x8                               0         50
## 12 mutualist     x16                              0         50
## # i 1 more variable: reps_with_extinct_hosts <int>
```

7.4 Host function performance

First, quantify which functions have been performed over the course of an entire run.

```
# Pivot task data long, selecting only relevant columns
host_over_time_parent_task_data <- time_data %>%
  select(
    SEED,
    sym_condition,
    environment,
    update,
    CurUpdate_AND_NOT_in_host_parent_org_counts,
    CurUpdate_AND_in_host_parent_org_counts,
    CurUpdate_EQU_in_host_parent_org_counts,
    CurUpdate_NAND_in_host_parent_org_counts,
    CurUpdate_NOR_in_host_parent_org_counts,
    CurUpdate_NOT_in_host_parent_org_counts,
    CurUpdate_OR_NOT_in_host_parent_org_counts,
    CurUpdate_OR_in_host_parent_org_counts,
    CurUpdate_XOR_in_host_parent_org_counts
  ) %>%
  pivot_longer(
    c(
      CurUpdate_AND_NOT_in_host_parent_org_counts,
      CurUpdate_AND_in_host_parent_org_counts,
      CurUpdate_EQU_in_host_parent_org_counts,
      CurUpdate_NAND_in_host_parent_org_counts,
      CurUpdate_NOR_in_host_parent_org_counts,
      CurUpdate_NOT_in_host_parent_org_counts,
      CurUpdate_OR_NOT_in_host_parent_org_counts,
      CurUpdate_OR_in_host_parent_org_counts,
      CurUpdate_XOR_in_host_parent_org_counts
    ),
    names_to = "task",
    values_to = "task_in_parent_count"
  ) %>%
  mutate(
    task = str_replace(task, "CurUpdate_", "")
  ) %>%
  mutate(
    task = str_replace(task, "_in_host_parent_org_counts", "")
  ) %>%
  mutate(
    task_completed_by_parent = task_in_parent_count >= threshold,
```

```
    task = factor(
      task,
      levels = tasks,
      labels = tasks
    )
  ) %>%
  mutate(
    task_complexity = case_when(
      task == "NOT" ~ 1,
      task == "NAND" ~ 1,
      task == "OR_NOT" ~ 2,
      task == "AND" ~ 2,
      task == "OR" ~ 3,
      task == "AND_NOT" ~ 3,
      task == "NOR" ~ 4,
      task == "XOR" ~ 4,
      task == "EQU" ~ 5,
      .default = -1
    )
  )

host_parent_tasks_all_time_by_task <- host_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    environment,
    task
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  group_by(
    sym_condition,
    environment,
    task
  ) %>%
  summarize(
    n = n(),
    parent_tasks_completed_ever = sum(parent_task_completed_ever == TRUE)
  )
```

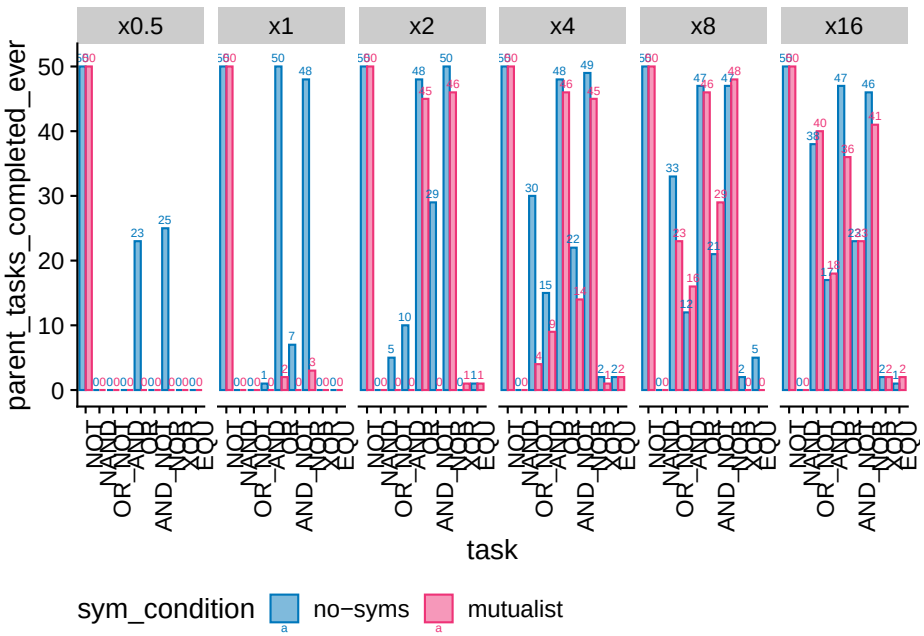
Next, summarize function complexity level.

```
host_max_task_complexity_data <- host_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    environment,
    task,
    task_complexity
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  # This filter works fine because all replicates start with a task
  filter(
    parent_task_completed_ever
  ) %>%
  group_by(
    SEED,
    sym_condition,
    environment
  ) %>%
  summarize(
    max_task_complexity = max(task_complexity)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c(
        "no-syms",
        "parasite",
        "mutualist"
      )
    )
  )
```

7.4.1 Host function performance - all time

```
host_parent_tasks_all_time_by_task_plt <- host_parent_tasks_all_time_by_task %>%
  ggplot(
    aes(
      x = task,
      color = sym_condition,
```

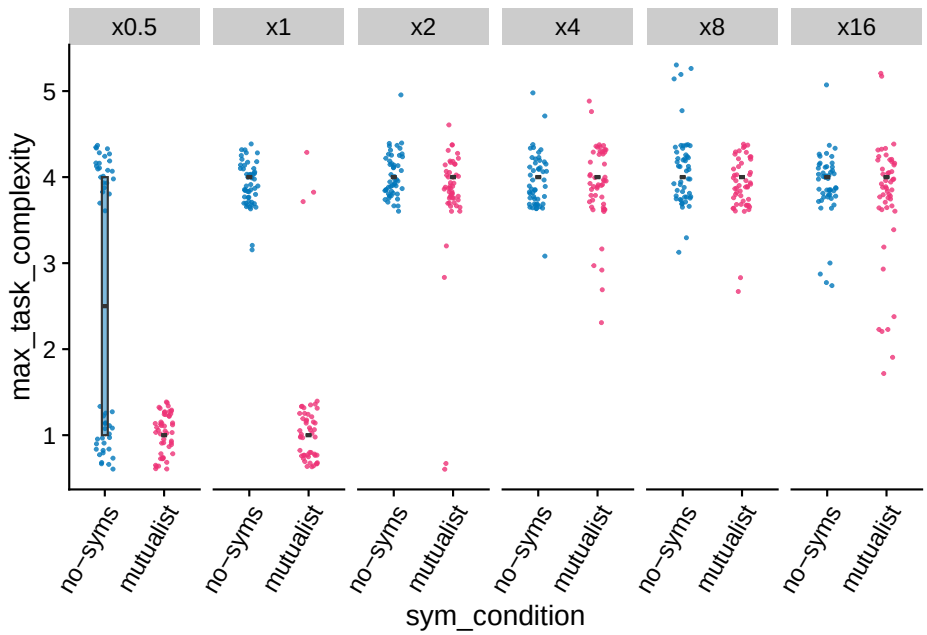
```
      fill = sym_condition,
      y = parent_tasks_completed_ever
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = parent_tasks_completed_ever),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  facet_wrap(~environment, nrow = 1) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(
      angle = 90,
      hjust = 1
    )
  )
ggsave(
  file = paste(
    plot_dir,
    "host_parent_tasks_all_time_by_task_plt.pdf",
    sep = "/"
  ),
  plot = host_parent_tasks_all_time_by_task_plt,
  width = 10,
  height = 7
)
host_parent_tasks_all_time_by_task_plt
```



7.4.2 Host maximum function complexity evolved

```
host_max_task_complexity_plt <- host_max_task_complexity_data %>%
  ggplot(
    aes(x = sym_condition, fill = sym_condition, y = max_task_complexity)
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, nrow = 1) +
  theme(
    legend.position = "none",
```

```
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_max_task_complexity_plt.pdf", sep = "/"),
  plot = host_max_task_complexity_plt,
  width = 4,
  height = 4
)
host_max_task_complexity_plt
```



7.5 Symbiont function performance

```
# Pivot task data long, selecting only relevant columns
sym_over_time_parent_task_data <- time_data %>%
  select(
    SEED,
    sym_condition,
    environment,
```

```
    update,
    CurUpdate_AND_NOT_in_sym_parent_org_counts,
    CurUpdate_AND_in_sym_parent_org_counts,
    CurUpdate_EQU_in_sym_parent_org_counts,
    CurUpdate_NAND_in_sym_parent_org_counts,
    CurUpdate_NOR_in_sym_parent_org_counts,
    CurUpdate_NOT_in_sym_parent_org_counts,
    CurUpdate_OR_NOT_in_sym_parent_org_counts,
    CurUpdate_OR_in_sym_parent_org_counts,
    CurUpdate_XOR_in_sym_parent_org_counts
  ) %>%
  pivot_longer(
    c(
      CurUpdate_AND_NOT_in_sym_parent_org_counts,
      CurUpdate_AND_in_sym_parent_org_counts,
      CurUpdate_EQU_in_sym_parent_org_counts,
      CurUpdate_NAND_in_sym_parent_org_counts,
      CurUpdate_NOR_in_sym_parent_org_counts,
      CurUpdate_NOT_in_sym_parent_org_counts,
      CurUpdate_OR_NOT_in_sym_parent_org_counts,
      CurUpdate_OR_in_sym_parent_org_counts,
      CurUpdate_XOR_in_sym_parent_org_counts
    ),
    names_to = "task",
    values_to = "task_in_parent_count"
  ) %>%
  mutate(
    task = str_replace(task, "CurUpdate_", "")
  ) %>%
  mutate(
    task = str_replace(task, "_in_sym_parent_org_counts", "")
  ) %>%
  mutate(
    task_completed_by_parent = task_in_parent_count >= threshold,
    task = factor(
      task,
      levels = tasks,
      labels = tasks
    )
  ) %>%
  mutate(
    task_complexity = case_when(
      task == "NOT" ~ 1,
      task == "NAND" ~ 1,
      task == "OR_NOT" ~ 2,
```

```
      task == "AND" ~ 2,
      task == "OR" ~ 3,
      task == "AND_NOT" ~ 3,
      task == "NOR" ~ 4,
      task == "XOR" ~ 4,
      task == "EQU" ~ 5,
      .default = -1
    )
  )

sym_parent_tasks_all_time_by_task <- sym_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    environment,
    task
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  group_by(
    sym_condition,
    environment,
    task
  ) %>%
  summarize(
    n = n(),
    parent_tasks_completed_ever = sum(parent_task_completed_ever == TRUE)
  )
```

Next, summarize function complexity level.

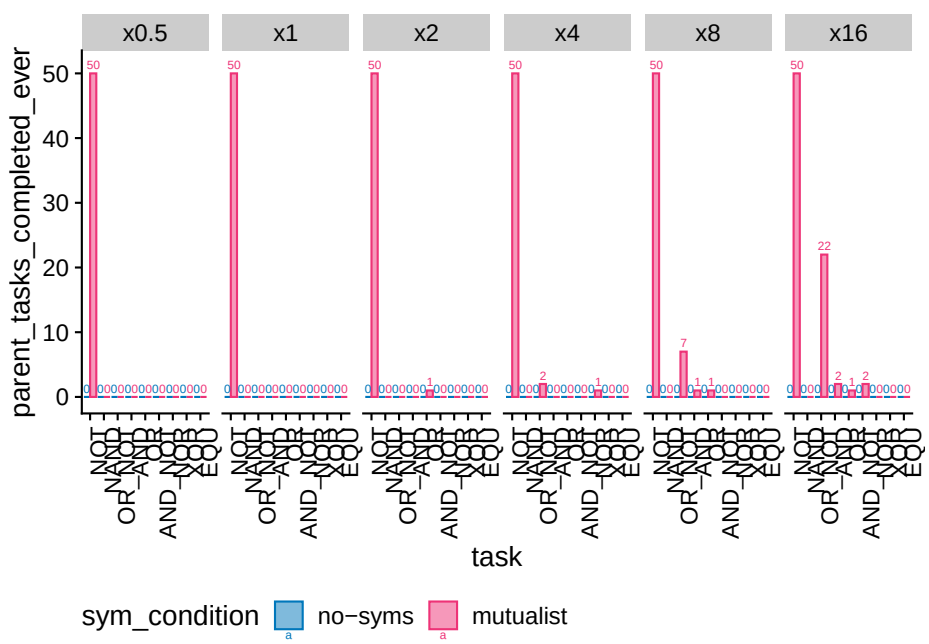
```
sym_max_task_complexity_data <- sym_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    environment,
    task,
    task_complexity
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
```

```
  ) %>%
  ungroup() %>%
  # This filter works fine because all replicates start with a task
  filter(
    parent_task_completed_ever
  ) %>%
  group_by(
    SEED,
    sym_condition,
    environment
  ) %>%
  summarize(
    max_task_complexity = max(task_complexity)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c(
        "no-syms",
        "parasite",
        "mutualist"
      )
    )
  )
```

7.5.1 Symbiont function performance - all time

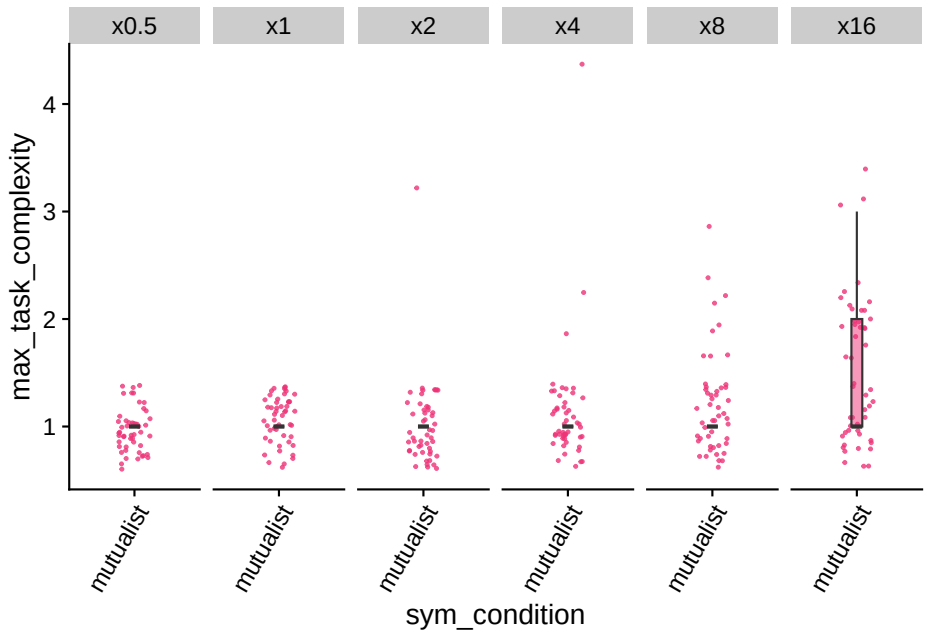
```
sym_parent_tasks_all_time_by_task_plt <- sym_parent_tasks_all_time_by_task %>%
  ggplot(
    aes(
      x = task,
      color = sym_condition,
      fill = sym_condition,
      y = parent_tasks_completed_ever
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = parent_tasks_completed_ever),
    position = position_dodge(width = 1),
```

```
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, nrow = 1) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(
      angle = 90,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_parent_tasks_all_time_by_task_plt.pdf", sep = "/"),
  plot = sym_parent_tasks_all_time_by_task_plt,
  width = 8,
  height = 8
)
sym_parent_tasks_all_time_by_task_plt
```



7.5.2 Symbiont maximum function complexity evolved

```
sym_max_task_complexity_plt <- sym_max_task_complexity_data %>%
  ggplot(
    aes(x = sym_condition, fill = sym_condition, y = max_task_complexity)
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = c(color_pal[2], color_pal[3])) +
  scale_fill_manual(values = c(color_pal[2], color_pal[3])) +
  facet_wrap(~environment, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_max_task_complexity_plt.pdf", sep = "/"),
  plot = sym_max_task_complexity_plt,
  width = 4,
  height = 4
)
sym_max_task_complexity_plt
```



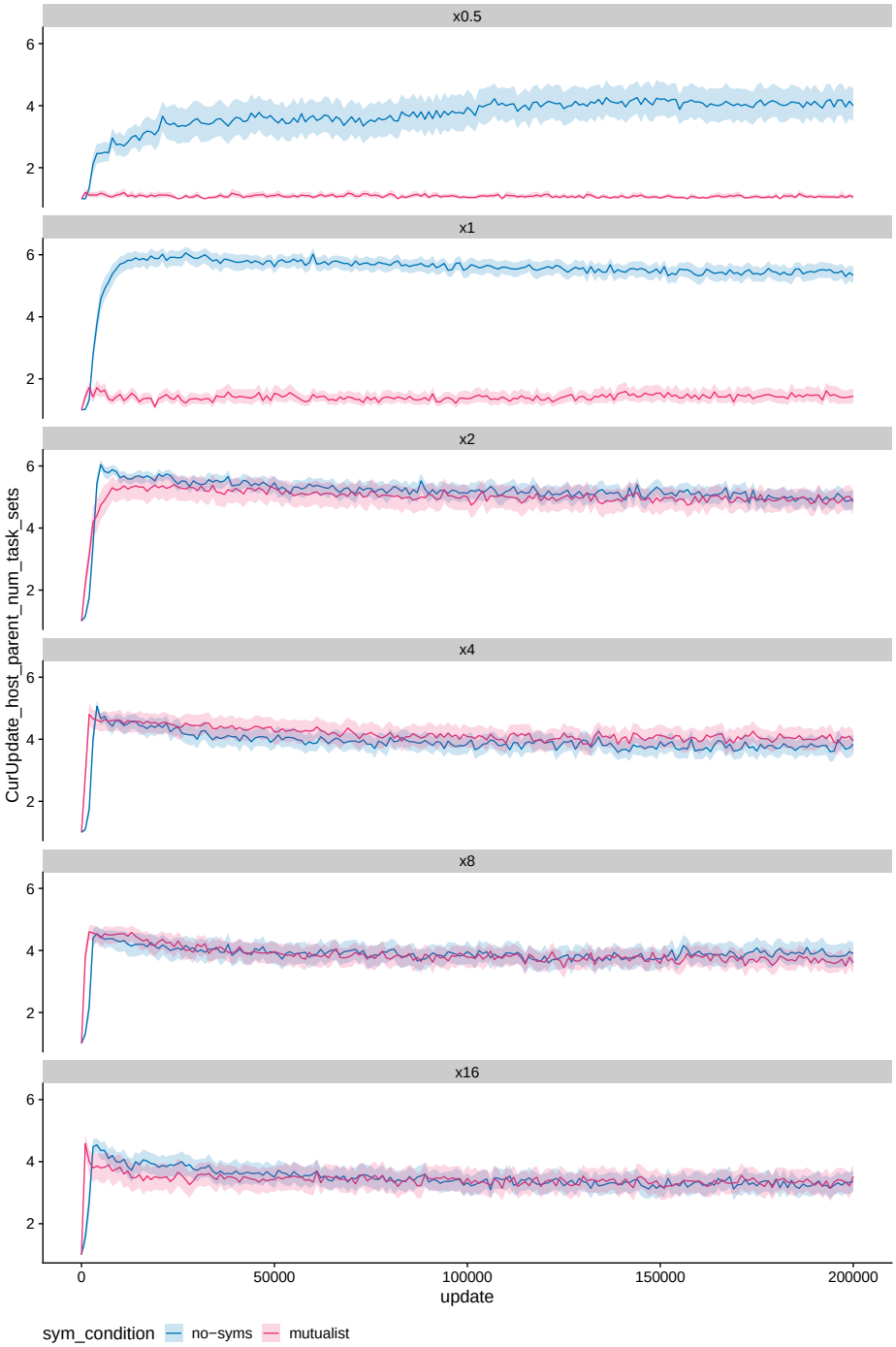
7.6 Host phenotypic diversity

7.6.1 Number of distinct function profiles

Over time:

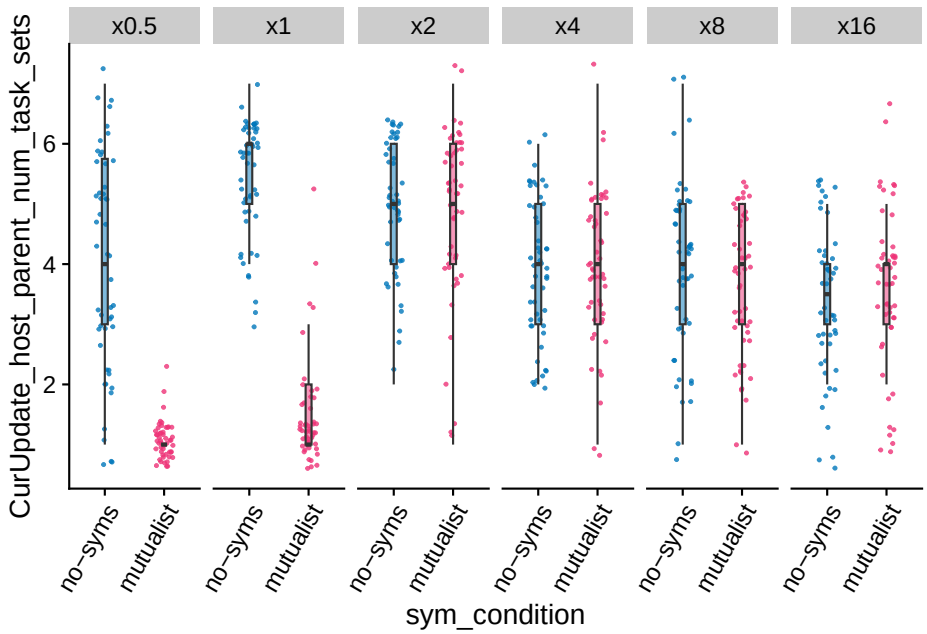
```
host_parent_num_task_sets_plt <- time_data %>%
  # sample to make plotting faster
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_parent_num_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
```

```
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_parent_num_task_sets_plt.pdf", sep = "/"),
  plot = host_parent_num_task_sets_plt,
  width = 8,
  height = 8
)
host_parent_num_task_sets_plt
```



At the end of the experiment:

```
host_final_num_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_parent_num_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_final_num_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = host_final_num_task_sets_elapsed_plt,
  width = 4,
  height = 4
)
host_final_num_task_sets_elapsed_plt
```

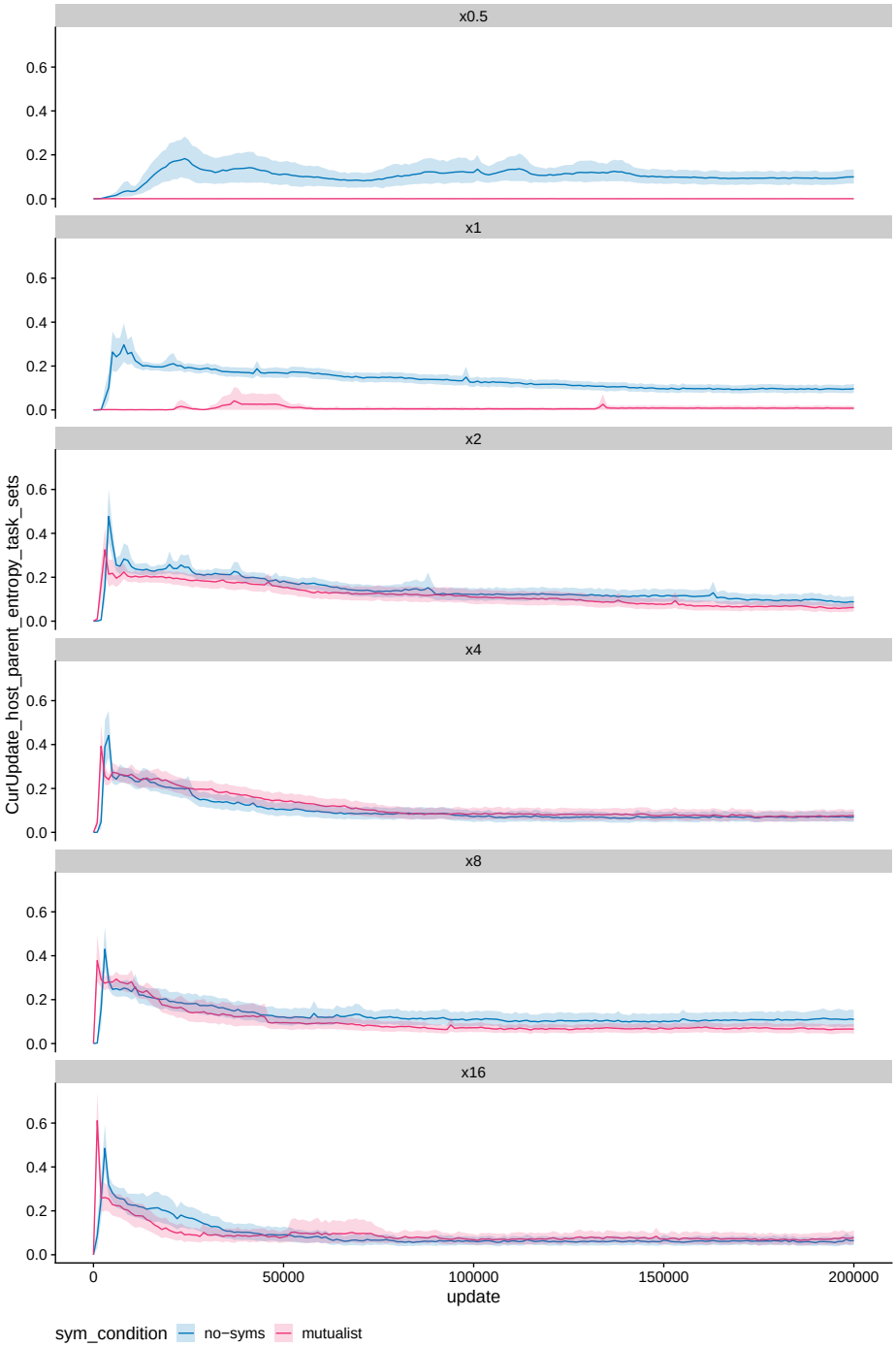


7.6.2 Shannon diversity of function profiles

Over time:

```
host_parent_entropy_task_sets_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
```

```
  facet_wrap(~environment, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_parent_entropy_task_sets_plt.pdf", sep = "/"),
  plot = host_parent_entropy_task_sets_plt,
  width = 8,
  height = 8
)
host_parent_entropy_task_sets_plt
```



At the end of the experiment:

```
host_final_entopy_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_parent_entropy_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  facet_wrap(~environment, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_final_entopy_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = host_final_entopy_task_sets_elapsed_plt,
  width = 10,
  height = 10
)
```

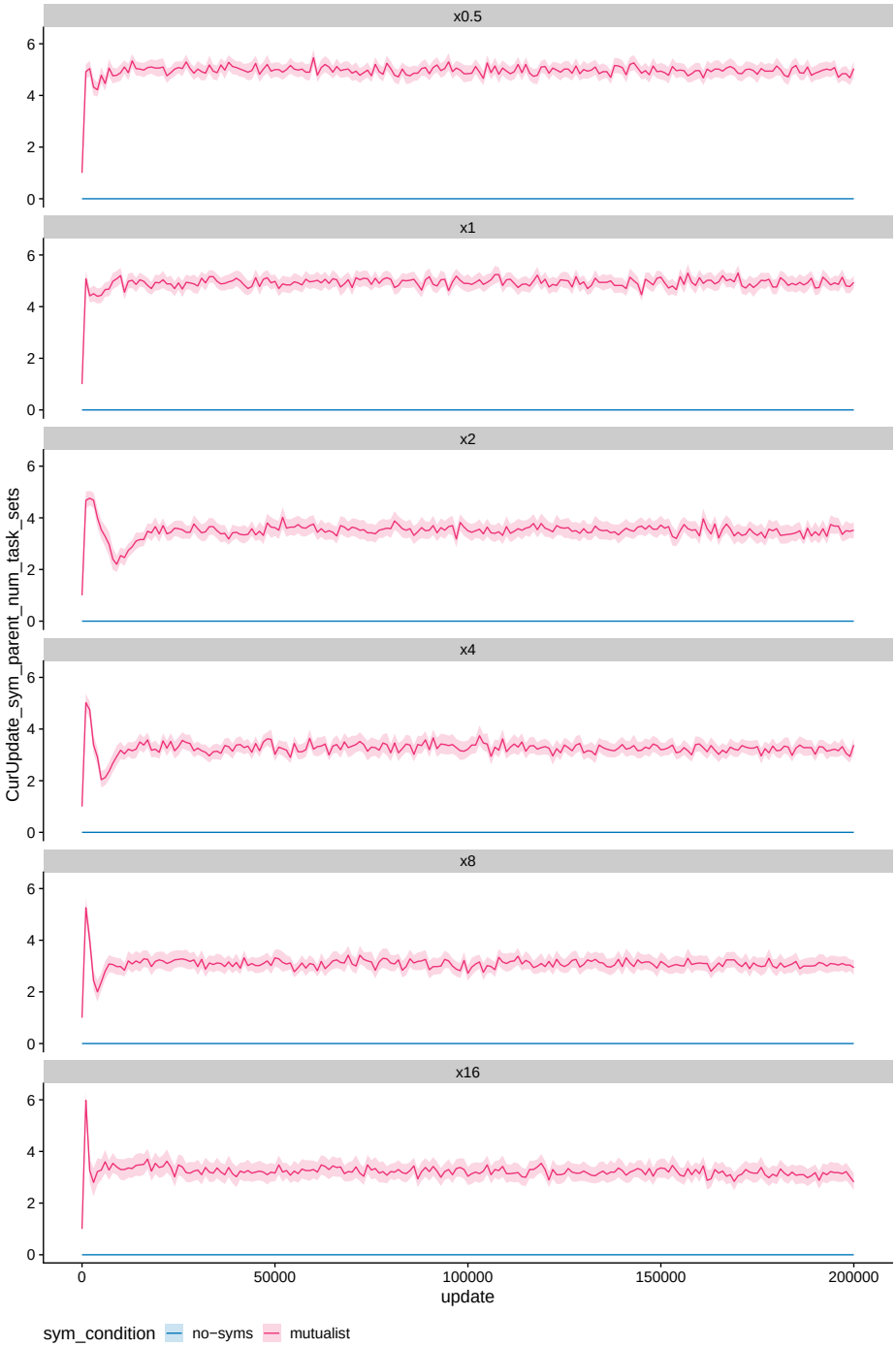
7.7 Symbiont phenotypic diversity

7.7.1 Number of distinct function profiles

Over time:

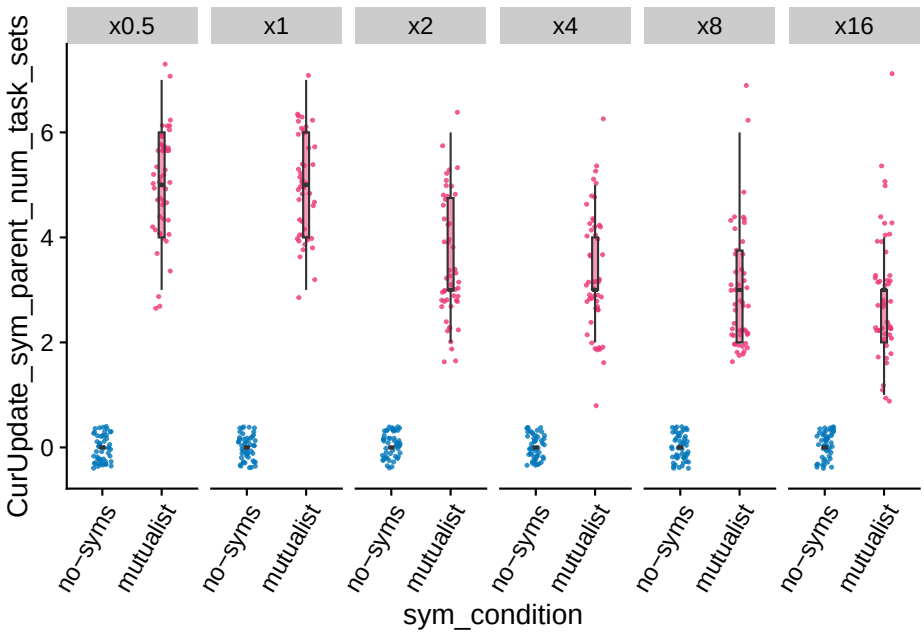
```
sym_parent_num_task_sets_plt <- time_data %>%
  # sample to make plotting faster
```

```
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_sym_parent_num_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "sym_parent_num_task_sets_plt.pdf", sep = "/"),
  plot = sym_parent_num_task_sets_plt,
  width = 8,
  height = 8
)
sym_parent_num_task_sets_plt
```



At the end of the experiment:

```
sym_final_num_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_parent_num_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_final_num_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = sym_final_num_task_sets_elapsed_plt,
  width = 4,
  height = 4
)
sym_final_num_task_sets_elapsed_plt
```

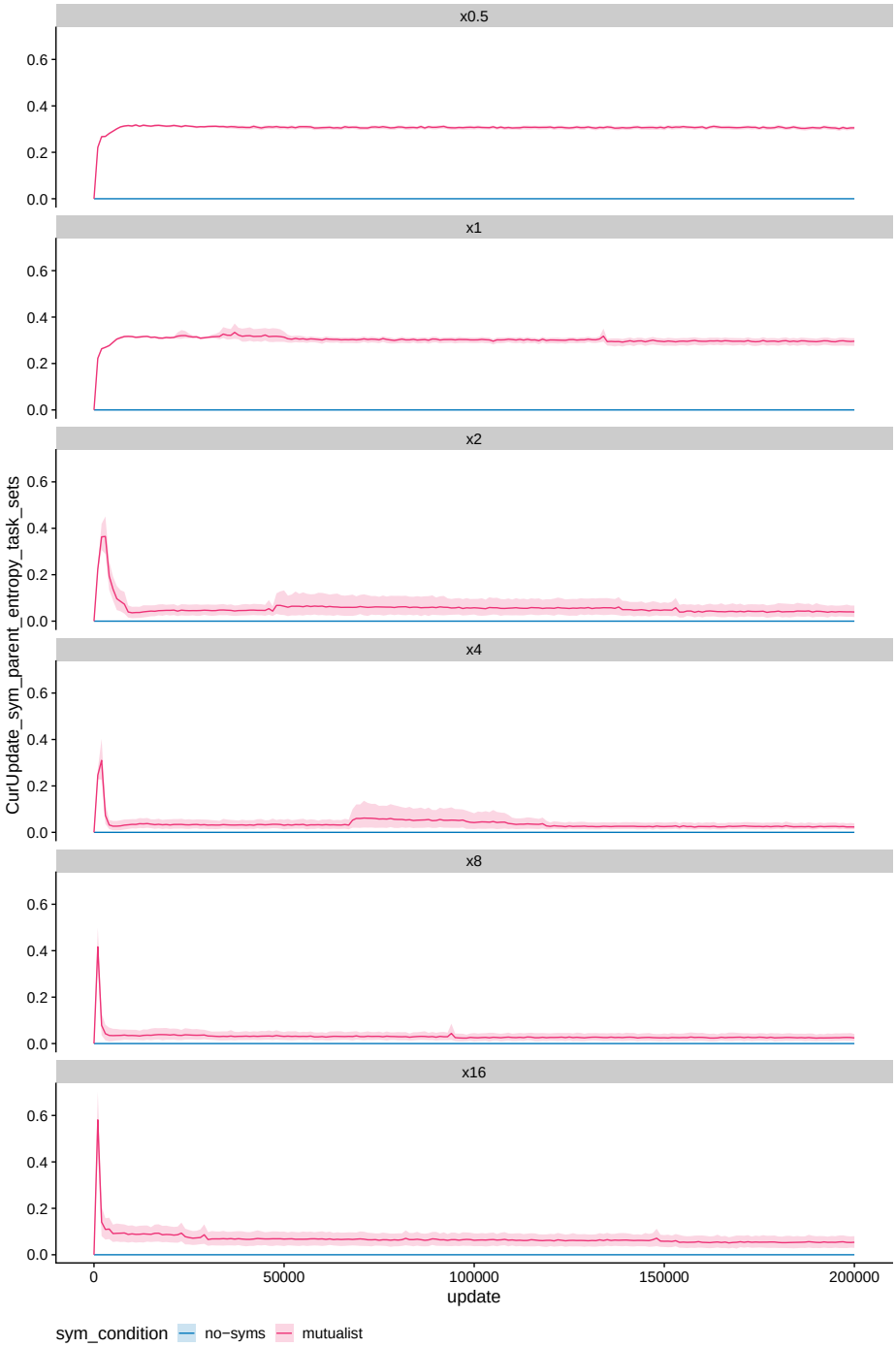


7.7.2 Shannon diversity of function profiles

Over time:

```
sym_parent_entropy_task_sets_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_sym_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
```

```
  facet_wrap(~environment, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "sym_parent_entropy_task_sets_plt.pdf", sep = "/"),
  plot = sym_parent_entropy_task_sets_plt,
  width = 8,
  height = 8
)
sym_parent_entropy_task_sets_plt
```



At the end of the experiment:

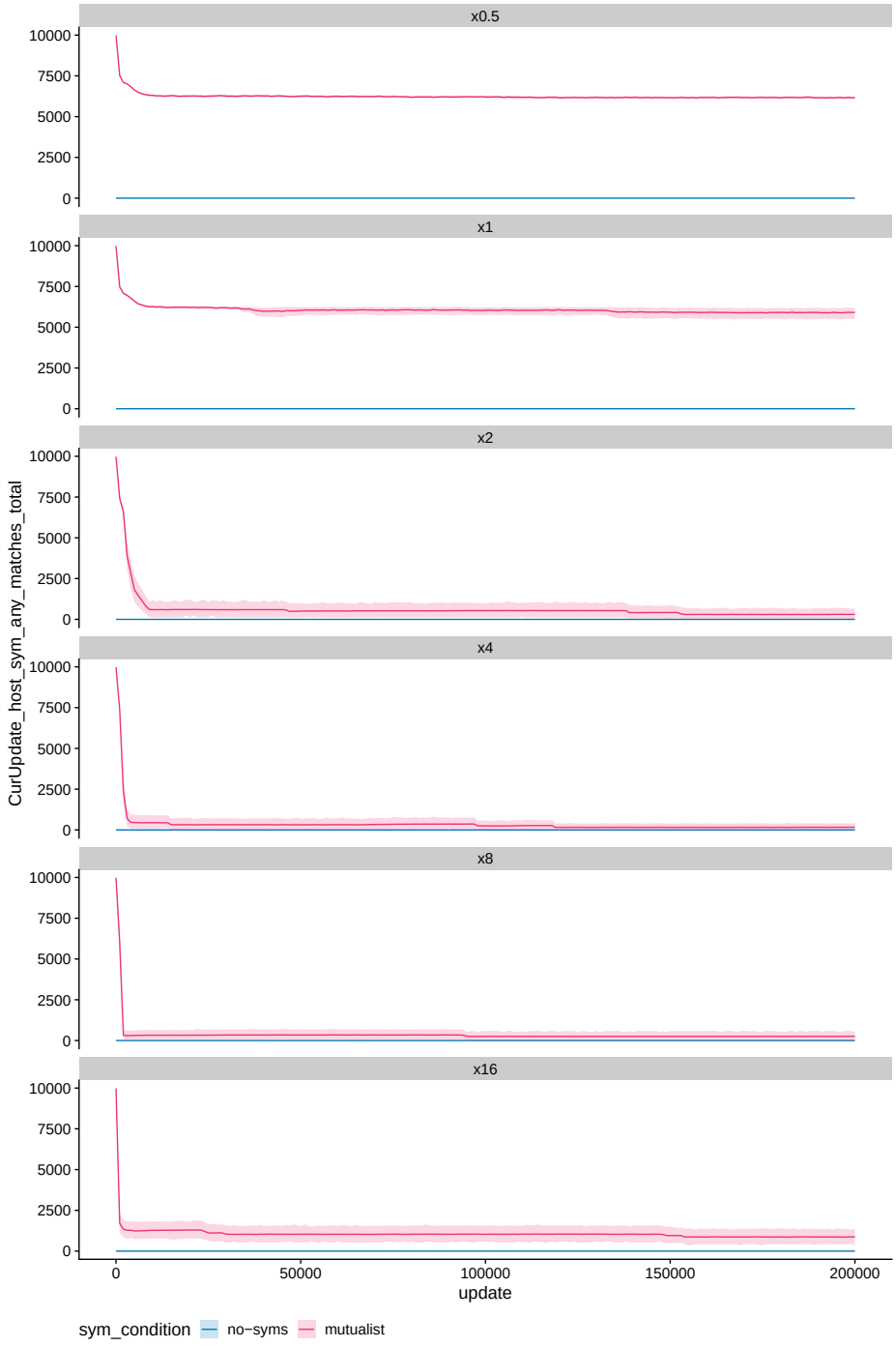
```
sym_final_entopy_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_parent_entropy_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  facet_wrap(~environment, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_final_entopy_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = sym_final_entopy_task_sets_elapsed_plt,
  width = 10,
  height = 10
)
```

7.8 Trait matches over time

Counts of host-symbiont pairs with any trait matches. Note that organisms that were just born will not have had time to express any Boolean logic functions, so they will appear as not matching.

```
host_sym_any_matches_total_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
```

```
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_sym_any_matches_total,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_sym_any_matches_total_plt.pdf", sep = "/"),
  plot = host_sym_any_matches_total_plt,
  width = 8,
  height = 8
)
host_sym_any_matches_total_plt
```


x0.5
x1
x2
x4
x8
x16
10000
7500
5000
2500
0
CurUpdate_host_sym_any_matches_total
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

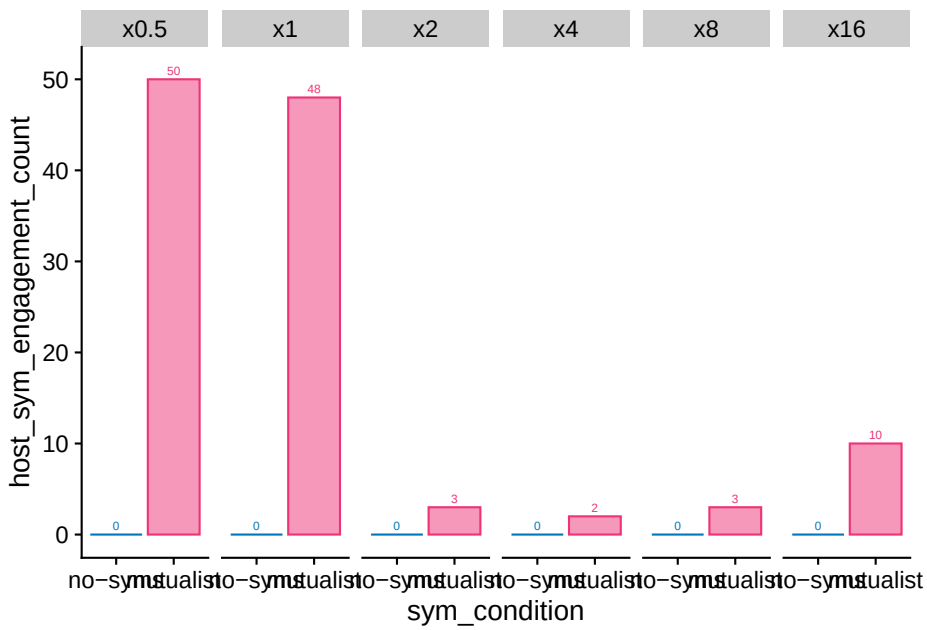
7.9 Evolutionary engagement

Use trait matching metrics to determine if host-symbiont populations are evolutionarily engaged.

```
engagement <- summary_data %>%
  mutate(
    host_sym_engagement = CurUpdate_host_sym_any_matches_total > threshold,
  ) %>%
  group_by(
    sym_condition,
    environment
  ) %>%
  summarise(
    host_sym_engagement_count = sum(host_sym_engagement)
  )
```

```
host_sym_engagement_count_plt <- engagement %>%
  ggplot(
    aes(
      x = sym_condition,
      color = sym_condition,
      fill = sym_condition,
      y = host_sym_engagement_count
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = host_sym_engagement_count),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  facet_wrap(~environment, nrow = 1) +
  theme(
    legend.position = "none"
```

```
  )

ggsave(
  file = paste(plot_dir, "host_sym_engagement_count_plt.pdf", sep = "/"),
  plot = host_sym_engagement_count_plt,
  width = 4,
  height = 4
)
host_sym_engagement_count_plt
```

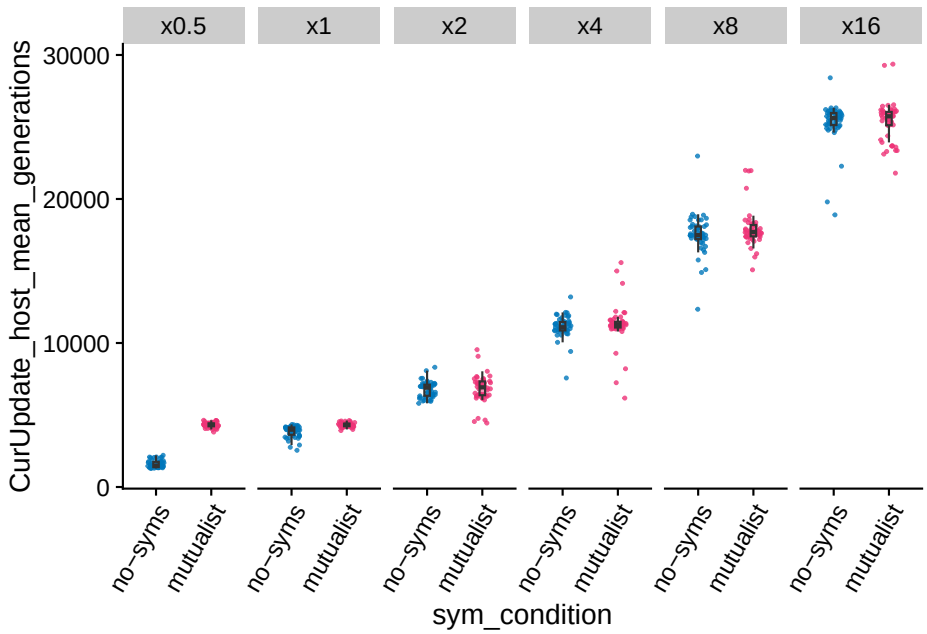


7.10 Generations of evolution elapsed

7.10.1 Hosts

```
host_avg_gens_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_mean_generations
    )
```

```
  ) +
  geom_point(
    mapping = aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_avg_gens_elapsed_plt.pdf", sep = "/"),
  plot = host_avg_gens_elapsed_plt,
  width = 4,
  height = 4
)
host_avg_gens_elapsed_plt
```



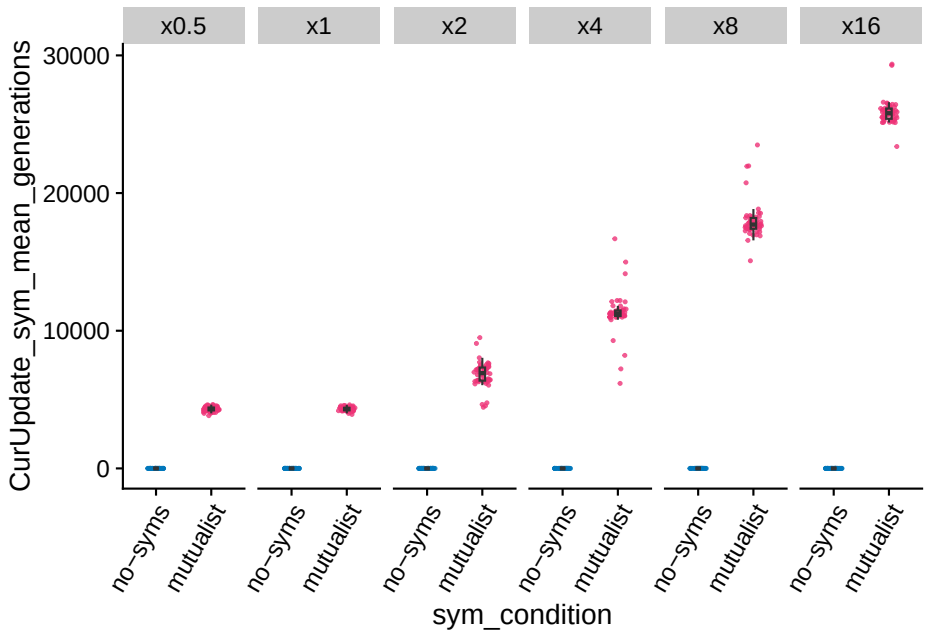

```
summary_data %>%
  group_by(
    sym_condition,
    environment
  ) %>%
  summarise(
    mean_gens = mean(CurUpdate_host_mean_generations),
    median_gens = median(CurUpdate_host_mean_generations),
    var_gens = var(CurUpdate_host_mean_generations),
    sdev_gens = sd(CurUpdate_host_mean_generations)
  )
```

```
## # A tibble: 12 x 6
## # Groups:   sym_condition [2]
##    sym_condition environment mean_gens median_gens var_gens sdev_gens
##    <fct>         <fct>           <dbl>       <dbl>    <dbl>     <dbl>
##  1 no-syms       x0.5            1587.       1485.   63978.      253.
##  2 no-syms       x1              3853.       3981.  160521.      401.
##  3 no-syms       x2              6768.       6899.  298842.      547.
##  4 no-syms       x4             11124.      11091.  605033.      778.
##  5 no-syms       x8             17544.      17492. 1888392.     1374.
##  6 no-syms       x16            25321.      25622. 2115792.     1455.
##  7 mutualist     x0.5            4307.       4318.   30094.      173.
##  8 mutualist     x1              4315.       4316.   21408.      146.
##  9 mutualist     x2              6818.       6921.  902392.      950.
```

```
## 10 mutualist     x4             11246.      11247. 2053291.     1433.
## 11 mutualist     x8             17936.      17703. 1679746.     1296.
## 12 mutualist     x16            25412.      25741. 1848285.     1360.
```

7.10.2 Symbionts

```
sym_avg_gens_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_mean_generations
    )
  ) +
  geom_point(
    mapping = aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_avg_gens_elapsed_plt.pdf", sep = "/"),
  plot = sym_avg_gens_elapsed_plt,
  width = 4,
  height = 4
)
sym_avg_gens_elapsed_plt
```




```
summary_data %>%
  group_by(
    sym_condition,
    environment
  ) %>%
  summarise(
    mean_gens = mean(CurUpdate_sym_mean_generations),
    median_gens = median(CurUpdate_sym_mean_generations),
    var_gens = var(CurUpdate_sym_mean_generations),
    sdev_gens = sd(CurUpdate_sym_mean_generations)
  )
```

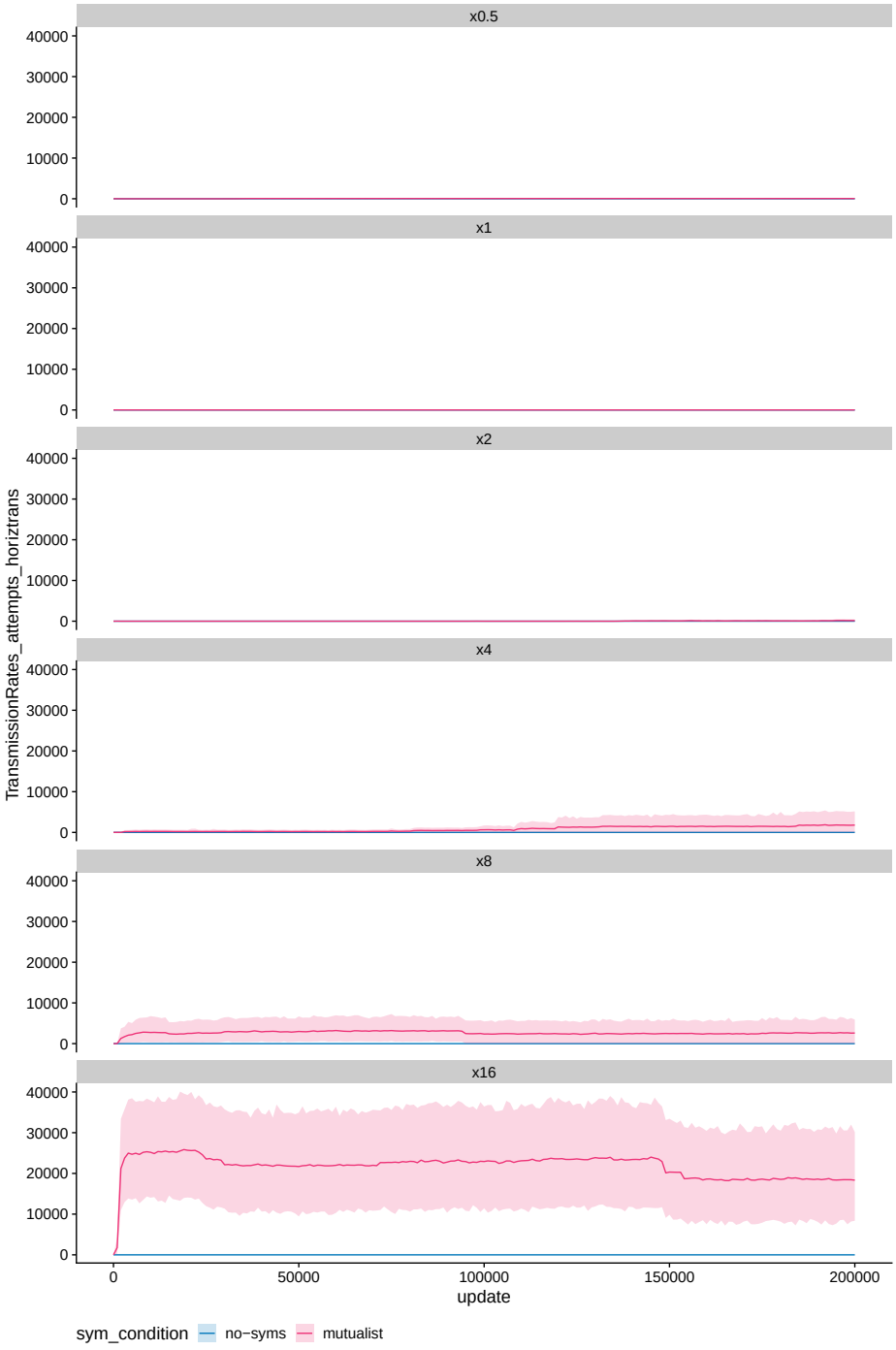
```
## # A tibble: 12 x 6
## # Groups:   sym_condition [2]
##    sym_condition environment mean_gens median_gens var_gens sdev_gens
##    <fct>         <fct>           <dbl>       <dbl>    <dbl>     <dbl>
##  1 no-syms       x0.5                0           0        0         0
##  2 no-syms       x1                  0           0        0         0
##  3 no-syms       x2                  0           0        0         0
##  4 no-syms       x4                  0           0        0         0
##  5 no-syms       x8                  0           0        0         0
##  6 no-syms       x16                 0           0        0         0
##  7 mutualist     x0.5             4307.       4316.   29730.      172.
##  8 mutualist     x1               4314.       4316.   21100.      145.
##  9 mutualist     x2               6817.       6921.  898102.      948.
```

```
## 10 mutualist     x4             11278.      11247. 2286634.     1512.
## 11 mutualist     x8             18006.      17712. 1871427.     1368.
## 12 mutualist     x16            25883.      25802.  791925.      890.
```

7.11 Symbiont transmission rates

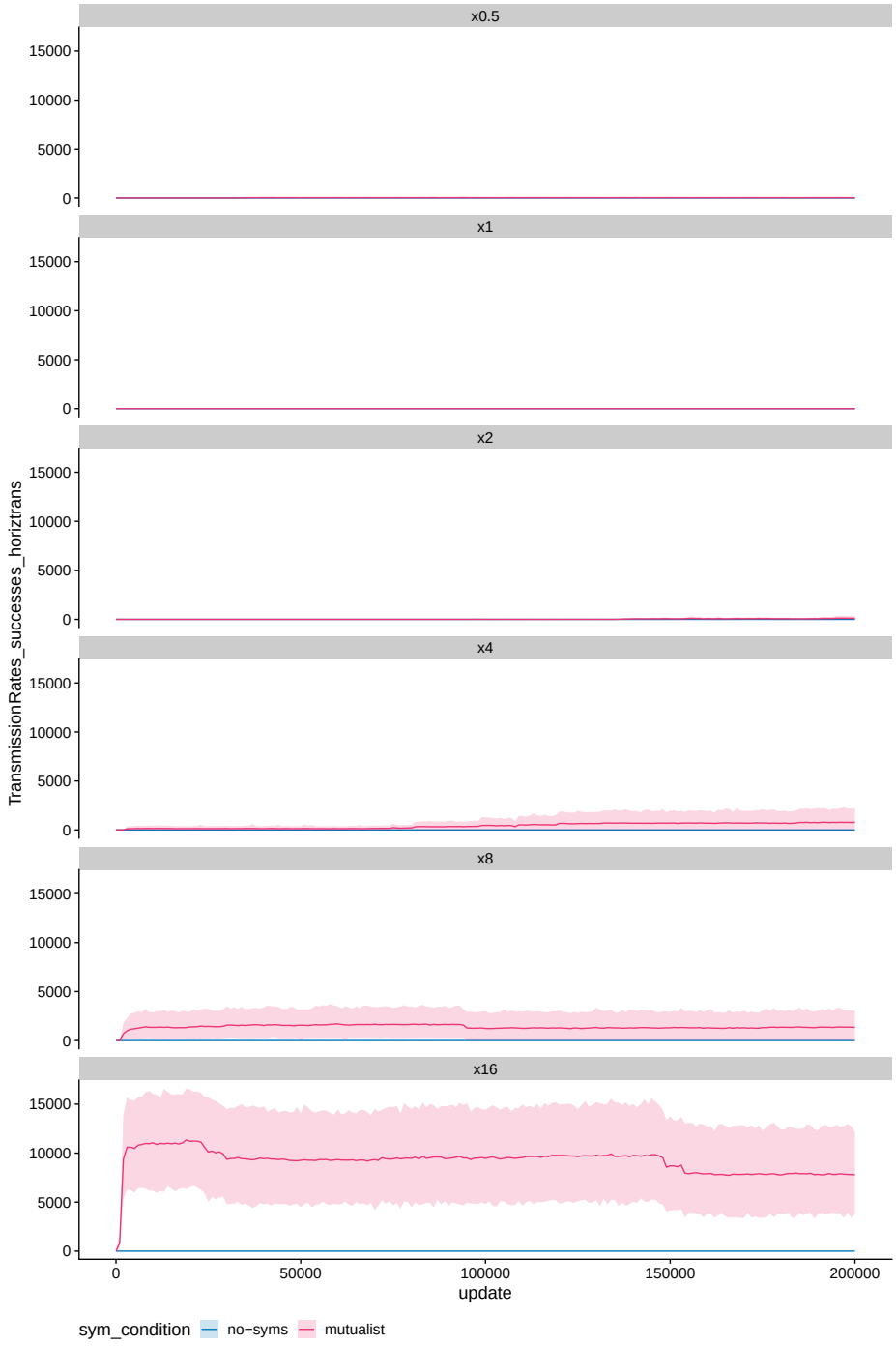
7.11.1 Horizontal transmission attempts

```
TransmissionRates_attempts_horiztrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_attempts_horiztrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_attempts_horiztrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_attempts_horiztrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_attempts_horiztrans_plt
```


x0.5
x1
x2
x4
x8
x16
TransmissionRates_attempts_horiztrans
40000
30000
20000
10000
0
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

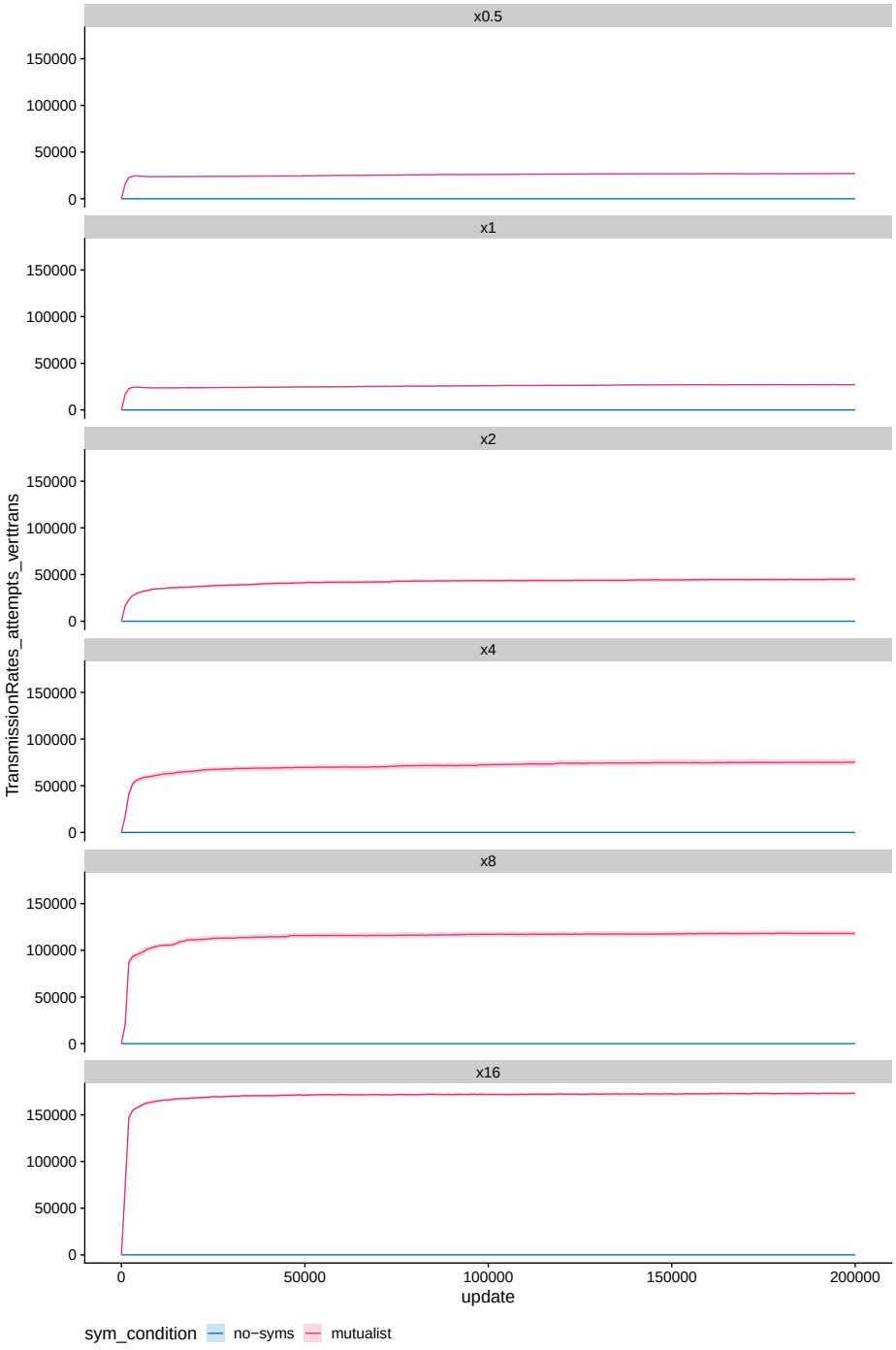
7.11.2 Horizontal transmission successes

```
TransmissionRates_successes_horiztrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_successes_horiztrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_successes_horiztrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_successes_horiztrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_successes_horiztrans_plt
```


x0.5
x1
x2
x4
x8
x16
15000
10000
5000
0
TransmissionRates_successes_horiztrans
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

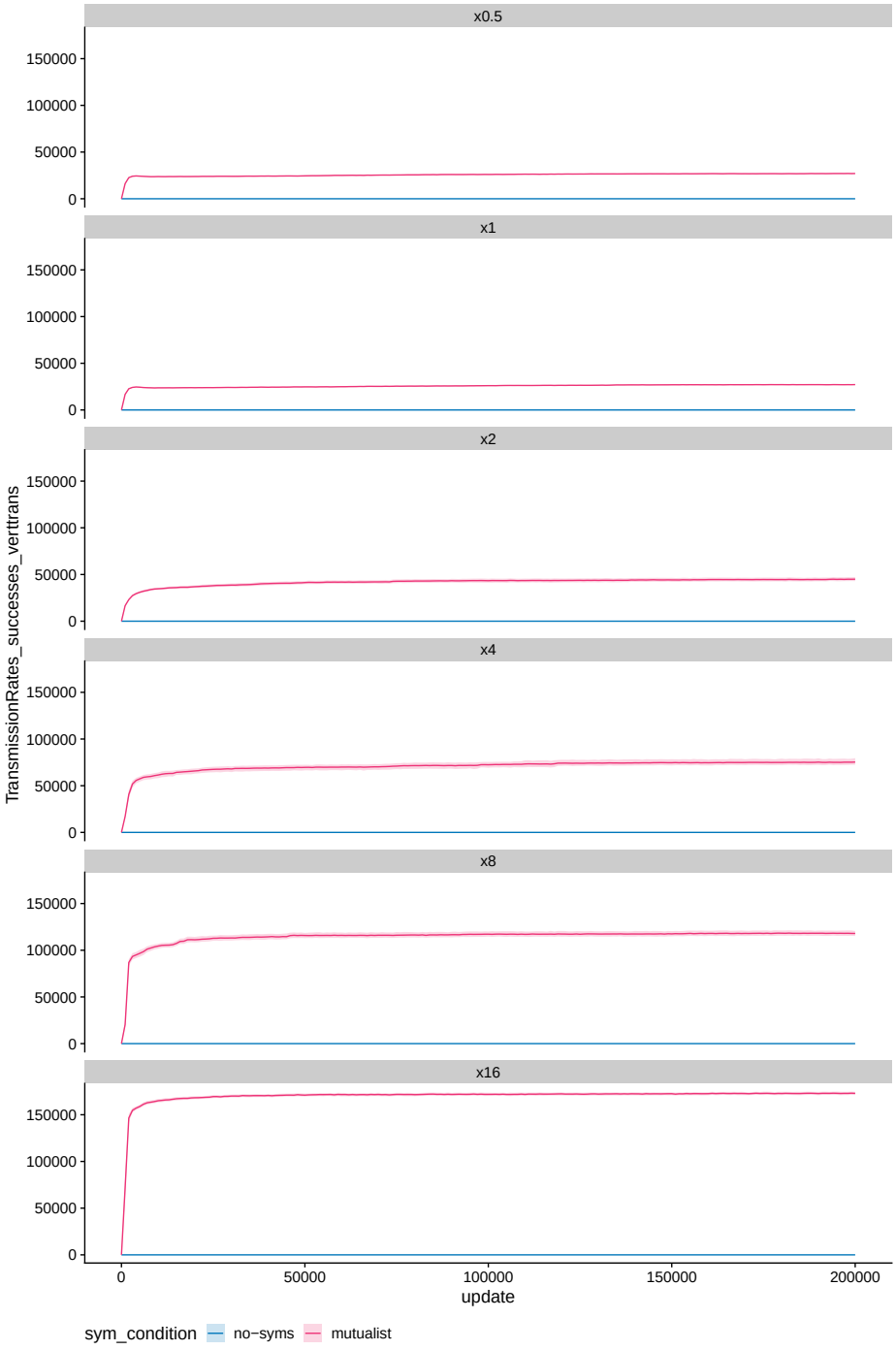
7.11.3 Vertical transmission attempts

```
TransmissionRates_attempts_verttrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_attempts_verttrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_attempts_verttrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_attempts_verttrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_attempts_verttrans_plt
```


x0.5
x1
x2
x4
x8
x16
150000
100000
50000
0
TransmissionRates_attempts_verttrans
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

7.11.4 Successes vertical transmission

```
TransmissionRates_successes_verttrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_successes_verttrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_successes_verttrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_successes_verttrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_successes_verttrans_plt
```


x0.5
x1
x2
x4
x8
x16
150000
100000
50000
0
TransmissionRates_successes_verttrans
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

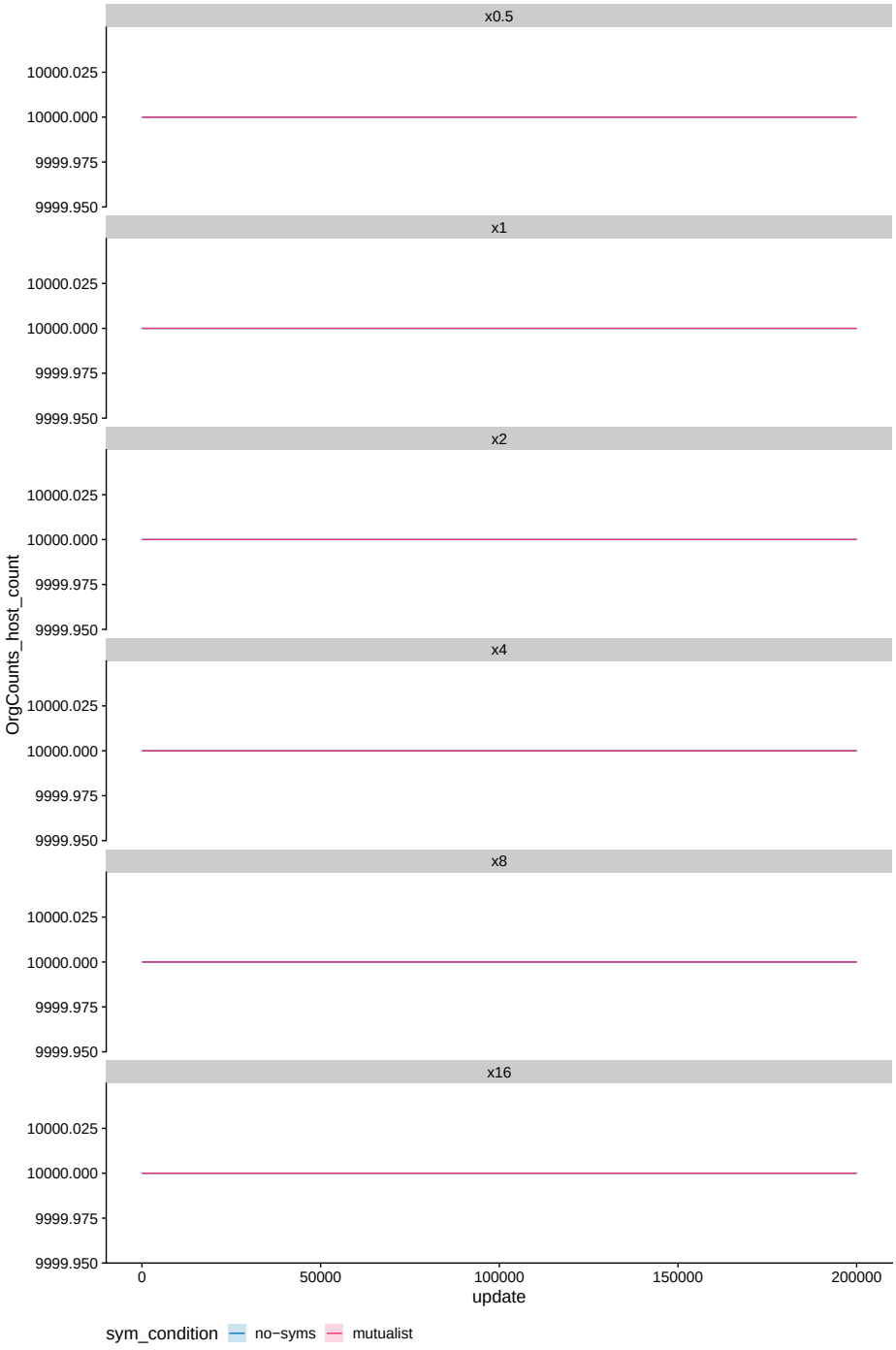
7.12 Population size

```
pop_size_summary <- time_data %>%
  group_by(
    SEED,
    sym_condition,
    environment
  ) %>%
  summarize(
    host_pop_size_avg = mean(OrgCounts_host_count),
    host_pop_size_median = median(OrgCounts_host_count),
    host_pop_size_variance = var(OrgCounts_host_count),
    sym_pop_size_avg = mean(OrgCounts_hosted_sym_count),
    sym_pop_size_median = median(OrgCounts_hosted_sym_count),
    sym_pop_size_variance = var(OrgCounts_hosted_sym_count)
  )
```

7.12.1 Host population size

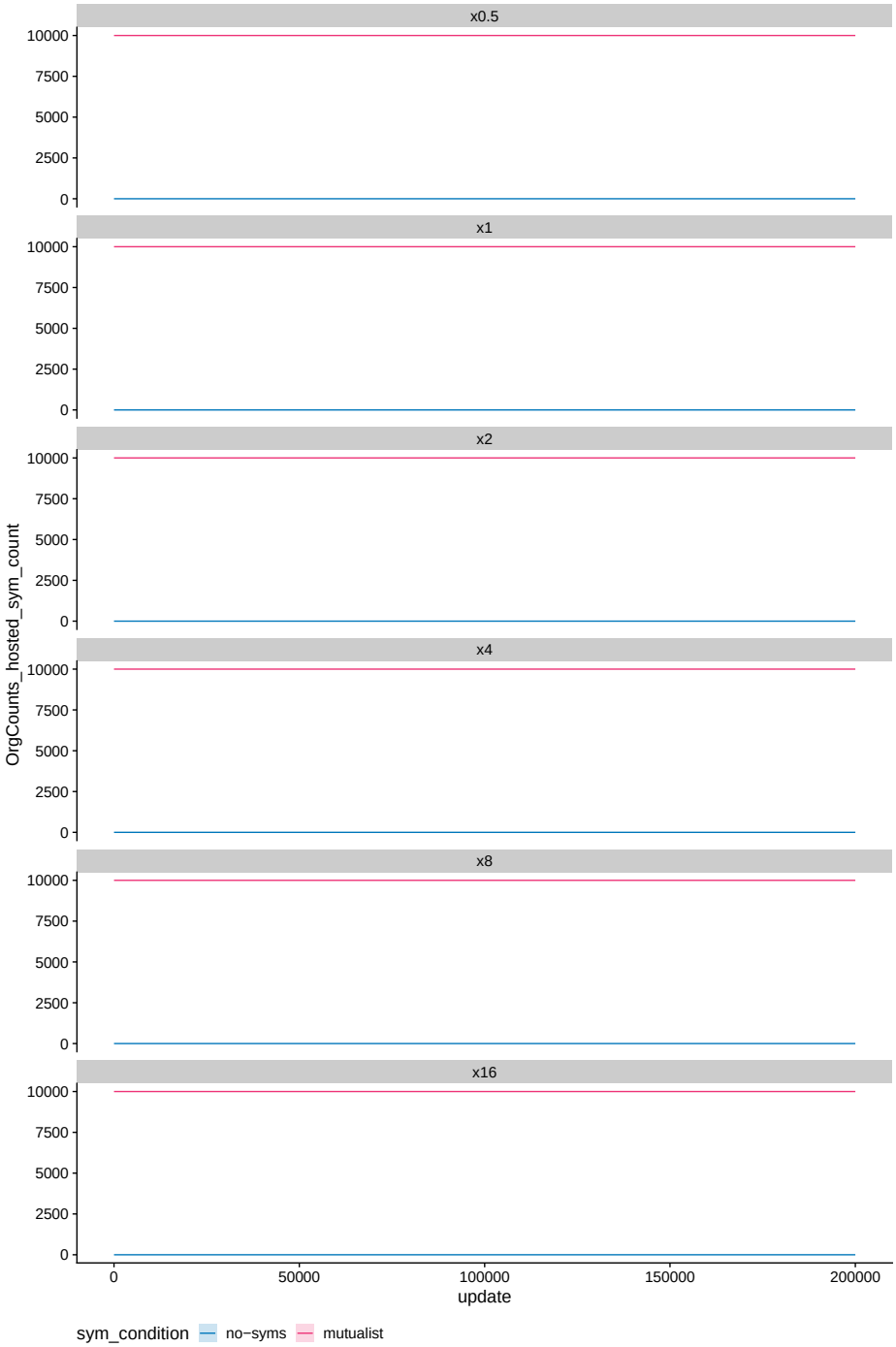
```
host_pop_size_ot_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = OrgCounts_host_count,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, ncol = 1) +
  theme(
    legend.position = "bottom"
```

```
  )
ggsave(
  file = paste(
    plot_dir,
    "host_pop_size_ot_plt.pdf",
    sep = "/"
  ),
  plot = host_pop_size_ot_plt,
  width = 8,
  height = 8
)
host_pop_size_ot_plt
```


x0.5
x1
x2
x4
x8
x16
10000.025
10000.000
9999.975
9999.950
OrgCounts_host_count
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

7.12.2 Symbiont population size

```
sym_pop_size_ot_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = OrgCounts_hosted_sym_count,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~environment, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "sym_pop_size_ot_plt.pdf",
    sep = "/"
  ),
  plot = sym_pop_size_ot_plt,
  width = 8,
  height = 8
)
sym_pop_size_ot_plt
```


x0.5
x1
x2
x4
x8
x16
10000
7500
5000
2500
0
OrgCounts_hosted_sym_count
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

Chapter 8

Mutualist donate proportion experiment

Repeat differential function rewards experiment, except vary mutualist donate proportion.

8.1 Dependencies and setup

Load R library dependencies.

```
library(tidyverse)
library(cowplot)
library(khroma)
library(rstatix)
library(knitr)
library(kableExtra)
```

Define relevant file paths.

```
# Check if Rmd is being compiled using bookdown
bookdown <- exists("bookdown_build")

# Build relative path from R's working directory to what is actually the working
# directory for this experiment.
experiment_slug <- "2026-02-24-health-donate-prop"
supp_file_prefix <- "donate-prop"
exp_directory <- paste(
  "experiments",
```

```
  experiment_slug,
  "analysis",
  sep = "/"
)

# Get path to the supplemental material analysis directory
supp_directory <- paste(
  "experiments",
  "manuscript-analyses",
  sep = "/"
)

# Adjust experiment directory if being knitted for bookdown build
if (bookdown) {
  exp_directory <- paste0(
    bookdown_wd_prefix,
    exp_directory
  )
  supp_directory <- paste0(
    bookdown_wd_prefix,
    supp_directory
  )
}
```

Define default plot theme, plot output directory.

```
# Use the cowplot theme as our default for ggplots
theme_set(theme_cowplot())
# Create directory to save plots
plot_dir <- paste(
  supp_directory,
  "plots",
  supp_file_prefix,
  sep = "/"
)
dir.create(
  plot_dir,
  showWarnings = FALSE
)

# Use vibrant theme for plot color palettes
vibrant_colors <- color("vibrant")(7)
color_pal <- c(
  vibrant_colors[["blue"]],
  vibrant_colors[["magenta"]]
```

```
)
```

8.2 Load and preprocess data

```
tasks <- c("NOT", "NAND", "OR_NOT", "AND", "OR", "AND_NOT", "NOR", "XOR", "EQU")
max_pop_size <- 100 * 100
threshold <- 0.025 * max_pop_size
```

Load summary data.

```
summary_data_path <- paste(
  exp_directory,
  "data",
  "summary.csv",
  sep = "/"
)
summary_data <- read_csv(summary_data_path)
# Do any universal preprocessing
summary_data <- summary_data %>%
  mutate(
    SEED = as.factor(SEED),
    task_credit = paste0(
      "H", HOST_ONLY_FIRST_TASK_CREDIT,
      "S", SYM_ONLY_FIRST_TASK_CREDIT
    ),
    sym_condition = case_when(
      START_MOI == "0" ~ "no-syms",
      START_MOI == "1" & HEALTH_TYPE == "parasite" ~ "parasite",
      START_MOI == "1" & HEALTH_TYPE == "mutualist" ~ "mutualist",
      .default = "unknown" # precaution against unexpected parameterizations
    ),
    interaction_multiplier = as.factor(PARASITE_CYCLE_STEAL_MULTIPLIER),
    MUTUALIST_CYCLE_GAIN_PROP = as.factor(MUTUALIST_CYCLE_GAIN_PROP)
  ) %>%
  mutate(
    task_credit = as.factor(task_credit),
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist")
    ),
    donate_prop = MUTUALIST_CYCLE_GAIN_PROP
  )
```

Load time series data.

```
time_data_path <- paste(
  exp_directory,
  "data",
  "time_series.csv",
  sep = "/"
)
time_data <- read_csv(time_data_path)

# Do any universal preprocessing
time_data <- time_data %>%
  mutate(
    SEED = as.factor(SEED),
    sym_condition = case_when(
      START_MOI == "0" ~ "no-syms",
      START_MOI == "1" & HEALTH_TYPE == "parasite" ~ "parasite",
      START_MOI == "1" & HEALTH_TYPE == "mutualist" ~ "mutualist",
      .default = "unknown" # precaution against unexpected parameterizations
    ),
    MUTUALIST_CYCLE_GAIN_PROP = as.factor(MUTUALIST_CYCLE_GAIN_PROP)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist")
    ),
    donate_prop = MUTUALIST_CYCLE_GAIN_PROP
  )
```

8.3 Extinctions

Did host or symbiont populations go extinct in any runs?

```
sym_min_pop_sizes <- time_data %>%
  group_by(SEED) %>%
  slice_min(
    OrgCounts_hosted_sym_count,
    n = 1
  ) %>%
  slice_min(
    update,
    n = 1
  )
```

```
reps_with_extinct_syms <- sym_min_pop_sizes %>%
  group_by(
    sym_condition,
    donate_prop
  ) %>%
  summarize(
    reps_with_extinct_syms = sum(OrgCounts_hosted_sym_count == 0),
    total_reps = n()
  )

host_min_pop_sizes <- time_data %>%
  group_by(SEED) %>%
  slice_min(
    OrgCounts_host_count,
    n = 1
  ) %>%
  slice_min(
    update,
    n = 1
  )

reps_with_extinct_hosts <- host_min_pop_sizes %>%
  group_by(
    sym_condition,
    donate_prop
  ) %>%
  summarize(
    reps_with_extinct_hosts = sum(OrgCounts_host_count == 0),
    total_reps = n()
  )

extinctions <- full_join(
  reps_with_extinct_syms,
  reps_with_extinct_hosts
)
extinctions <- extinctions %>%
  arrange(
    desc(reps_with_extinct_hosts), desc(reps_with_extinct_syms)
  )
extinction_table <- kable(extinctions) %>%
  kable_styling(latex_options = "striped")
save_kable(
  extinction_table,
  paste(
    plot_dir,
```

```
    "extinctions.pdf",
    sep = "/"
  )
)

extinctions
```

```
## # A tibble: 12 x 5
## # Groups:   sym_condition [2]
##    sym_condition donate_prop reps_with_extinct_syms total_reps
##    <fct>         <fct>                        <int>      <int>
##  1 no-syms       0.125                           50         50
##  2 no-syms       0.25                            50         50
##  3 no-syms       0.5                             50         50
##  4 no-syms       0.75                            50         50
##  5 no-syms       0.875                           50         50
##  6 no-syms       1                               50         50
##  7 mutualist     0.125                            0         50
##  8 mutualist     0.25                             0         50
##  9 mutualist     0.5                              0         50
## 10 mutualist     0.75                             0         50
## 11 mutualist     0.875                            0         50
## 12 mutualist     1                                0         50
## # i 1 more variable: reps_with_extinct_hosts <int>
```

8.4 Host function performance

First, quantify which functions have been performed over the course of an entire run.

```
# Pivot task data long, selecting only relevant columns
host_over_time_parent_task_data <- time_data %>%
  select(
    SEED,
    sym_condition,
    donate_prop,
    update,
    CurUpdate_AND_NOT_in_host_parent_org_counts,
    CurUpdate_AND_in_host_parent_org_counts,
    CurUpdate_EQU_in_host_parent_org_counts,
    CurUpdate_NAND_in_host_parent_org_counts,
    CurUpdate_NOR_in_host_parent_org_counts,
    CurUpdate_NOT_in_host_parent_org_counts,
```

```
    CurUpdate_OR_NOT_in_host_parent_org_counts,
    CurUpdate_OR_in_host_parent_org_counts,
    CurUpdate_XOR_in_host_parent_org_counts
  ) %>%
  pivot_longer(
    c(
      CurUpdate_AND_NOT_in_host_parent_org_counts,
      CurUpdate_AND_in_host_parent_org_counts,
      CurUpdate_EQU_in_host_parent_org_counts,
      CurUpdate_NAND_in_host_parent_org_counts,
      CurUpdate_NOR_in_host_parent_org_counts,
      CurUpdate_NOT_in_host_parent_org_counts,
      CurUpdate_OR_NOT_in_host_parent_org_counts,
      CurUpdate_OR_in_host_parent_org_counts,
      CurUpdate_XOR_in_host_parent_org_counts
    ),
    names_to = "task",
    values_to = "task_in_parent_count"
  ) %>%
  mutate(
    task = str_replace(task, "CurUpdate_", "")
  ) %>%
  mutate(
    task = str_replace(task, "_in_host_parent_org_counts", "")
  ) %>%
  mutate(
    task_completed_by_parent = task_in_parent_count >= threshold,
    task = factor(
      task,
      levels = tasks,
      labels = tasks
    )
  ) %>%
  mutate(
    task_complexity = case_when(
      task == "NOT" ~ 1,
      task == "NAND" ~ 1,
      task == "OR_NOT" ~ 2,
      task == "AND" ~ 2,
      task == "OR" ~ 3,
      task == "AND_NOT" ~ 3,
      task == "NOR" ~ 4,
      task == "XOR" ~ 4,
      task == "EQU" ~ 5,
      .default = -1
```

```
    )
  )

host_parent_tasks_all_time_by_task <- host_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    donate_prop,
    task
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  group_by(
    sym_condition,
    donate_prop,
    task
  ) %>%
  summarize(
    n = n(),
    parent_tasks_completed_ever = sum(parent_task_completed_ever == TRUE)
  )
```

Next, summarize function complexity level.

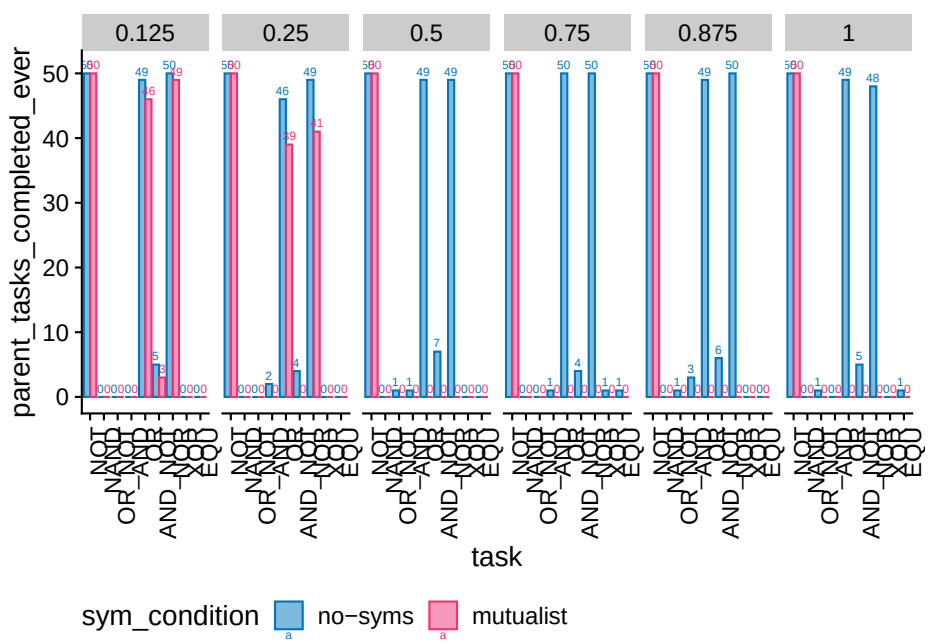
```
host_max_task_complexity_data <- host_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    donate_prop,
    task,
    task_complexity
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  # This filter works fine because all replicates start with a task
  filter(
    parent_task_completed_ever
  ) %>%
  group_by(
```

```
    SEED,
    sym_condition,
    donate_prop
  ) %>%
  summarize(
    max_task_complexity = max(task_complexity)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c(
        "no-syms",
        "parasite",
        "mutualist"
      )
    )
  )
```

8.4.1 Host function performance - all time

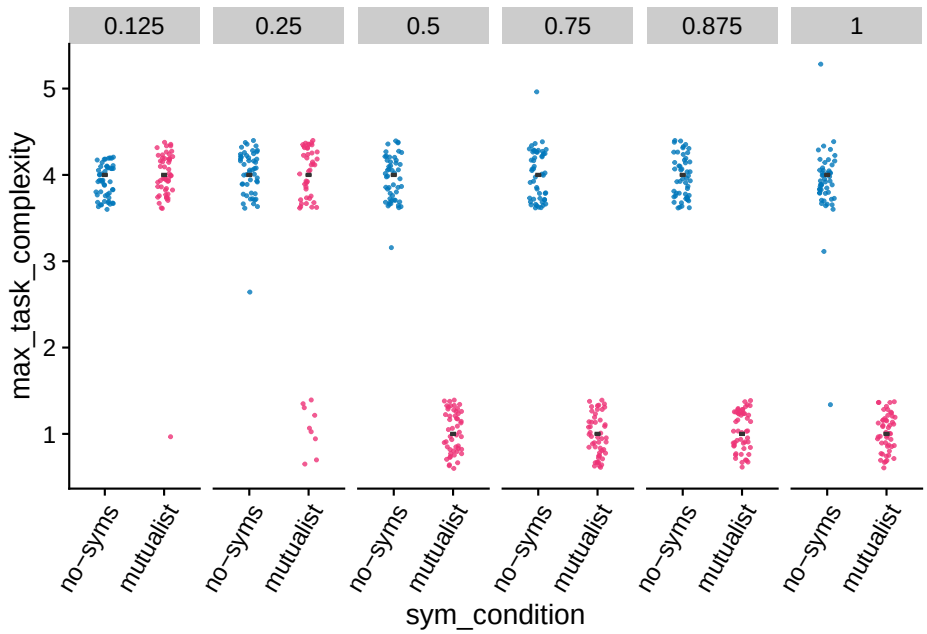
```
host_parent_tasks_all_time_by_task_plt <- host_parent_tasks_all_time_by_task %>%
  ggplot(
    aes(
      x = task,
      color = sym_condition,
      fill = sym_condition,
      y = parent_tasks_completed_ever
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = parent_tasks_completed_ever),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
```

```
    limits = c(0, 51)
  ) +
  facet_wrap(~donate_prop, nrow = 1) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(
      angle = 90,
      hjust = 1
    )
  )
ggsave(
  file = paste(
    plot_dir,
    "host_parent_tasks_all_time_by_task_plt.pdf",
    sep = "/"
  ),
  plot = host_parent_tasks_all_time_by_task_plt,
  width = 10,
  height = 7
)
host_parent_tasks_all_time_by_task_plt
```



8.4.2 Host maximum function complexity evolved

```
host_max_task_complexity_plt <- host_max_task_complexity_data %>%
  ggplot(
    aes(x = sym_condition, fill = sym_condition, y = max_task_complexity)
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_max_task_complexity_plt.pdf", sep = "/"),
  plot = host_max_task_complexity_plt,
  width = 4,
  height = 4
)
host_max_task_complexity_plt
```



8.5 Symbiont function performance

```
# Pivot task data long, selecting only relevant columns
sym_over_time_parent_task_data <- time_data %>%
  select(
    SEED,
    sym_condition,
    donate_prop,
    update,
    CurUpdate_AND_NOT_in_sym_parent_org_counts,
    CurUpdate_AND_in_sym_parent_org_counts,
    CurUpdate_EQU_in_sym_parent_org_counts,
    CurUpdate_NAND_in_sym_parent_org_counts,
    CurUpdate_NOR_in_sym_parent_org_counts,
    CurUpdate_NOT_in_sym_parent_org_counts,
    CurUpdate_OR_NOT_in_sym_parent_org_counts,
    CurUpdate_OR_in_sym_parent_org_counts,
    CurUpdate_XOR_in_sym_parent_org_counts
  ) %>%
  pivot_longer(
    c(
      CurUpdate_AND_NOT_in_sym_parent_org_counts,
```

```
      CurUpdate_AND_in_sym_parent_org_counts,
      CurUpdate_EQU_in_sym_parent_org_counts,
      CurUpdate_NAND_in_sym_parent_org_counts,
      CurUpdate_NOR_in_sym_parent_org_counts,
      CurUpdate_NOT_in_sym_parent_org_counts,
      CurUpdate_OR_NOT_in_sym_parent_org_counts,
      CurUpdate_OR_in_sym_parent_org_counts,
      CurUpdate_XOR_in_sym_parent_org_counts
    ),
    names_to = "task",
    values_to = "task_in_parent_count"
  ) %>%
  mutate(
    task = str_replace(task, "CurUpdate_", "")
  ) %>%
  mutate(
    task = str_replace(task, "_in_sym_parent_org_counts", "")
  ) %>%
  mutate(
    task_completed_by_parent = task_in_parent_count >= threshold,
    task = factor(
      task,
      levels = tasks,
      labels = tasks
    )
  ) %>%
  mutate(
    task_complexity = case_when(
      task == "NOT" ~ 1,
      task == "NAND" ~ 1,
      task == "OR_NOT" ~ 2,
      task == "AND" ~ 2,
      task == "OR" ~ 3,
      task == "AND_NOT" ~ 3,
      task == "NOR" ~ 4,
      task == "XOR" ~ 4,
      task == "EQU" ~ 5,
      .default = -1
    )
  )

sym_parent_tasks_all_time_by_task <- sym_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
```

```
    donate_prop,
    task
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  group_by(
    sym_condition,
    donate_prop,
    task
  ) %>%
  summarize(
    n = n(),
    parent_tasks_completed_ever = sum(parent_task_completed_ever == TRUE)
  )
```

Next, summarize function complexity level.

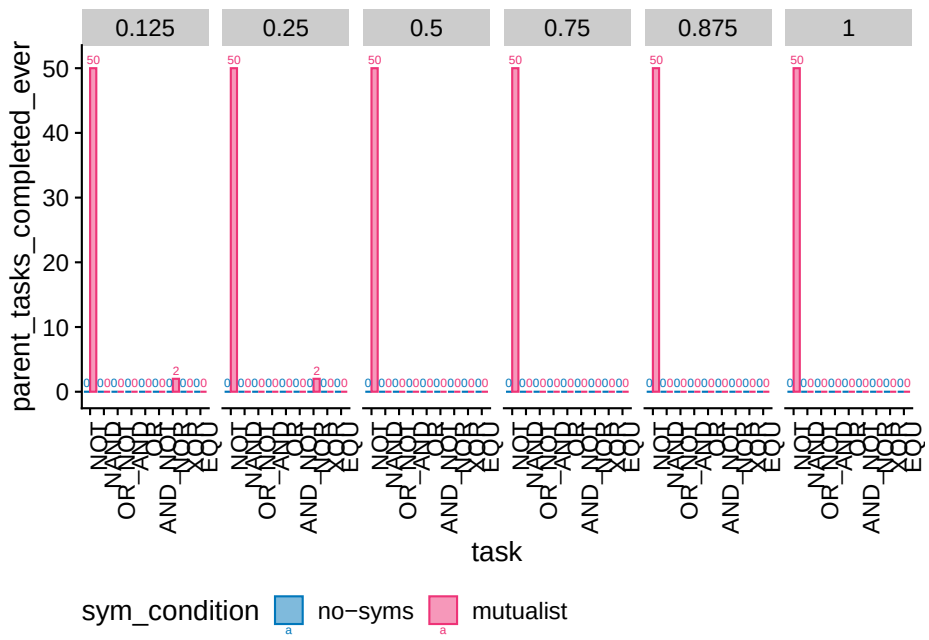
```
sym_max_task_complexity_data <- sym_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    donate_prop,
    task,
    task_complexity
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  # This filter works fine because all replicates start with a task
  filter(
    parent_task_completed_ever
  ) %>%
  group_by(
    SEED,
    sym_condition,
    donate_prop
  ) %>%
  summarize(
    max_task_complexity = max(task_complexity)
  ) %>%
```

```
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c(
        "no-syms",
        "parasite",
        "mutualist"
      )
    )
  )
```

8.5.1 Symbiont function performance - all time

```
sym_parent_tasks_all_time_by_task_plt <- sym_parent_tasks_all_time_by_task %>%
  ggplot(
    aes(
      x = task,
      color = sym_condition,
      fill = sym_condition,
      y = parent_tasks_completed_ever
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = parent_tasks_completed_ever),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, nrow = 1) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(
      angle = 90,
```

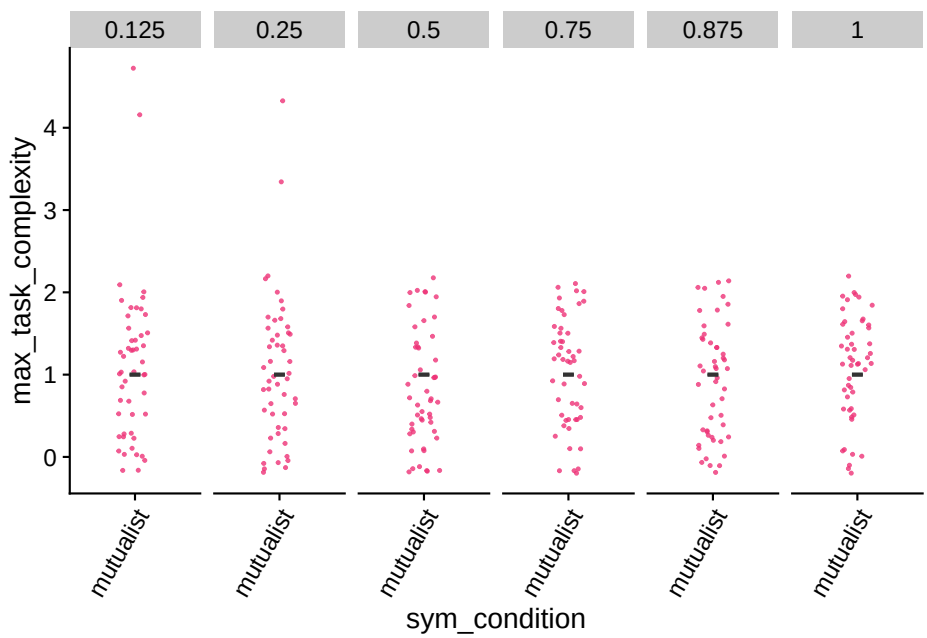
```
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_parent_tasks_all_time_by_task_plt.pdf", sep = "/"),
  plot = sym_parent_tasks_all_time_by_task_plt,
  width = 8,
  height = 8
)
sym_parent_tasks_all_time_by_task_plt
```



8.5.2 Symbiont maximum function complexity evolved

```
sym_max_task_complexity_plt <- sym_max_task_complexity_data %>%
  ggplot(
    aes(x = sym_condition, fill = sym_condition, y = max_task_complexity)
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
```

```
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = c(color_pal[2], color_pal[3])) +
  scale_fill_manual(values = c(color_pal[2], color_pal[3])) +
  facet_wrap(~donate_prop, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_max_task_complexity_plt.pdf", sep = "/"),
  plot = sym_max_task_complexity_plt,
  width = 4,
  height = 4
)
sym_max_task_complexity_plt
```

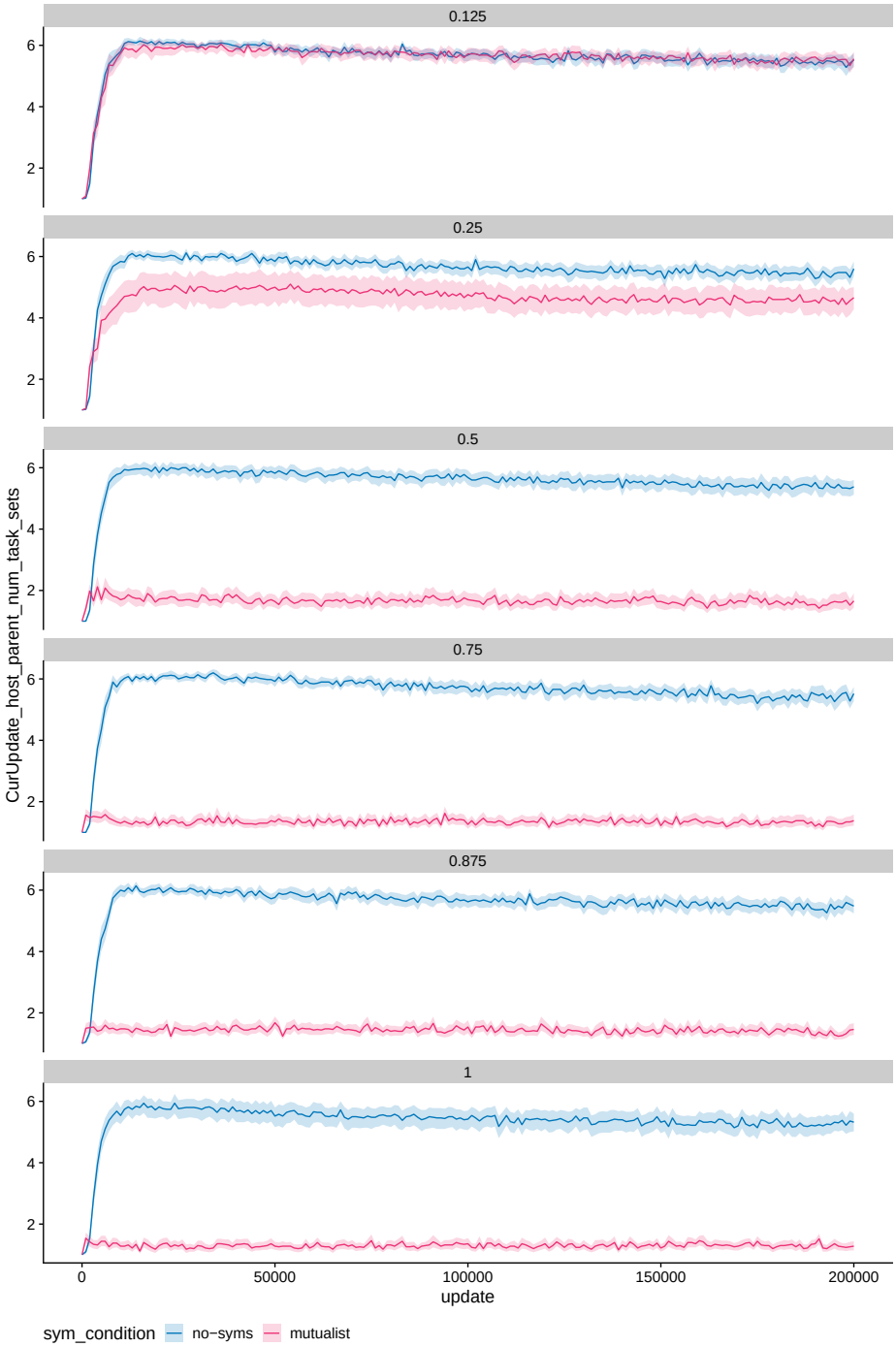


8.6 Host phenotypic diversity

8.6.1 Number of distinct function profiles

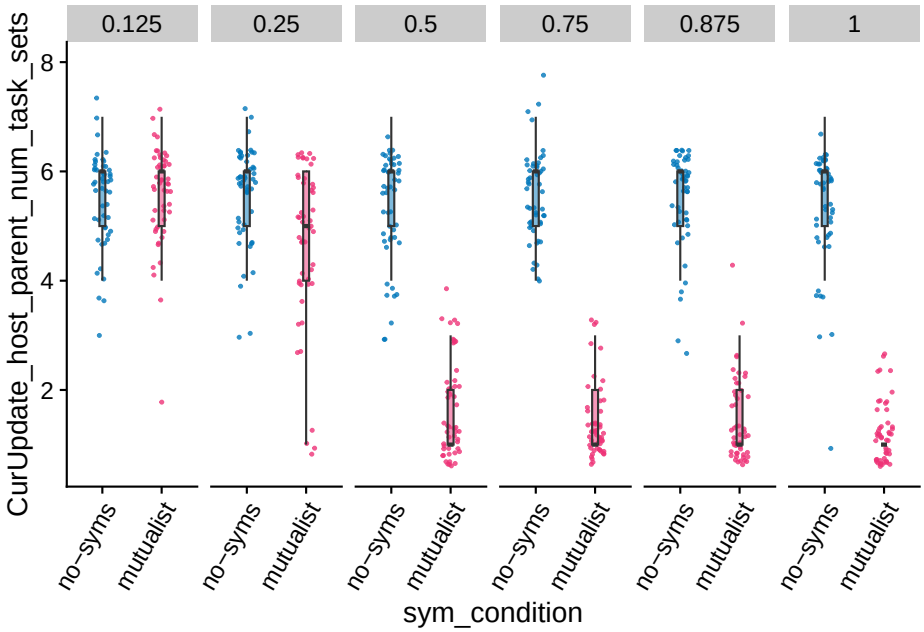
Over time:

```
host_parent_num_task_sets_plt <- time_data %>%
  # sample to make plotting faster
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_parent_num_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_parent_num_task_sets_plt.pdf", sep = "/"),
  plot = host_parent_num_task_sets_plt,
  width = 8,
  height = 8
)
host_parent_num_task_sets_plt
```



At the end of the experiment:

```
host_final_num_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_parent_num_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_final_num_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = host_final_num_task_sets_elapsed_plt,
  width = 4,
  height = 4
)
host_final_num_task_sets_elapsed_plt
```

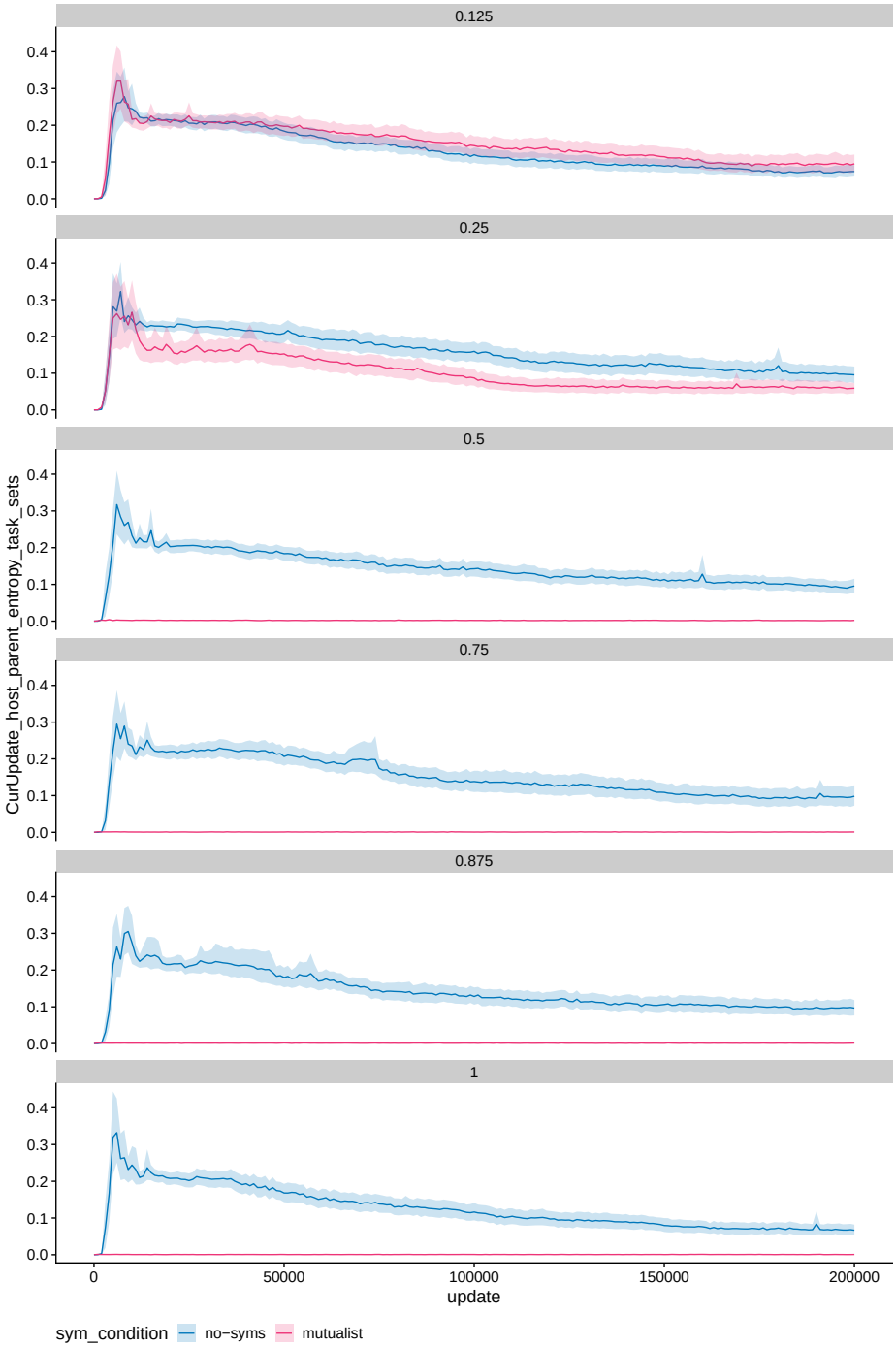


8.6.2 Shannon diversity of function profiles

Over time:

```
host_parent_entropy_task_sets_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
```

```
  facet_wrap(~donate_prop, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_parent_entropy_task_sets_plt.pdf", sep = "/"),
  plot = host_parent_entropy_task_sets_plt,
  width = 8,
  height = 8
)
host_parent_entropy_task_sets_plt
```



At the end of the experiment:

```
host_final_entopy_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_parent_entropy_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  facet_wrap(~donate_prop, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_final_entopy_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = host_final_entopy_task_sets_elapsed_plt,
  width = 10,
  height = 10
)
```

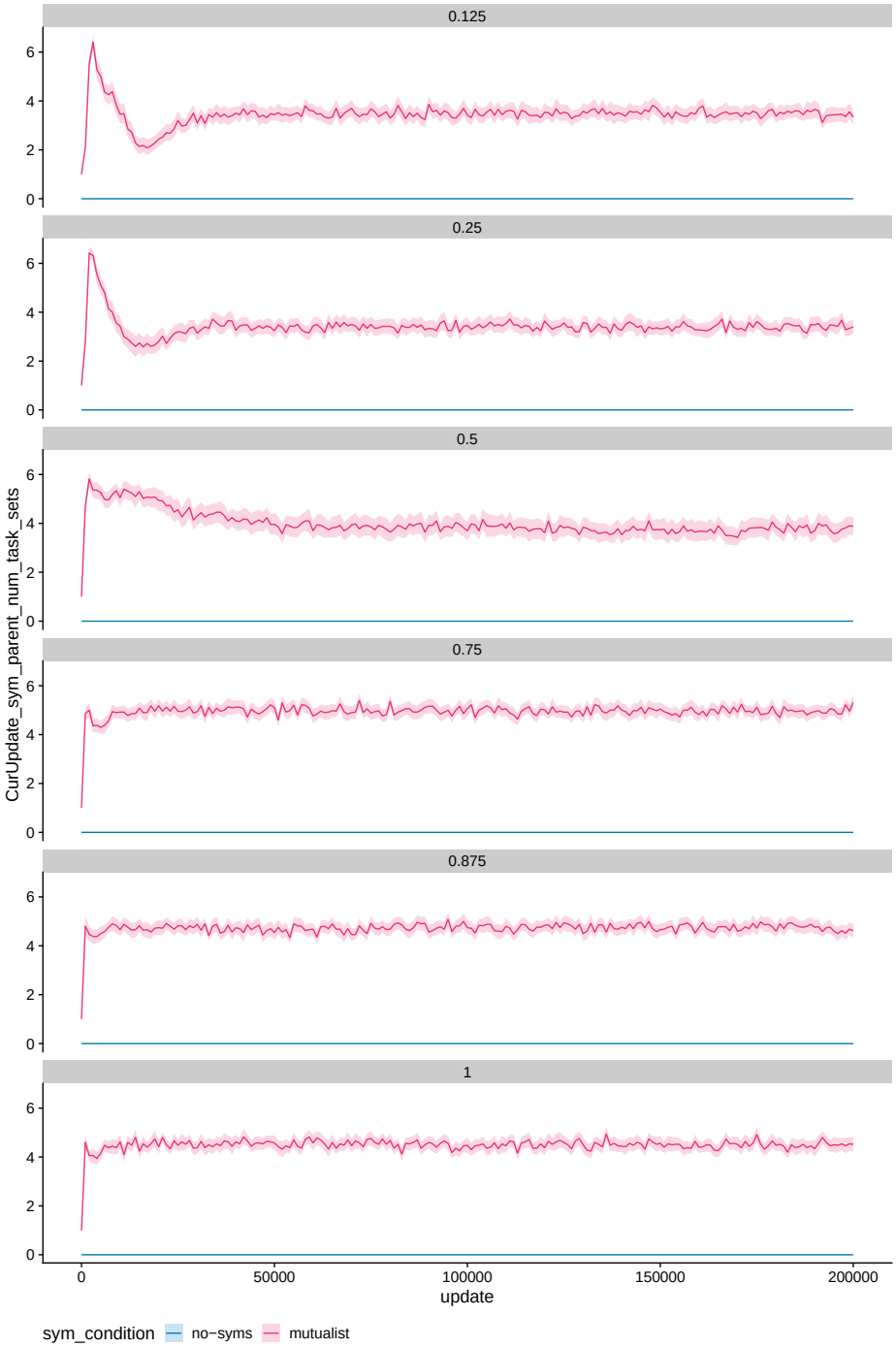
8.7 Symbiont phenotypic diversity

8.7.1 Number of distinct function profiles

Over time:

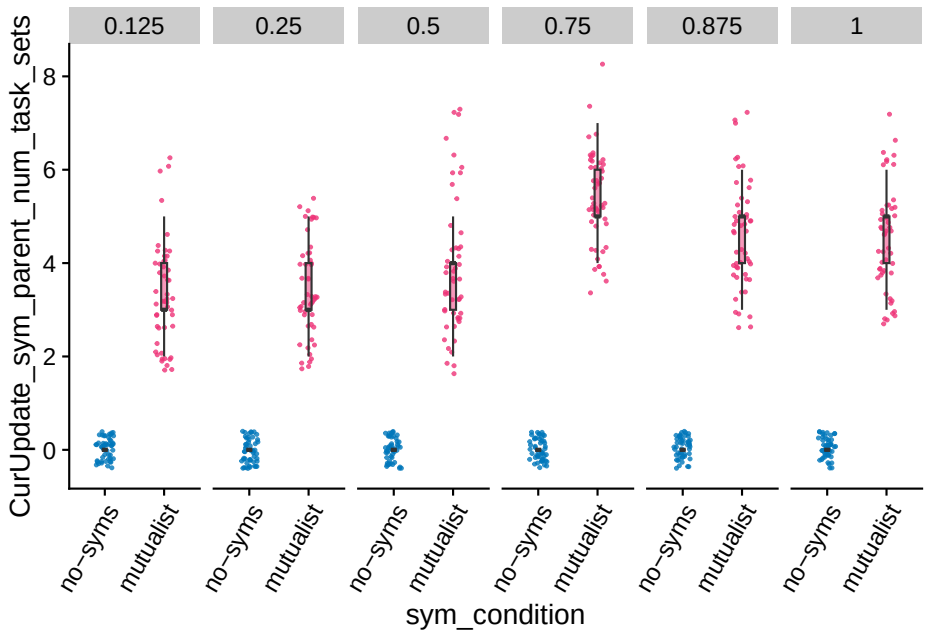
```
sym_parent_num_task_sets_plt <- time_data %>%
  # sample to make plotting faster
```

```
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_sym_parent_num_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "sym_parent_num_task_sets_plt.pdf", sep = "/"),
  plot = sym_parent_num_task_sets_plt,
  width = 8,
  height = 8
)
sym_parent_num_task_sets_plt
```



At the end of the experiment:

```
sym_final_num_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_parent_num_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_final_num_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = sym_final_num_task_sets_elapsed_plt,
  width = 4,
  height = 4
)
sym_final_num_task_sets_elapsed_plt
```

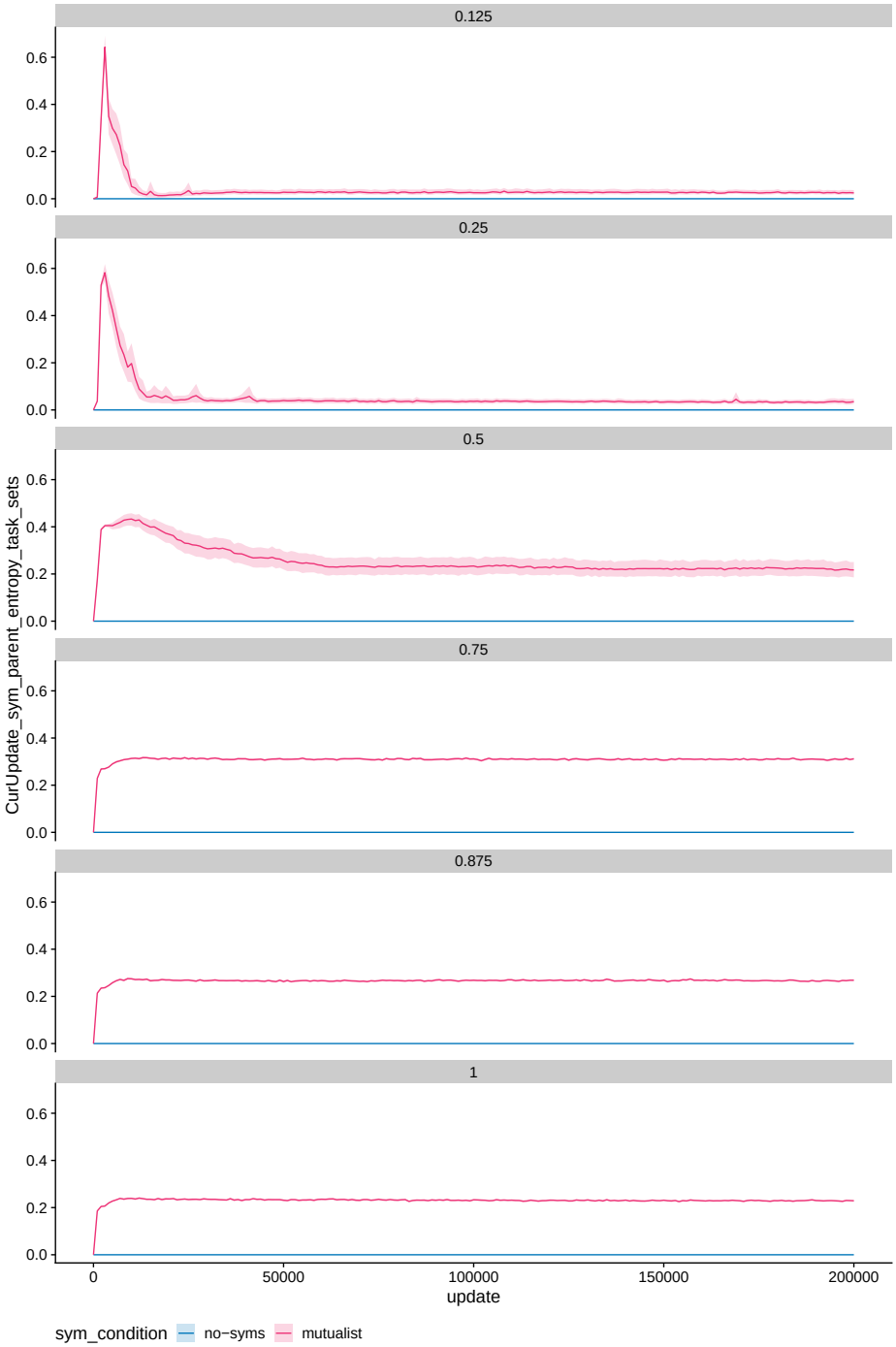


8.7.2 Shannon diversity of function profiles

Over time:

```
sym_parent_entropy_task_sets_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_sym_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
```

```
  facet_wrap(~donate_prop, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "sym_parent_entropy_task_sets_plt.pdf", sep = "/"),
  plot = sym_parent_entropy_task_sets_plt,
  width = 8,
  height = 8
)
sym_parent_entropy_task_sets_plt
```



At the end of the experiment:

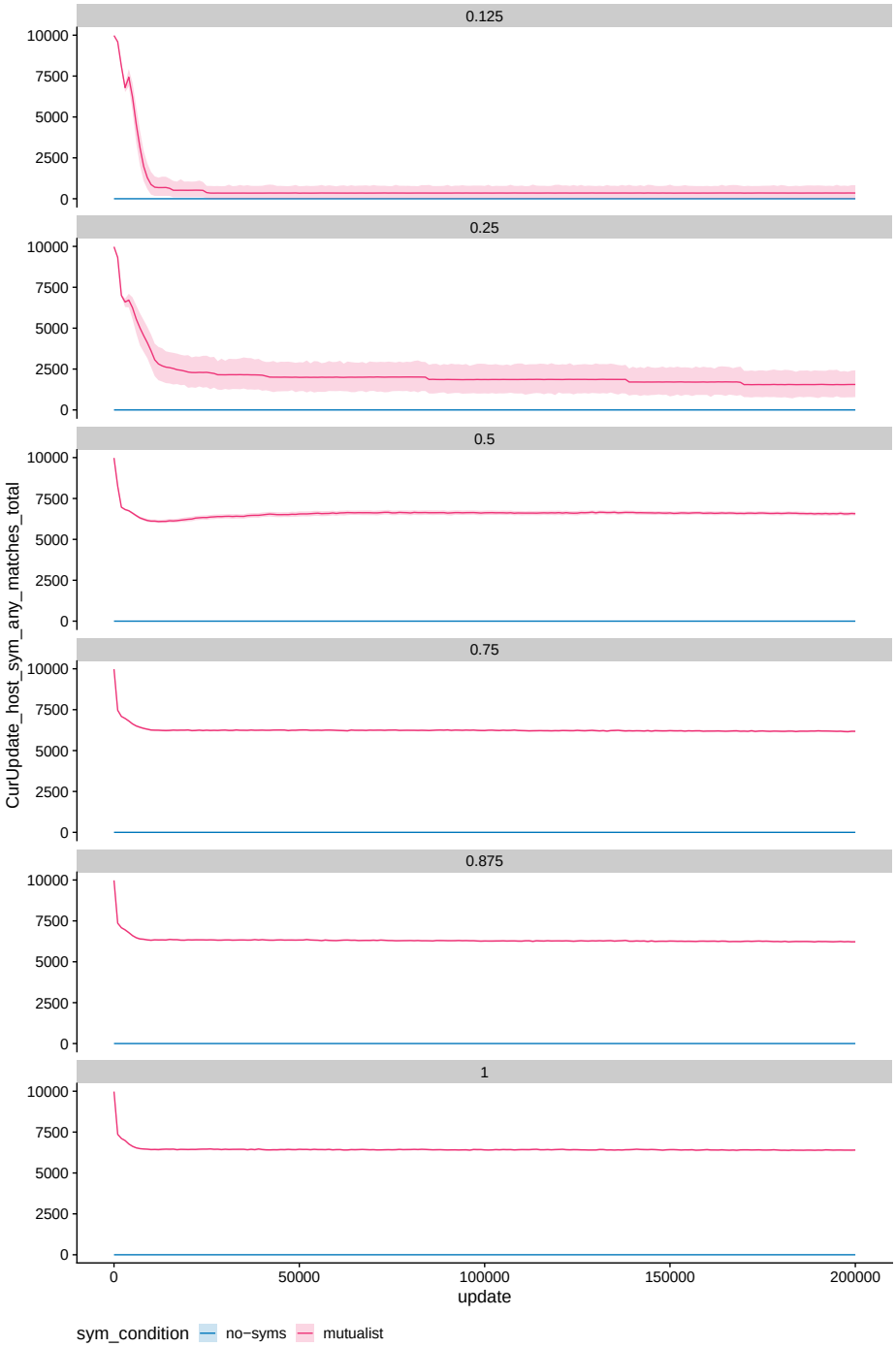
```
sym_final_entopy_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_parent_entropy_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  facet_wrap(~donate_prop, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_final_entopy_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = sym_final_entopy_task_sets_elapsed_plt,
  width = 10,
  height = 10
)
```

8.8 Trait matches over time

Counts of host-symbiont pairs with any trait matches. Note that organisms that were just born will not have had time to express any Boolean logic functions, so they will appear as not matching.

```
host_sym_any_matches_total_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
```

```
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_sym_any_matches_total,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_sym_any_matches_total_plt.pdf", sep = "/"),
  plot = host_sym_any_matches_total_plt,
  width = 8,
  height = 8
)
host_sym_any_matches_total_plt
```


0.125
0.25
0.5
0.75
0.875
1
10000
7500
5000
2500
0
CurUpdate_host_sym_any_matches_total
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

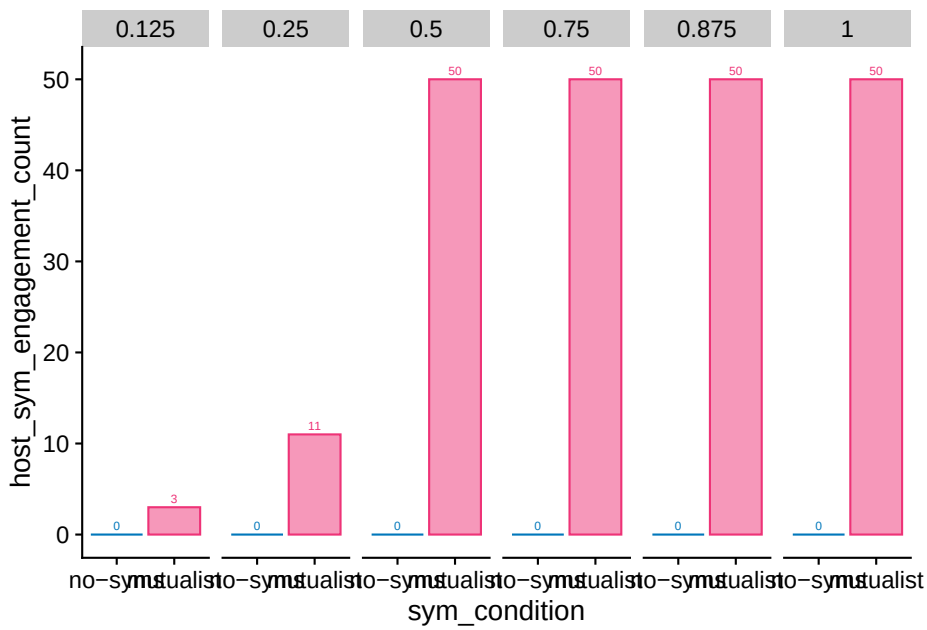
8.9 Evolutionary engagement

Use trait matching metrics to determine if host-symbiont populations are evolutionarily engaged.

```
engagement <- summary_data %>%
  mutate(
    host_sym_engagement = CurUpdate_host_sym_any_matches_total > threshold,
  ) %>%
  group_by(
    sym_condition,
    donate_prop
  ) %>%
  summarise(
    host_sym_engagement_count = sum(host_sym_engagement)
  )
```

```
host_sym_engagement_count_plt <- engagement %>%
  ggplot(
    aes(
      x = sym_condition,
      color = sym_condition,
      fill = sym_condition,
      y = host_sym_engagement_count
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = host_sym_engagement_count),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  facet_wrap(~donate_prop, nrow = 1) +
  theme(
    legend.position = "none"
```

```
  )

ggsave(
  file = paste(plot_dir, "host_sym_engagement_count_plt.pdf", sep = "/"),
  plot = host_sym_engagement_count_plt,
  width = 4,
  height = 4
)
host_sym_engagement_count_plt
```

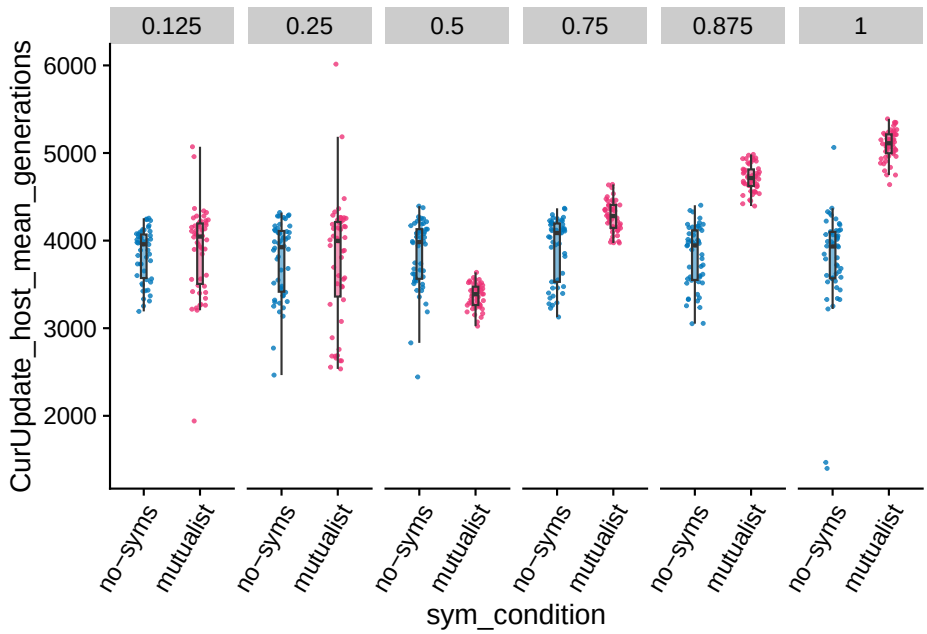


8.10 Generations of evolution elapsed

8.10.1 Hosts

```
host_avg_gens_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_mean_generations
    )
```

```
  ) +
  geom_point(
    mapping = aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_avg_gens_elapsed_plt.pdf", sep = "/"),
  plot = host_avg_gens_elapsed_plt,
  width = 4,
  height = 4
)
host_avg_gens_elapsed_plt
```



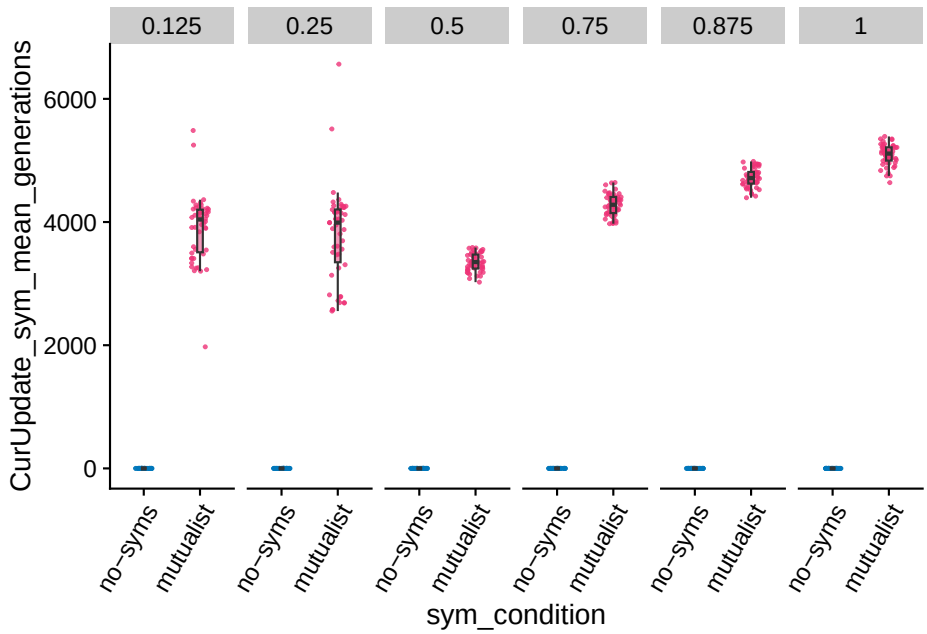

```
summary_data %>%
  group_by(
    sym_condition,
    donate_prop
  ) %>%
  summarise(
    mean_gens = mean(CurUpdate_host_mean_generations),
    median_gens = median(CurUpdate_host_mean_generations),
    var_gens = var(CurUpdate_host_mean_generations),
    sdev_gens = sd(CurUpdate_host_mean_generations)
  )
```

```
## # A tibble: 12 x 6
## # Groups:   sym_condition [2]
##    sym_condition donate_prop mean_gens median_gens var_gens sdev_gens
##    <fct>         <fct>           <dbl>       <dbl>    <dbl>     <dbl>
##  1 no-syms       0.125           3838.       3959.   88787.      298.
##  2 no-syms       0.25            3776.       3925.  191454.      438.
##  3 no-syms       0.5             3835.       3983.  164677.      406.
##  4 no-syms       0.75            3899.       4087.  139120.      373.
##  5 no-syms       0.875           3817.       3945.  127364.      357.
##  6 no-syms       1               3776.       3933.  358573.      599.
##  7 mutualist     0.125           3892.       4046.  261986.      512.
##  8 mutualist     0.25            3768.       3997.  510987.      715.
##  9 mutualist     0.5             3367.       3388.   19722.      140.
```

```
## 10 mutualist     0.75            4281.       4281.   30901.      176.
## 11 mutualist     0.875           4714.       4714.   23590.      154.
## 12 mutualist     1               5093.       5111.   28977.      170.
```

8.10.2 Symbionts

```
sym_avg_gens_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_mean_generations
    )
  ) +
  geom_point(
    mapping = aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_avg_gens_elapsed_plt.pdf", sep = "/"),
  plot = sym_avg_gens_elapsed_plt,
  width = 4,
  height = 4
)
sym_avg_gens_elapsed_plt
```




```
summary_data %>%
  group_by(
    sym_condition,
    donate_prop
  ) %>%
  summarise(
    mean_gens = mean(CurUpdate_sym_mean_generations),
    median_gens = median(CurUpdate_sym_mean_generations),
    var_gens = var(CurUpdate_sym_mean_generations),
    sdev_gens = sd(CurUpdate_sym_mean_generations)
  )
```

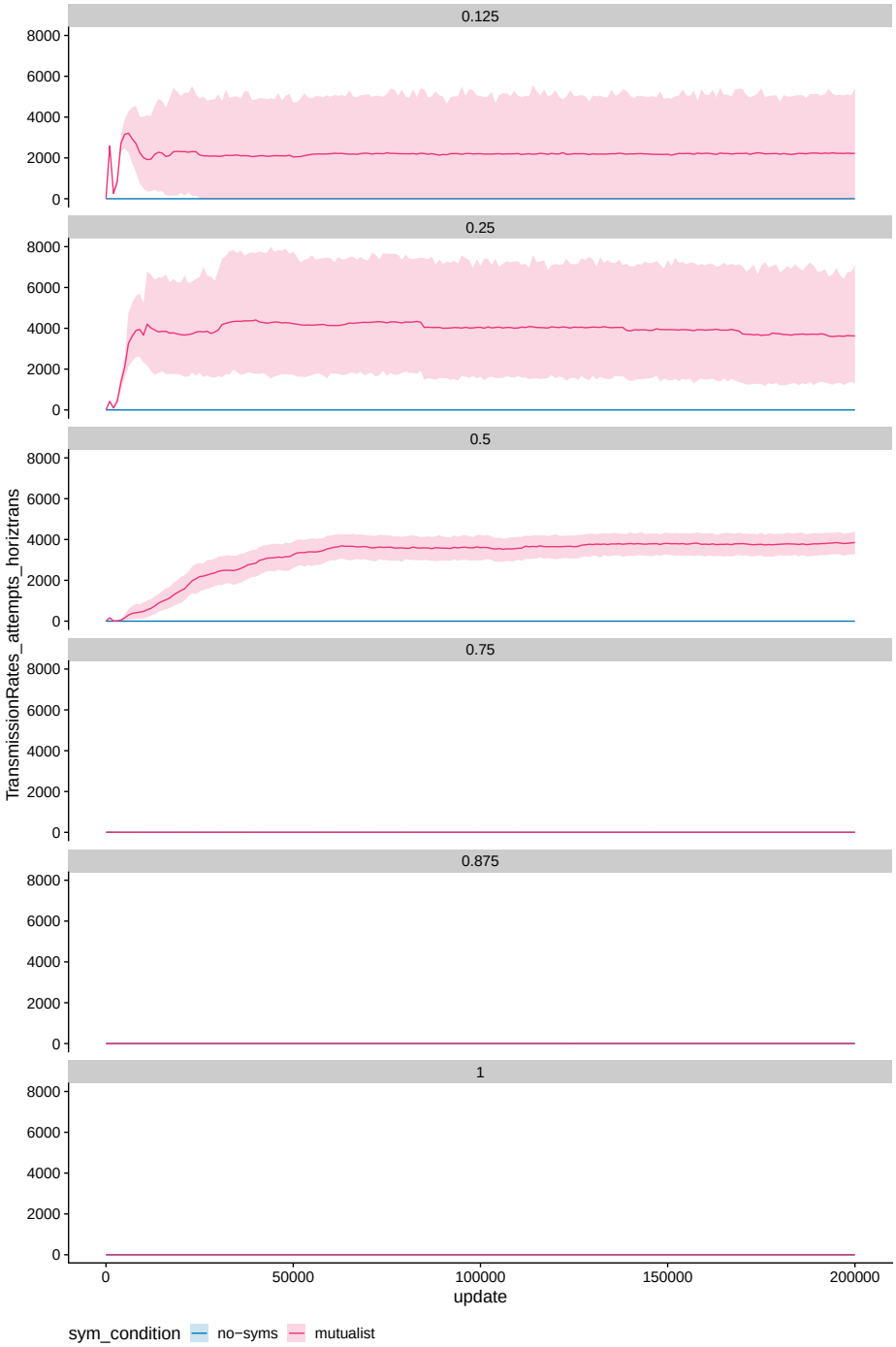
```
## # A tibble: 12 x 6
## # Groups:   sym_condition [2]
##    sym_condition donate_prop mean_gens median_gens var_gens sdev_gens
##    <fct>         <fct>           <dbl>       <dbl>    <dbl>     <dbl>
##  1 no-syms       0.125               0           0        0         0
##  2 no-syms       0.25                0           0        0         0
##  3 no-syms       0.5                 0           0        0         0
##  4 no-syms       0.75                0           0        0         0
##  5 no-syms       0.875               0           0        0         0
##  6 no-syms       1                   0           0        0         0
##  7 mutualist     0.125            3904.       4042.  297240.      545.
##  8 mutualist     0.25             3790.       3990.  578584.      761.
##  9 mutualist     0.5              3345.       3350.   20790.      144.
```

```
## 10 mutualist     0.75            4281.       4281.   30752.      175.
## 11 mutualist     0.875           4714.       4714.   23593.      154.
## 12 mutualist     1               5093.       5111.   28978.      170.
```

8.11 Symbiont transmission rates

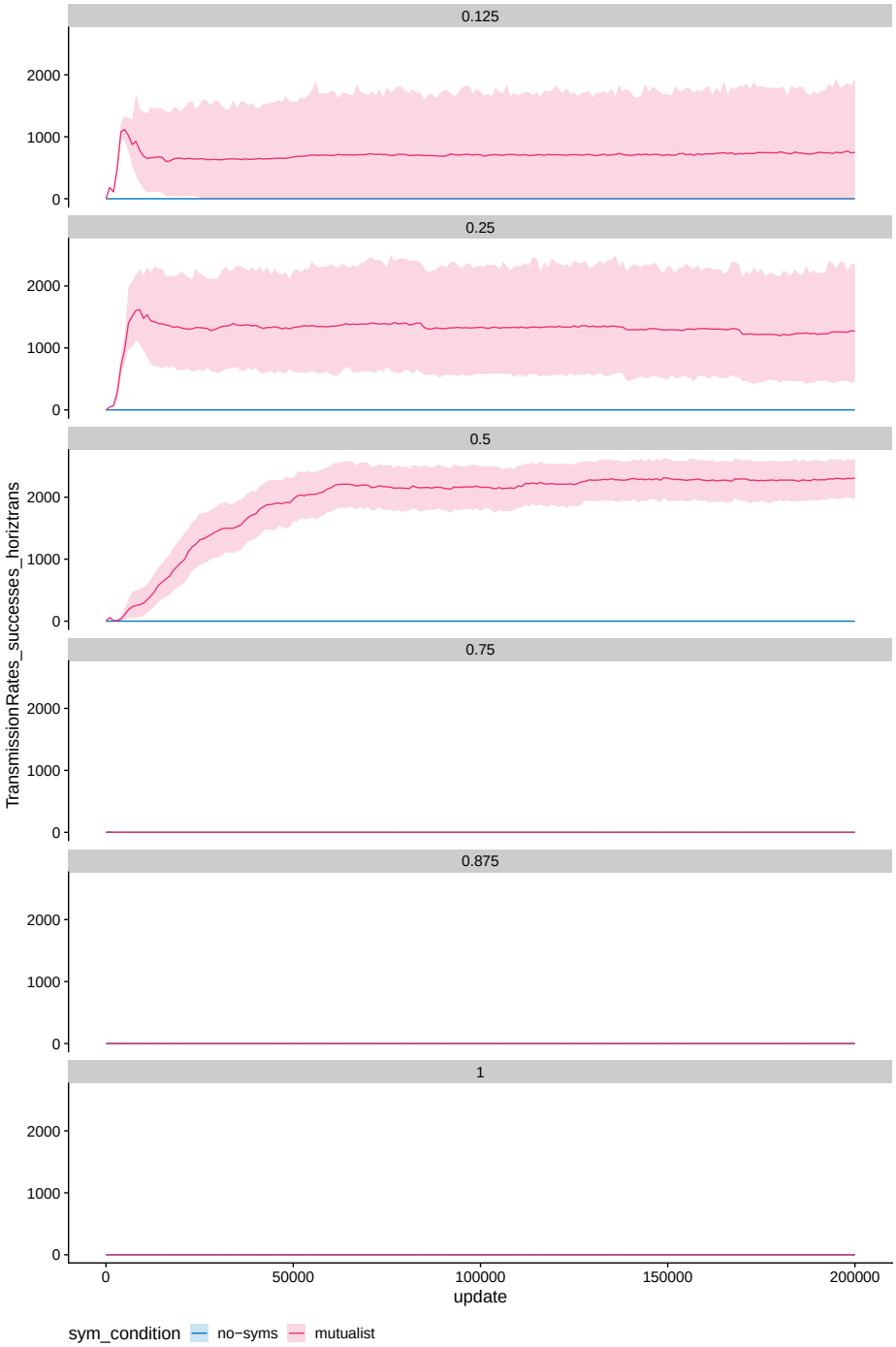
8.11.1 Horizontal transmission attempts

```
TransmissionRates_attempts_horiztrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_attempts_horiztrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_attempts_horiztrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_attempts_horiztrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_attempts_horiztrans_plt
```


0.125
0.25
0.5
0.75
0.875
1
8000
6000
4000
2000
0
TransmissionRates_attempts_horiztrans
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

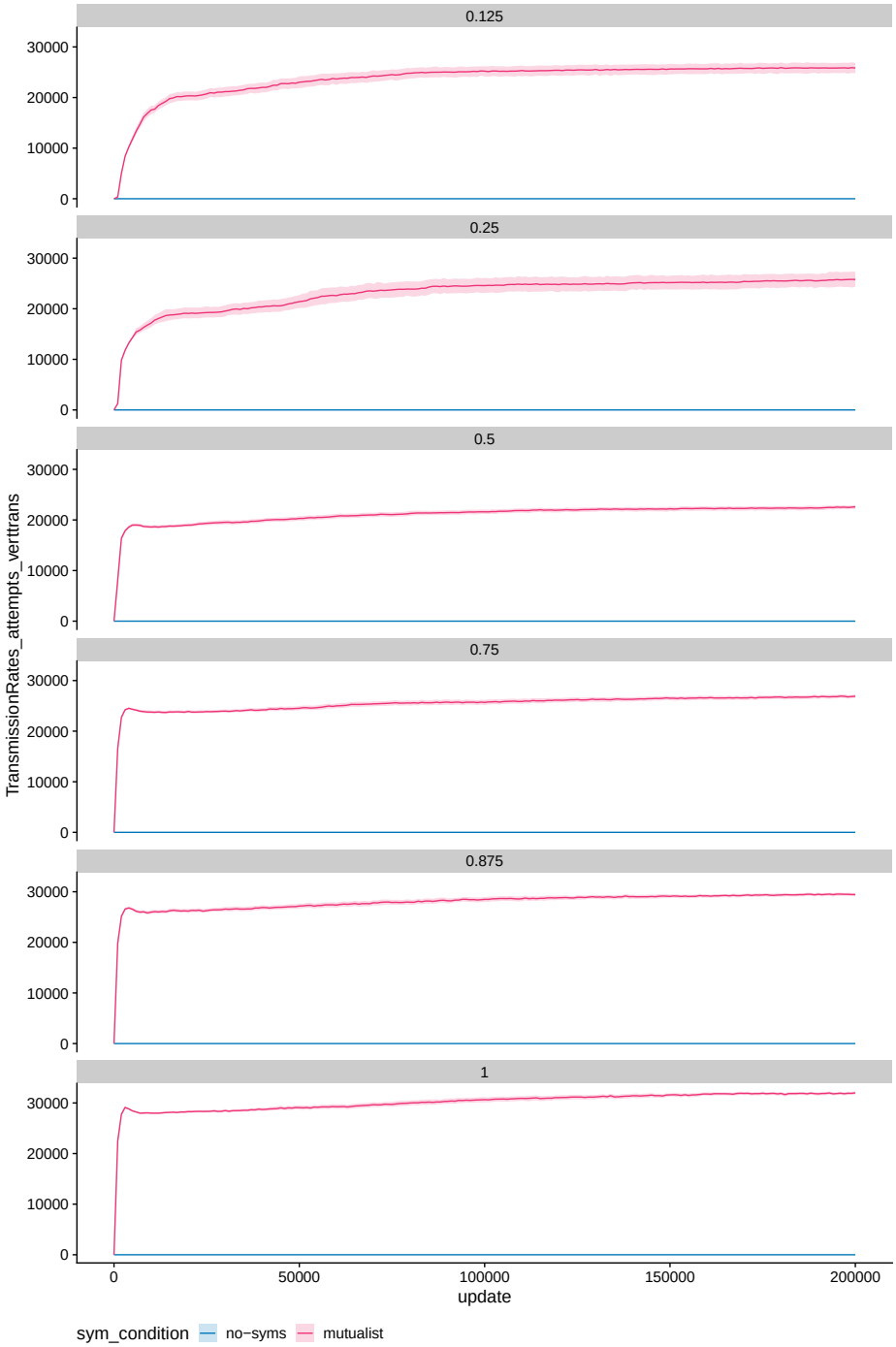
8.11.2 Horizontal transmission successes

```
TransmissionRates_successes_horiztrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_successes_horiztrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_successes_horiztrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_successes_horiztrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_successes_horiztrans_plt
```


0.125
0.25
0.5
0.75
0.875
1
2000
1000
0
TransmissionRates_successes_horiztrans
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

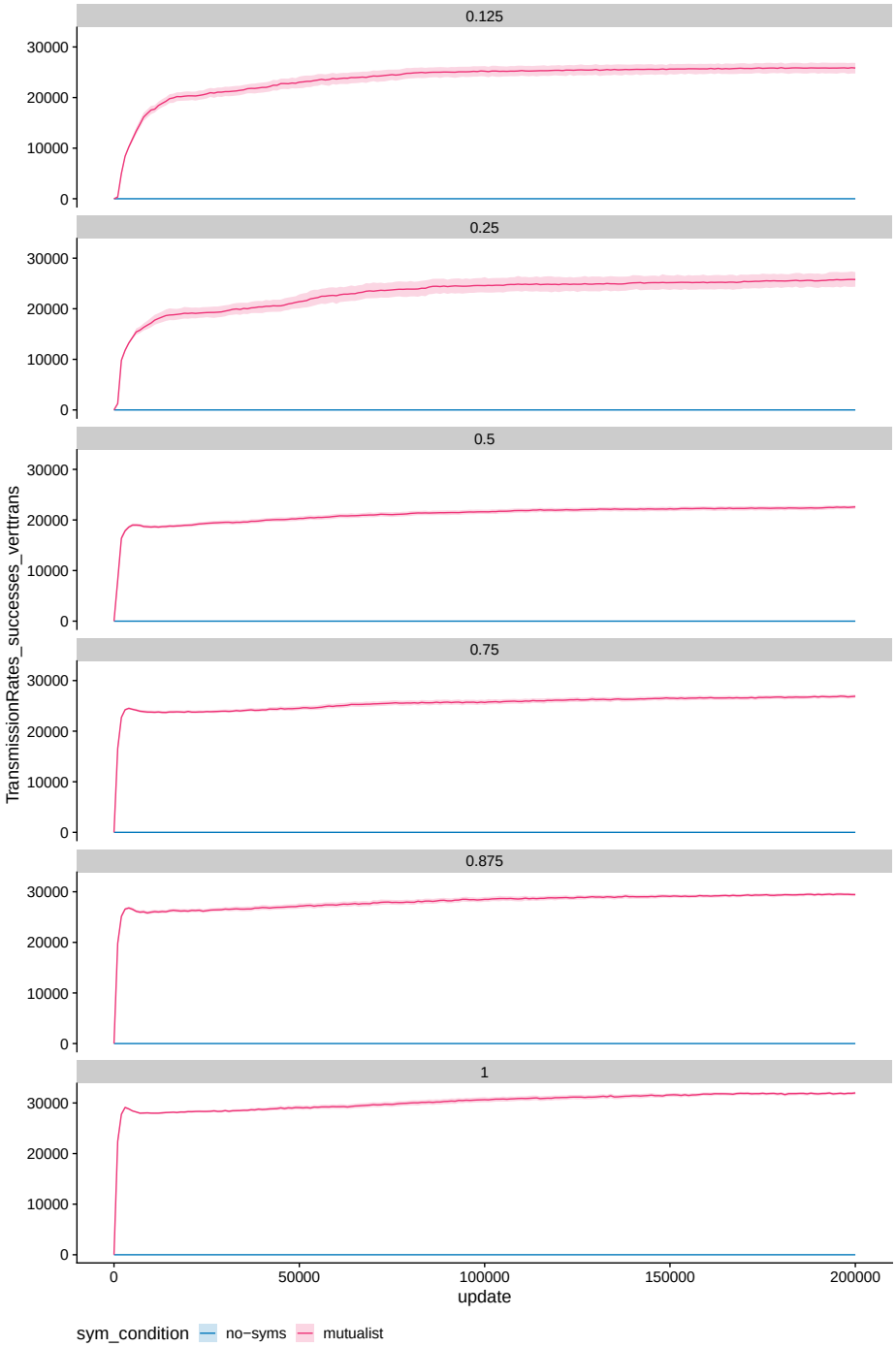
8.11.3 Vertical transmission attempts

```
TransmissionRates_attempts_verttrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_attempts_verttrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_attempts_verttrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_attempts_verttrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_attempts_verttrans_plt
```


0.125
0.25
0.5
0.75
0.875
1
30000
20000
10000
0
TransmissionRates_attempts_verttrans
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

8.11.4 Successes vertical transmission

```
TransmissionRates_successes_verttrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_successes_verttrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_successes_verttrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_successes_verttrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_successes_verttrans_plt
```


0.125
0.25
0.5
0.75
0.875
1
30000
20000
10000
0
TransmissionRates_successes_vertrans
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

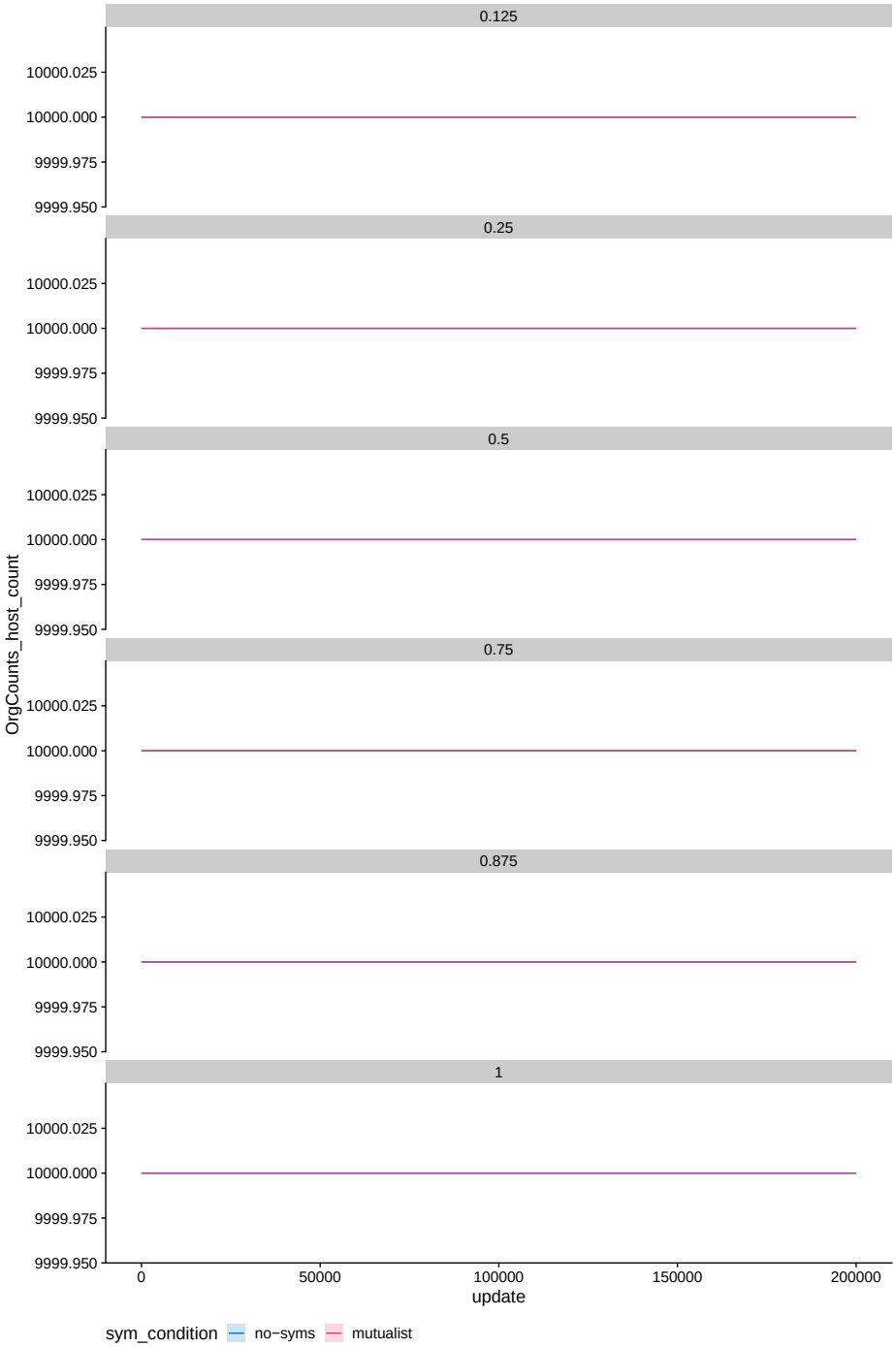
8.12 Population size

```
pop_size_summary <- time_data %>%
  group_by(
    SEED,
    sym_condition,
    donate_prop
  ) %>%
  summarize(
    host_pop_size_avg = mean(OrgCounts_host_count),
    host_pop_size_median = median(OrgCounts_host_count),
    host_pop_size_variance = var(OrgCounts_host_count),
    sym_pop_size_avg = mean(OrgCounts_hosted_sym_count),
    sym_pop_size_median = median(OrgCounts_hosted_sym_count),
    sym_pop_size_variance = var(OrgCounts_hosted_sym_count)
  )
```

8.12.1 Host population size

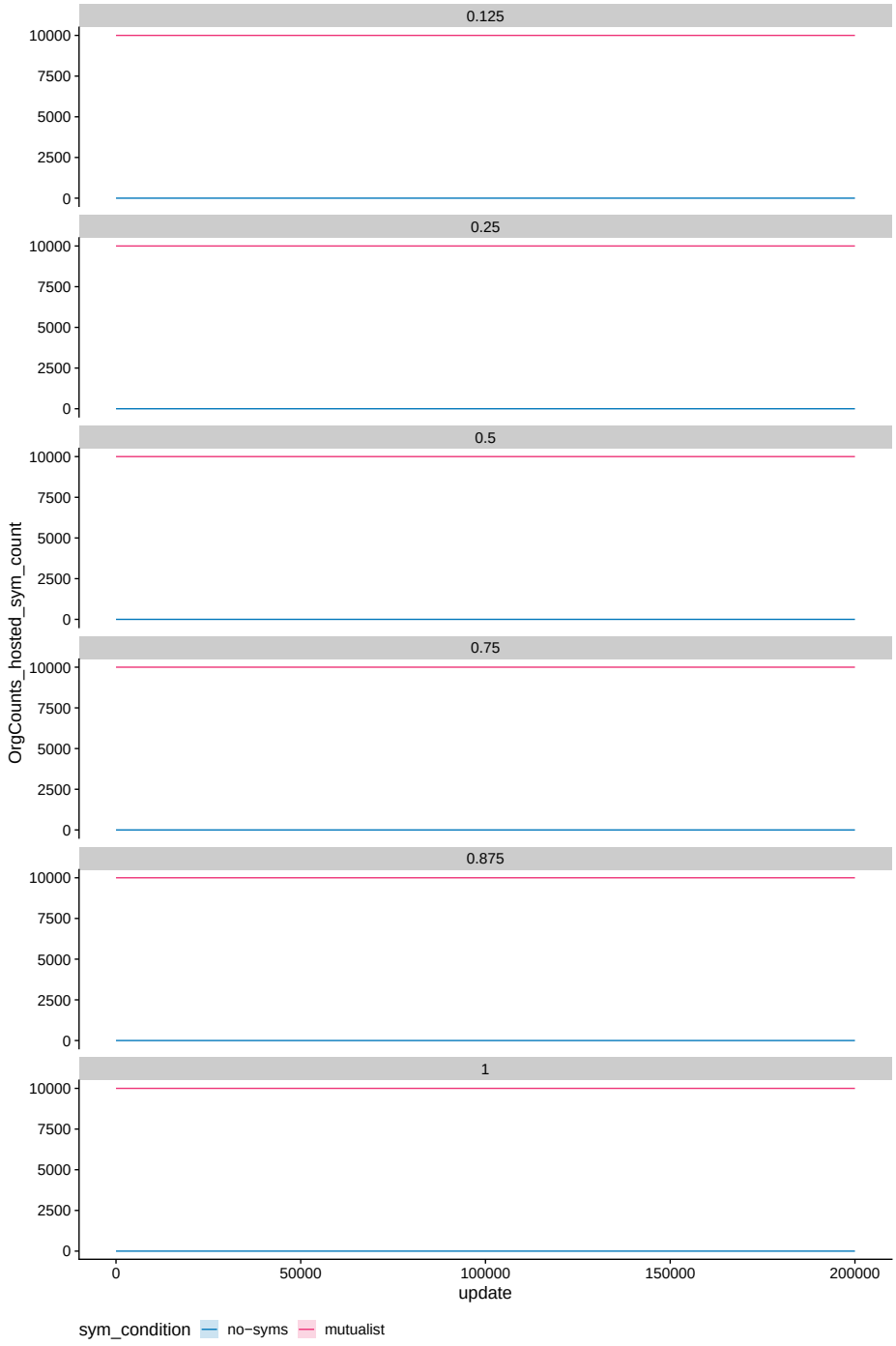
```
host_pop_size_ot_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = OrgCounts_host_count,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, ncol = 1) +
  theme(
    legend.position = "bottom"
```

```
  )
ggsave(
  file = paste(
    plot_dir,
    "host_pop_size_ot_plt.pdf",
    sep = "/"
  ),
  plot = host_pop_size_ot_plt,
  width = 8,
  height = 8
)
host_pop_size_ot_plt
```


0.125
0.25
0.5
0.75
0.875
1
10000.025
10000.000
9999.975
9999.950
OrgCounts_host_count
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

8.12.2 Symbiont population size

```
sym_pop_size_ot_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = OrgCounts_hosted_sym_count,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~donate_prop, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "sym_pop_size_ot_plt.pdf",
    sep = "/"
  ),
  plot = sym_pop_size_ot_plt,
  width = 8,
  height = 8
)
sym_pop_size_ot_plt
```


0.125
0.25
0.5
0.75
0.875
1
10000
7500
5000
2500
0
OrgCounts_hosted_sym_count
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

Chapter 9

Synergy factor

Repeat differential function rewards experiment, except vary the synergy factor for host-symbiont interactions.

9.1 Dependencies and setup

Load R library dependencies.

```
library(tidyverse)
library(cowplot)
library(khroma)
library(rstatix)
library(knitr)
library(kableExtra)
```

Define relevant file paths.

```
# Check if Rmd is being compiled using bookdown
bookdown <- exists("bookdown_build")

# Build relative path from R's working directory to what is actually the working
# directory for this experiment.
experiment_slug <- "2026-02-24-health-synergy"
supp_file_prefix <- "synergy"
exp_directory <- paste(
  "experiments",
  experiment_slug,
  "analysis",
```

```
  sep = "/"
)

# Get path to the supplemental material analysis directory
supp_directory <- paste(
  "experiments",
  "manuscript-analyses",
  sep = "/"
)

# Adjust experiment directory if being knitted for bookdown build
if (bookdown) {
  exp_directory <- paste0(
    bookdown_wd_prefix,
    exp_directory
  )
  supp_directory <- paste0(
    bookdown_wd_prefix,
    supp_directory
  )
}
```

Define default plot theme, plot output directory.

```
# Use the cowplot theme as our default for ggplots
theme_set(theme_cowplot())
# Create directory to save plots
plot_dir <- paste(
  supp_directory,
  "plots",
  supp_file_prefix,
  sep = "/"
)
dir.create(
  plot_dir,
  showWarnings = FALSE
)

# Use vibrant theme for plot color palettes
vibrant_colors <- color("vibrant")(7)
color_pal <- c(
  vibrant_colors[["blue"]],
  vibrant_colors[["magenta"]]
)
```

9.2 Load and preprocess data

```
tasks <- c("NOT", "NAND", "OR_NOT", "AND", "OR", "AND_NOT", "NOR", "XOR", "EQU")
max_pop_size <- 100 * 100
threshold <- 0.025 * max_pop_size
```

Load summary data.

```
summary_data_path <- paste(
  exp_directory,
  "data",
  "summary.csv",
  sep = "/"
)
summary_data <- read_csv(summary_data_path)
# Do any universal preprocessing
summary_data <- summary_data %>%
  mutate(
    SEED = as.factor(SEED),
    sym_condition = case_when(
      START_MOI == "0" ~ "no-syms",
      START_MOI == "1" & HEALTH_TYPE == "parasite" ~ "parasite",
      START_MOI == "1" & HEALTH_TYPE == "mutualist" ~ "mutualist",
      .default = "unknown" # precaution against unexpected parameterizations
    ),
    interaction_multiplier = as.factor(MUTUALIST_CYCLE_DONATE_MULTIPLIER)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist")
    )
  )
```

Load time series data.

```
time_data_path <- paste(
  exp_directory,
  "data",
  "time_series.csv",
  sep = "/"
)
time_data <- read_csv(time_data_path)
```

```
# Do any universal preprocessing
time_data <- time_data %>%
  mutate(
    SEED = as.factor(SEED),
    sym_condition = case_when(
      START_MOI == "0" ~ "no-syms",
      START_MOI == "1" & HEALTH_TYPE == "parasite" ~ "parasite",
      START_MOI == "1" & HEALTH_TYPE == "mutualist" ~ "mutualist",
      .default = "unknown" # precaution against unexpected parameterizations
    ),
    interaction_multiplier = as.factor(MUTUALIST_CYCLE_DONATE_MULTIPLIER)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist")
    )
  )
```

9.3 Extinctions

Did host or symbiont populations go extinct in any runs?

```
sym_min_pop_sizes <- time_data %>%
  group_by(SEED) %>%
  slice_min(
    OrgCounts_hosted_sym_count,
    n = 1
  ) %>%
  slice_min(
    update,
    n = 1
  )

reps_with_extinct_syms <- sym_min_pop_sizes %>%
  group_by(
    sym_condition,
    interaction_multiplier
  ) %>%
  summarize(
    reps_with_extinct_syms = sum(OrgCounts_hosted_sym_count == 0),
    total_reps = n()
  )
```

```
host_min_pop_sizes <- time_data %>%
  group_by(SEED) %>%
  slice_min(
    OrgCounts_host_count,
    n = 1
  ) %>%
  slice_min(
    update,
    n = 1
  )

reps_with_extinct_hosts <- host_min_pop_sizes %>%
  group_by(
    sym_condition,
    interaction_multiplier
  ) %>%
  summarize(
    reps_with_extinct_hosts = sum(OrgCounts_host_count == 0),
    total_reps = n()
  )

extinctions <- full_join(
  reps_with_extinct_syms,
  reps_with_extinct_hosts
)
extinctions <- extinctions %>%
  arrange(
    desc(reps_with_extinct_hosts), desc(reps_with_extinct_syms)
  )
extinction_table <- kable(extinctions) %>%
  kable_styling(latex_options = "striped")
save_kable(
  extinction_table,
  paste(
    plot_dir,
    "extinctions.pdf",
    sep = "/"
  )
)

extinctions
```

```
## # A tibble: 10 x 5
## # Groups:   sym_condition [2]
##    sym_condition interaction_multiplier reps_with_extinct_syms total_reps
```

```
##    <fct>         <fct>                                    <int>      <int>
##  1 no-syms       0.5                                         50         50
##  2 no-syms       1                                           50         50
##  3 no-syms       2                                           50         50
##  4 no-syms       4                                           50         50
##  5 no-syms       8                                           50         50
##  6 mutualist     0.5                                          0         50
##  7 mutualist     1                                            0         50
##  8 mutualist     2                                            0         50
##  9 mutualist     4                                            0         50
## 10 mutualist     8                                            0         50
## # i 1 more variable: reps_with_extinct_hosts <int>
```

9.4 Host function performance

First, quantify which functions have been performed over the course of an entire run.

```
# Pivot task data long, selecting only relevant columns
host_over_time_parent_task_data <- time_data %>%
  select(
    SEED,
    sym_condition,
    interaction_multiplier,
    update,
    CurUpdate_AND_NOT_in_host_parent_org_counts,
    CurUpdate_AND_in_host_parent_org_counts,
    CurUpdate_EQU_in_host_parent_org_counts,
    CurUpdate_NAND_in_host_parent_org_counts,
    CurUpdate_NOR_in_host_parent_org_counts,
    CurUpdate_NOT_in_host_parent_org_counts,
    CurUpdate_OR_NOT_in_host_parent_org_counts,
    CurUpdate_OR_in_host_parent_org_counts,
    CurUpdate_XOR_in_host_parent_org_counts
  ) %>%
  pivot_longer(
    c(
      CurUpdate_AND_NOT_in_host_parent_org_counts,
      CurUpdate_AND_in_host_parent_org_counts,
      CurUpdate_EQU_in_host_parent_org_counts,
      CurUpdate_NAND_in_host_parent_org_counts,
      CurUpdate_NOR_in_host_parent_org_counts,
      CurUpdate_NOT_in_host_parent_org_counts,
      CurUpdate_OR_NOT_in_host_parent_org_counts,
```

```
      CurUpdate_OR_in_host_parent_org_counts,
      CurUpdate_XOR_in_host_parent_org_counts
    ),
    names_to = "task",
    values_to = "task_in_parent_count"
  ) %>%
  mutate(
    task = str_replace(task, "CurUpdate_", "")
  ) %>%
  mutate(
    task = str_replace(task, "_in_host_parent_org_counts", "")
  ) %>%
  mutate(
    task_completed_by_parent = task_in_parent_count >= threshold,
    task = factor(
      task,
      levels = tasks,
      labels = tasks
    )
  ) %>%
  mutate(
    task_complexity = case_when(
      task == "NOT" ~ 1,
      task == "NAND" ~ 1,
      task == "OR_NOT" ~ 2,
      task == "AND" ~ 2,
      task == "OR" ~ 3,
      task == "AND_NOT" ~ 3,
      task == "NOR" ~ 4,
      task == "XOR" ~ 4,
      task == "EQU" ~ 5,
      .default = -1
    )
  )

host_parent_tasks_all_time_by_task <- host_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier,
    task
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
```

```
  ) %>%
  ungroup() %>%
  group_by(
    sym_condition,
    interaction_multiplier,
    task
  ) %>%
  summarize(
    n = n(),
    parent_tasks_completed_ever = sum(parent_task_completed_ever == TRUE)
  )
```

Next, summarize function complexity level.

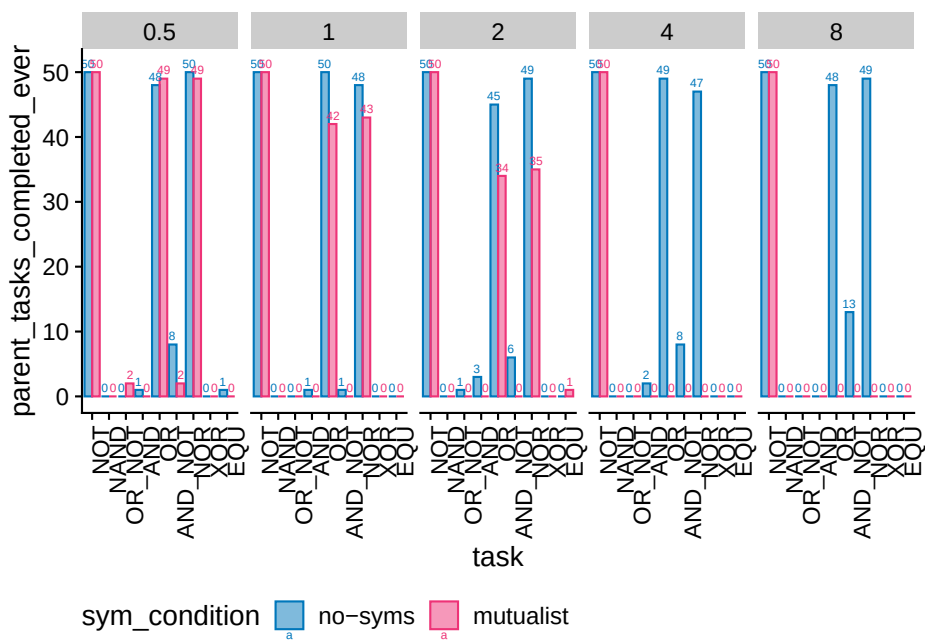
```
host_max_task_complexity_data <- host_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier,
    task,
    task_complexity
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  # This filter works fine because all replicates start with a task
  filter(
    parent_task_completed_ever
  ) %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier
  ) %>%
  summarize(
    max_task_complexity = max(task_complexity)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c(
        "no-syms",
        "parasite",
```

```
        "mutualist"
      )
    )
  )
```

9.4.1 Host function performance - all time

```
host_parent_tasks_all_time_by_task_plt <- host_parent_tasks_all_time_by_task %>%
  ggplot(
    aes(
      x = task,
      color = sym_condition,
      fill = sym_condition,
      y = parent_tasks_completed_ever
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = parent_tasks_completed_ever),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  facet_wrap(~interaction_multiplier, nrow = 1) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(
      angle = 90,
      hjust = 1
    )
  )
ggsave(
  file = paste(
    plot_dir,
```

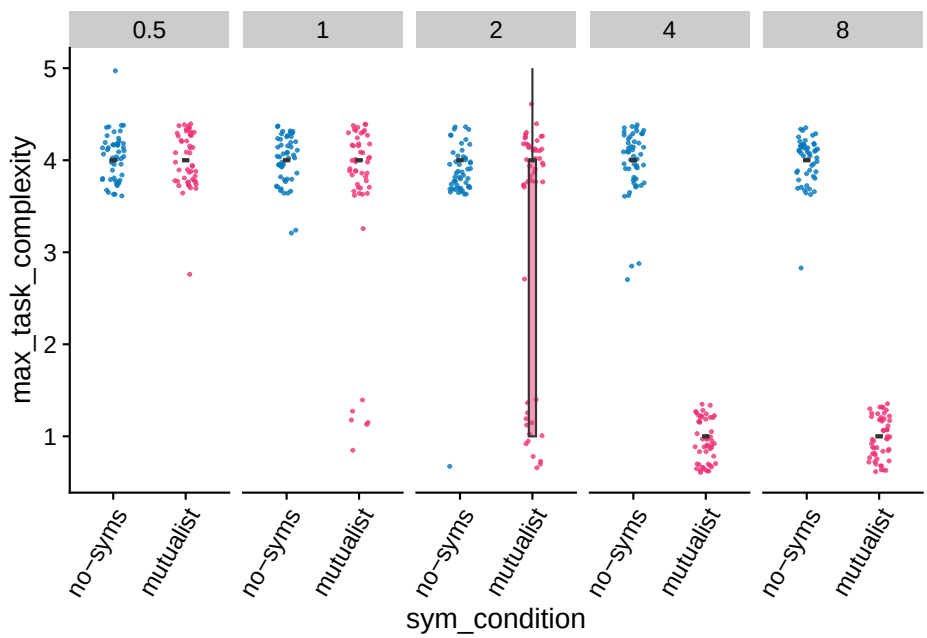
```
    "host_parent_tasks_all_time_by_task_plt.pdf",
    sep = "/"
  ),
  plot = host_parent_tasks_all_time_by_task_plt,
  width = 10,
  height = 7
)
host_parent_tasks_all_time_by_task_plt
```



9.4.2 Host maximum function complexity evolved

```
host_max_task_complexity_plt <- host_max_task_complexity_data %>%
  ggplot(
    aes(x = sym_condition, fill = sym_condition, y = max_task_complexity)
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
```

```
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_max_task_complexity_plt.pdf", sep = "/"),
  plot = host_max_task_complexity_plt,
  width = 4,
  height = 4
)
host_max_task_complexity_plt
```



9.5 Symbiont function performance

```
# Pivot task data long, selecting only relevant columns
sym_over_time_parent_task_data <- time_data %>%
  select(
    SEED,
    sym_condition,
    interaction_multiplier,
    update,
    CurUpdate_AND_NOT_in_sym_parent_org_counts,
    CurUpdate_AND_in_sym_parent_org_counts,
    CurUpdate_EQU_in_sym_parent_org_counts,
    CurUpdate_NAND_in_sym_parent_org_counts,
    CurUpdate_NOR_in_sym_parent_org_counts,
    CurUpdate_NOT_in_sym_parent_org_counts,
    CurUpdate_OR_NOT_in_sym_parent_org_counts,
    CurUpdate_OR_in_sym_parent_org_counts,
    CurUpdate_XOR_in_sym_parent_org_counts
  ) %>%
  pivot_longer(
    c(
      CurUpdate_AND_NOT_in_sym_parent_org_counts,
      CurUpdate_AND_in_sym_parent_org_counts,
      CurUpdate_EQU_in_sym_parent_org_counts,
      CurUpdate_NAND_in_sym_parent_org_counts,
      CurUpdate_NOR_in_sym_parent_org_counts,
      CurUpdate_NOT_in_sym_parent_org_counts,
      CurUpdate_OR_NOT_in_sym_parent_org_counts,
      CurUpdate_OR_in_sym_parent_org_counts,
      CurUpdate_XOR_in_sym_parent_org_counts
    ),
    names_to = "task",
    values_to = "task_in_parent_count"
  ) %>%
  mutate(
    task = str_replace(task, "CurUpdate_", "")
  ) %>%
  mutate(
    task = str_replace(task, "_in_sym_parent_org_counts", "")
  ) %>%
  mutate(
    task_completed_by_parent = task_in_parent_count >= threshold,
    task = factor(
      task,
```

```
      levels = tasks,
      labels = tasks
    )
  ) %>%
  mutate(
    task_complexity = case_when(
      task == "NOT" ~ 1,
      task == "NAND" ~ 1,
      task == "OR_NOT" ~ 2,
      task == "AND" ~ 2,
      task == "OR" ~ 3,
      task == "AND_NOT" ~ 3,
      task == "NOR" ~ 4,
      task == "XOR" ~ 4,
      task == "EQU" ~ 5,
      .default = -1
    )
  )

sym_parent_tasks_all_time_by_task <- sym_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier,
    task
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  group_by(
    sym_condition,
    interaction_multiplier,
    task
  ) %>%
  summarize(
    n = n(),
    parent_tasks_completed_ever = sum(parent_task_completed_ever == TRUE)
  )
```

Next, summarize function complexity level.

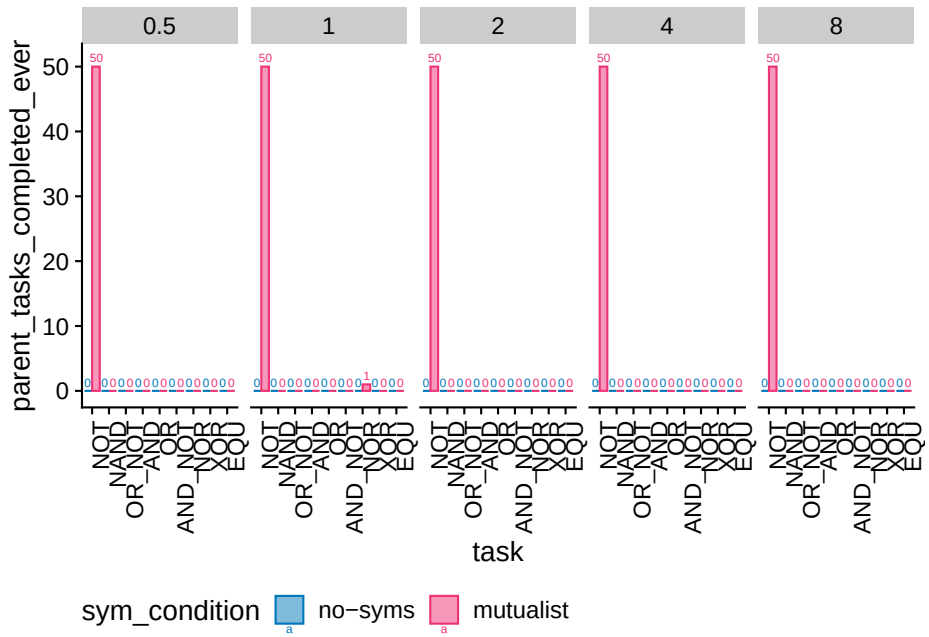
```
sym_max_task_complexity_data <- sym_over_time_parent_task_data %>%
  group_by(
```

```
    SEED,
    sym_condition,
    interaction_multiplier,
    task,
    task_complexity
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  # This filter works fine because all replicates start with a task
  filter(
    parent_task_completed_ever
  ) %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier
  ) %>%
  summarize(
    max_task_complexity = max(task_complexity)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c(
        "no-syms",
        "parasite",
        "mutualist"
      )
    )
  )
```

9.5.1 Symbiont function performance - all time

```
sym_parent_tasks_all_time_by_task_plt <- sym_parent_tasks_all_time_by_task %>%
  ggplot(
    aes(
      x = task,
      color = sym_condition,
      fill = sym_condition,
      y = parent_tasks_completed_ever
```

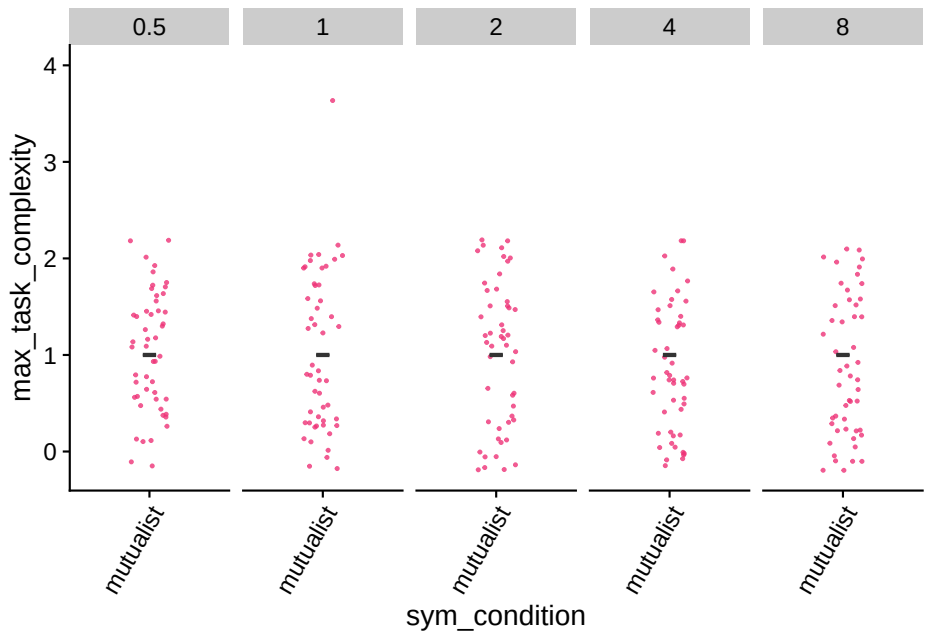
```
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = parent_tasks_completed_ever),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, nrow = 1) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(
      angle = 90,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_parent_tasks_all_time_by_task_plt.pdf", sep = "/"),
  plot = sym_parent_tasks_all_time_by_task_plt,
  width = 8,
  height = 8
)
sym_parent_tasks_all_time_by_task_plt
```



9.5.2 Symbiont maximum function complexity evolved

```
sym_max_task_complexity_plt <- sym_max_task_complexity_data %>%
  ggplot(
    aes(x = sym_condition, fill = sym_condition, y = max_task_complexity)
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = c(color_pal[2], color_pal[3])) +
  scale_fill_manual(values = c(color_pal[2], color_pal[3])) +
  facet_wrap(~interaction_multiplier, nrow = 1) +
  theme(
    legend.position = "none",
```

```
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_max_task_complexity_plt.pdf", sep = "/"),
  plot = sym_max_task_complexity_plt,
  width = 4,
  height = 4
)
sym_max_task_complexity_plt
```



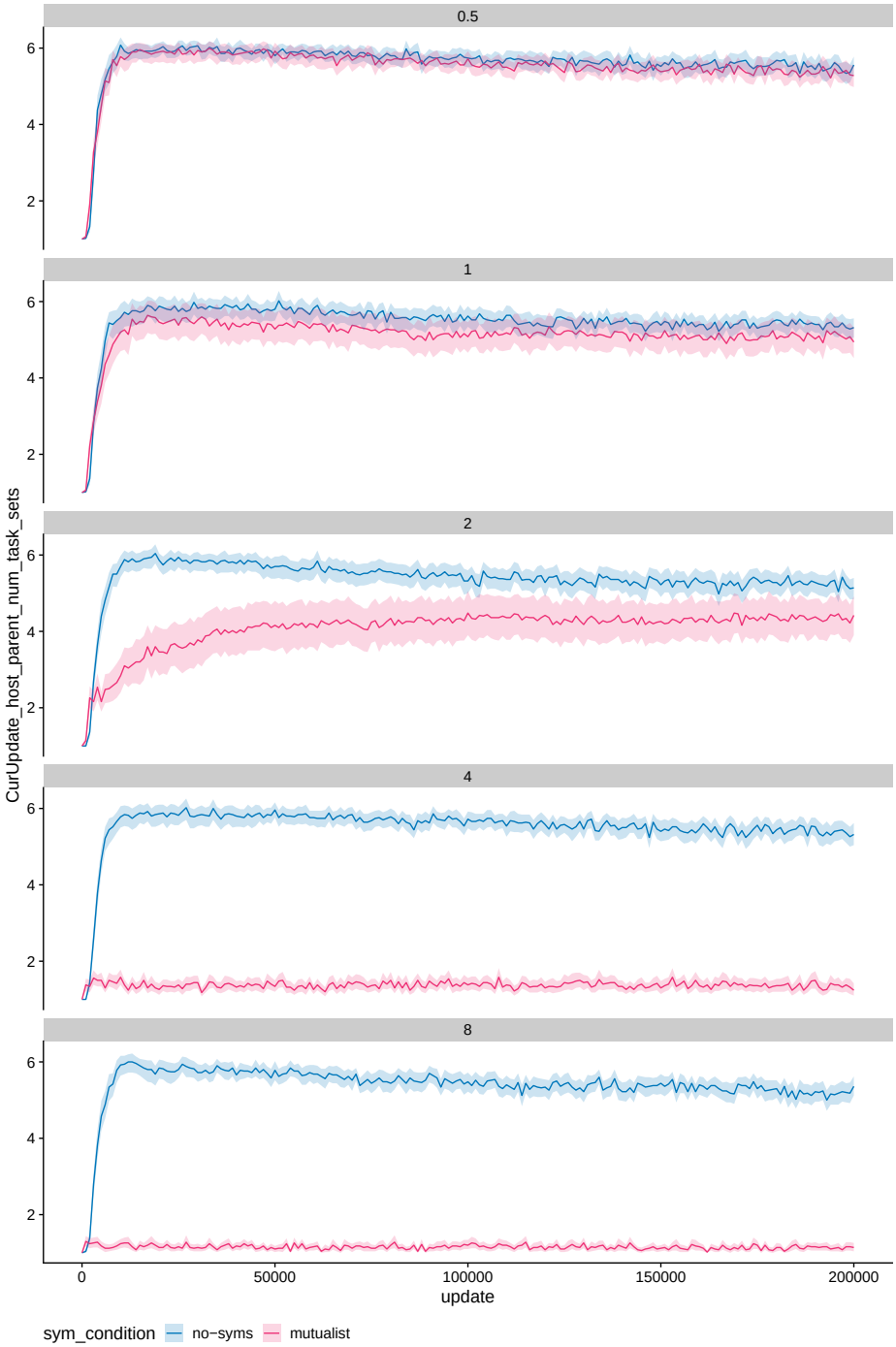
9.6 Host phenotypic diversity

9.6.1 Number of distinct function profiles

Over time:

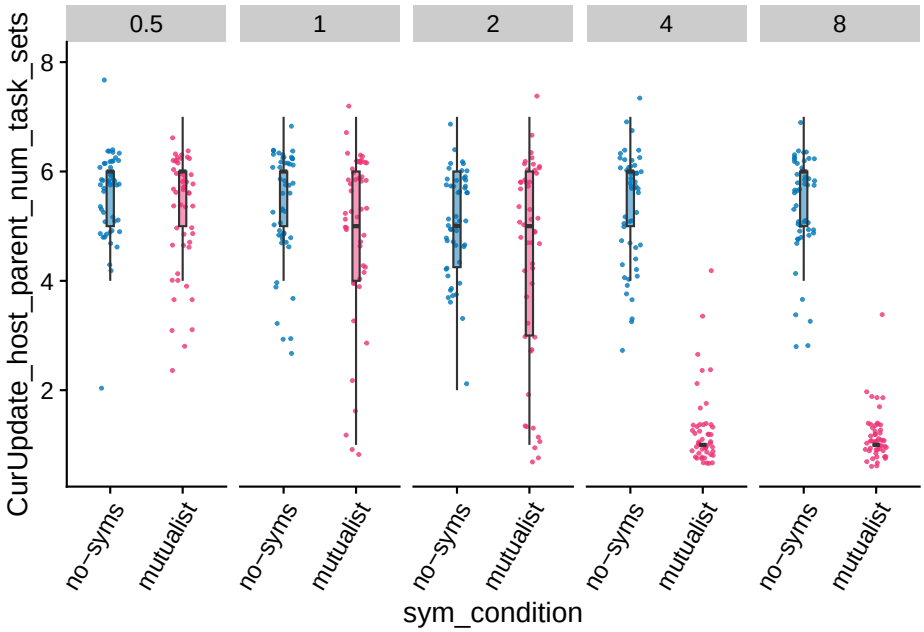
```
host_parent_num_task_sets_plt <- time_data %>%
  # sample to make plotting faster
  filter(update %% 1000 == 0) %>%
```

```
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_parent_num_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_parent_num_task_sets_plt.pdf", sep = "/"),
  plot = host_parent_num_task_sets_plt,
  width = 8,
  height = 8
)
host_parent_num_task_sets_plt
```



At the end of the experiment:

```
host_final_num_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_parent_num_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_final_num_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = host_final_num_task_sets_elapsed_plt,
  width = 4,
  height = 4
)
host_final_num_task_sets_elapsed_plt
```

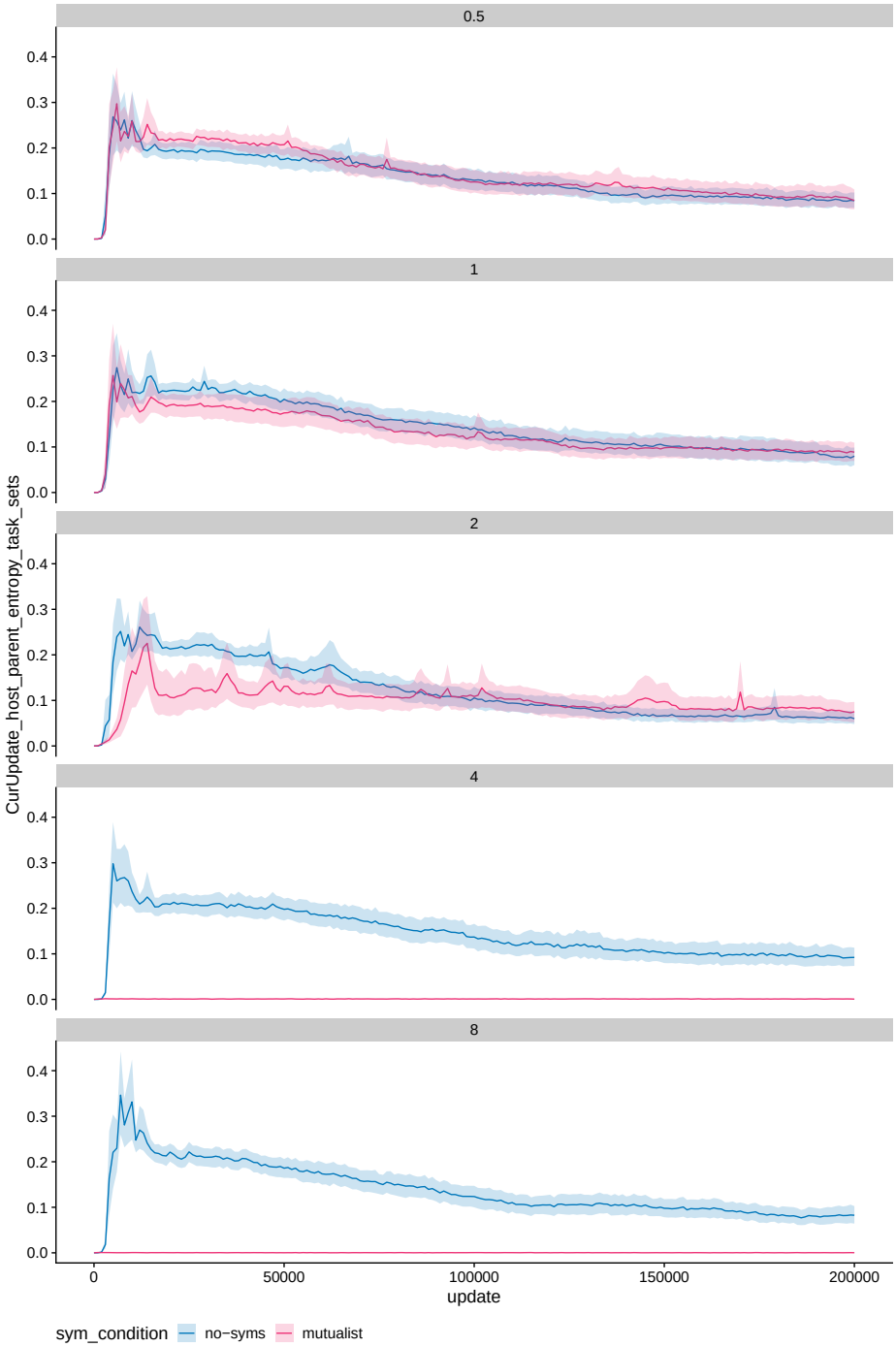


9.6.2 Shannon diversity of function profiles

Over time:

```
host_parent_entropy_task_sets_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
```

```
  facet_wrap(~interaction_multiplier, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_parent_entropy_task_sets_plt.pdf", sep = "/"),
  plot = host_parent_entropy_task_sets_plt,
  width = 8,
  height = 8
)
host_parent_entropy_task_sets_plt
```



At the end of the experiment:

```
host_final_entopy_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_parent_entropy_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  facet_wrap(~interaction_multiplier, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_final_entopy_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = host_final_entopy_task_sets_elapsed_plt,
  width = 10,
  height = 10
)
```

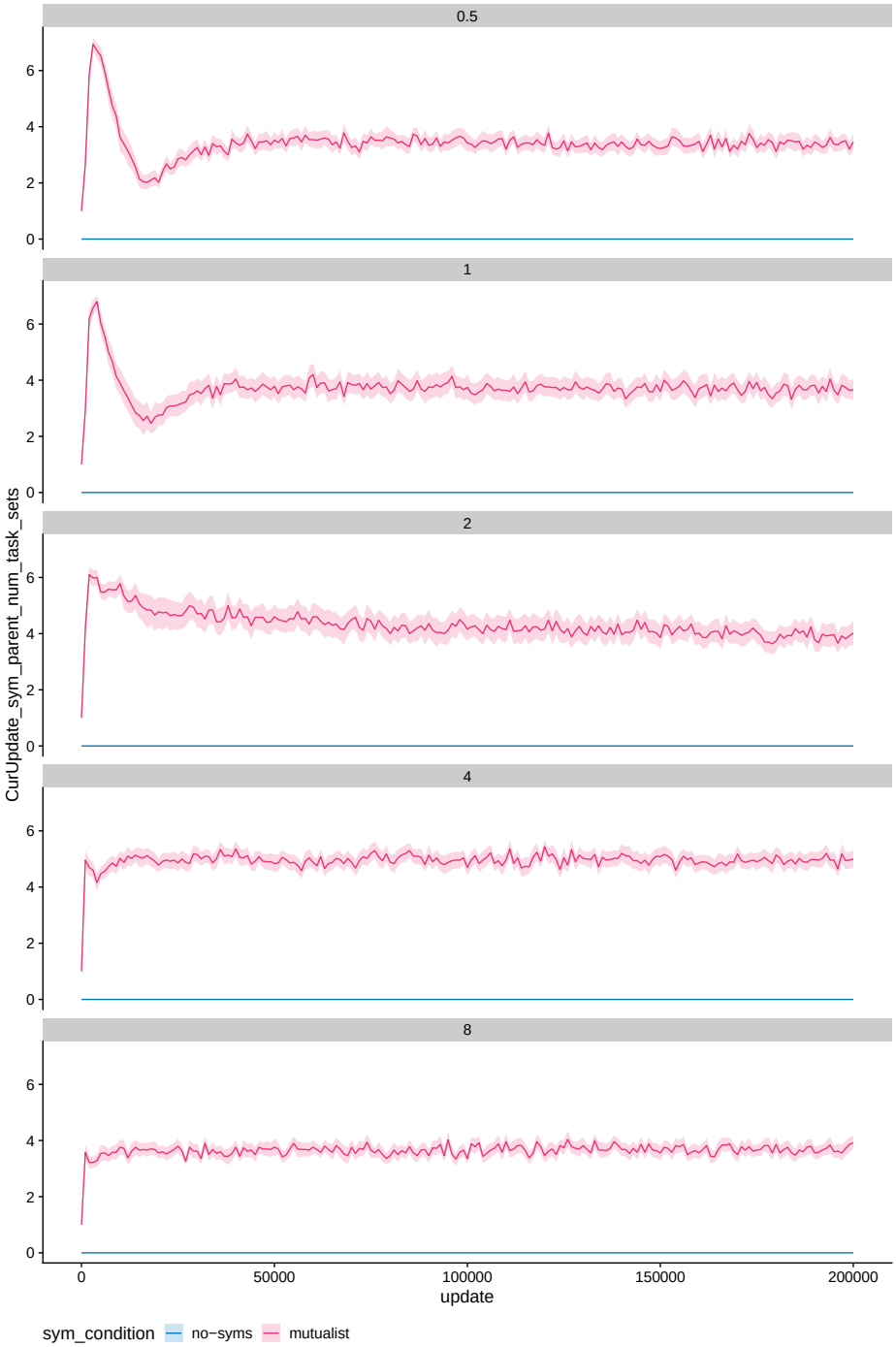
9.7 Symbiont phenotypic diversity

9.7.1 Number of distinct function profiles

Over time:

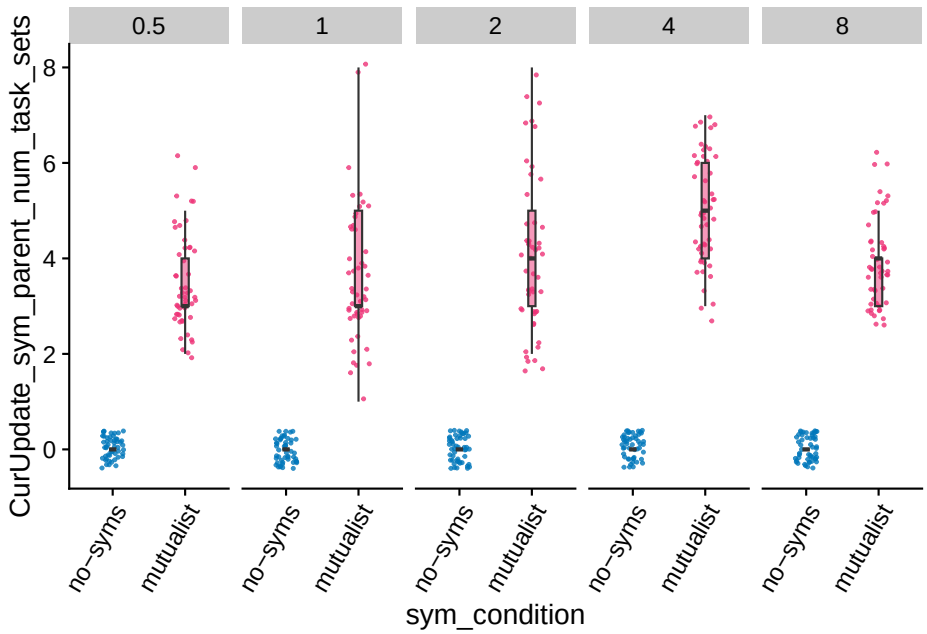
```
sym_parent_num_task_sets_plt <- time_data %>%
  # sample to make plotting faster
```

```
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_sym_parent_num_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "sym_parent_num_task_sets_plt.pdf", sep = "/"),
  plot = sym_parent_num_task_sets_plt,
  width = 8,
  height = 8
)
sym_parent_num_task_sets_plt
```



At the end of the experiment:

```
sym_final_num_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_parent_num_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_final_num_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = sym_final_num_task_sets_elapsed_plt,
  width = 4,
  height = 4
)
sym_final_num_task_sets_elapsed_plt
```

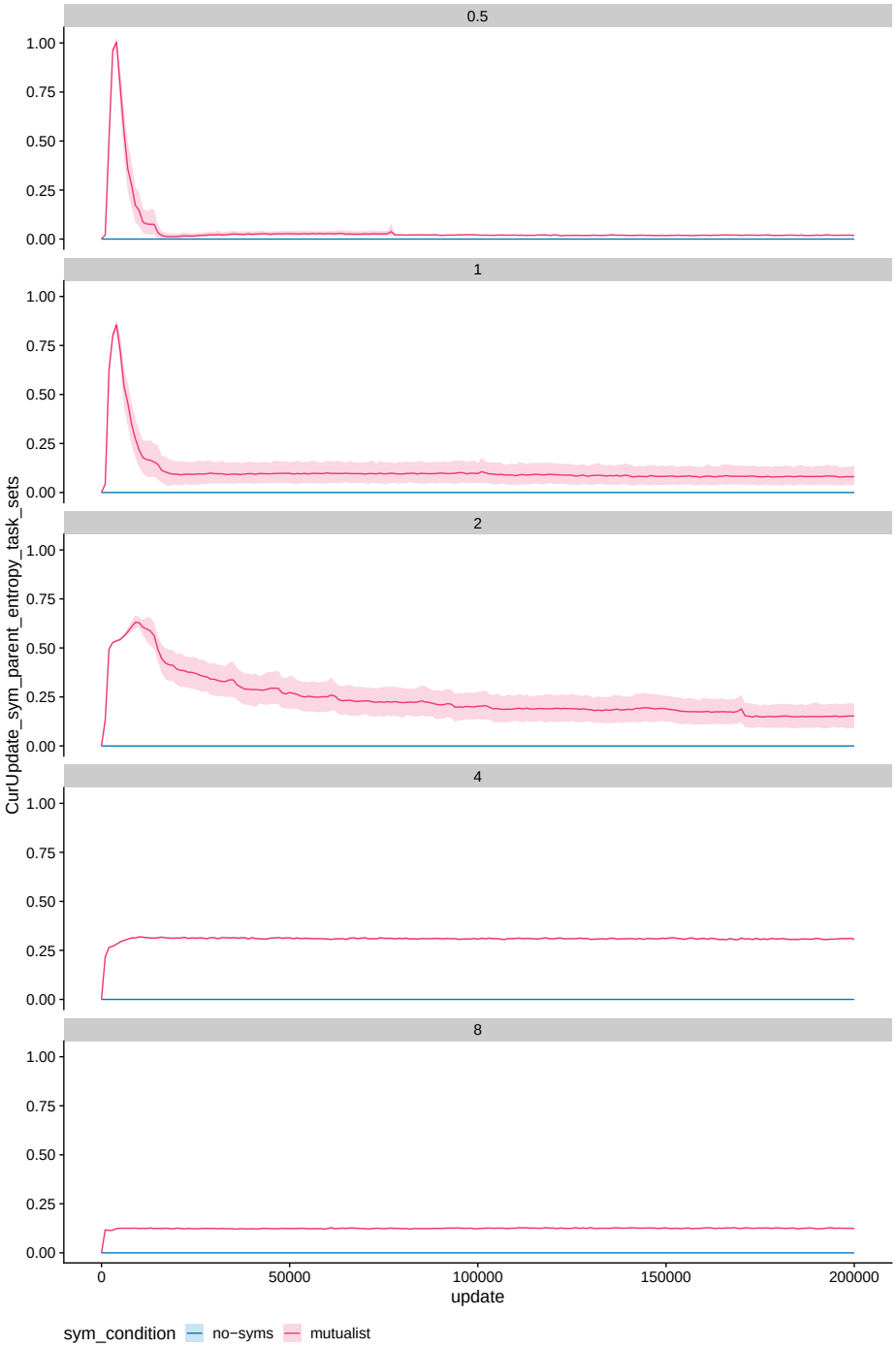


9.7.2 Shannon diversity of function profiles

Over time:

```
sym_parent_entropy_task_sets_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_sym_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
```

```
  facet_wrap(~interaction_multiplier, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "sym_parent_entropy_task_sets_plt.pdf", sep = "/"),
  plot = sym_parent_entropy_task_sets_plt,
  width = 8,
  height = 8
)
sym_parent_entropy_task_sets_plt
```



At the end of the experiment:

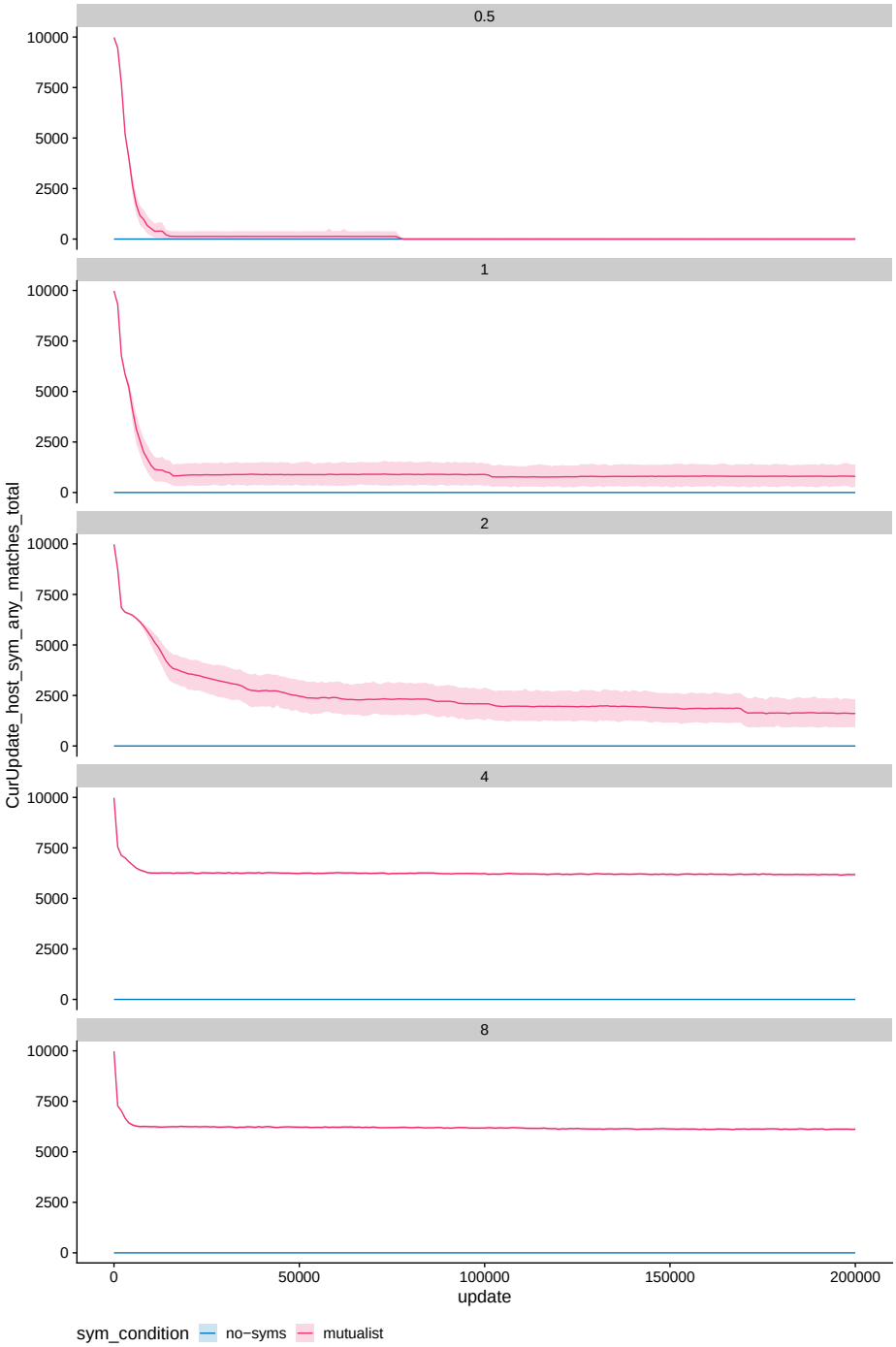
```
sym_final_entopy_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_parent_entropy_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  facet_wrap(~interaction_multiplier, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_final_entopy_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = sym_final_entopy_task_sets_elapsed_plt,
  width = 10,
  height = 10
)
```

9.8 Trait matches over time

Counts of host-symbiont pairs with any trait matches. Note that organisms that were just born will not have had time to express any Boolean logic functions, so they will appear as not matching.

```
host_sym_any_matches_total_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
```

```
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_sym_any_matches_total,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_sym_any_matches_total_plt.pdf", sep = "/"),
  plot = host_sym_any_matches_total_plt,
  width = 8,
  height = 8
)
host_sym_any_matches_total_plt
```


0.5
1
2
4
8
10000
7500
5000
2500
0
CurUpdate_host_sym_any_matches_total
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

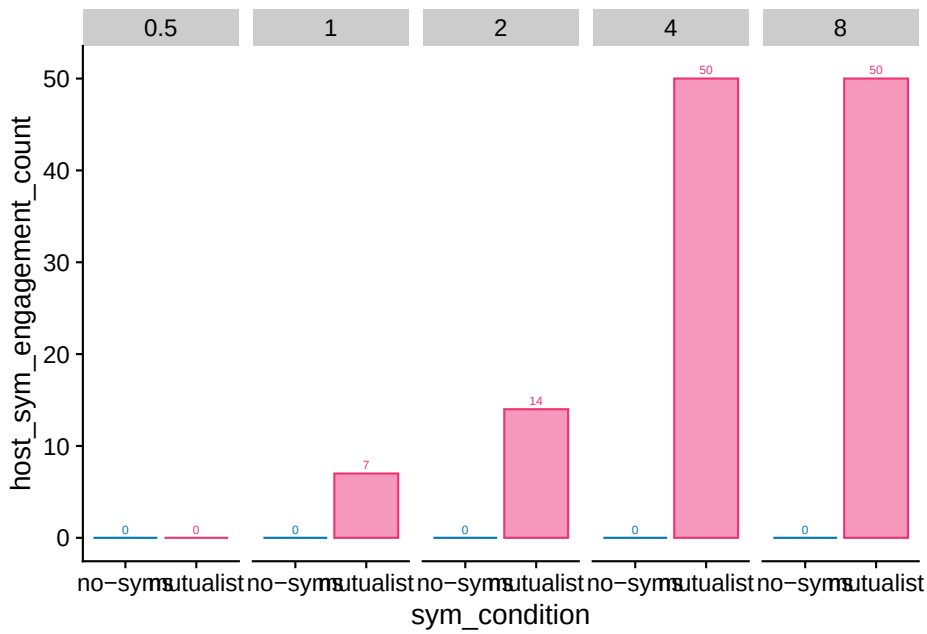
9.9 Evolutionary engagement

Use trait matching metrics to determine if host-symbiont populations are evolutionarily engaged.

```
engagement <- summary_data %>%
  mutate(
    host_sym_engagement = CurUpdate_host_sym_any_matches_total > threshold,
  ) %>%
  group_by(
    sym_condition,
    interaction_multiplier
  ) %>%
  summarise(
    host_sym_engagement_count = sum(host_sym_engagement)
  )
```

```
host_sym_engagement_count_plt <- engagement %>%
  ggplot(
    aes(
      x = sym_condition,
      color = sym_condition,
      fill = sym_condition,
      y = host_sym_engagement_count
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = host_sym_engagement_count),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  facet_wrap(~interaction_multiplier, nrow = 1) +
  theme(
    legend.position = "none"
```

```
  )

ggsave(
  file = paste(plot_dir, "host_sym_engagement_count_plt.pdf", sep = "/"),
  plot = host_sym_engagement_count_plt,
  width = 4,
  height = 4
)
host_sym_engagement_count_plt
```

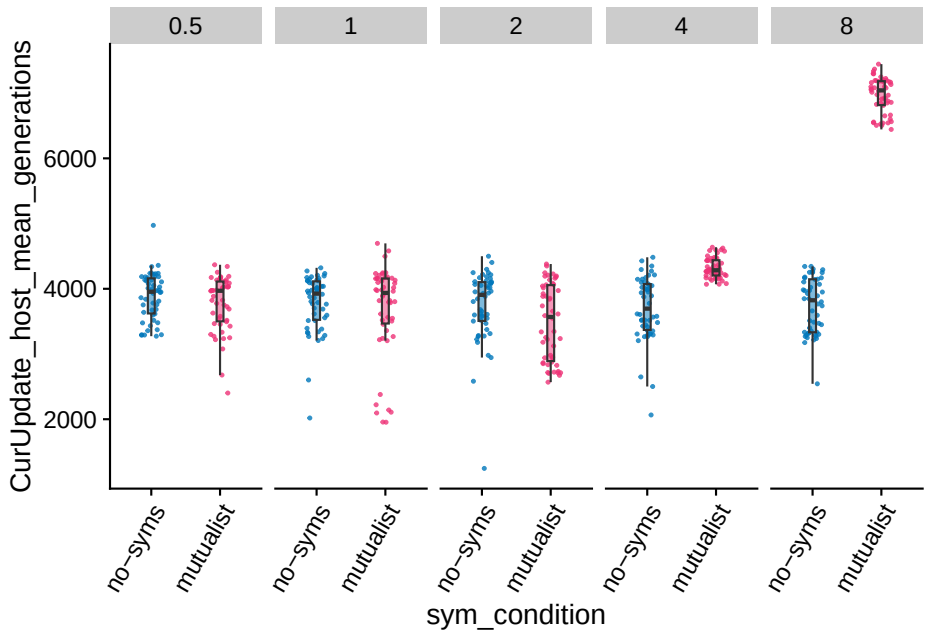


9.10 Generations of evolution elapsed

9.10.1 Hosts

```
host_avg_gens_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_mean_generations
    )
```

```
  ) +
  geom_point(
    mapping = aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_avg_gens_elapsed_plt.pdf", sep = "/"),
  plot = host_avg_gens_elapsed_plt,
  width = 4,
  height = 4
)
host_avg_gens_elapsed_plt
```



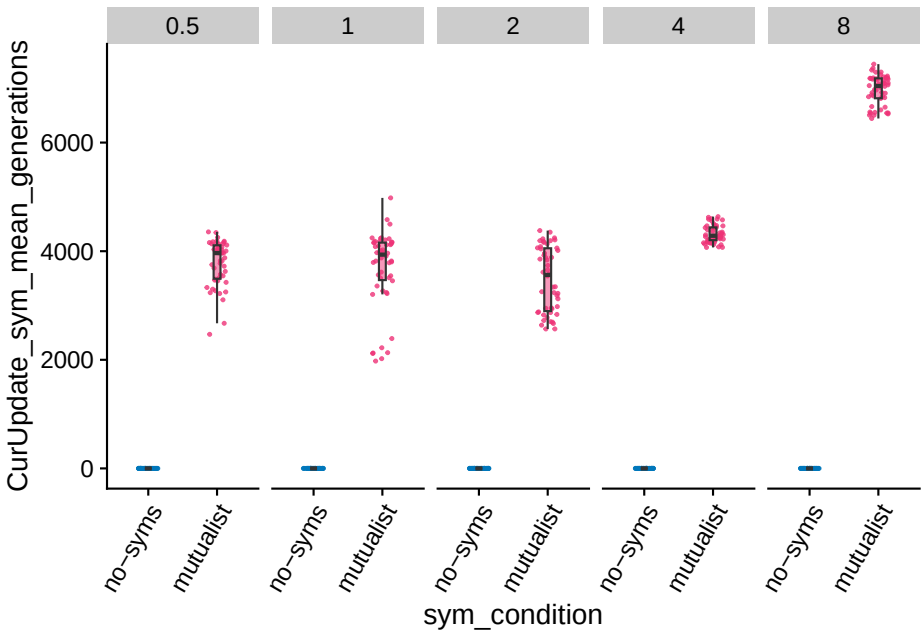

```
summary_data %>%
  group_by(
    sym_condition,
    interaction_multiplier
  ) %>%
  summarise(
    mean_gens = mean(CurUpdate_host_mean_generations),
    median_gens = median(CurUpdate_host_mean_generations),
    var_gens = var(CurUpdate_host_mean_generations),
    sdev_gens = sd(CurUpdate_host_mean_generations)
  )
```

```
## # A tibble: 10 x 6
## # Groups:   sym_condition [2]
##    sym_condition interaction_multiplier mean_gens median_gens var_gens sdev_gens
##    <fct>         <fct>                      <dbl>       <dbl>    <dbl>     <dbl>
##  1 no-syms       0.5                        3884.       3955.  130062.      361.
##  2 no-syms       1                          3776.       3925.  196832.      444.
##  3 no-syms       2                          3736.       3907.  304829.      552.
##  4 no-syms       4                          3698.       3696.  246404.      496.
##  5 no-syms       8                          3747.       3824.  183406.      428.
##  6 mutualist     0.5                        3782.       3969.  181053.      426.
##  7 mutualist     1                          3673.       3936.  516975.      719.
##  8 mutualist     2                          3487.       3568.  346244.      588.
##  9 mutualist     4                          4321.       4284.   25333.      159.
```

```
## 10 mutualist     8                          6966.       7042.   75634.      275.
```

9.10.2 Symbionts

```
sym_avg_gens_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_mean_generations
    )
  ) +
  geom_point(
    mapping = aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_avg_gens_elapsed_plt.pdf", sep = "/"),
  plot = sym_avg_gens_elapsed_plt,
  width = 4,
  height = 4
)
sym_avg_gens_elapsed_plt
```




```
summary_data %>%
  group_by(
    sym_condition,
    interaction_multiplier
  ) %>%
  summarise(
    mean_gens = mean(CurUpdate_sym_mean_generations),
    median_gens = median(CurUpdate_sym_mean_generations),
    var_gens = var(CurUpdate_sym_mean_generations),
    sdev_gens = sd(CurUpdate_sym_mean_generations)
  )
```

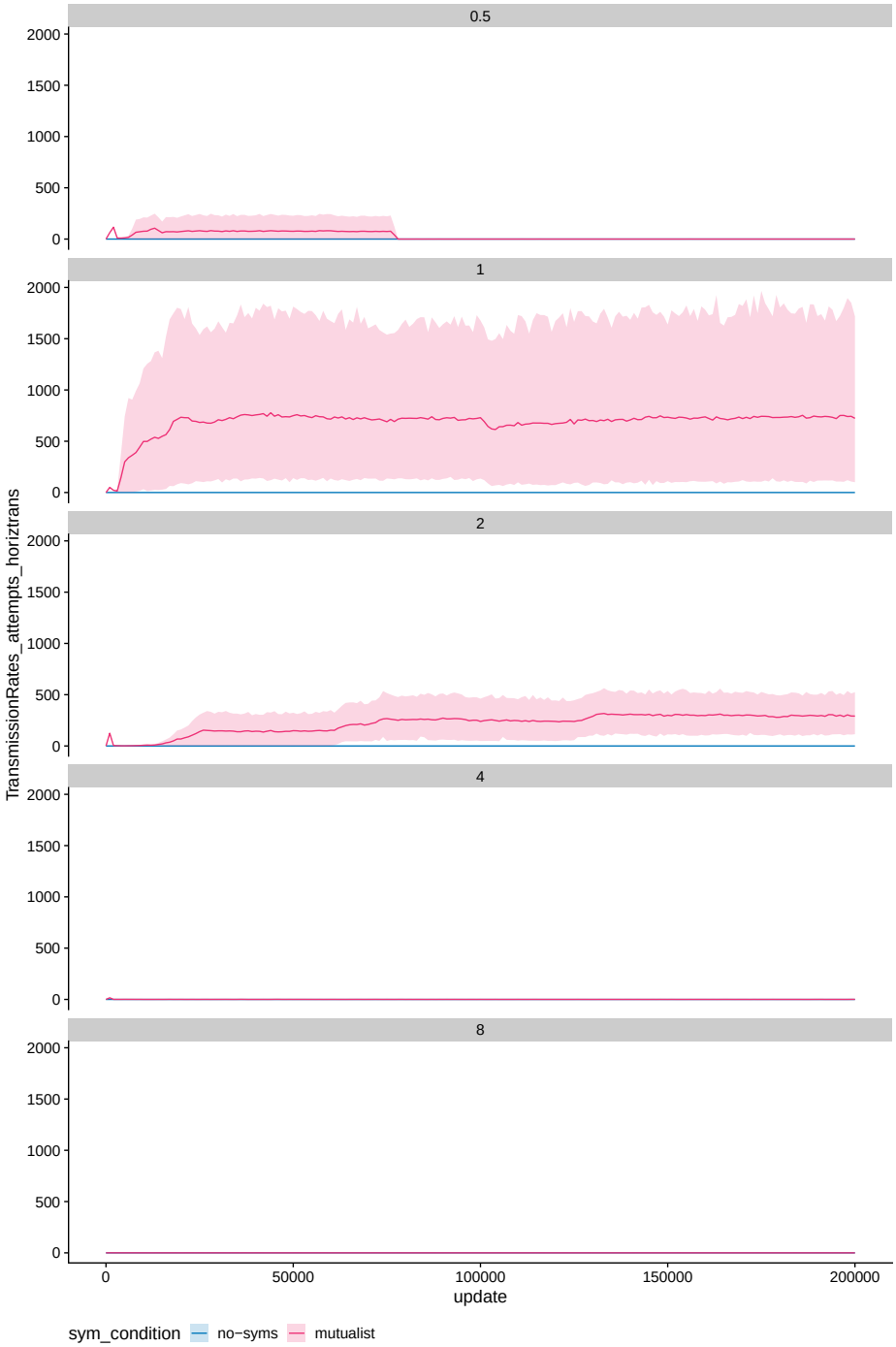
```
## # A tibble: 10 x 6
## # Groups:   sym_condition [2]
##    sym_condition interaction_multiplier mean_gens median_gens var_gens sdev_gens
##    <fct>         <fct>                       <dbl>       <dbl>    <dbl>     <dbl>
##  1 no-syms       0.5                             0           0        0         0
##  2 no-syms       1                               0           0        0         0
##  3 no-syms       2                               0           0        0         0
##  4 no-syms       4                               0           0        0         0
##  5 no-syms       8                               0           0        0         0
##  6 mutualist     0.5                          3781.       3965.  175970.      419.
##  7 mutualist     1                            3680.       3936.  521472.      722.
##  8 mutualist     2                            3483.       3563.  351486.      593.
##  9 mutualist     4                            4319.       4284.   25780.      161.
```

```
## 10 mutualist     8                          6966.       7042.   75588.      275.
```

9.11 Symbiont transmission rates

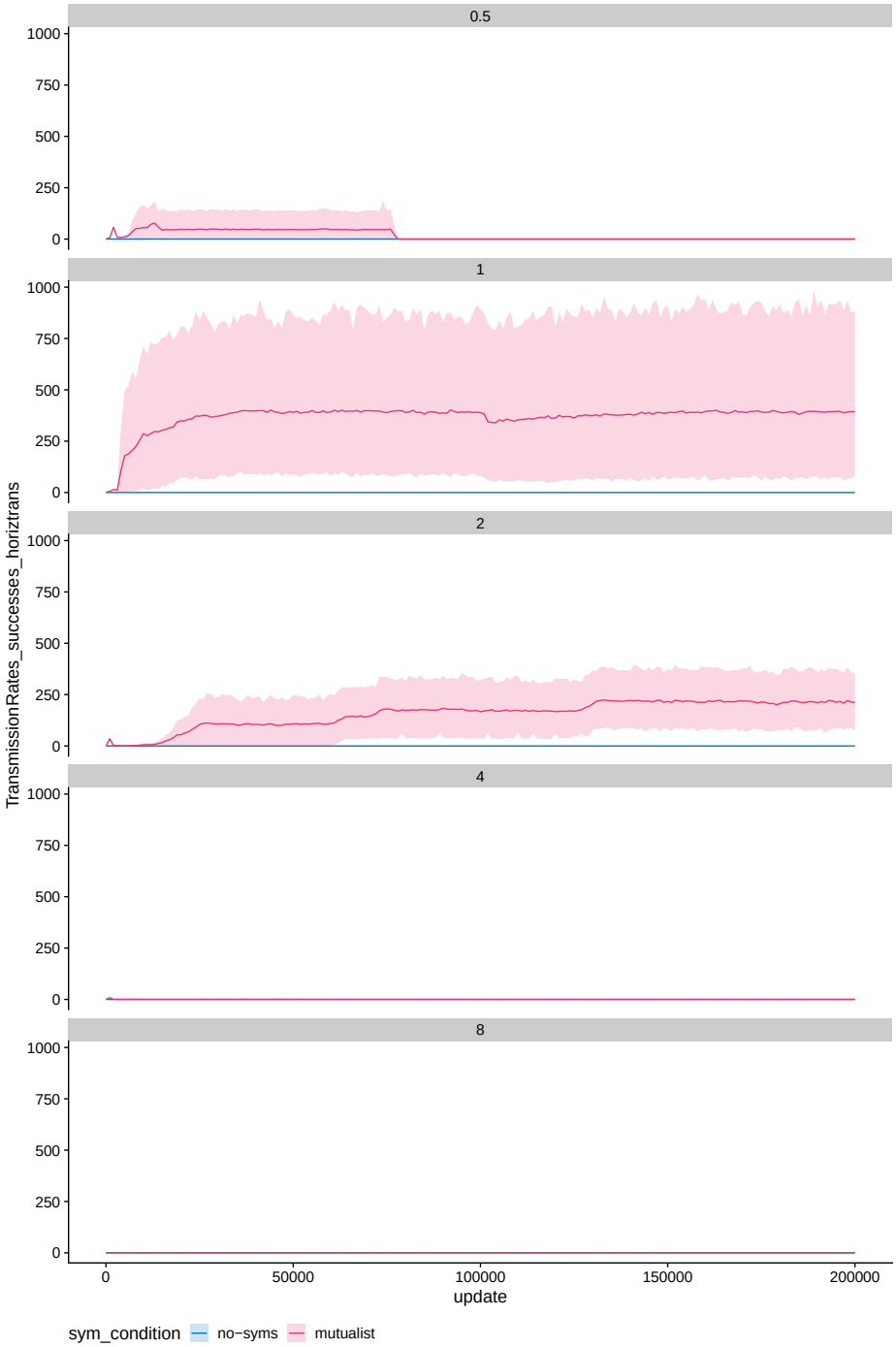
9.11.1 Horizontal transmission attempts

```
TransmissionRates_attempts_horiztrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_attempts_horiztrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_attempts_horiztrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_attempts_horiztrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_attempts_horiztrans_plt
```


0.5
1
2
4
8
TransmissionRates_attempts_horiztrans
2000
1500
1000
500
0
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

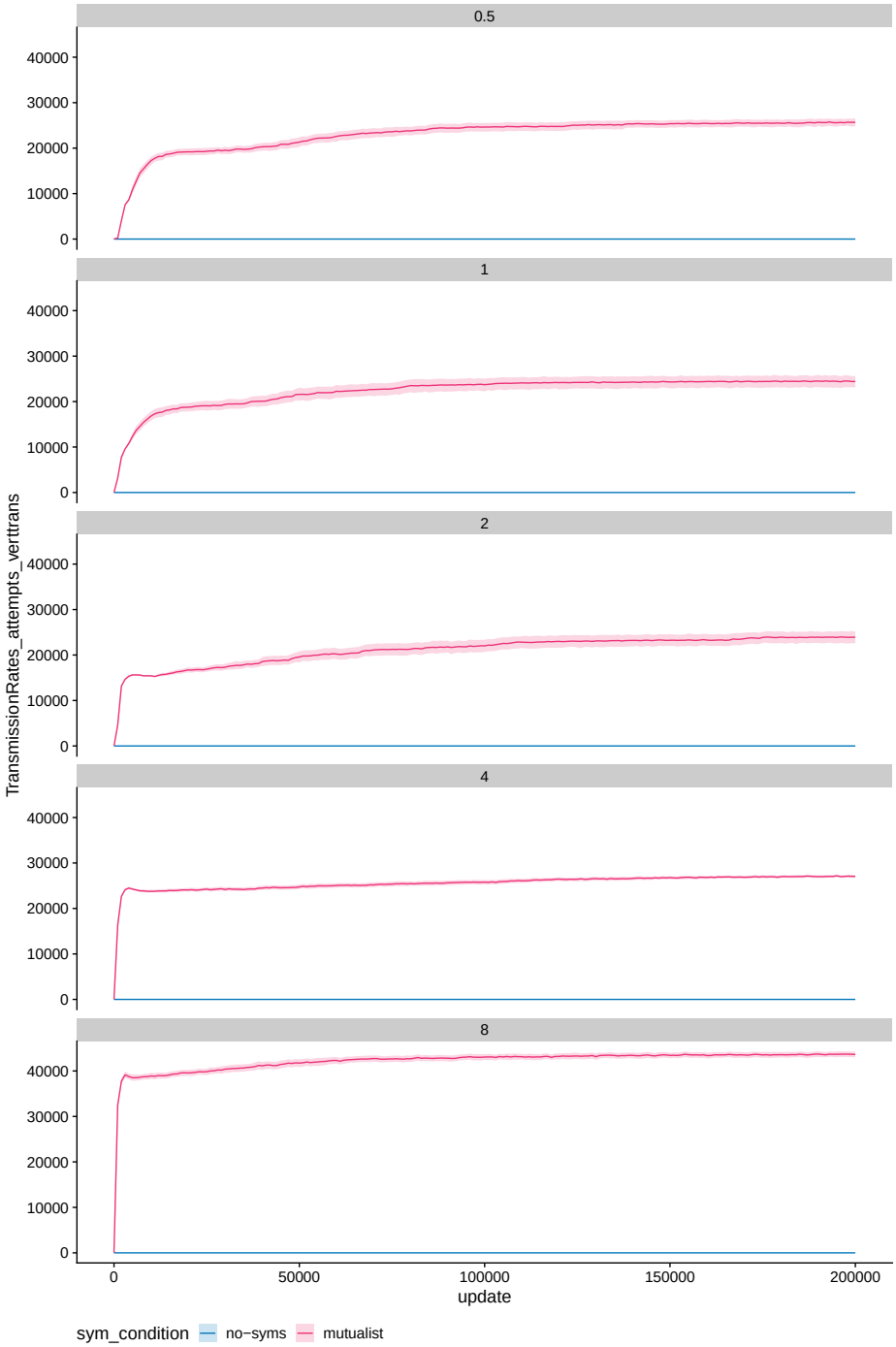
9.11.2 Horizontal transmission successes

```
TransmissionRates_successes_horiztrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_successes_horiztrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_successes_horiztrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_successes_horiztrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_successes_horiztrans_plt
```


0.5
1
2
4
8
1000
750
500
250
0
TransmissionRates_successes_horiztrans
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

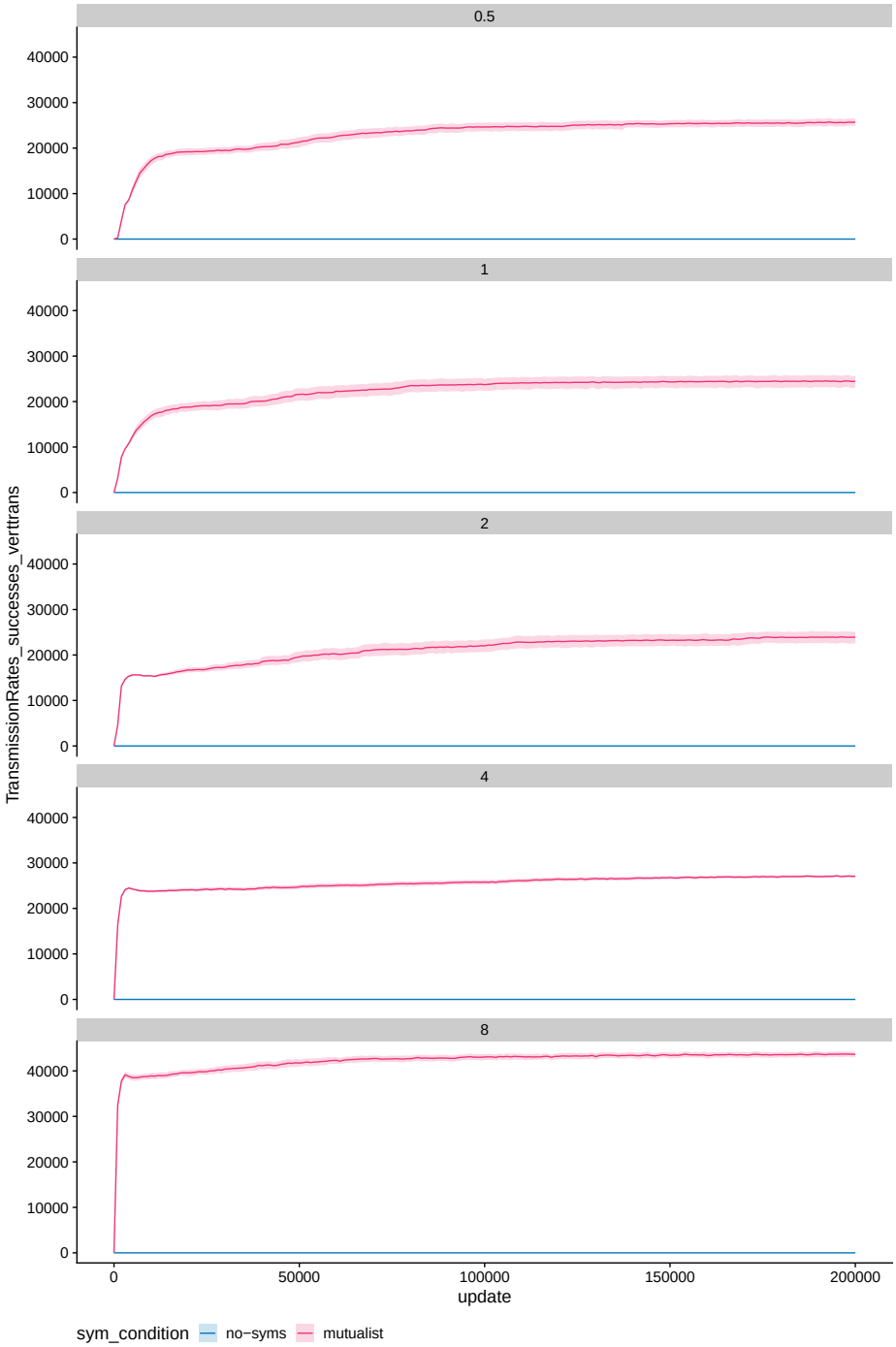
9.11.3 Vertical transmission attempts

```
TransmissionRates_attempts_verttrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_attempts_verttrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_attempts_verttrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_attempts_verttrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_attempts_verttrans_plt
```


0.5
1
2
4
8
TransmissionRates_attempts_verttrans
40000
30000
20000
10000
0
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

9.11.4 Successes vertical transmission

```
TransmissionRates_successes_verttrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_successes_verttrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_successes_verttrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_successes_verttrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_successes_verttrans_plt
```


0.5
1
2
4
8
TransmissionRates_successes_verttrans
40000
30000
20000
10000
0
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

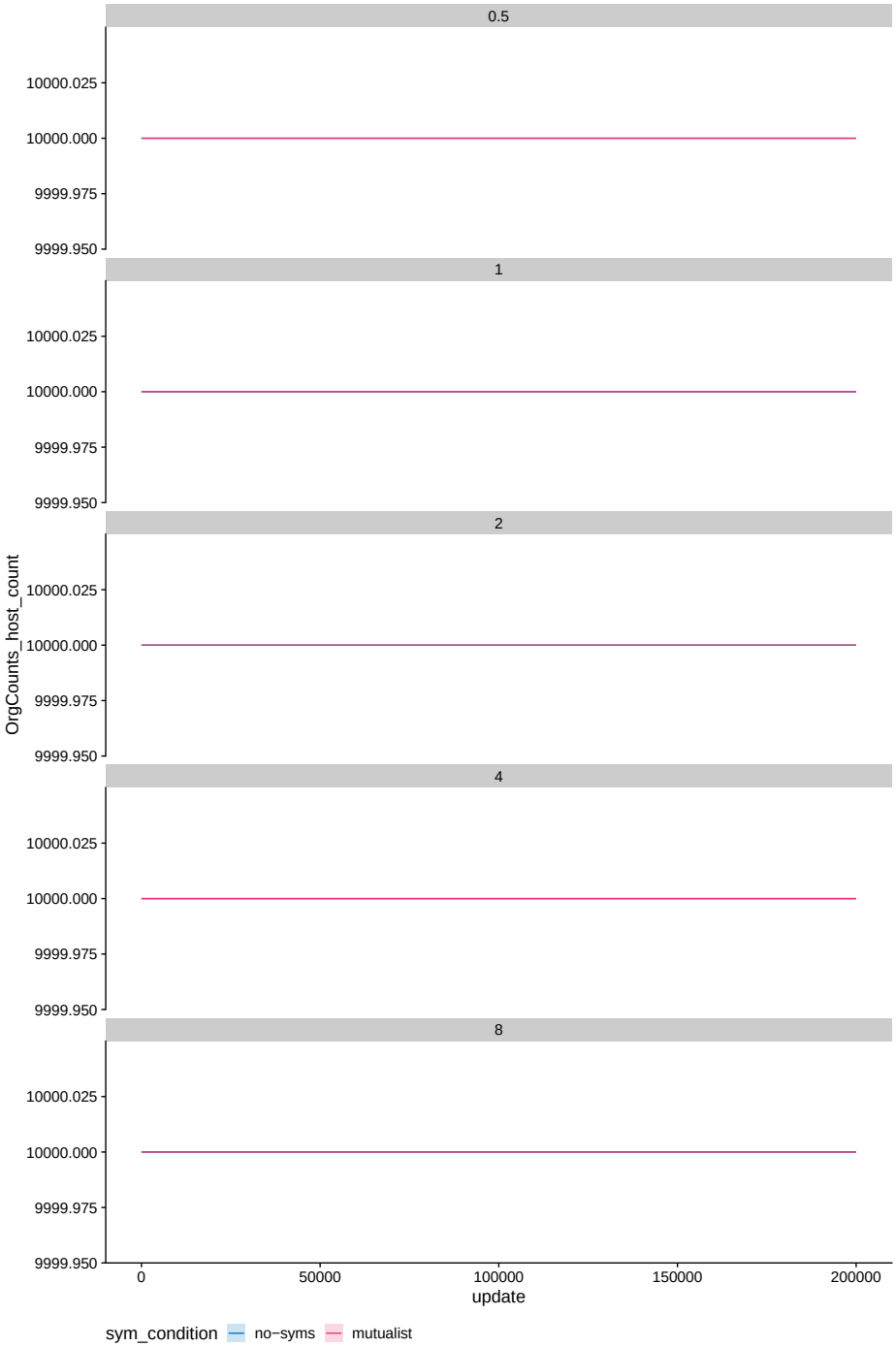
9.12 Population size

```
pop_size_summary <- time_data %>%
  group_by(
    SEED,
    sym_condition,
    interaction_multiplier
  ) %>%
  summarize(
    host_pop_size_avg = mean(OrgCounts_host_count),
    host_pop_size_median = median(OrgCounts_host_count),
    host_pop_size_variance = var(OrgCounts_host_count),
    sym_pop_size_avg = mean(OrgCounts_hosted_sym_count),
    sym_pop_size_median = median(OrgCounts_hosted_sym_count),
    sym_pop_size_variance = var(OrgCounts_hosted_sym_count)
  )
```

9.12.1 Host population size

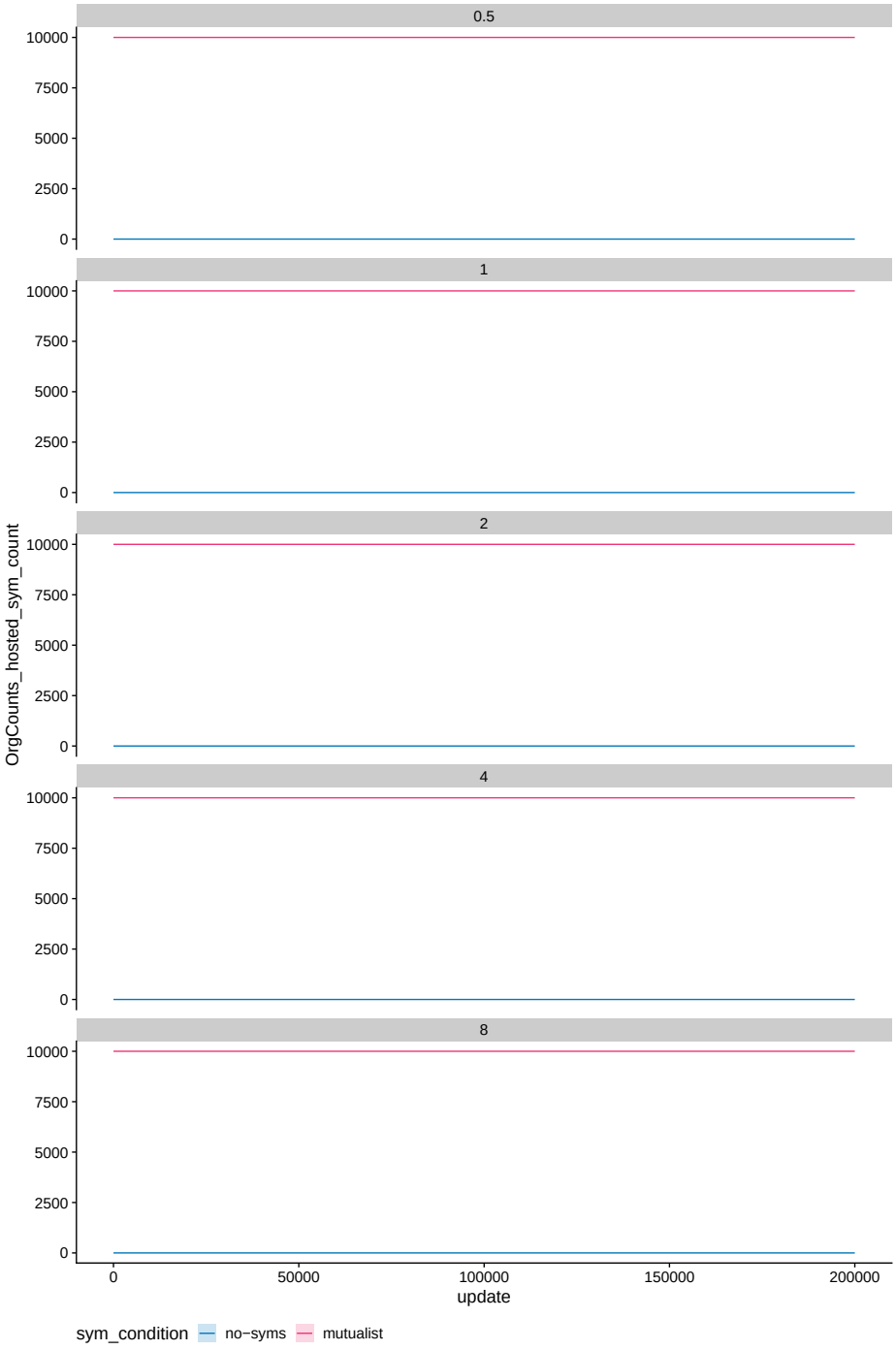
```
host_pop_size_ot_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = OrgCounts_host_count,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, ncol = 1) +
  theme(
    legend.position = "bottom"
```

```
  )
ggsave(
  file = paste(
    plot_dir,
    "host_pop_size_ot_plt.pdf",
    sep = "/"
  ),
  plot = host_pop_size_ot_plt,
  width = 8,
  height = 8
)
host_pop_size_ot_plt
```


0.5
1
2
4
8
10000.025
10000.000
9999.975
9999.950
OrgCounts_host_count
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

9.12.2 Symbiont population size

```
sym_pop_size_ot_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = OrgCounts_hosted_sym_count,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~interaction_multiplier, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "sym_pop_size_ot_plt.pdf",
    sep = "/"
  ),
  plot = sym_pop_size_ot_plt,
  width = 8,
  height = 8
)
sym_pop_size_ot_plt
```


0.5
1
2
4
8
10000
7500
5000
2500
0
OrgCounts_hosted_sym_count
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

Chapter 10

Host-symbiont trait permissibility

Repeat differential function rewards experiment, except vary whether hosts and symbionts are restricted to expressing a single type of function or can express any number of different functions. This controls whether an organism is required to specialize on a function type or is allowed to be a generalist.

10.1 Dependencies and setup

Load R library dependencies.

```
library(tidyverse)
library(cowplot)
library(khroma)
library(rstatix)
library(knitr)
library(kableExtra)
```

Define relevant file paths.

```
# Check if Rmd is being compiled using bookdown
bookdown <- exists("bookdown_build")

# Build relative path from R's working directory to what is actually the working
# directory for this experiment.
experiment_slug <- "2026-02-23-health-task-credit"
supp_file_prefix <- "task-credit"
```

```
exp_directory <- paste(
  "experiments",
  experiment_slug,
  "analysis",
  sep = "/"
)

# Get path to the supplemental material analysis directory
supp_directory <- paste(
  "experiments",
  "manuscript-analyses",
  sep = "/"
)

# Adjust experiment directory if being knitted for bookdown build
if (bookdown) {
  exp_directory <- paste0(
    bookdown_wd_prefix,
    exp_directory
  )
  supp_directory <- paste0(
    bookdown_wd_prefix,
    supp_directory
  )
}
```

Define default plot theme, plot output directory.

```
# Use the cowplot theme as our default for ggplots
theme_set(theme_cowplot())
# Create directory to save plots
plot_dir <- paste(
  supp_directory,
  "plots",
  supp_file_prefix,
  sep = "/"
)
dir.create(
  plot_dir,
  showWarnings = FALSE
)

# Use vibrant theme for plot color palettes
vibrant_colors <- color("vibrant")(7)
color_pal <- c(
```

```
  vibrant_colors[["blue"]],
  vibrant_colors[["magenta"]]
)
```

10.2 Load and preprocess data

```
tasks <- c("NOT", "NAND", "OR_NOT", "AND", "OR", "AND_NOT", "NOR", "XOR", "EQU")
max_pop_size <- 100 * 100
threshold <- 0.025 * max_pop_size
```

Load summary data.

```
summary_data_path <- paste(
  exp_directory,
  "data",
  "summary.csv",
  sep = "/"
)
summary_data <- read_csv(summary_data_path)
# Do any universal preprocessing
summary_data <- summary_data %>%
  mutate(
    SEED = as.factor(SEED),
    task_credit = paste0(
      "H", HOST_ONLY_FIRST_TASK_CREDIT,
      "S", SYM_ONLY_FIRST_TASK_CREDIT
    ),
    sym_condition = case_when(
      START_MOI == "0" ~ "no-syms",
      START_MOI == "1" & HEALTH_TYPE == "parasite" ~ "parasite",
      START_MOI == "1" & HEALTH_TYPE == "mutualist" ~ "mutualist",
      .default = "unknown" # precaution against unexpected parameterizations
    )
  ) %>%
  mutate(
    task_credit = as.factor(task_credit),
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist")
    )
  )
```

Load time series data.

```
time_data_path <- paste(
  exp_directory,
  "data",
  "time_series.csv",
  sep = "/"
)
time_data <- read_csv(time_data_path)

# Do any universal preprocessing
time_data <- time_data %>%
  mutate(
    SEED = as.factor(SEED),
    sym_condition = case_when(
      START_MOI == "0" ~ "no-syms",
      START_MOI == "1" & HEALTH_TYPE == "parasite" ~ "parasite",
      START_MOI == "1" & HEALTH_TYPE == "mutualist" ~ "mutualist",
      .default = "unknown" # precaution against unexpected parameterizations
    ),
    task_credit = paste0(
      "H", HOST_ONLY_FIRST_TASK_CREDIT,
      "S", SYM_ONLY_FIRST_TASK_CREDIT
    )
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c("no-syms", "parasite", "mutualist")
    ),
    task_credit = as.factor(task_credit)
  )
```

10.3 Extinctions

Did host or symbiont populations go extinct in any runs?

```
sym_min_pop_sizes <- time_data %>%
  group_by(SEED) %>%
  slice_min(
    OrgCounts_hosted_sym_count,
    n = 1
  ) %>%
  slice_min(
```

```
    update,
    n = 1
  )

reps_with_extinct_syms <- sym_min_pop_sizes %>%
  group_by(
    sym_condition,
    task_credit
  ) %>%
  summarize(
    reps_with_extinct_syms = sum(OrgCounts_hosted_sym_count == 0),
    total_reps = n()
  )

host_min_pop_sizes <- time_data %>%
  group_by(SEED) %>%
  slice_min(
    OrgCounts_host_count,
    n = 1
  ) %>%
  slice_min(
    update,
    n = 1
  )

reps_with_extinct_hosts <- host_min_pop_sizes %>%
  group_by(
    sym_condition,
    task_credit
  ) %>%
  summarize(
    reps_with_extinct_hosts = sum(OrgCounts_host_count == 0),
    total_reps = n()
  )

extinctions <- full_join(
  reps_with_extinct_syms,
  reps_with_extinct_hosts
)
extinctions <- extinctions %>%
  arrange(
    desc(reps_with_extinct_hosts), desc(reps_with_extinct_syms)
  )
extinction_table <- kable(extinctions) %>%
  kable_styling(latex_options = "striped")
```

```
save_kable(
  extinction_table,
  paste(
    plot_dir,
    "extinctions.pdf",
    sep = "/"
  )
)

extinctions
```

```
## # A tibble: 8 x 5
## # Groups:   sym_condition [2]
##   sym_condition task_credit reps_with_extinct_syms total_reps
##   <fct>         <fct>                        <int>      <int>
## 1 no-syms       H0S0                            50         50
## 2 no-syms       H0S1                            50         50
## 3 no-syms       H1S0                            50         50
## 4 no-syms       H1S1                            50         50
## 5 mutualist     H0S0                             0         50
## 6 mutualist     H0S1                             0         50
## 7 mutualist     H1S0                             0         50
## 8 mutualist     H1S1                             0         50
## # i 1 more variable: reps_with_extinct_hosts <int>
```

10.4 Host function performance

First, quantify which functions have been performed over the course of an entire run.

```
# Pivot task data long, selecting only relevant columns
host_over_time_parent_task_data <- time_data %>%
  select(
    SEED,
    sym_condition,
    task_credit,
    update,
    CurUpdate_AND_NOT_in_host_parent_org_counts,
    CurUpdate_AND_in_host_parent_org_counts,
    CurUpdate_EQU_in_host_parent_org_counts,
    CurUpdate_NAND_in_host_parent_org_counts,
    CurUpdate_NOR_in_host_parent_org_counts,
    CurUpdate_NOT_in_host_parent_org_counts,
```

```
    CurUpdate_OR_NOT_in_host_parent_org_counts,
    CurUpdate_OR_in_host_parent_org_counts,
    CurUpdate_XOR_in_host_parent_org_counts
  ) %>%
  pivot_longer(
    c(
      CurUpdate_AND_NOT_in_host_parent_org_counts,
      CurUpdate_AND_in_host_parent_org_counts,
      CurUpdate_EQU_in_host_parent_org_counts,
      CurUpdate_NAND_in_host_parent_org_counts,
      CurUpdate_NOR_in_host_parent_org_counts,
      CurUpdate_NOT_in_host_parent_org_counts,
      CurUpdate_OR_NOT_in_host_parent_org_counts,
      CurUpdate_OR_in_host_parent_org_counts,
      CurUpdate_XOR_in_host_parent_org_counts
    ),
    names_to = "task",
    values_to = "task_in_parent_count"
  ) %>%
  mutate(
    task = str_replace(task, "CurUpdate_", "")
  ) %>%
  mutate(
    task = str_replace(task, "_in_host_parent_org_counts", "")
  ) %>%
  mutate(
    task_completed_by_parent = task_in_parent_count >= threshold,
    task = factor(
      task,
      levels = tasks,
      labels = tasks
    )
  ) %>%
  mutate(
    task_complexity = case_when(
      task == "NOT" ~ 1,
      task == "NAND" ~ 1,
      task == "OR_NOT" ~ 2,
      task == "AND" ~ 2,
      task == "OR" ~ 3,
      task == "AND_NOT" ~ 3,
      task == "NOR" ~ 4,
      task == "XOR" ~ 4,
      task == "EQU" ~ 5,
      .default = -1
```

```
    )
  )

host_parent_tasks_all_time_by_task <- host_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    task_credit,
    task
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  group_by(
    sym_condition,
    task_credit,
    task
  ) %>%
  summarize(
    n = n(),
    parent_tasks_completed_ever = sum(parent_task_completed_ever == TRUE)
  )
```

Next, summarize function complexity level.

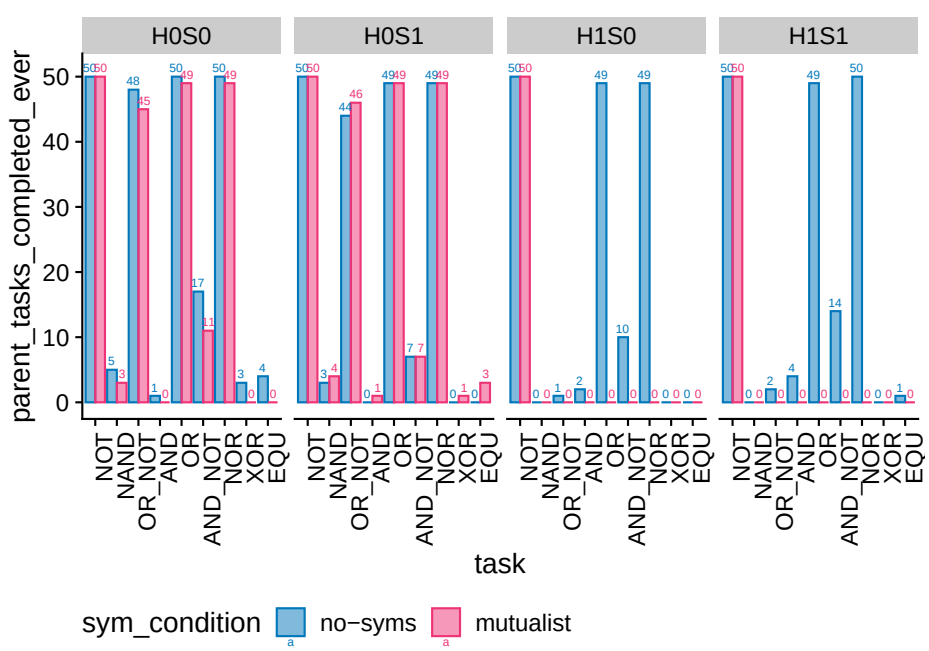
```
host_max_task_complexity_data <- host_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    task_credit,
    task,
    task_complexity
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  # This filter works fine because all replicates start with a task
  filter(
    parent_task_completed_ever
  ) %>%
  group_by(
```

```
    SEED,
    sym_condition,
    task_credit
  ) %>%
  summarize(
    max_task_complexity = max(task_complexity)
  ) %>%
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c(
        "no-syms",
        "parasite",
        "mutualist"
      )
    )
  )
```

10.4.1 Host function performance - all time

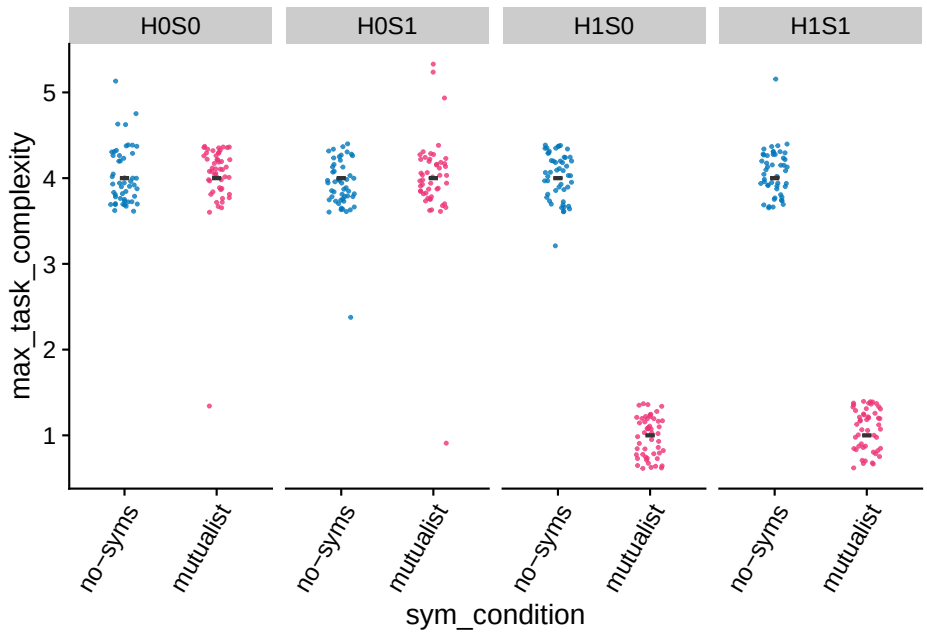
```
host_parent_tasks_all_time_by_task_plt <- host_parent_tasks_all_time_by_task %>%
  ggplot(
    aes(
      x = task,
      color = sym_condition,
      fill = sym_condition,
      y = parent_tasks_completed_ever
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = parent_tasks_completed_ever),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
```

```
    limits = c(0, 51)
  ) +
  facet_wrap(~task_credit, nrow = 1) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(
      angle = 90,
      hjust = 1
    )
  )
ggsave(
  file = paste(
    plot_dir,
    "host_parent_tasks_all_time_by_task_plt.pdf",
    sep = "/"
  ),
  plot = host_parent_tasks_all_time_by_task_plt,
  width = 10,
  height = 7
)
host_parent_tasks_all_time_by_task_plt
```



10.4.2 Host maximum function complexity evolved

```
host_max_task_complexity_plt <- host_max_task_complexity_data %>%
  ggplot(
    aes(x = sym_condition, fill = sym_condition, y = max_task_complexity)
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_max_task_complexity_plt.pdf", sep = "/"),
  plot = host_max_task_complexity_plt,
  width = 4,
  height = 4
)
host_max_task_complexity_plt
```



10.5 Symbiont function performance

```
# Pivot task data long, selecting only relevant columns
sym_over_time_parent_task_data <- time_data %>%
  select(
    SEED,
    sym_condition,
    task_credit,
    update,
    CurUpdate_AND_NOT_in_sym_parent_org_counts,
    CurUpdate_AND_in_sym_parent_org_counts,
    CurUpdate_EQU_in_sym_parent_org_counts,
    CurUpdate_NAND_in_sym_parent_org_counts,
    CurUpdate_NOR_in_sym_parent_org_counts,
    CurUpdate_NOT_in_sym_parent_org_counts,
    CurUpdate_OR_NOT_in_sym_parent_org_counts,
    CurUpdate_OR_in_sym_parent_org_counts,
    CurUpdate_XOR_in_sym_parent_org_counts
  ) %>%
  pivot_longer(
    c(
      CurUpdate_AND_NOT_in_sym_parent_org_counts,
```

```
      CurUpdate_AND_in_sym_parent_org_counts,
      CurUpdate_EQU_in_sym_parent_org_counts,
      CurUpdate_NAND_in_sym_parent_org_counts,
      CurUpdate_NOR_in_sym_parent_org_counts,
      CurUpdate_NOT_in_sym_parent_org_counts,
      CurUpdate_OR_NOT_in_sym_parent_org_counts,
      CurUpdate_OR_in_sym_parent_org_counts,
      CurUpdate_XOR_in_sym_parent_org_counts
    ),
    names_to = "task",
    values_to = "task_in_parent_count"
  ) %>%
  mutate(
    task = str_replace(task, "CurUpdate_", "")
  ) %>%
  mutate(
    task = str_replace(task, "_in_sym_parent_org_counts", "")
  ) %>%
  mutate(
    task_completed_by_parent = task_in_parent_count >= threshold,
    task = factor(
      task,
      levels = tasks,
      labels = tasks
    )
  ) %>%
  mutate(
    task_complexity = case_when(
      task == "NOT" ~ 1,
      task == "NAND" ~ 1,
      task == "OR_NOT" ~ 2,
      task == "AND" ~ 2,
      task == "OR" ~ 3,
      task == "AND_NOT" ~ 3,
      task == "NOR" ~ 4,
      task == "XOR" ~ 4,
      task == "EQU" ~ 5,
      .default = -1
    )
  )

sym_parent_tasks_all_time_by_task <- sym_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
```

```
    task_credit,
    task
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  group_by(
    sym_condition,
    task_credit,
    task
  ) %>%
  summarize(
    n = n(),
    parent_tasks_completed_ever = sum(parent_task_completed_ever == TRUE)
  )
```

Next, summarize function complexity level.

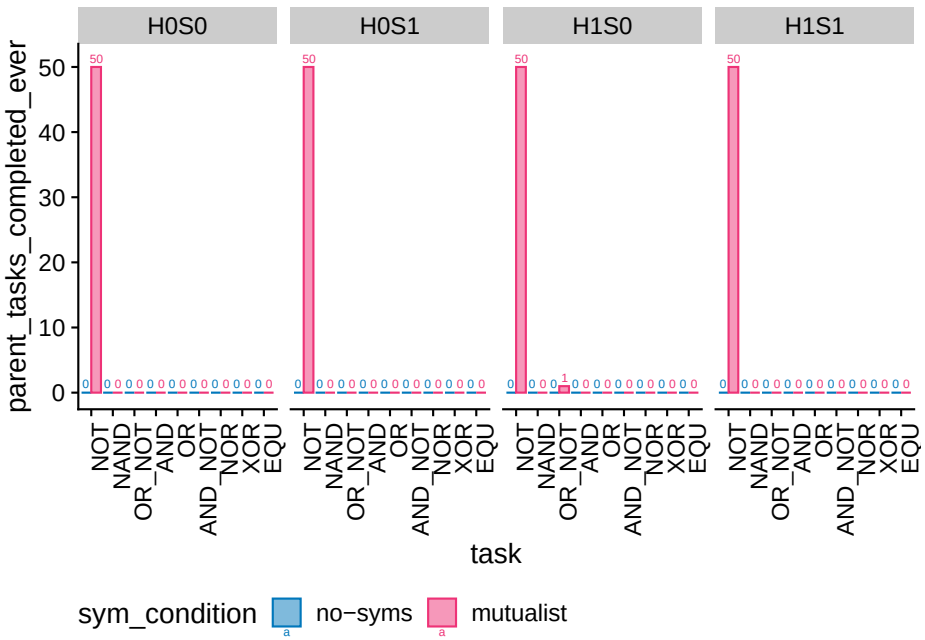
```
sym_max_task_complexity_data <- sym_over_time_parent_task_data %>%
  group_by(
    SEED,
    sym_condition,
    task_credit,
    task,
    task_complexity
  ) %>%
  summarize(
    n = n(),
    parent_task_completed_ever = any(task_completed_by_parent)
  ) %>%
  ungroup() %>%
  # This filter works fine because all replicates start with a task
  filter(
    parent_task_completed_ever
  ) %>%
  group_by(
    SEED,
    sym_condition,
    task_credit
  ) %>%
  summarize(
    max_task_complexity = max(task_complexity)
  ) %>%
```

```
  mutate(
    sym_condition = factor(
      sym_condition,
      levels = c(
        "no-syms",
        "parasite",
        "mutualist"
      )
    )
  )
```

10.5.1 Symbiont function performance - all time

```
sym_parent_tasks_all_time_by_task_plt <- sym_parent_tasks_all_time_by_task %>%
  ggplot(
    aes(
      x = task,
      color = sym_condition,
      fill = sym_condition,
      y = parent_tasks_completed_ever
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = parent_tasks_completed_ever),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, nrow = 1) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(
      angle = 90,
```

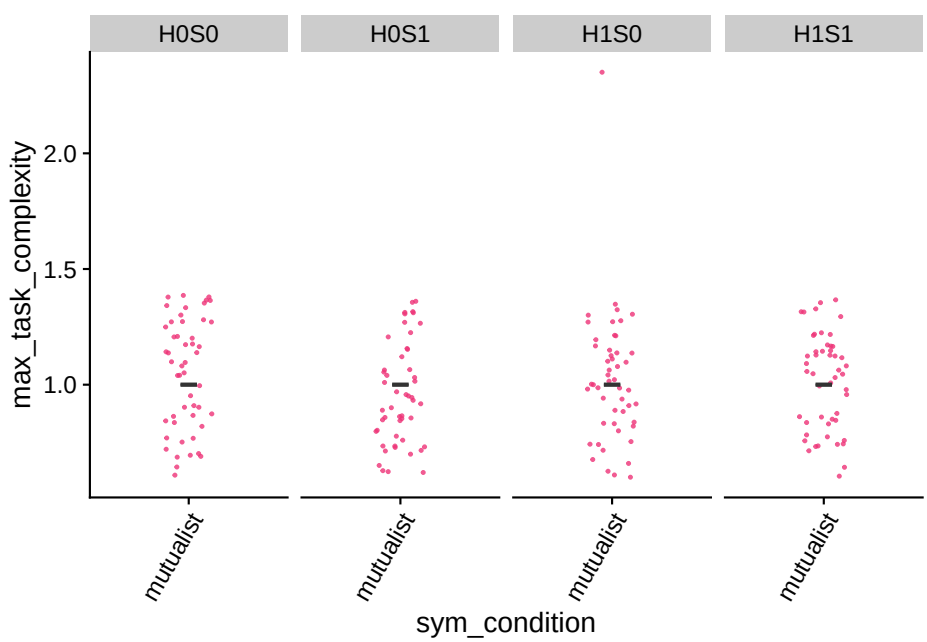
```
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_parent_tasks_all_time_by_task_plt.pdf", sep = "/"),
  plot = sym_parent_tasks_all_time_by_task_plt,
  width = 8,
  height = 8
)
sym_parent_tasks_all_time_by_task_plt
```



10.5.2 Symbiont maximum function complexity evolved

```
sym_max_task_complexity_plt <- sym_max_task_complexity_data %>%
  ggplot(
    aes(x = sym_condition, fill = sym_condition, y = max_task_complexity)
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
```

```
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = c(color_pal[2], color_pal[3])) +
  scale_fill_manual(values = c(color_pal[2], color_pal[3])) +
  facet_wrap(~task_credit, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_max_task_complexity_plt.pdf", sep = "/"),
  plot = sym_max_task_complexity_plt,
  width = 4,
  height = 4
)
sym_max_task_complexity_plt
```

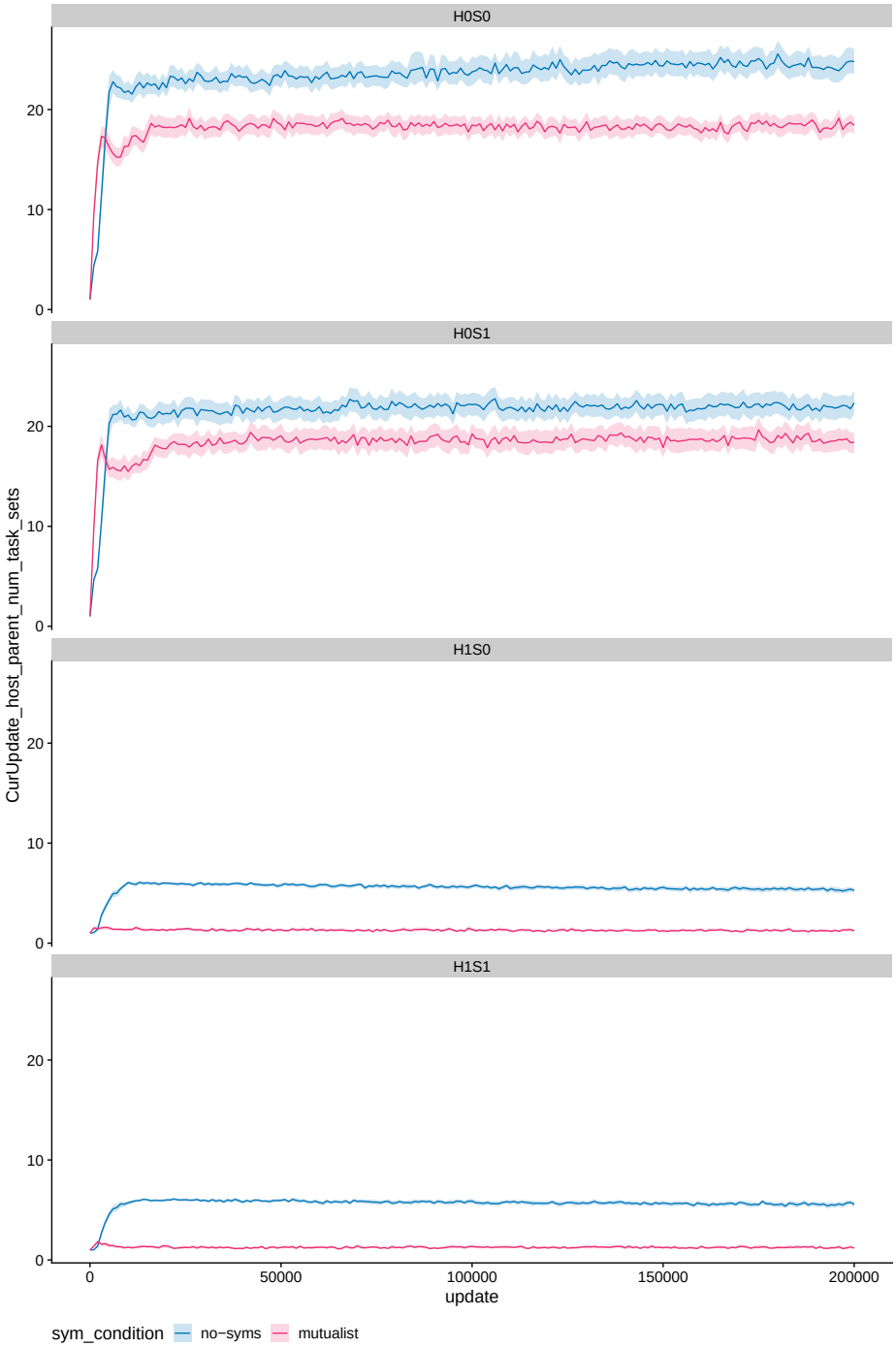


10.6 Host phenotypic diversity

10.6.1 Number of distinct function profiles

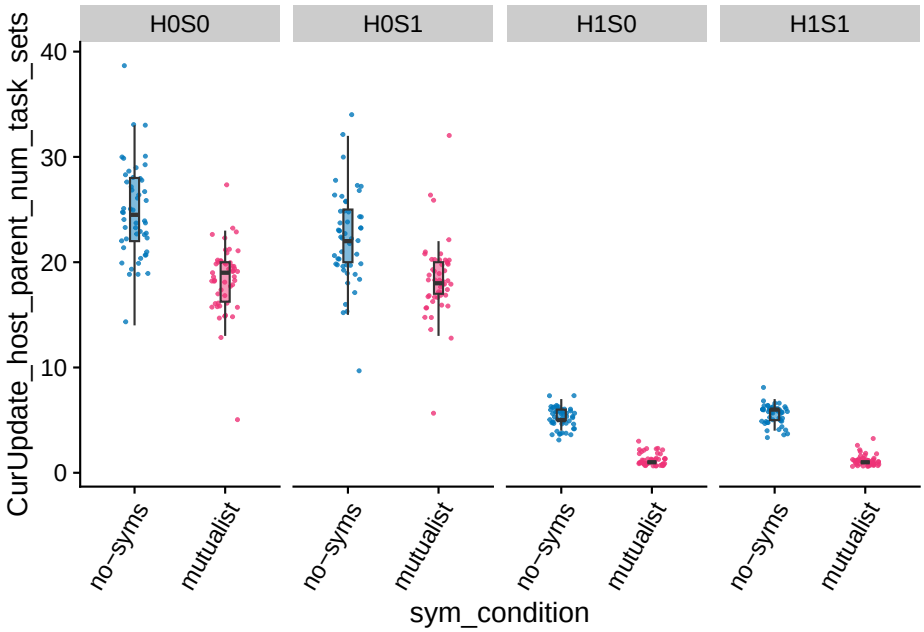
Over time:

```
host_parent_num_task_sets_plt <- time_data %>%
  # sample to make plotting faster
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_parent_num_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_parent_num_task_sets_plt.pdf", sep = "/"),
  plot = host_parent_num_task_sets_plt,
  width = 8,
  height = 8
)
host_parent_num_task_sets_plt
```



At the end of the experiment:

```
host_final_num_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_parent_num_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_final_num_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = host_final_num_task_sets_elapsed_plt,
  width = 4,
  height = 4
)
host_final_num_task_sets_elapsed_plt
```

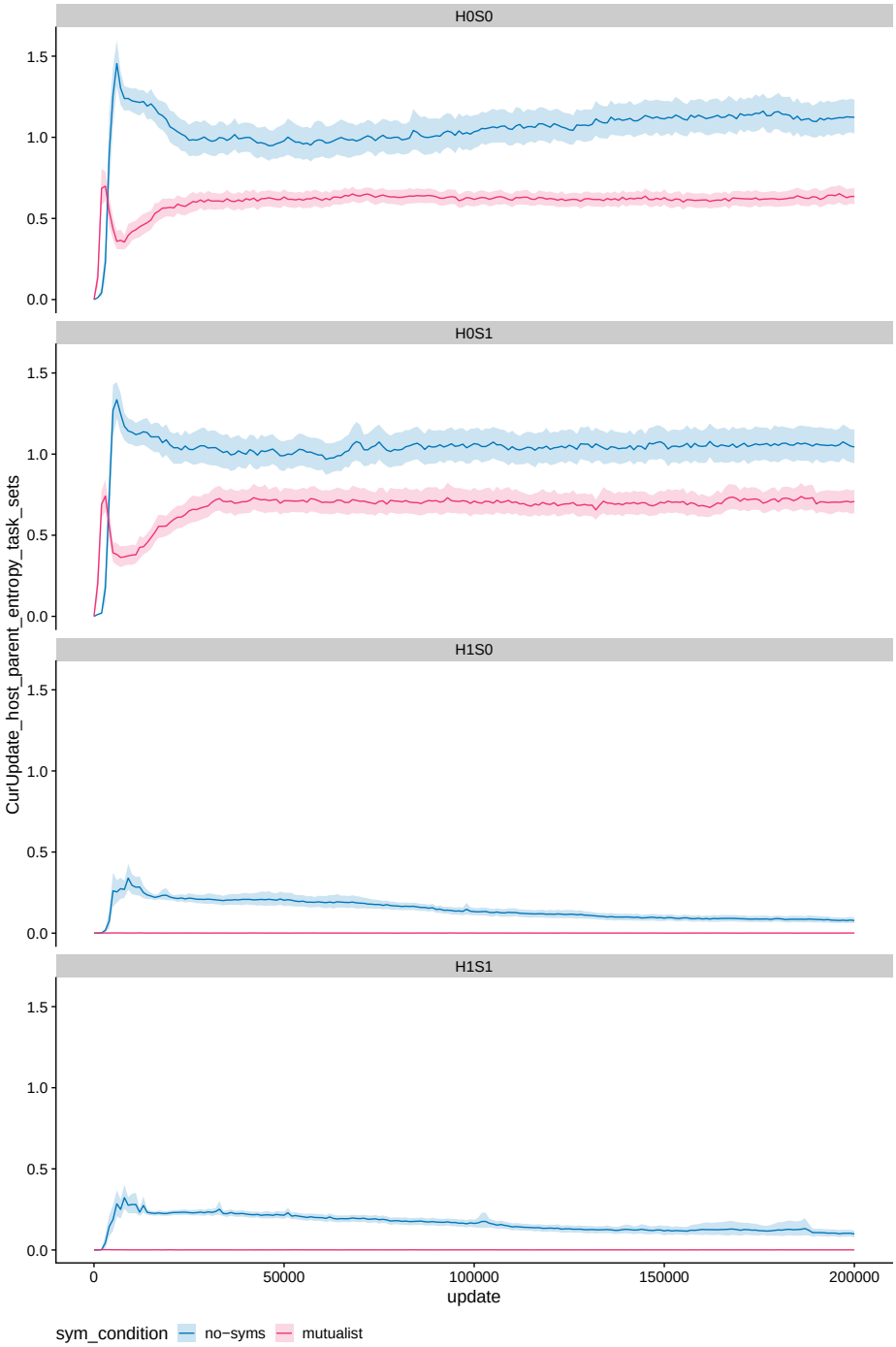


10.6.2 Shannon diversity of function profiles

Over time:

```
host_parent_entropy_task_sets_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
```

```
  facet_wrap(~task_credit, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_parent_entropy_task_sets_plt.pdf", sep = "/"),
  plot = host_parent_entropy_task_sets_plt,
  width = 8,
  height = 8
)
host_parent_entropy_task_sets_plt
```



At the end of the experiment:

```
host_final_entopy_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_parent_entropy_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  facet_wrap(~task_credit, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_final_entopy_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = host_final_entopy_task_sets_elapsed_plt,
  width = 10,
  height = 10
)
```

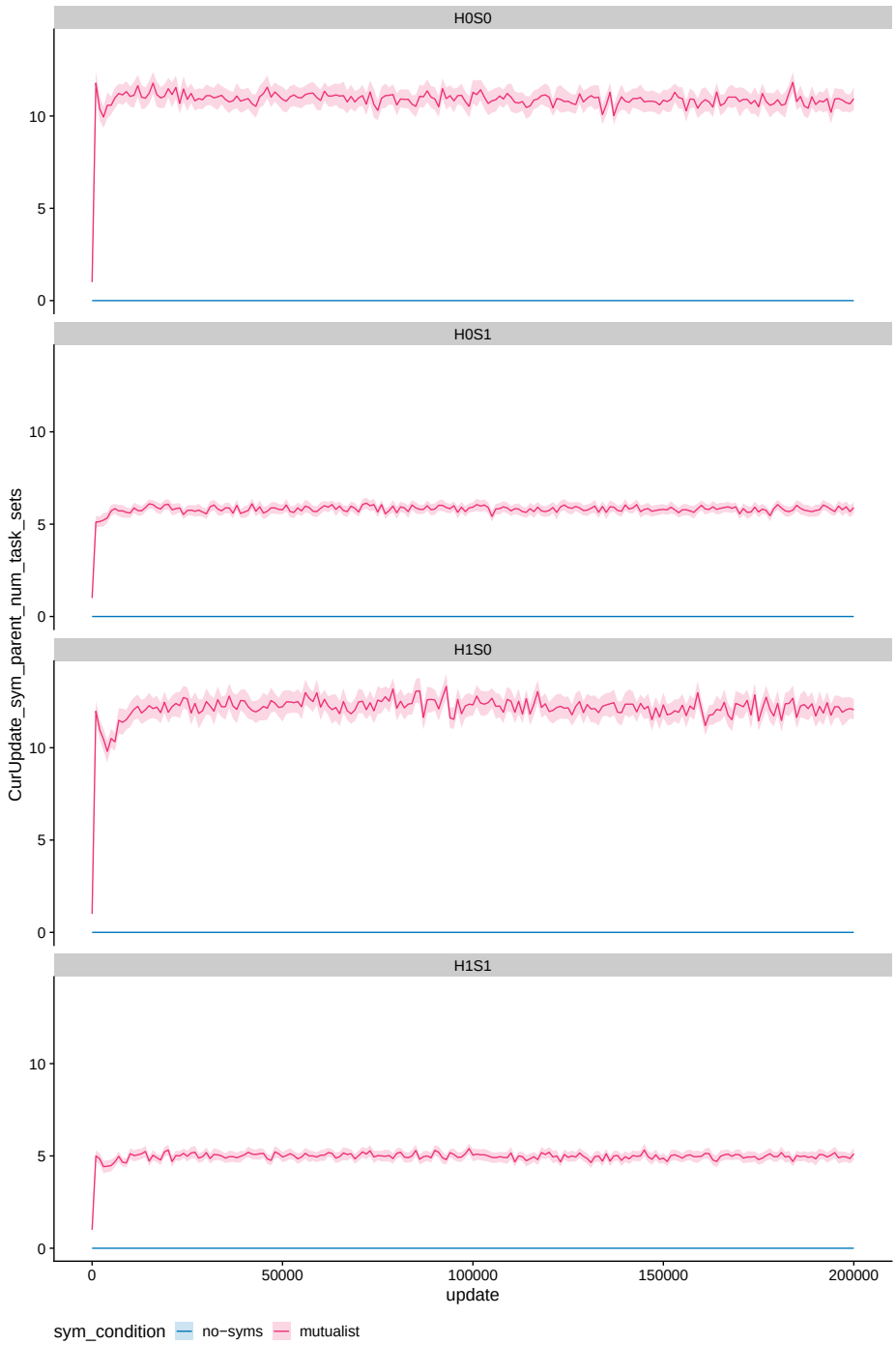
10.7 Symbiont phenotypic diversity

10.7.1 Number of distinct function profiles

Over time:

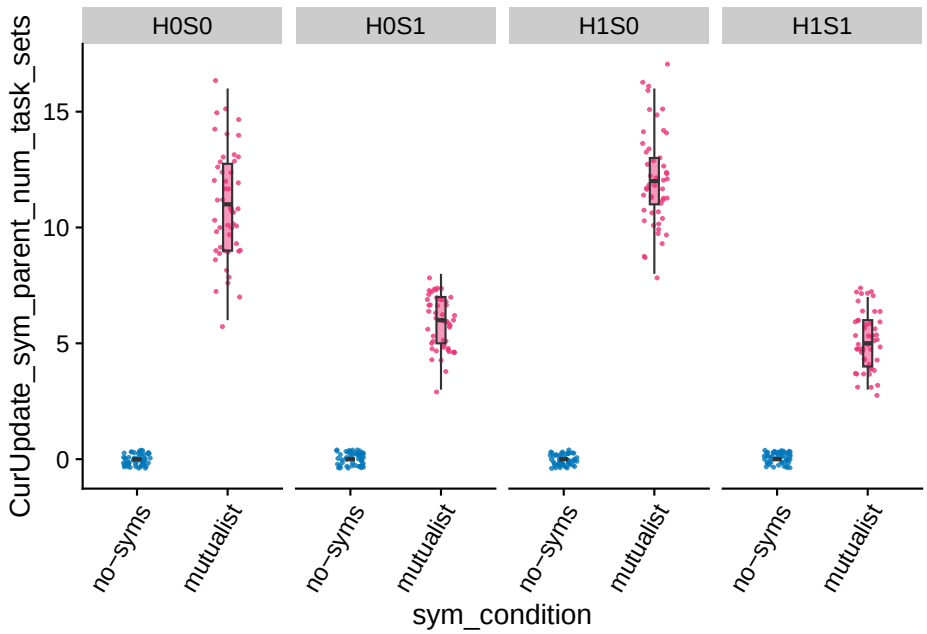
```
sym_parent_num_task_sets_plt <- time_data %>%
  # sample to make plotting faster
```

```
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_sym_parent_num_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "sym_parent_num_task_sets_plt.pdf", sep = "/"),
  plot = sym_parent_num_task_sets_plt,
  width = 8,
  height = 8
)
sym_parent_num_task_sets_plt
```



At the end of the experiment:

```
sym_final_num_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_parent_num_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_final_num_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = sym_final_num_task_sets_elapsed_plt,
  width = 4,
  height = 4
)
sym_final_num_task_sets_elapsed_plt
```

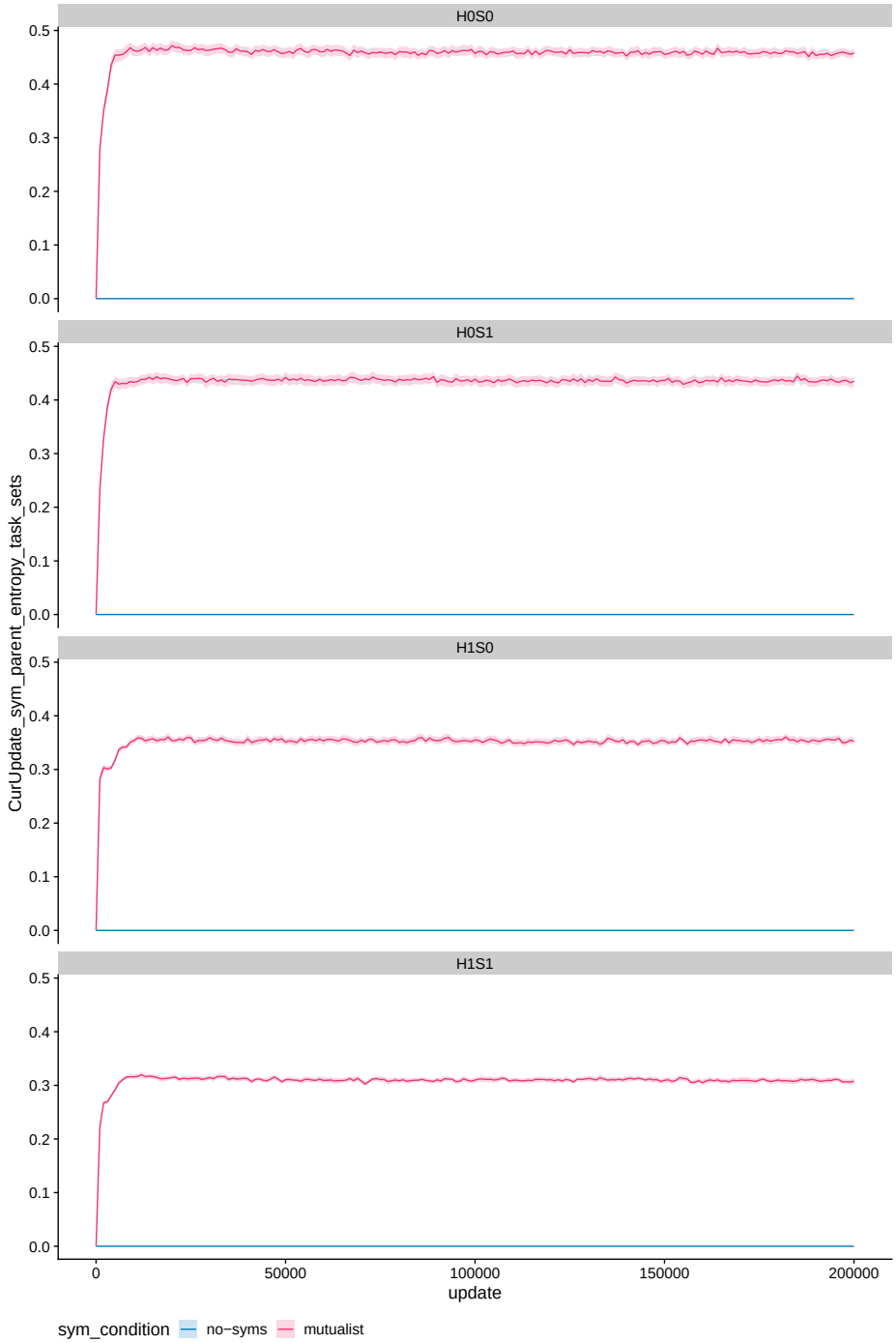


10.7.2 Shannon diversity of function profiles

Over time:

```
sym_parent_entropy_task_sets_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = CurUpdate_sym_parent_entropy_task_sets,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
```

```
  facet_wrap(~task_credit, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "sym_parent_entropy_task_sets_plt.pdf", sep = "/"),
  plot = sym_parent_entropy_task_sets_plt,
  width = 8,
  height = 8
)
sym_parent_entropy_task_sets_plt
```



At the end of the experiment:

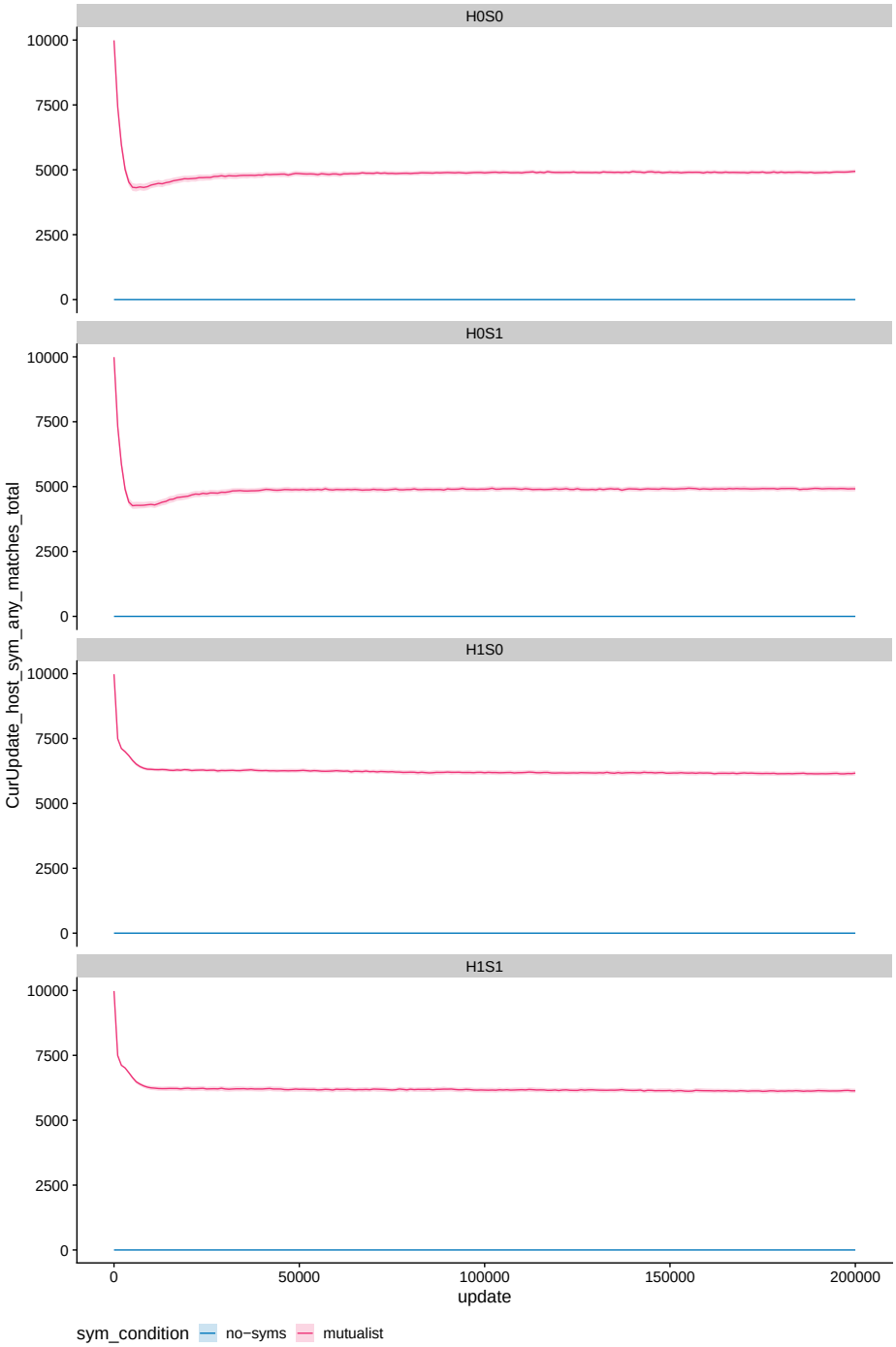
```
sym_final_entopy_task_sets_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_parent_entropy_task_sets
    )
  ) +
  geom_point(
    mapping=aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  facet_wrap(~task_credit, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_final_entopy_task_sets_elapsed_plt.pdf", sep = "/"),
  plot = sym_final_entopy_task_sets_elapsed_plt,
  width = 10,
  height = 10
)
```

10.8 Trait matches over time

Counts of host-symbiont pairs with any trait matches. Note that organisms that were just born will not have had time to express any Boolean logic functions, so they will appear as not matching.

```
host_sym_any_matches_total_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
```

```
  ggplot(
    aes(
      x = update,
      y = CurUpdate_host_sym_any_matches_total,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(plot_dir, "host_sym_any_matches_total_plt.pdf", sep = "/"),
  plot = host_sym_any_matches_total_plt,
  width = 8,
  height = 8
)
host_sym_any_matches_total_plt
```


H0S0
H0S1
H1S0
H1S1
10000
7500
5000
2500
0
CurUpdate_host_sym_any_matches_total
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

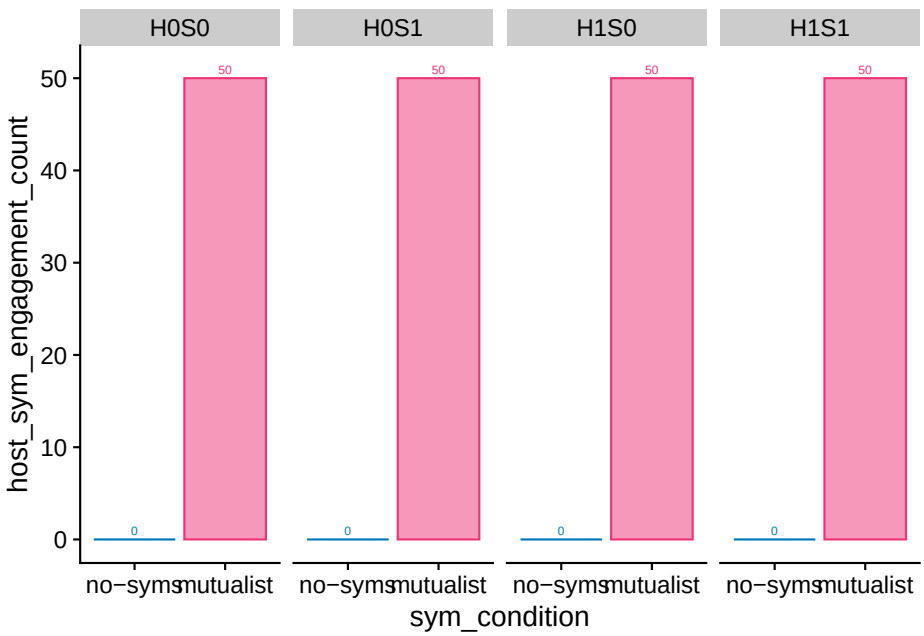
10.9 Evolutionary engagement

Use trait matching metrics to determine if host-symbiont populations are evolutionarily engaged.

```
engagement <- summary_data %>%
  mutate(
    host_sym_engagement = CurUpdate_host_sym_any_matches_total > threshold,
  ) %>%
  group_by(
    sym_condition,
    task_credit
  ) %>%
  summarise(
    host_sym_engagement_count = sum(host_sym_engagement)
  )
```

```
host_sym_engagement_count_plt <- engagement %>%
  ggplot(
    aes(
      x = sym_condition,
      color = sym_condition,
      fill = sym_condition,
      y = host_sym_engagement_count
    )
  ) +
  geom_col(
    position = "dodge",
    alpha = 0.5
  ) +
  geom_text(
    aes(label = host_sym_engagement_count),
    position = position_dodge(width = 1),
    vjust = -0.5,
    size = 2
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  scale_x_discrete() +
  scale_y_continuous(
    limits = c(0, 51)
  ) +
  facet_wrap(~task_credit, nrow = 1) +
  theme(
    legend.position = "none"
```

```
  )

ggsave(
  file = paste(plot_dir, "host_sym_engagement_count_plt.pdf", sep = "/"),
  plot = host_sym_engagement_count_plt,
  width = 4,
  height = 4
)
host_sym_engagement_count_plt
```

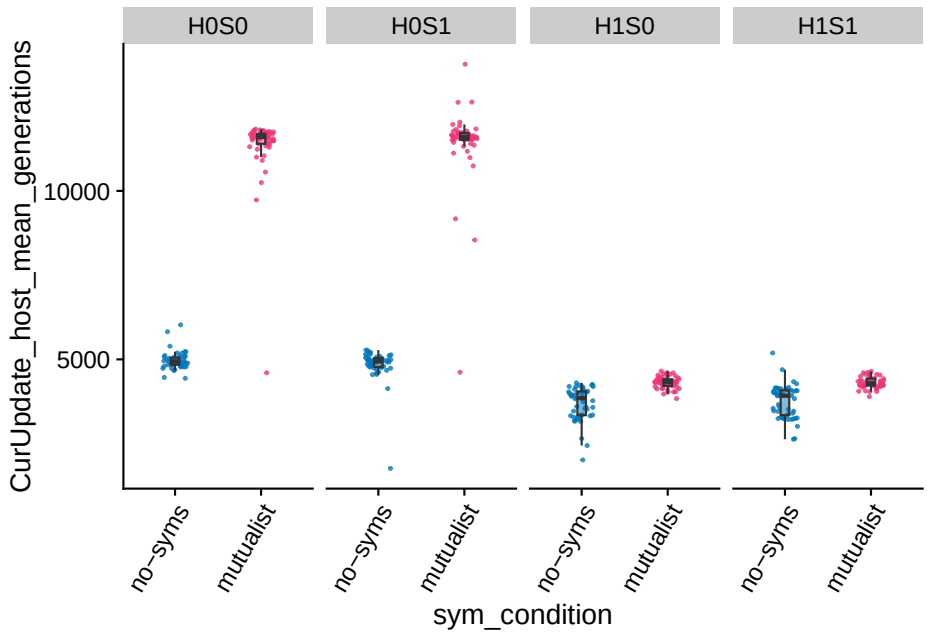


10.10 Generations of evolution elapsed

10.10.1 Hosts

```
host_avg_gens_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_host_mean_generations
    )
```

```
  ) +
  geom_point(
    mapping = aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "host_avg_gens_elapsed_plt.pdf", sep = "/"),
  plot = host_avg_gens_elapsed_plt,
  width = 4,
  height = 4
)
host_avg_gens_elapsed_plt
```

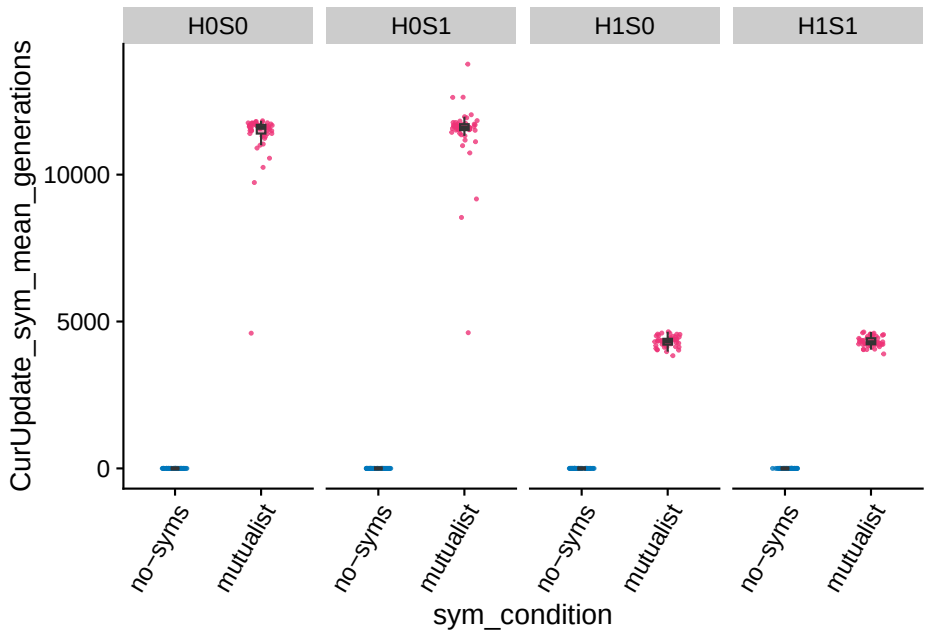



```
summary_data %>%
  group_by(
    sym_condition,
    task_credit
  ) %>%
  summarise(
    mean_gens = mean(CurUpdate_host_mean_generations),
    median_gens = median(CurUpdate_host_mean_generations),
    var_gens = var(CurUpdate_host_mean_generations),
    sdev_gens = sd(CurUpdate_host_mean_generations)
  )
```

```
## # A tibble: 8 x 6
## # Groups:   sym_condition [2]
##   sym_condition task_credit mean_gens median_gens var_gens sdev_gens
##   <fct>         <fct>           <dbl>       <dbl>    <dbl>     <dbl>
## 1 no-syms       H0S0            4969.       4936.   70763.      266.
## 2 no-syms       H0S1            4850.       4942.  242065.      492.
## 3 no-syms       H1S0            3696.       3850.  244382.      494.
## 4 no-syms       H1S1            3756.       3921.  242352.      492.
## 5 mutualist     H0S0           11343.      11598. 1105918.     1052.
## 6 mutualist     H0S1           11417.      11602. 1460871.     1209.
## 7 mutualist     H1S0            4315.       4326.   31537.      178.
## 8 mutualist     H1S1            4315.       4292.   28352.      168.
```

10.10.2 Symbionts

```
sym_avg_gens_elapsed_plt <- summary_data %>%
  ggplot(
    aes(
      x = sym_condition,
      fill = sym_condition,
      y = CurUpdate_sym_mean_generations
    )
  ) +
  geom_point(
    mapping = aes(color = sym_condition),
    position = position_jitter(width = .15),
    size = .5,
    alpha = 0.8
  ) +
  geom_boxplot(
    width = .1,
    outlier.shape = NA,
    alpha = 0.5
  ) +
  scale_x_discrete() +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, nrow = 1) +
  theme(
    legend.position = "none",
    axis.text.x = element_text(
      angle = 60,
      hjust = 1
    )
  )
ggsave(
  file = paste(plot_dir, "sym_avg_gens_elapsed_plt.pdf", sep = "/"),
  plot = sym_avg_gens_elapsed_plt,
  width = 4,
  height = 4
)
sym_avg_gens_elapsed_plt
```



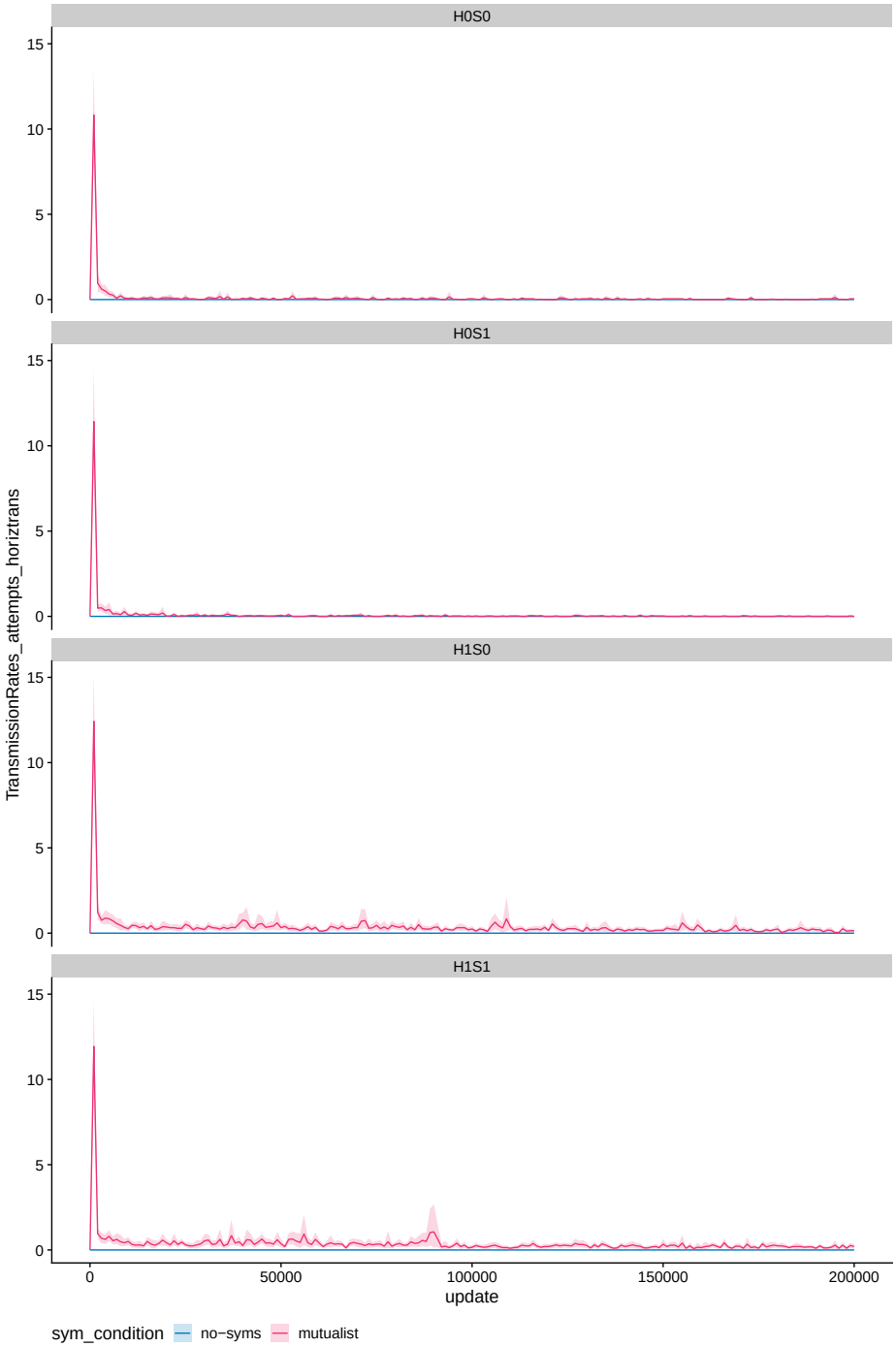

```
summary_data %>%
  group_by(
    sym_condition,
    task_credit
  ) %>%
  summarise(
    mean_gens = mean(CurUpdate_sym_mean_generations),
    median_gens = median(CurUpdate_sym_mean_generations),
    var_gens = var(CurUpdate_sym_mean_generations),
    sdev_gens = sd(CurUpdate_sym_mean_generations)
  )
```

```
## # A tibble: 8 x 6
## # Groups:   sym_condition [2]
##   sym_condition task_credit mean_gens median_gens var_gens sdev_gens
##   <fct>         <fct>           <dbl>       <dbl>    <dbl>     <dbl>
## 1 no-syms       H0S0                0           0        0         0
## 2 no-syms       H0S1                0           0        0         0
## 3 no-syms       H1S0                0           0        0         0
## 4 no-syms       H1S1                0           0        0         0
## 5 mutualist     H0S0            11343.      11598. 1105738.     1052.
## 6 mutualist     H0S1            11417.      11602. 1460849.     1209.
## 7 mutualist     H1S0             4315.       4326.   31577.      178.
## 8 mutualist     H1S1             4315.       4292.   28327.      168.
```

10.11 Symbiont transmission rates

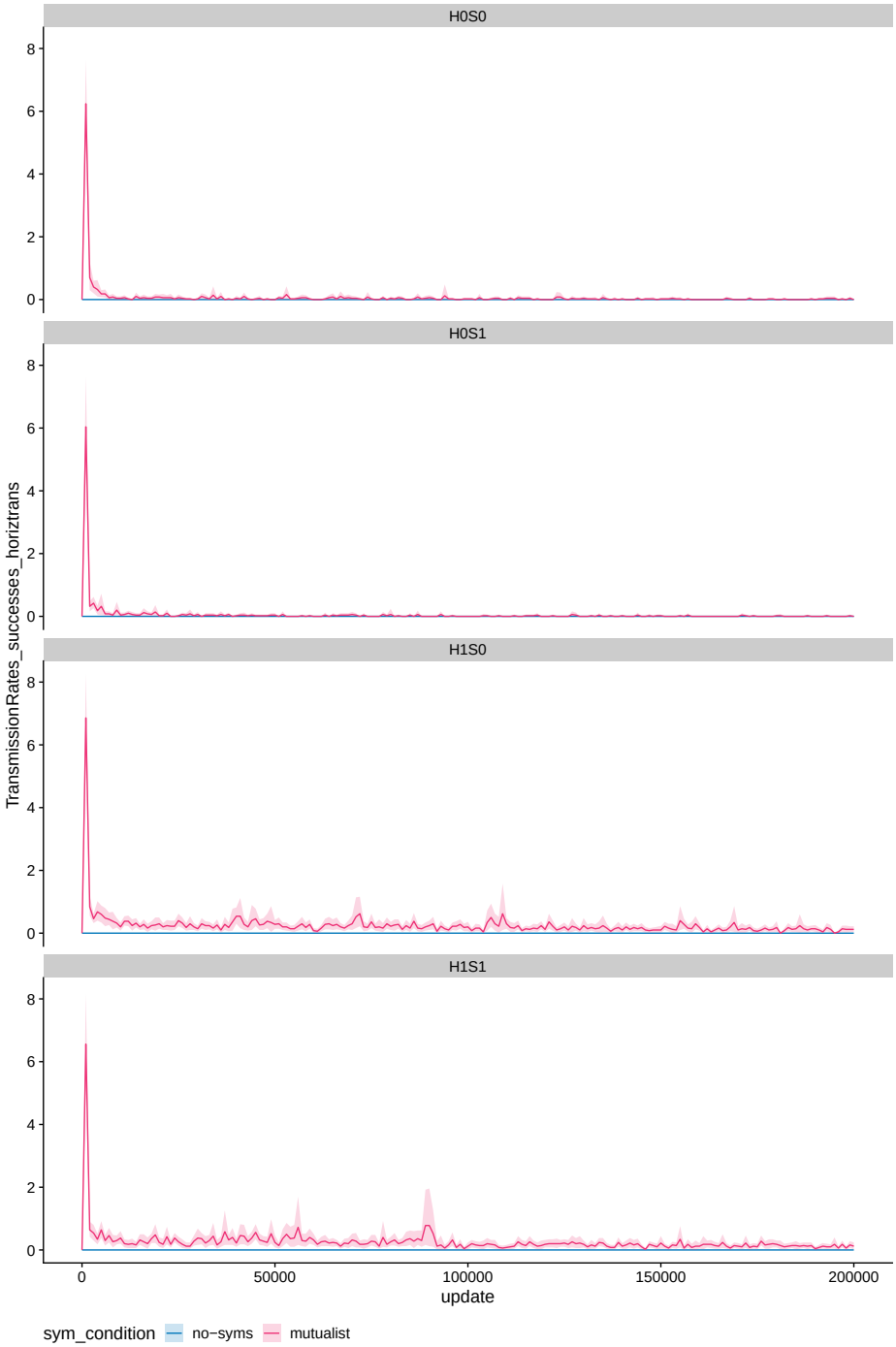
10.11.1 Horizontal transmission attempts

```
TransmissionRates_attempts_horiztrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_attempts_horiztrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_attempts_horiztrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_attempts_horiztrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_attempts_horiztrans_plt
```


H0S0
H0S1
H1S0
H1S1
TransmissionRates_attempts_horiztrans
15
10
5
0
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

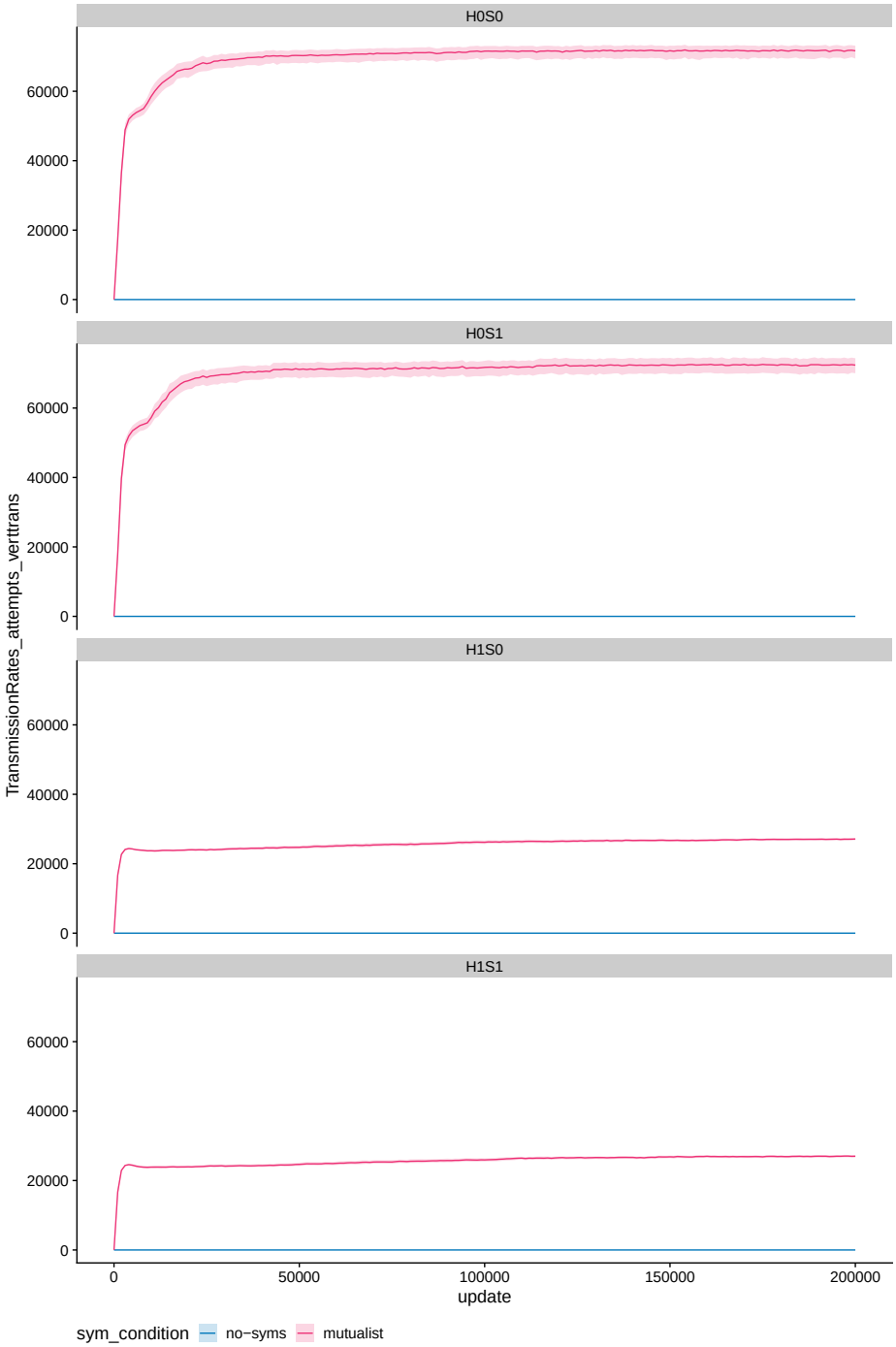
10.11.2 Horizontal transmission successes

```
TransmissionRates_successes_horiztrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_successes_horiztrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_successes_horiztrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_successes_horiztrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_successes_horiztrans_plt
```


H0S0
H0S1
H1S0
H1S1
TransmissionRates_successes_horiztrans
update
0
50000
100000
150000
200000
sym_condition
no−syms
mutualist

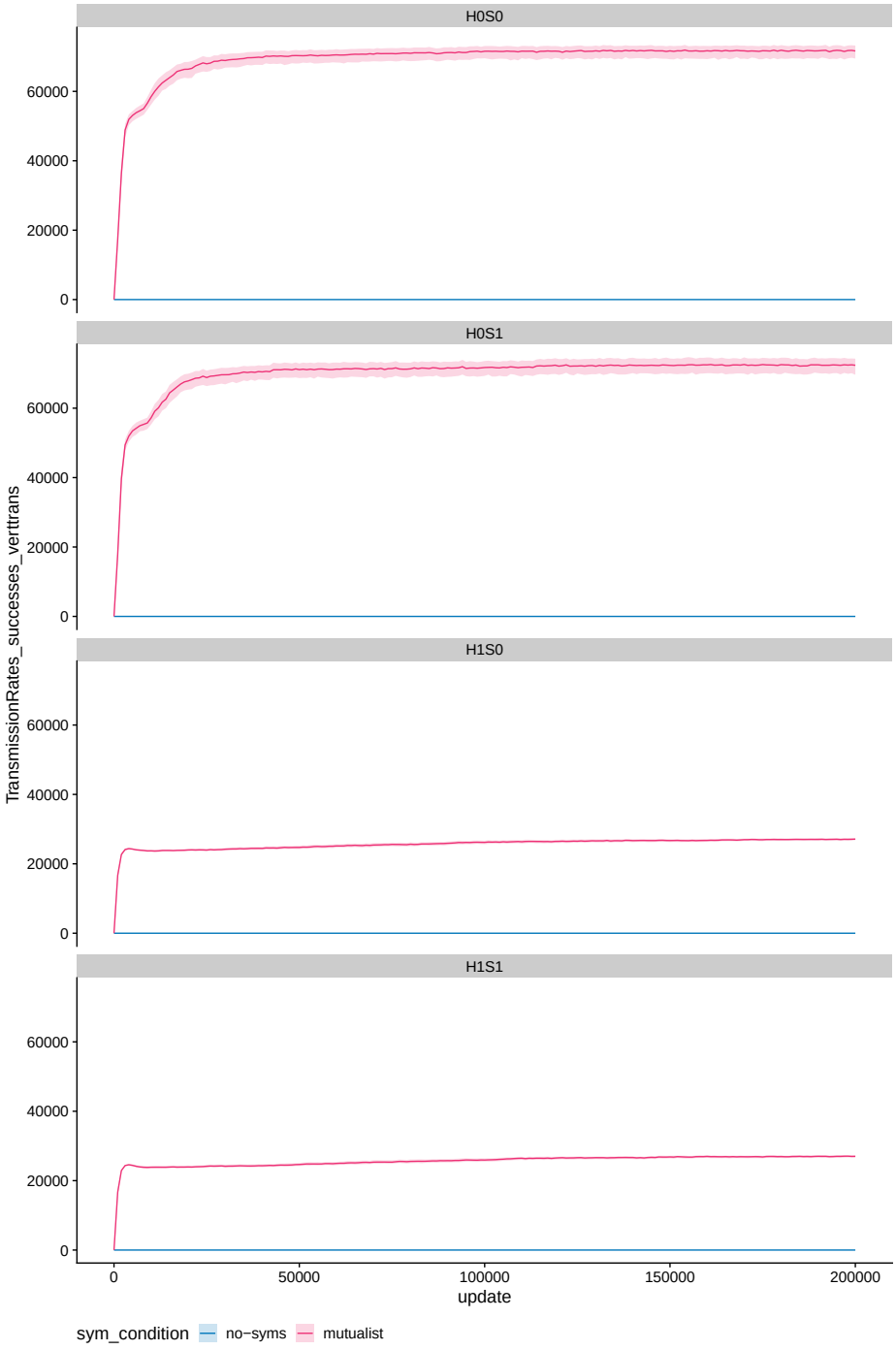
10.11.3 Vertical transmission attempts

```
TransmissionRates_attempts_verttrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_attempts_verttrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_attempts_verttrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_attempts_verttrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_attempts_verttrans_plt
```


H0S0
H0S1
H1S0
H1S1
TransmissionRates_attempts_verttrans
60000
40000
20000
0
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

10.11.4 Successes vertical transmission

```
TransmissionRates_successes_verttrans_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = TransmissionRates_successes_verttrans,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "TransmissionRates_successes_verttrans_plt.pdf",
    sep = "/"
  ),
  plot = TransmissionRates_successes_verttrans_plt,
  width = 8,
  height = 8
)
TransmissionRates_successes_verttrans_plt
```


H0S0
H0S1
H1S0
H1S1
TransmissionRates_successes_vertrans
0
20000
40000
60000
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

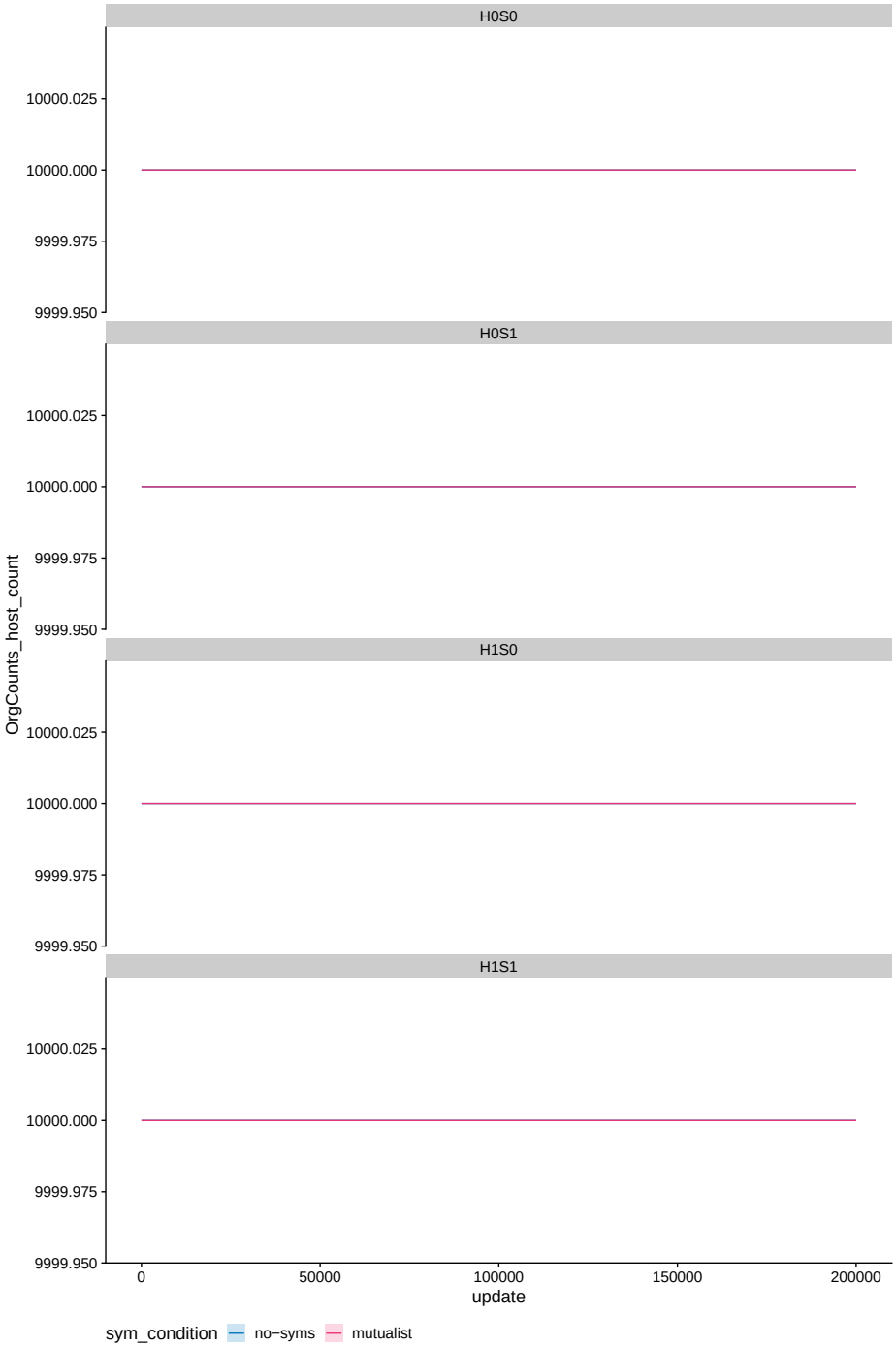
10.12 Population size

```
pop_size_summary <- time_data %>%
  group_by(
    SEED,
    sym_condition,
    task_credit
  ) %>%
  summarize(
    host_pop_size_avg = mean(OrgCounts_host_count),
    host_pop_size_median = median(OrgCounts_host_count),
    host_pop_size_variance = var(OrgCounts_host_count),
    sym_pop_size_avg = mean(OrgCounts_hosted_sym_count),
    sym_pop_size_median = median(OrgCounts_hosted_sym_count),
    sym_pop_size_variance = var(OrgCounts_hosted_sym_count)
  )
```

10.12.1 Host population size

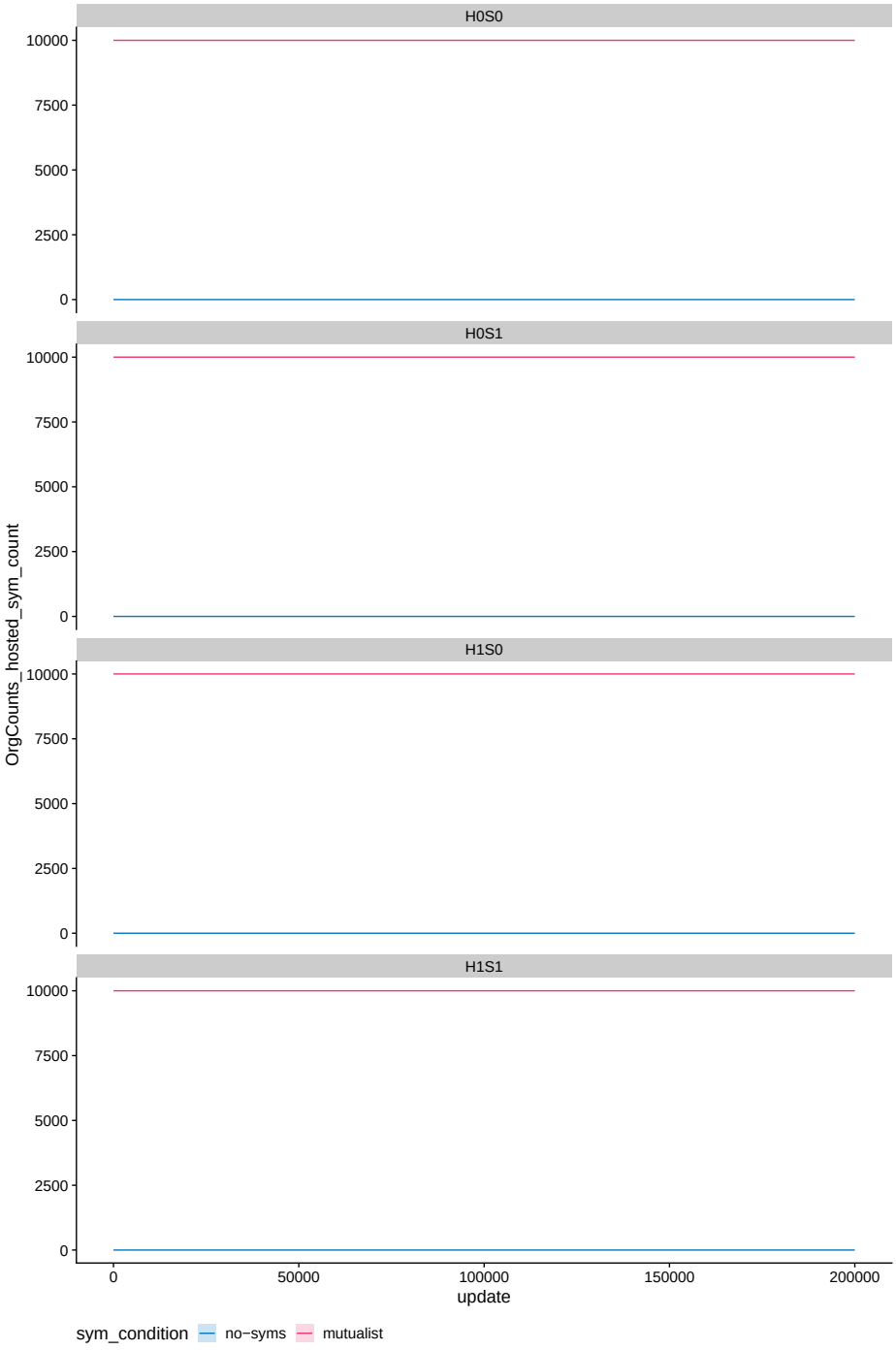
```
host_pop_size_ot_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = OrgCounts_host_count,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, ncol = 1) +
  theme(
    legend.position = "bottom"
```

```
  )
ggsave(
  file = paste(
    plot_dir,
    "host_pop_size_ot_plt.pdf",
    sep = "/"
  ),
  plot = host_pop_size_ot_plt,
  width = 8,
  height = 8
)
host_pop_size_ot_plt
```


H0S0
H0S1
H1S0
H1S1
10000.025
10000.000
9999.975
9999.950
OrgCounts_host_count
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist

10.12.2 Symbiont population size

```
sym_pop_size_ot_plt <- time_data %>%
  filter(update %% 1000 == 0) %>%
  ggplot(
    aes(
      x = update,
      y = OrgCounts_hosted_sym_count,
      color = sym_condition,
      fill = sym_condition
    )
  ) +
  stat_summary(fun = "mean", geom = "line") +
  stat_summary(
    fun.data = "mean_cl_boot",
    fun.args = list(conf.int = 0.95),
    geom = "ribbon",
    alpha = 0.2,
    linetype = 0
  ) +
  scale_color_manual(values = color_pal) +
  scale_fill_manual(values = color_pal) +
  facet_wrap(~task_credit, ncol = 1) +
  theme(
    legend.position = "bottom"
  )
ggsave(
  file = paste(
    plot_dir,
    "sym_pop_size_ot_plt.pdf",
    sep = "/"
  ),
  plot = sym_pop_size_ot_plt,
  width = 8,
  height = 8
)
sym_pop_size_ot_plt
```


H0S0
H0S1
H1S0
H1S1
10000
7500
5000
2500
0
OrgCounts_hosted_sym_count
0
50000
100000
150000
200000
update
sym_condition
no−syms
mutualist